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[illegible]
Rubin [illegible] 1974 [illegible] Potential Outcomes Framework [illegible]
[illegible]
[illegible]
[illegible] Peng Ding [illegible] A First Course in Causal Inference [illegible]

- [illegible]
- Fisher [illegible] Neyman [illegible]
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- [illegible]
- [illegible]

[illegible]

[illegible]

(1) [illegible] **K**Peng Ding [illegible]
(2) [illegible] **K**`https://arxiv.org/abs/2305.18793`
(3) [illegible] **K**2023
(4) [illegible] **K**490
(5) [illegible] **K**
   - [illegible] Fisher [illegible] Neyman [illegible]
   - [illegible] CRE [illegible]
   - [illegible]
   - [illegible]

(6) [illegible] **K**This book provides an introduction to causal inference for students and researchers interested in estimating treatment effects from randomized experiments and observational studies. It unifies the potential outcomes framework with a focus on randomization-based inference, covering topics from basic concepts like the Yule-Simpson paradox to advanced methods including propensity score matching, instrumental variables, and sensitivity analysis. The book emphasizes both finite-sample exact inference (Fisher's approach) and asymptotic inference (Neyman's approach), providing readers with a comprehensive toolkit for causal analysis.

(7) [illegible] Causal Inference§ Potential Outcomes§ Randomization Test§ Neyman Inference§ Observational Studies§ Propensity Score

[illegible]

- [illegible] *Causal Inference for Statistics, Social, and Biomedical Sciences.* Cambridge University Press. [illegible]
- [illegible] "Estimating Causal Effects of Treatments in Randomized and Non-Randomized Studies." *Journal of Educational Psychology.* [illegible]
- [illegible] "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika.* [illegible]
- [illegible] *Mastering 'Metrics.* Princeton University Press. [illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible]

- Markov [illegible] Chebyshev [illegible]
- Hoeffding [illegible] Chernoff [illegible]
- CLT [illegible]

[illegible]

(1) [illegible] 1-2 [illegible]
(2) [illegible] 3-4 [illegible] Fisher [illegible] Neyman [illegible]
(3) [illegible] 5 [illegible]
(4) [illegible] 6-8 [illegible]
(5) [illegible] 9 [illegible]
(6) [illegible] 10-18 [illegible]

[illegible]

[illegible]

- [illegible] [illegible] Fisher [illegible] Neyman [illegible]
- [illegible] [illegible]
- [illegible] [illegible]
- [illegible] [illegible]

CHAPTER 1

[illegible]

[illegible]

[illegible]

[illegible]

- ” [illegible] ”
- ” [illegible] ”
- ” [illegible] ”

[illegible] correlation [illegible]

[illegible]

[illegible]

[illegible] 1.1. 一家公司想要知道某种新药是否有效。他们给一组患者服用新药，另一组服用安慰剂。结果显示服药组的康复率更高。这能否说明药物有效？

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible] causal inference [illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible] 相关性可以是因果的，也可以是非因果的。

[illegible]

[illegible] Z [illegible] Y [illegible]

	$Y = 1$	$Y = 0$
$Z = 1$	p_{11}	p_{10}
$Z = 0$	p_{01}	p_{00}

[illegible]

$$\mathrm{RD} = \mathbb{P}(Y = 1 \mid Z = 1) - \mathbb{P}(Y = 1 \mid Z = 0) = \frac{p_{11}}{p_{11} + p_{10}} - \frac{p_{01}}{p_{01} + p_{00}}$$

[illegible]

$$\mathrm{RR} = \frac{\mathbb{P}(Y = 1 \mid Z = 1)}{\mathbb{P}(Y = 1 \mid Z = 0)} = \frac{p_{11}}{p_{11} + p_{10}} \Big/ \frac{p_{01}}{p_{01} + p_{00}}$$

[illegible]

$$\mathrm{OR} = \frac{\mathbb{P}(Y = 1 \mid Z = 1)/\mathbb{P}(Y = 0 \mid Z = 1)}{\mathbb{P}(Y = 1 \mid Z = 0)/\mathbb{P}(Y = 0 \mid Z = 0)} = \frac{p_{11}p_{00}}{p_{10}p_{01}}$$

[illegible] $\perp\!\!\!\perp$ [illegible]

[illegible] (1) *以下命题等价：$Z \perp\!\!\!\perp Y$、$RD = 0$、$RR = 1$、$OR = 1$。*

(2) *若所有 $p_{zy} > 0$，则 $RD > 0$ 等价于 $RR > 1$，也等价于 $OR > 1$。*

(3) *若 $\mathbb{P}(Y = 1 \mid Z = 1)$ 和 $\mathbb{P}(Y = 1 \mid Z = 0)$ 都很小，则 $OR \approx RR$。*

[illegible]

[illegible]

[illegible] 1.2 ([illegible] Yule-Simpson [illegible]). 考虑一个大学录取数据：

院系	男生录取率	女生录取率
A	*62% (341/550)*	*82% (137/167)*
B	*63% (28/44)*	*68% (24/35)*
	45% (369/816)	***30% (161/525)***

有趣的现象：

- 在 *A* 院系，男女生录取率分别是 *62%* 和 *82%*
- 在 *B* 院系，男女生录取率分别是 *63%* 和 *68%*
- 但总体来看，男生的录取率反而更高！

这就是 *Yule-Simpson* 悖论：[illegible]。

[illegible] 如果我们不控制混杂因素（*confounding factors*），就可能得出完全错误的因果结论

[illegible]

[illegible]

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{XY} & \rho_{XZ} \\ \rho_{XY} & 1 & \rho_{YZ} \\ \rho_{XZ} & \rho_{YZ} & 1 \end{pmatrix}\right)$$

Y [illegible] Z [illegible] $\rho_{YZ|X}$ [illegible] $(Y, Z) \mid X$ [illegible]

[illegible]

$$\rho_{YZ|X} = \frac{\rho_{YZ} - \rho_{YX}\rho_{ZX}}{\sqrt{1 - \rho_{YX}^2}\sqrt{1 - \rho_{ZX}^2}}$$

[illegible] $\rho_{YZ} > 0$ [illegible] $\rho_{YZ|X}$ [illegible] Yule-Simpson [illegible]

[illegible]

LaLonde [illegible] National Supported Work Demonstration [illegible] $2^{10} = 1024$ [illegible] treatment [illegible]

[illegible]

Bertrand and Mullainathan (2004) [illegible]

[illegible] ” [illegible] ” [illegible]

[illegible]

[illegible]

(1) [illegible] $\neq$ [illegible]
(2) [illegible] RD、RR、OR [illegible]
(3) 《 [illegible]
(4) [illegible]
(5) [illegible]

[illegible]

[illegible]

[illegible] 1.1. { {独立性命题的证明 证明 *Proposition* { {中的命题 *(1)* 和 *(2)*。

[illegible] 1.2. { {*Yule-Simpson* 悖论的更多例子

(1) 给出一个 $2 \times 2 \times 2$ 列联表的数值例子，其中出现 *Yule-Simpson* 悖论。
(2) 找一个现实生活中的 *Yule-Simpson* 悖论例子。

[illegible] 1.3. { {相关与偏相关 考虑三维正态随机向量 $(X, Y, Z)^\top \sim N(\mathbf{0}, \Sigma)$。
证明偏相关系数公式（见 *Proposition* { }{。
给出一个 $\rho_{YZ} > 0$ 但 $\rho_{YZ|X} < 0$ 的数值例子。
：这是正态随机向量情形下的 *Yule-Simpson* 悖论。

[illegible] 1.4. { {规格搜索 重做 *Hainmueller (2012)* 的数据分析。使用结局 ***re78*** 和治疗 ***treat***。数据集还包含 *10* 个协变量，共有 $2^{10} = 1024$ 种可能的协变量子集组合。运行所有可能的线性回归子集，并报告 *treatment* 系数的分布：有多少是正显著、负显著、不显著的？

[illegible] 1.5. { {种族歧视的进一步分析 重做 *Bertrand and Mullainathan (2004)* 的简历数据分析。分别对男性和女性进行分析，讨论亚组分析的结果与总体结论的差异。

ΰ́ 1.6. { {推荐阅读 ┤ḓ : { 是关于 *Berkeley* 录取数据悖论的原始论文。{ 从因果推断角度重新审视了该研究。

α̈ɥ e : { 是统计因果推断领域的经典综述文章，它推广了 ”*Rubin* 因果模型 ” 这一名称。在 *UC Berkeley*，我们称之为 ”*Neyman* 模型 ”——原因显而易见。

CHAPTER 2

[illegible]

[illegible]

[illegible] Yule-Simpson [illegible]

[illegible]

[illegible]

[illegible] Neyman [illegible] 1923 [illegible] Rubin [illegible] 1970 [illegible] counterfactual [illegible]

[illegible]

[illegible] i [illegible]

- $T_i = 1$ [illegible]
- $T_i = 0$ [illegible]

[illegible]

- $Y_i(1)$ [illegible]
- $Y_i(0)$ [illegible]

[illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible]

2.1 (2.1 [illegible]). [illegible] (*counterfactual*) 指的是未被观测到的潜在结果。具体而言：

- 对于治疗组个体 i（$T_i = 1$），我们观察到 $Y_i(1)$，而 $Y_i(0)$ 是反事实
- 对于对照组个体 i（$T_i = 0$），我们观察到 $Y_i(0)$，而 $Y_i(1)$ 是反事实

正因为潜在结果框架讨论的是“如果接受相反治疗会怎样”，所以它也被称为[illegible] (*counterfactual framework*)。

2.2. [illegible] “[illegible]” [illegible]**：**因为它们描述的是“如果... 会怎样”的情形，这种结果是“潜在的”而非“现实的”。这正是 *counterfactual*（反事实）思维的本质。

[illegible]

[illegible] $n \times 2$ [illegible]

i	$Y_i(1)$	$Y_i(0)$
1	$Y_1(1)$	$Y_1(0)$
2	$Y_2(1)$	$Y_2(0)$
$\vdots$	$\vdots$	$\vdots$
n	$Y_n(1)$	$Y_n(0)$

[illegible]

[illegible]

[illegible] i [illegible]

$$\tau_i = Y_i(1) - Y_i(0)$$

[illegible] “[illegible]” [illegible] 1923 [illegible] Jerzy Neyman [illegible]

对于每个个体，我们只能观察到一种潜在结果——实际发生的那种。另一种潜在结果是潜在的、永不可见的。

[illegible]

2.3. [illegible]：因为科学上感兴趣的大多数因果问题，都涉及将现实与反事实进行比较。我们可以问“这种药对这个患者有效吗？”——但这个问题要求我们同时知道患者吃药和不吃药的结果，而这是不可能的。

[illegible]

[illegible]

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \bar{Y}(1) - \bar{Y}(0) \tag{2.1}$$

[illegible][1]

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\} = \frac{1}{n}\sum_{i=1}^{n} Y_i(1) - \frac{1}{n}\sum_{i=1}^{n} Y_i(0) \tag{2.2}$$

[illegible]

$$\tau = \bar{Y}(1) - \bar{Y}(0) = \frac{1}{n}\sum_{i=1}^{n}\tau_i \tag{2.3}$$

---

[1] [illegible] **??**

[illegible]

[illegible] X_i [illegible]

- [illegible] $X_i = 1$ [illegible] $X_i = 0$ [illegible]
- [illegible] / [illegible]
- [illegible] / [illegible]

[illegible][2]

$$\tau_x = \frac{\sum_{i=1}^{n} \mathbb{I}(X_i = x)\{Y_i(1) - Y_i(0)\}}{\sum_{i=1}^{n} \mathbb{I}(X_i = x)}, \quad (x = 0, 1) \tag{2.4}$$

[illegible]

[illegible][3]

$$\tau = \pi_1\tau_1 + \pi_0\tau_0 \tag{2.5}$$

[illegible] $\pi_x = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(X_i = x)$ [illegible] $X_i = x$ [illegible] $\tau_1 > 0$ [illegible] $\tau_0 > 0$ [illegible] $\tau > 0$ [illegible]

[illegible]

[illegible]

[illegible] T_i [illegible] i [illegible] Y_i [illegible]

$$\begin{aligned}
Y_i &= \begin{cases} Y_i(1), & \text{if } T_i = 1 \\ Y_i(0), & \text{if } T_i = 0 \end{cases} && (1)\\
&= T_i Y_i(1) + (1 - T_i) Y_i(0) && (2)\\
&= Y_i(0) + T_i\{Y_i(1) - Y_i(0)\} && (3)\\
&= Y_i(0) + T_i\tau_i && (4)
\end{aligned}$$

[illegible] $\tau_i = Y_i(1) - Y_i(0)$ [illegible]

[illegible]

[illegible]

[illegible] 2.1 ([illegible]). 单元 i 的潜在结果不依赖于其他单元的治疗分配。这有时被称为[illegible]假设。

[illegible] 2.2 ( ). 治疗水平没有其他版本。等价地，我们要求治疗水平是良好定义的，至少对关注的结局是如此。这有时被称为 假设。

[illegible] 2.3 (SUTVA). 假设 [illegible]和 [illegible]同时成立。

---

[2] [illegible] ??

[3] [illegible] ??

¹I¹ alho ÂN×1$ ÂT†í-Wi Ä €Í çl,ü' *¹.
²T $Y_i(1)$ „ $Y_i(0)$ ü,^ r2HH@/¥}iMñ2ì¥´||%¿ ð¦¥›¹ Í ¥
8™T œñ•Oüö)ïb

´Q alho ¥è0

íÄ L! MQ@BMi2`72`2M+2 ´bQ¥mf™KTiBQM

UR.V™hÐfì¤Õ BM72+iBQmb /Bb2 b2b M/'%þh++µBM™ŸiBQM
H Bñ¦^Ž¤Õfì,Ç•Yð1-¥Ñh„B Î•YðÛ¶¦$Ë™¥
ÀqbèÂ ÂTá¥]Y¤Õ @Ëfì áÑ@Ë¥Àq9ö†®ii'P
á1-Àµ¤Õb ƒÕfƒ/ †í i ¥. ²T $Y_i(1)$ „ $Y_i(0)$ ûG ¿ ð
¦¥Í s¥b

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2z2+ib

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óÂTƒñÑ€ù 0 ²Tö8"$ŒCL^ ðV?Àµù0 XXXô

ƒÕ±»ZTV?ž¦$¤ `.À}ÐÄ Œñ^áÌ ,yT¥·BøÈZTb.
²T•O¥N´ ¿ ñ|ƒÕ}Ðó³ Ä¹b ¥"ÐÔýb

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\delta_1 = \text{median}\{Y_i(1)\}_{i=1}^n - \text{median}\{Y_i(0)\}_{i=1}^n \tag{2.6}$$

[illegible]

$$\delta_2 = \text{median}\{Y_i(1) - Y_i(0)\}_{i=1}^n \tag{2.7}$$

[illegible][4]

[illegible]

[illegible]

(1) [illegible] $Y_i(1)$ [illegible] $Y_i(0)$
(2) [illegible] $n \times 2$ [illegible] $\{Y_i(1), Y_i(0)\}_{i=1}^n$
(3) [illegible] $\tau_i = Y_i(1) - Y_i(0)$
(4) [illegible] $\tau = \bar{Y}(1) - \bar{Y}(0) = n^{-1}\sum_{i=1}^n \tau_i$
(5) [illegible]
(6) [illegible] $Y_i = T_iY_i(1) + (1 - T_i)Y_i(0)$
(7) [illegible] + [illegible]
(8) [illegible]

[illegible]

[illegible]

[illegible] 2.1. [illegible] 完美的医生 假设潜在结果由以下模型生成：

$$Y_i(0) \sim N(0,1), \quad \tau_i = -0.5 + Y_i(0), \quad Y_i(1) = Y_i(0) + \tau_i$$

治疗分配为 $T_i = \mathbb{I}(\tau_i \geq 0)$，观测结果为 $Y_i = T_iY_i(1) + (1 - T_i)Y_i(0)$。

计算条件均值差 $\mathbb{E}(Y_i \mid T_i = 1) - \mathbb{E}(Y_i \mid T_i = 0)$。

[illegible]：需要使用截断正态分布的均值公式。当 $X \sim N(\mu, \sigma^2)$ 时，

$$\mathbb{E}(X \mid a < X < b) = \mu - \sigma\frac{\phi\left(\frac{b-\mu}{\sigma}\right) - \phi\left(\frac{a-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)}$$

---

[4] [illegible] ??

其中 ϕ 和 Φ 分别是标准正态分布的密度函数和分布函数。

ΰ 2.2. { {非线性因果参数 证明平均因果效应是线性因果参数的例子，即 $\bar{Y}(1) - \bar{Y}(0) = n^{-1}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\}$。

其他参数可能不是线性的。例如，定义中位数治疗效应为：

$$\delta_1 = median\{Y_i(1)\} - median\{Y_i(0)\}$$

这通常不同于个体治疗效应的中位数：

$$\delta_2 = median\{Y_i(1) - Y_i(0)\}$$

(1) 给出数值例子分别满足 $\delta_1 = \delta_2$、$\delta_1 > \delta_2$ 和 $\delta_1 < \delta_2$。
(2) 哪个参数更有意义？为什么？请用例子来证明你的结论。

ΰ 2.3. { {平均效应与个体效应 给出一个数值例子，其中 $\tau = n^{-1}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\} > 0$，但有不到一半的单元满足 $Y_i(1) > Y_i(0)$。即平均因果效应为正，但治疗对不到一半的单元有益。

ΰ 2.4. { {推荐阅读 ⊣ḓ : { 是统计因果推断领域的经典综述文章，它推广了 *“Rubin* 因果模型*”* 这一名称。在 *UC Berkeley*，我们称之为 *“Neyman* 模型*”*——原因显而易见。

CHAPTER 3

[illegible]

[illegible]

[illegible]

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \bar{Y}(1) - \bar{Y}(0) \tag{5}$$

[illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible][1] [illegible]

[illegible] X [illegible] Y [illegible] X [illegible] Y [illegible]

[illegible]

Ronald Fisher 1935 [illegible][3] Design of Experiments [illegible]

(1) [illegible] treatment [illegible] control

(2) [illegible]

[illegible] treatment [illegible] control [illegible]

Fisher [illegible][2,3] [illegible]

[illegible]

[illegible] n [illegible] n_1 [illegible] n_0 [illegible] $n = n_1 + n_0$

3.1 ([illegible] Complete Randomized Experiment, CRE). 固定 n_1 和 n_0 (其中 $n = n_1 + n_0$)。完全随机实验的治疗 *assignment* 机制为:

$$\mathbb{P}(\mathbf{Z} = \mathbf{z}) = \frac{1}{\binom{n}{n_1}}, \tag{6}$$

其中 $\mathbf{z} = (z_1, \ldots, z_n)$ 满足 $\sum_{i=1}^{n} z_i = n_1$ 和 $\sum_{i=1}^{n}(1 - z_i) = n_0$。

---

[1] [illegible] **??**

[2] [illegible] Fisher Randomization Test [illegible] **??**

[3] [illegible] Fisher [illegible] **??**

[illegible] $\mathbf{Y}(1)$ [illegible] $\mathbf{Y}(0)$ [illegible]

$$\mathbb{P}\{\mathbf{Z}=\mathbf{z} \mid \mathbf{Y}(1),\mathbf{Y}(0)\} = \frac{1}{\binom{n}{n_1}} \tag{7}$$

CRE $\mathbf{Z} \perp\!\!\!\perp \{\mathbf{Y}(1),\mathbf{Y}(0)\}$ [illegible] CRE [illegible] $\mathbf{Z}$ [illegible] n_1 [illegible] 1 [illegible] n_0 [illegible] 0 [illegible]

[illegible]

[illegible]

$$H_{0F}: Y_i(1) = Y_i(0) \quad [illegible] \quad i = 1,\dots,n. \tag{8}$$

[illegible] $\mathbf{Y}(1) = \mathbf{Y}(0) = \mathbf{Y} = (Y_1,\dots,Y_n)$ [illegible][4] [illegible] H_{0F} [illegible] FRT [illegible]

$$T = T(\mathbf{Z},\mathbf{Y}), \tag{9}$$

[illegible] $\mathbf{Y}$ [illegible] H_{0F} [illegible] T [illegible] $\mathbf{Z}$ [illegible] $\mathbf{Z}$ [illegible] T [illegible] p [illegible] CRE [illegible] $\mathbf{Z}$ [illegible] $\{\mathbf{z}^1,\dots,\mathbf{z}^M\}$ [illegible] $M = \binom{n}{n_1}$ [illegible] $\mathbf{z}^m$ [illegible] n_1 [illegible] 1 [illegible] n_0 [illegible] 0 [illegible][5]

[illegible]

[illegible] T [illegible]

$$p_{\text{frt}} = M^{-1}\sum_{m=1}^{M} \mathbb{I}\{T(\mathbf{z}^m,\mathbf{Y}) \ge T(\mathbf{Z},\mathbf{Y})\}. \tag{10}$$

[illegible] Fisher [illegible] p [illegible] p_{frt} [illegible] H_{0F} [illegible] finite-sample exact[illegible][6]

$$\mathbb{P}(p_{\text{frt}} \le u) \le u \quad \text{for all} \quad 0 \le u \le 1. \tag{11}$$

3.1. [illegible] ” [illegible] ” [illegible]：这意味着 p 值的分布（虽然在离散情况下是阶梯函数）满足上式，因此 p 值在 $[0,1]$ 上均匀分布（对于连续情况）。

---

[4] [illegible] H_{0F} [illegible] $\mathbf{Y}$ [illegible] ??

[5] [illegible] $\{\mathbf{z}^1,\dots,\mathbf{z}^M\}$ [illegible] ??

[6] [illegible] ??

[illegible]

[illegible] M [illegible] $n = 100, n_1 = 50$ [illegible] $M > 10^{29}$ [illegible] Monte Carlo [illegible] p_{frt} [illegible] $\tilde{p}_{\text{frt}}$ [illegible] [7][8]

$$\hat{p}_{\text{frt}} = R^{-1} \sum_{r=1}^{R} \mathbb{I}\{T(\mathbf{z}^r, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}, \tag{12}$$

[illegible] $\mathbf{z}^r$ [illegible] $\mathbf{Z}$ [illegible] R [illegible] p_{frt} [illegible] $\mathbf{Z}$ [illegible] FRT [illegible] CRE [illegible]

$$\tilde{p}_{\text{frt}} = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\{T(\mathbf{z}^r, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}}{1 + R}. \tag{13}$$

[illegible] R [illegible] p

[illegible]

FRT [illegible] p [illegible] H_{0F} [illegible]

[illegible]

$$\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0), \tag{14}$$

[illegible]

$$\hat{\bar{Y}}(1) = n_1^{-1} \sum_{Z_i=1} Y_i = n_1^{-1} \sum_{i=1}^{n} Z_i Y_i \tag{15}$$

[illegible] treatment [illegible]

$$\hat{\bar{Y}}(0) = n_0^{-1} \sum_{Z_i=0} Y_i = n_0^{-1} \sum_{i=1}^{n} (1 - Z_i) Y_i \tag{16}$$

[illegible] control [illegible] H_{0F} [illegible] $\hat{\tau}$ [illegible] 0 [illegible]

$$\text{var}(\hat{\tau}) = \frac{n}{n_1 n_0} s^2, \tag{17}$$

[illegible] $s^2 = (n-1)^{-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$ [illegible] $\bar{Y} = n^{-1} \sum_{i=1}^{n} Y_i$ [9] [illegible] Finite Population Central Limit Theorem [illegible]

$$\frac{\hat{\tau}}{\sqrt{\frac{n}{n_1 n_0} s^2}} \rightarrow \mathcal{N}(0, 1) \tag{18}$$

in distribution.

---

[7] [illegible] Monte Carlo [illegible] **??**

[8] [illegible] $\mathbf{Z}$ [illegible] **??**

[9] [illegible] **??**

[illegible] H_{0F} [illegible] p [illegible] t [illegible]
p
[illegible] **: k o “ :** [illegible][10] [illegible]

$$(n-1)s^2 = \sum_{Z_i=1}\{Y_i - \hat{\bar{Y}}(1)\}^2 + \sum_{Z_i=0}\{Y_i - \hat{\bar{Y}}(0)\}^2 + \frac{n_1 n_0}{n}\hat{\tau}^2. \tag{19}$$

[illegible] ANOVA [illegible] $Y_i - \bar{Y}$ [illegible] $(Y_i - \hat{\bar{Y}}(Z_i))$ [illegible] $(\hat{\bar{Y}}(Z_i) - \bar{Y})$ [illegible] treatment [illegible] control [illegible] $\frac{n_1 n_0}{n}\hat{\tau}^2$ [11]

3.2. [illegible] ***FRT*** [illegible] ***t*** [illegible] **:** 传统 *t* 检验需要正态性假设。但 *FRT* 框架表明，只要随机化分配正确执行，*t* 检验程序在有限总体 + 随机化情境下可以使用——渐近理论只依赖于有限总体 *CLT*，而不依赖于数据的原始分布。

[illegible]

$$t = \frac{\hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)}{\sqrt{\frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}}}, \tag{20}$$

[illegible]

$$\hat{S}^2(1) = (n_1 - 1)^{-1}\sum_{Z_i=1}[Y_i - \hat{\bar{Y}}(1)]^2, \quad \hat{S}^2(0) = (n_0 - 1)^{-1}\sum_{Z_i=0}[Y_i - \hat{\bar{Y}}(0)]^2 \tag{21}$$

[illegible] treatment [illegible] control [illegible] H_{0F} [illegible] CLT [illegible] t [illegible] $t \to \mathcal{N}(0,1)$ [illegible] `t.test(var.equal = FALSE)` [illegible] p [illegible]

3.3. [illegible] **:** 当 *treatment* 组和 *control* 组的方差差异较大时，差值均值的检验可能失效。学生化通过分别估计两组的方差，能够更好地处理异方差情况。

[illegible] Chi-Square Test [illegible]

[illegible]

Insight [illegible] t [illegible]

[illegible] Karl Pearson [illegible] 1900 [illegible]

---

[10] [illegible] ??

[11] ANOVA [illegible] ??

7.3.1. 拟合优度检验. [illegible] k [illegible] $O_1, O_2, \ldots, O_k$ [illegible] n [illegible] H_0 [illegible] i [illegible] $E_i = np_{i0}$ [illegible] p_{i0} [illegible] H_0 [illegible]

[illegible]

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}. \tag{22}$$

[illegible] $(O_i - E_i)^2/E_i$ [illegible] E_i [illegible] 100 [illegible] 100 [illegible]

[illegible] H_0 [illegible] n [illegible]

$$\chi^2 \xrightarrow{d} \chi^2_{k-1}, \tag{23}$$

[illegible] $k-1$ [illegible] $\sum O_i = \sum E_i = n$

Insight [illegible] $k-1$ [illegible] k [illegible] n [illegible] k [illegible] $k-1$ [illegible]

[illegible] 3.1 ([illegible]). [illegible]：投掷一枚骰子 *120* 次，想检验它是否均匀（各面概率均为 1/6）。

[illegible]：

面	*1*	*2*	*3*	*4*	*5*	*6*
观测频数 O_i	*15*	*22*	*18*	*25*	*17*	*23*
期望频数 E_i	*20*	*20*	*20*	*20*	*20*	*20*

[illegible]：

$$\chi^2 = \frac{(15-20)^2}{20} + \frac{(22-20)^2}{20} + \frac{(18-20)^2}{20} + \frac{(25-20)^2}{20} + \frac{(17-20)^2}{20} + \frac{(23-20)^2}{20} = 4.3$$

自由度 $k-1=5$，查表得 $\chi^2_{0.05,5} = 11.07$。

由于 $4.3 < 11.07$，在 $\alpha = 0.05$ 水平下[illegible] 均匀性假设。

R [illegible]：

```
> obs <- c(15, 22, 18, 25, 17, 23)
> chisq.test(obs, p = rep(1/6, 6))

	Chi-squared test for given probabilities

X-squared = 4.3, df = 5, p-value = 0.507
```

p 值为 0.507，远大于 0.05，与手工计算一致。

□

Insight [illegible] $E_i = 20$ [illegible] < 5 [illegible] exact test[illegible]

[illegible] 3.2 ([illegible]). [illegible]：孟德尔杂交黄色圆粒豌豆与绿色皱粒豌豆，预期后代比例为：

$$\text{黄圆} : \text{黄皱} : \text{绿圆} : \text{绿皱} = 9 : 3 : 3 : 1$$

[illegible] (共 *556* 株)：

$$O = (315, 101, 108, 32), \quad E = (556 \times \tfrac{9}{16}, \dots) = (312.75, 104.25, 104.25, 34.75)$$

[illegible]：

$$\chi^2 = \frac{(315-312.75)^2}{312.75} + \frac{(101-104.25)^2}{104.25} + \frac{(108-104.25)^2}{104.25} + \frac{(32-34.75)^2}{34.75} \approx 0.47$$

自由度 $k-1=3$，查表得 $\chi^2_{0.05,3}=7.81$。由于 $0.47<7.81$，[illegible] 孟德尔比例。

R [illegible]：

```
> obs <- c(315, 101, 108, 32)
> prob <- c(9/16, 3/16, 3/16, 1/16)
> chisq.test(obs, p = prob)

    Chi-squared test for given probabilities

X-squared = 0.47, df = 3, p-value = 0.925
```

p 值为 0.925——数据与孟德尔比例高度吻合。

□

7.3.2. 独立性检验. [illegible]

[illegible] $r \times c$ [illegible] X [illegible] Y [illegible] O_{ij} [illegible] (i,j) [illegible] $O_{i\cdot} = \sum_j O_{ij}$ [illegible] i [illegible] $O_{\cdot j} = \sum_i O_{ij}$ [illegible] j [illegible] n

[illegible] $H_0 : X \perp Y$ [illegible]

$$E_{ij} = \frac{O_{i\cdot} \cdot O_{\cdot j}}{n}. \tag{24}$$

[illegible]

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}. \tag{25}$$

[illegible] H_0 [illegible] $(r-1)(c-1)$

Insight [illegible] 2×2 [illegible] 1 [illegible] p [illegible]

[illegible] 2×2 [illegible] R [illegible] `chisq.test` [illegible] Yates [illegible]

$$\chi^2_{\text{Yates}} = \sum \frac{(|O-E| - 0.5)^2}{E}.$$

[illegible] 3.3 (2×2 [illegible]). 考虑二元结局的随机实验，数据汇总为：

	成功 $(Y=1)$	失败 $(Y=0)$	合计
治疗 $(Z=1)$	$n_{11}=30$	$n_{10}=20$	$n_1=50$
对照 $(Z=0)$	$n_{01}=20$	$n_{00}=30$	$n_0=50$

1 :

$$E_{11}=\frac{50\times 50}{100}=25,\quad E_{10}=\frac{50\times 50}{100}=25,\quad E_{01}=\frac{50\times 50}{100}=25,\quad E_{00}=25$$

2 χ^2:

$$\chi^2=\frac{(30-25)^2}{25}+\frac{(20-25)^2}{25}+\frac{(20-25)^2}{25}+\frac{(30-25)^2}{25}=4$$

自由度 $(2-1)(2-1)=1$，临界值 $\chi^2_{0.05,1}=3.84$。由于 $4>3.84$， ——治疗与结局存在关联。

R :

```
> tb <- matrix(c(30, 20, 20, 30), nrow = 2, byrow = TRUE)
> chisq.test(tb)

    Pearson's Chi-squared test with Yates' continuity correction

X-squared = 4, df = 1, p-value = 0.0455
```

注意 *Yates* 修正后的 χ^2 从 4 降到 3.24，p 值 $=0.072$——这说明连续性修正在小样本时影响显著。

□

3.4 (3×2). : 研究吸烟状态与肺癌诊断的关联，数据如下：

	患肺癌	未患肺癌	合计
重度吸烟	*60*	*40*	*100*
轻度吸烟	*20*	*30*	*50*
不吸烟	*10*	*40*	*50*

1 :

$$E_{11}=\frac{100\times 90}{200}=45,\quad E_{12}=\frac{100\times 110}{200}=55$$
$$E_{21}=\frac{50\times 90}{200}=22.5,\quad E_{22}=\frac{50\times 110}{200}=27.5$$
$$E_{31}=\frac{50\times 90}{200}=22.5,\quad E_{32}=\frac{50\times 110}{200}=27.5$$

2 χ^2:

$$\chi^2=\frac{(60-45)^2}{45}+\frac{(40-55)^2}{55}+\frac{(20-22.5)^2}{22.5}+\frac{(30-27.5)^2}{27.5}+\frac{(10-22.5)^2}{22.5}+\frac{(40-27.5)^2}{27.5}\approx 20.4$$

自由度 $(3-1)(2-1)=2$，查表 $\chi^2_{0.05,2}=5.99$。由于 $20.4>5.99$， ——吸烟与肺癌存在显著关联。

R :

```
> tb <- matrix(c(60, 40, 20, 30, 10, 40), nrow = 3, byrow = TRUE)
> rownames(tb) <- c("重度吸烟", "轻度吸烟", "不吸烟")
> colnames(tb) <- c("患肺癌", "未患肺癌")
> chisq.test(tb)

    Pearson's Chi-squared test
```

```
X-squared = 20.4, df = 2, p-value = 3.7e-05
```

p 值极小，结论稳健。

□

[illegible] 3.5 (4×3 [illegible]). [illegible]：调查 *500* 人的教育程度与收入水平：

	低收入	中等收入	高收入	合计
初中及以下	*80*	*50*	*20*	*150*
高中	*60*	*80*	*30*	*170*
本科	*30*	*70*	*40*	*140*
研究生	*10*	*30*	*100*	*140*
合计	*180*	*230*	*190*	*500*

R [illegible]：

```
> edu <- c(80, 50, 20, 60, 80, 30, 30, 70, 40, 10, 30, 100)
> tb <- matrix(edu, nrow = 4, byrow = TRUE)
> rownames(tb) <- c("初中及以下", "高中", "本科", "研究生")
> colnames(tb) <- c("低收入", "中等收入", "高收入")
> chisq.test(tb)

    Pearson's Chi-squared test

X-squared = 186.7, df = 6, p-value < 2.2e-16
```

p 值极小，拒绝独立性——教育程度与收入水平存在强关联。

[illegible]：对于 $r \times c$ 表，还可以计算 *Cramér* V 来衡量关联强度：

$$V = \sqrt{\frac{\chi^2}{n \times \min(r-1, c-1)}}. \tag{26}$$

对于本例，$V = \sqrt{186.7/(500 \times 2)} \approx 0.43$，表示中等偏强的关联。

□

7.3.3. 二元结局下的卡方检验：与 *Fisher* 精确检验的对比. [illegible]

[illegible] $Z \perp Y$ [illegible]

U a。[illegible] **Q** [illegible]

	Chi-square test	Fisher exact test
[illegible]	[illegible]	[illegible]
p	[illegible]	[illegible]
[illegible]	[illegible] > 5	[illegible] < 5

Insight [illegible] Fisher [illegible] p [illegible]
[illegible] Chi-square [illegible]

[illegible] Bertrand & Mullainathan, 2004[illegible]
[illegible]
[illegible]

```
resume = read.csv("resume.csv")
Alltable = table(resume$race, resume$call)
Alltable
```

```
#          0    1
# black 2278  157
# white 2200  235
```

[illegible] 2×2 [illegible]

		[illegible] $(Y=1)$	[illegible] $(Y=0)$	[illegible]
[illegible]	$(Z=1)$	157	2278	2435
[illegible]	$(Z=0)$	235	2200	2435

[illegible]

$$E_{11} = \frac{2435 \times 392}{4870} \approx 195.7, \quad E_{12} = \frac{2435 \times 4478}{4870} \approx 2239.3$$

$$E_{21} = \frac{2435 \times 392}{4870} \approx 196.3, \quad E_{22} = \frac{2435 \times 4478}{4870} \approx 2238.7$$

[illegible] 5[illegible]

```
> chisq.test(Alltable)

	Pearson's Chi-squared test with Yates' continuity correction

X-squared = 17.6, df = 1, p-value = 2.7e-05
```

p [illegible]

[illegible] [illegible] $H_0 : \tau = 0$[illegible] $\widehat{\mathrm{RD}} = \hat{p}_1 - \hat{p}_0$ [illegible]

$$\mathrm{var}(\widehat{\mathrm{RD}}) \approx \frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_0(1-\hat{p}_0)}{n_0}$$

[illegible] χ^2 [illegible] $(\widehat{\mathrm{RD}})^2 / \mathrm{var}(\widehat{\mathrm{RD}})$ [illegible]

[illegible]

(1) [illegible] < 5[illegible] 2×2 [illegible] Fisher [illegible]

(2) [illegible]

(3) [illegible] `chisq.test(tb, simulate.p.value = TRUE)`

Insight [illegible] ” [illegible] ” [illegible] > 5 [illegible] < 5[illegible] p [illegible] Fisher [illegible]

[illegible] [illegible]

[illegible] Wilcoxon [illegible]

$$W = \sum_{i=1}^{n} Z_i R_i,$$

[illegible] R_i [illegible] Y_i [illegible][12][illegible] H_{0F} [illegible] **Z**[illegible] R_i [illegible]

$$\mathbb{E}(W) = \frac{n_1(n+1)}{2}, \quad \mathrm{var}(W) = \frac{n_1 n_0 (n+1)}{12}.$$

[illegible]

---

[12] [illegible] ??

- [illegible]
- [illegible]
- [illegible]

[illegible] KS [illegible]

[illegible]

$$D = \max_y |\hat{F}_1(y) - \hat{F}_0(y)|,$$

[illegible] $\hat{F}_1(y) = n_1^{-1} \sum_{i=1}^n Z_i \mathbb{I}(Y_i \le y)$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible] t [illegible] FRT [illegible] t [illegible] + [illegible] CLT [illegible]

[illegible]

1986 [illegible] Robert LaLonde [illegible] National Supported Work Demonstration [illegible]

[illegible] R [illegible] FRT [illegible]

```
library(Matching)
data(lalonde)
z <- lalonde$treat
y <- lalonde$re78
```

[illegible]

[illegible]	[illegible]	[illegible]
$\hat{\tau}$	2.84	[illegible] t [illegible]
t	2.67	[illegible]
W	27402.5	Wilcoxon [illegible]
D	0.132	KS [illegible]

[illegible] Monte Carlo [illegible] Monte Carlo [illegible] $R = 10^4$ [illegible]

```
> exact.pv = c(mean(Tauhat >= tauhat),
+              mean(Student >= student),
+              mean(Wilcox >= W),
+              mean(Ks >= D))
> round(exact.pv, 3)
[1] 0.002 0.002 0.006 0.040
```

[illegible] p [illegible] 0.05 [illegible]

[illegible]

FRT [illegible]

[illegible] James Lind (1716-1794) [illegible] 12 [illegible] 6 [illegible]

[illegible] 2×2 [illegible] H_{0F} [illegible]

$$p_{\text{frt}} = \frac{1}{\binom{12}{2}} = \frac{1}{66} = 0.015.$$

[illegible] [illegible] 8 [illegible] 4 [illegible] 4 [illegible] $X = 4$ [illegible] p [illegible]

$$p_{\text{frt}} = \frac{1}{70} = 0.014.$$

[illegible][13] [illegible] H_{0F} [illegible] X [illegible] Hypergeometric$(N = 8, K = 4, n = 4)$ [illegible] "[illegible]" [illegible] $\mathbb{P}(X = 4) = 1/70$ [illegible] p $p_{\text{frt}} = 1/70 \approx 0.014$ [illegible] Fisher [illegible]

(1) [illegible] p [illegible]
(2) [illegible] FRT [illegible]

[illegible]

FRT [illegible] H_{0F} [illegible] sharp null [illegible]

$$H_0(\boldsymbol{\tau}) : Y_i(1) - Y_i(0) = \tau_i \quad \text{[illegible]} \quad i. \tag{27}$$

[illegible] $\boldsymbol{\tau}$ [illegible] $(1 - \alpha)$ [illegible]

[illegible]

[illegible]

(1) [illegible] $\mathbb{P}(\mathbf{Z} = \mathbf{z}) = 1/\binom{n}{n_1}$
(2) [illegible] + [illegible]
(3) [illegible] $H_{0F}: Y_i(1) = Y_i(0)$ [illegible] i
(4) [illegible] $p_{\text{frt}} = M^{-1} \sum_{m=1}^{M} \mathbb{I}\{T(\mathbf{z}^m, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}$
(5) [illegible] $\hat{\tau}$ [illegible] t [illegible] Wilcoxon W [illegible] KS D
(6) [illegible] FRT [illegible] t [illegible]
(7) [illegible] Lind [illegible] Fisher [illegible]

[illegible] Neymanian [illegible]

[illegible]

[illegible] 3.1. *3.1 —p_{frt} 的精确性 证明在 sharp null 下，p_{frt} 满足：*

$$\mathbb{P}(p_{frt} \leq u) \leq u \quad \textit{for all} \quad 0 \leq u \leq 1.$$

---

[13] [illegible] **??**

3.2. *3.2 —Monte Carlo* 误差 设 p_{frt} 是固定值，$\hat{p}_{frt}$ 是其 *Monte Carlo* 估计（见 { }。证明：

$$\mathbb{E}_{mc}(\hat{p}_{frt}) = p_{frt}, \quad \mathrm{var}_{mc}(\hat{p}_{frt}) \le \frac{1}{4R},$$

其中下标”*mc*”表示由于 *Monte Carlo* 的随机性。

3.3. *3.3* —有限样本有效的 *Monte Carlo* 近似 证明修正的 *Monte Carlo* 近似

$$\tilde{p}_{frt} = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\{T(\mathbf{z}^r, \mathbf{Y}) \ge T(\mathbf{Z}, \mathbf{Y})\}}{1+R}$$

在任意有限的 R 下都是有限样本有效的 p 值。

：可以利用以下两个基本概率结果：

1. (1) 对于两个二项分布 $X_1 \sim Binomial(R, p_1)$ 和 $X_2 \sim Binomial(R, p_2)$，若 $p_1 \ge p_2$，则 $\mathbb{P}(X_1 \le x) \le \mathbb{P}(X_2 \le x)$。
2. (2) 若 $p \sim Uniform(0, 1)$ 且 $X \mid p \sim Binomial(R, p)$，则 X 在 $\{0, 1, \ldots, R\}$ 上均匀分布。

3.4. *3.4* —*Fisher* 精确检验 考虑二元结局的 *CRE*，数据汇总为 2×2 列联表：

	$Y=1$	$Y=0$	合计
$Z=1$	n_{11}	n_{10}	n_1 着
$Z=0$	n_{01}	n_{00}	n_0

在 H_{0F} 下，证明任何检验统计量 $T(n_{11}, n_{10}, n_{01}, n_{00})$ 是 n_{11} 的函数，且 n_{11} 的精确分布是超几何分布。请说明超几何分布的参数。

3.5. *3.5* —女士品茶详细计算 回忆女士品茶实验（见 { }。计算 $\mathbb{P}(X = k)$，其中 X 是女士正确识别 k 杯先加牛奶的茶的个数，$k = 0, 1, 2, 3, 4$。

3.6. *3.6* —协变量调整的 *FRT* 使用 *LaLonde* 数据集（见 { }，添加协变量进行 *FRT* 分析。假设所有潜在结果和协变量都是固定数值。

有两种策略：

1. (1) 运行结局对协变量的线性回归，得到残差（即”伪结局”）。基于残差定义四种检验统计量，进行 *FRT* 并报告 p 值。
2. (2) 定义检验统计量为结局对治疗和协变量线性回归中治疗系数的 *FRT*。

解释为什么上述两种策略的五个 p 值都是有限样本精确的。

3.7. *3.7* —*FRT* 与广义线性模型 使用与 { }相同的数据集，但将结局改为 *re78 > 0* 的二元指示变量。运行结局对治疗和协变量的 *logistic* 回归。治疗系数是否显著？*FRT* p 值是多少？

3.8. *3.8* —代数细节 验证 { 。{

3.9. *3.9* —推荐阅读 { 是一篇倡导在复杂实验中同时报告 p 值及其相应随机化分布的论文。

CHAPTER 4

[illegible]

[illegible]

[illegible] Fisher's randomization test, FRT [illegible] sharp null [illegible] p [illegible]

[illegible] FRT [illegible] null [illegible] FRT [illegible]

[illegible] Fisher [illegible] p [illegible] " [illegible] " [illegible] " [illegible] " [illegible] Fisher [illegible] distribution-free [illegible]

Jerzy Neyman [illegible] 1923 [illegible] p [illegible] Fisherian [illegible] Neymanian [illegible]

4.1. [illegible] : *FRT* 适用于小样本精确推断和假设检验，*Neymanian* 适用于大样本渐近推断和置信区间构建。两者互为补充。

[illegible] 1

Neymanian [illegible] Neyman [illegible] Fisher [illegible]

[illegible] n [illegible] n_1 [illegible] $n_0 = n - n_1$ [illegible] $i = 1, \ldots, n$ [illegible]

- $Y_i(1)$ [illegible] i [illegible]
- $Y_i(0)$ [illegible] i [illegible]
- $\tau_i = Y_i(1) - Y_i(0)$ [illegible] i [illegible]

[illegible] " [illegible] " [illegible]

$$\bar{Y}(1) = \frac{1}{n}\sum_{i=1}^{n} Y_i(1), \quad \bar{Y}(0) = \frac{1}{n}\sum_{i=1}^{n} Y_i(0)$$

[illegible] $n - 1$ [illegible]

$$S^2(1) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(1) - \bar{Y}(1)\}^2, \quad S^2(0) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(0) - \bar{Y}(0)\}^2$$

[illegible]

$$S(1,0) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(1)-\bar{Y}(1)\}\{Y_i(0)-\bar{Y}(0)\}$$

[illegible]

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\tau_i = \bar{Y}(1)-\bar{Y}(0)$$

[illegible]

$$S^2(\tau) = \frac{1}{n-1}\sum_{i=1}^{n}(\tau_i-\tau)^2$$

[illegible]

4.2. [illegible]：$Y(1)$ 和 $Y(0)$ 之间可能有相关性。若治疗组和对照组的个体是匹配的（正相关），估计的方差会减小；若是负相关，方差会增大。

[illegible]

[illegible] Neymanian [illegible]

[illegible] (Lemma 4.1 [illegible])

$$2S(1,0) = S^2(1)+S^2(0)-S^2(\tau)$$

其中 $\tau_i = Y_i(1)-Y_i(0)$ 是个体因果效应。

[illegible] $S^2(\tau)=0$ [illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible] $S(1,0)$ [illegible] $[S^2(1)+S^2(0)]/2$ [illegible] $S^2(\tau)>0$ [illegible] $2S(1,0) < S^2(1)+S^2(0)$[1] [illegible]

[illegible]

[1] [illegible]

$$\hat{\bar{Y}}(1) = \frac{1}{n_1}\sum_{i=1}^{n}Z_iY_i, \quad \hat{\bar{Y}}(0) = \frac{1}{n_0}\sum_{i=1}^{n}(1-Z_i)Y_i$$

[illegible] $Z_i \in \{0,1\}$ [illegible]

$$\hat{S}^2(1) = \frac{1}{n_1-1}\sum_{i=1}^{n}Z_i\{Y_i-\hat{\bar{Y}}(1)\}^2, \quad \hat{S}^2(0) = \frac{1}{n_0-1}\sum_{i=1}^{n}(1-Z_i)\{Y_i-\hat{\bar{Y}}(0)\}^2$$

[illegible] $S(1,0)$ [illegible] $S^2(\tau)$ [illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible] Neyman [illegible]

[illegible] (Neyman (1923) [illegible]) 在完全随机化实验（*CRE*）下，令 $\hat{\tau} = \hat{\bar{Y}}(1)-\hat{\bar{Y}}(0)$。则：

---

[1] [illegible] **??**

(1) [illegible]：$\hat{\tau}$ 对 τ 是无偏的：

$$\mathbb{E}(\hat{\tau}) = \tau$$

(2) [illegible]：$\hat{\tau}$ 的方差为

$$\operatorname{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} \tag{28}$$

$$= \frac{n_0}{n_1 n}S^2(1) + \frac{n_1}{n_0 n}S^2(0) + \frac{2}{n}S(1,0) \tag{29}$$

(3) [illegible]：方差估计量

$$\hat{V} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}$$

对 $\operatorname{var}(\hat{\tau})$ 是保守的，即

$$\mathbb{E}(\hat{V}) - \operatorname{var}(\hat{\tau}) = \frac{S^2(\tau)}{n} \geq 0$$

等号当且仅当 $\tau_i \equiv \tau$（所有单元个体效应恒定）时成立。

[illegible] $\{Y_i(1), Y_i(0)\}$ [illegible] Z_i [illegible] n_1 [illegible] 1 [illegible] n_0 [illegible] 0 [illegible]

4.3. [illegible]：直觉上：若个体因果效应存在变异（$S^2(\tau) > 0$），治疗组和对照组之间的差异不仅来自随机抽样误差，还来自因果效应本身的真实变异。这种”真实变异”不应被算作估计误差。

Ww[illegible]

[illegible] $S^2(\tau)$

[illegible] $\hat{\tau}$ [illegible]

[illegible] $\{Y_i(1)\}$ [illegible]

- [illegible] $S(1,0) > 0$ [illegible] $\hat{\tau}$
- [illegible] $S(1,0) < 0$ [illegible] $\hat{\tau}$

[illegible] $\hat{\tau}$ [illegible]

[illegible] $S^2(\tau) = 0$ [illegible] $\operatorname{var}(\hat{\tau}) = S^2(1)/n_1 + S^2(0)/n_0$ [illegible] Neymanian [illegible]

[illegible]

[illegible][2]

[illegible][3]

[illegible][4]

---

[2] [illegible] ??

[3] [illegible] ??

[4] [illegible] ??

[illegible] $\hat{V}$ [illegible] HC2 [illegible] $\hat{V}$[6] [illegible] LaLonde [illegible]

```
> tauhat = mean(y[z==1]) - mean(y[z==0])
> vhat   = var(y[z==1])/n1 + var(y[z==0])/n0
> sehat  = sqrt(vhat)
> tauhat
[1] 1794.343
> sehat
[1] 670.9967
```

OLS [illegible]

```
> olsfit = lm(y ~ z)
> summary(olsfit)$coef[2, 1:2]
  Estimate Std. Error
 1794.3431   632.8536
```

[illegible] EHW [illegible] HC2 [illegible]

```
> sqrt(hccm(olsfit, type = "hc2")[2, 2])
[1] 670.9967
```

[illegible]

[illegible] CLT [illegible]

[illegible] Cauchy [illegible]

[illegible]

[illegible] Neymanian [illegible]

(1) [illegible] hat [illegible] $\{\bar{Y}(1), \bar{Y}(0), S^2(1), S^2(0), S(1,0), S^2(\tau)\}$

(2) [illegible] $2S(1,0) = S^2(1) + S^2(0) - S^2(\tau)$

(3) [illegible]

  - $\mathbb{E}(\hat{\tau}) = \tau$ [illegible]
  - $\mathrm{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$
  - $\mathbb{E}(\hat{V}) = \mathrm{var}(\hat{\tau}) + \frac{S^2(\tau)}{n}$ [illegible]

(4) [illegible] $\hat{V} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}$

(5) [illegible] $\hat{\tau} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$

(6) [illegible] $\frac{\hat{\tau}-\tau}{\sqrt{\mathrm{var}(\hat{\tau})}} \xrightarrow{d} N(0,1)$

(7) [illegible] $\hat{\beta} = \hat{\tau}$ [illegible] EHW [illegible]

---

[6] [illegible] ??

q q w

ǘ 4.1. { {引理的证明 证明 *Lemma* { :{ $2S(1,0) = S^2(1) + S^2(0) - S^2(\tau)$。

ǘ 4.2. { {*Neyman (1923)* 定理的另一种证明 在 *CRE* 下，计算：

$$\operatorname{var}\{\hat{\bar{Y}}(1)\}, \quad \operatorname{var}\{\hat{\bar{Y}}(0)\}, \quad \operatorname{cov}\{\hat{\bar{Y}}(1), \hat{\bar{Y}}(0)\}$$

并利用这些公式推导 $\operatorname{var}(\hat{\tau})$。

ǘ 4.3. { {*Neymanian* 推断与 *OLS* 证明 { {(*OLS* 系数等于差值均值)。证明 *EHW* 稳健方差估计量的 *HC2* 变体恰好等于 $\hat{V}$。
þ‖ȯh { ˳{

ǘ 4.4. { {治疗效应的异质性 证明 $S^2(\tau) = 0$ 蕴含 $S^2(1) = S^2(0)$。给出反例说明 $S^2(1) = S^2(0)$ 但 $S^2(\tau) \neq 0$。
证明 $S^2(1) < S^2(0)$ 蕴含 $S(Y(0), \tau) < 0$。给出反例说明 $S^2(1) > S^2(0)$ 但 $S(Y(0), \tau) < 0$。

ǘ 4.5. { {方差公式的改进上界 证明方差公式的改进上界：

$$\operatorname{var}(\hat{\tau}) \leq \frac{1}{n}\left\{\sqrt{\frac{n_0}{n_1}}S(1) + \sqrt{\frac{n_1}{n_0}}S(0)\right\}^2$$

并讨论等号成立的条件。

ǘ 4.6. { {*Neyman (1923)* 的向量版本 将 *Neyman (1923)* 的结果推广到多结局情形。考虑向量结局 $\mathbf{V} \in \mathbb{R}^K$ 的平均因果效应：

$$\tau_{\mathbf{V}} = \frac{1}{n}\sum_{i=1}^{n}\{\mathbf{V}_i(1) - \mathbf{V}_i(0)\}$$

证明 $\hat{\tau}_{\mathbf{V}}$ 对 $\tau_{\mathbf{V}}$ 是无偏的，并求出其协方差矩阵。

ǘ 4.7. { {完全随机化实验中的推断 回忆在 { {中定义的完全随机化实验。在这种设定下，推导平均因果效应的方差公式，并比较其与 *CRE* 中方差的差异。

ǘ 4.8. { {推荐阅读 ┤ḓ : { 重新审视了因果推断中的一些经典问题。{ 是关于因果推断的权威综述。
ä̤ɇ e : *Neyman 1923* 年的原始论文 { 奠定了有限总体推断的基础。当代统计学家将其视为因果推断范式的里程碑。

CHAPTER 5

[illegible]

[illegible]

"Block what you can and randomize what you cannot." — ([illegible] page 103)

[illegible] George Box [illegible] " [illegible]

[illegible] CRE [illegible] chance [illegible]

[illegible] covariate imbalance [illegible] treatment [illegible]

[illegible]

[illegible] n [illegible] $X_i \in \{1, \dots, K\}$ [illegible]

$$n_{[k]} = \#\{i : X_i = k\}, \quad \pi_{[k]} = n_{[k]}/n$$

[illegible] k [illegible] $k = 1, \dots, K$ [illegible] CRE [illegible] n_1 [illegible] n_0 [illegible] $n_{[k]1}$ [illegible] $n_{[k]0}$ [illegible] CRE [illegible][1]

$$\mathbb{E}\left(\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}\right) = 0, \quad \frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0} \neq 0 \quad .$$

[illegible]

[illegible] 5.1 ([illegible] (SRE)). 固定 $n_{[k]1}$ 或 $n_{[k]0}$（$k = 1, \dots, K$），在离散的协变量 X 的每一层内独立进行 *CRE*。

[illegible] blocks [illegible] block randomization [illegible]

$$e_{[k]} = \frac{n_{[k]1}}{n_{[k]}}.$$

[illegible] SRE CRE [illegible]

(1) SRE [illegible] CRE [illegible]

---

[1] [illegible] ??

(2) SRE $e_{[k]}$ [illegible] CRE [illegible]

[illegible]

[illegible] i [illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible] $\tau_i = Y_i(1) - Y_i(0)$ [illegible] k [illegible]

$$\tau_{[k]} = n_{[k]}^{-1} \sum_{X_i=k} \tau_i.$$

[illegible]

$$\tau = n^{-1} \sum_{i=1}^{n} \tau_i = \sum_{k=1}^{K} \pi_{[k]} \tau_{[k]}.$$

[illegible] τ [illegible]

$$\hat{\tau}_{\mathrm{SRE}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]},$$

[illegible]

$$\hat{\tau}_{[k]} = n_{[k]1}^{-1} \sum_{i=1}^{n} \mathbb{I}(X_i = k, Z_i = 1) Y_i - n_{[k]0}^{-1} \sum_{i=1}^{n} \mathbb{I}(X_i = k, Z_i = 0) Y_i$$

[illegible] k [illegible] $\tau_{[k]}$ [illegible] k [illegible] CRE [illegible] τ [illegible]

[illegible]

CRE [illegible] SRE [illegible] Fisher [illegible] FRT [illegible] Sharp null [illegible]

$$H_{0F} : Y_i(1) = Y_i(0) \quad i = 1, \ldots, n.$$

FRT [illegible] $T = T(\mathbf{Z}, \mathbf{Y}, \mathbf{X})$ [illegible] $\mathbf{Z}, \mathbf{Y}, \mathbf{X}$ [illegible] SRE [illegible] H_{0F} [illegible] T [illegible] $\mathbf{Z}$ SRE [illegible]

[illegible] FRT [illegible]

(1) [illegible] X [illegible]

(2) [illegible] SRE [illegible]

[illegible]

4.1.1. 分层差值均值估计量. [illegible] FRT [illegible] $\hat{\tau}_{\mathrm{SRE}}$ [illegible]

[illegible] 5.1 ([illegible]). 在 *FRT* 中，我们可以使用分层估计量作为检验统计量：

$$T = \hat{\tau}_{\mathrm{S}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]},$$

其中 $\hat{\tau}_{[k]}$ 由 [illegible] 定义。这个统计量直接衡量了跨层的平均因果效应。

4.1.2. 学生化分层估计量. [illegible] FRT [illegible]

$$t_{\mathrm{SRE}} = \frac{\hat{\tau}_{\mathrm{SRE}}}{\sqrt{\hat{V}_{\mathrm{SRE}}}},$$

[illegible]

$$\hat{V}_{\mathrm{SRE}} = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right).$$

[illegible] $\hat{S}_{[k]}^2(1)$ [illegible] $\hat{S}_{[k]}^2(0)$ [illegible] k [illegible][2]

[illegible] 5.2 ([illegible]). 对于 [illegible]中的学生化统计量 t_{SRE}，我们需要计算其方差估计量 $\hat{V}_{SRE}$（由 [illegible]给出）。在 *FRT* 中，我们通过模拟所有可行的 *treatment* 分配来计算 t_{SRE} 的精确分布。

4.1.3. 合并 *Wilcoxon* 秩和统计量. [illegible] k [illegible] Wilcoxon [illegible] $W_{[k]}$[illegible]

$$W_{\mathrm{SRE}} = \sum_{k=1}^{K} c_{[k]} W_{[k]}.$$

[illegible] $c_{[k]} = 1/(n_{[k]1} n_{[k]0})$[illegible] $c_{[k]} = 1/(n_{[k]} + 1)$

[illegible] 5.3 (Van Elteren [illegible]). 在第 k 层内，我们计算 *Wilcoxon* 秩和统计量 $W_{[k]}$（回忆第 [illegible]章的 *Wilcoxon* 秩和检验），然后使用 [illegible]合并各层统计量。*Van Elteren (1960)* 的两种加权方法在不同场景下各有优势。

4.1.4. *Hodges–Lehmann* 对齐秩统计量. Van Elteren (1960) [illegible]

$$\tilde{Y}_i = Y_i - \bar{Y}_{[k]}, \quad [illegible] \quad X_i = k,$$

[illegible] $\bar{Y}_{[k]} = n_{[k]}^{-1} \sum_{X_i = k} Y_i$ [illegible] k [illegible] $(\tilde{Y}_1, \ldots, \tilde{Y}_n)$ [illegible] $(\tilde{R}_1, \ldots, \tilde{R}_n)$[illegible]

$$\tilde{W} = \sum_{i=1}^{n} Z_i \tilde{R}_i.$$

[illegible] 5.4 (Hodges–Lehmann [illegible]). [illegible]的方法首先通过 [illegible]对齐结果，这消除了层间差异的影响。然后对池化结果求秩，并使用 [illegible]作为检验统计量。这种方法在多个小层情况下特别有效。

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[2] [illegible] ?? [illegible] ?? [illegible] ??

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µ Ö Û x M X ¥ * _ 1 Ï » k ª = ¤ s Í „ v ¥ † í ” ^ Û ¥ b
a _ 1Ï ƒ t ” 3 ^ % ç ¥ b Œ Â T á Ì ó ç $M = \{n_{[k]1}; n_{[k]0}\}_{k=1}^{K}$ ¥ H q / É › H q w
* ¹ * _ 1 ü M î a _ 1b
” Ð Â T M¥ î µ s û , ¹ , * ¹ [3]

$$S_{*\_1}(w = x_j \mid M) = \frac{S_{*\_1}(w = x_j M)}{S_{*\_1}(M)} = \prod_{k=1}^{K} \frac{1}{\frac{n_{[k]}}{n_{[k]1}}} :$$

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y N ó ç M¥ H q / á Ì V [^ s a _ 1B " s { µ Ö Û x M X ¥ * _ 1b
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$$\hat{S}_a = \sum_{k=1}^{K} {}_{[k]} \hat{}_{[k]} :$$

™ T Ð $\hat{}_{a\_1}$ M] b
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10. 习题

练习 5.1. *CRE 中的协变量平衡* 证明 ??：在 *CRE* 下，

$$\mathbb{E}\left(\frac{n_{[k]1}}{n_1}-\frac{n_{[k]0}}{n_0}\right)=0.$$

练习 5.2. *常数层别倾向得分的结果* 证明 ??：当所有层的层别倾向得分相等 $e_{[k]}=e$（对所有 k）时，

$$\hat{\tau}=\hat{\tau}_{SRE}.$$

练习 5.3. *常数个体因果效应的后果* 假设个体因果效应是常数 $\tau_i=\tau$（对所有 $i=1,\ldots,n$）。考虑以下 τ 的加权估计量类：

$$\hat{\tau}_w=\sum_{k=1}^{K}w_{[k]}\hat{\tau}_{[k]},$$

其中权重 $w_{[k]}$ 对所有 k 非负。

(1) 找出使 $\hat{\tau}_w$ 对 τ 无偏的 $w_{[k]}$ 条件。
(2) 在所有无偏估计量中，找出使 $\hat{\tau}_w$ 方差最小的权重。

练习 5.4. *比较 CRE 和 SRE* 证明 ??：当层样本量较大时，$var_{CRE}(\hat{\tau})$ 与 $var_{SRE}(\hat{\tau}_{SRE})$ 之差约为

$$\sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left\{\sqrt{\frac{n_0}{n_1}}\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\}+\sqrt{\frac{n_1}{n_0}}\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\}\right\}^2\geq 0.$$

练习 5.5. *从 CRE 到 SRE* 证明 ??：数学上，如果 $\mathbf{n}$ 的所有分量都不为零，那么

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z}\mid\mathbf{n})=\frac{1}{\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}}.$$

即，$\mathbf{Z}$ 在给定 $\mathbf{n}$ 条件下从 *CRE* 得到的条件分布与从 *SRE* 得到的分布相同。

练习 5.6. *更多 FRT 统计量* 在 ?? 中，我们介绍了多种检验统计量。选择另一种统计量（如 *Wilcoxon* 秩和统计量或 *Hodges–Lehmann* 对齐秩统计量），重新进行 *FRT* 分析。

练习 5.7. *Imbens and Rubin (2015) 中的 SRE 的 FRT* *Imbens and Rubin (2015)* 讨论了 *1985-1986* 年在田纳西州进行的 *SRE*（*Student/Teacher Achievement Ratio* 实验）。*kindergarten* 数据如下：

层对应于学校，分析单元是教师或班级。*treatment* 等于 *1* 表示小班（每教师 *13-17* 名学生），等于 *0* 表示普通班（*22-25* 名学生）。*outcome* 是标准化的平均数学成绩。

使用 *FRT* 重新分析 *Project STAR* 数据。使用 $\hat{\tau}_{SRE}$、W_{SRE} 和 $\tilde{W}$ 作为 *FRT* 的检验统计量。比较 p 值。

注：本书使用 Z 表示治疗指示变量，但 *Imbens and Rubin (2015)* 使用 W。

练习 5.8. *多中心试验* *Gould (1998, Table 1)* 报告了以下来自多中心试验的数据：

这是一个以中心为层的 *SRE*。该试验旨在研究非那雄胺（用于治疗良性前列腺增生的药物）的疗效和耐受性。在每个中心的 *29* 个中心内，患者被随机分配到三个组：对照组、非那雄胺 *1mg* 和非那雄胺 *5mg*。

由于没有报告个体水平的数据，我们无法实施 *FRT*。然而，*Neymanian* 推断只需要汇总统计。分别报告非那雄胺 *1mg* 和非那雄胺 *5mg* 与对照组比较的点估计量和方差估计。

练习 5.9. 数据分析重新分析 重新分析第 ?? 章 ?? 节中使用的 *LaLonde* 数据。同时进行 *Fisherian* 和 *Neymanian* 推断。

Mahalanobis 距离的一个优美性质是它对协变量的非奇异线性变换保持不变。

引理 6.1 (引理 6.1 —Mahalanobis 距离的不变性). *式* **??** *中的 M 在变换 $X_i \mapsto b_0 + BX_i$（其中 $b_0 \in \mathbb{R}^K$，$B \in \mathbb{R}^{K\times K}$ 可逆）下保持不变。*

为什么这个性质重要？因为它意味着在使用 Mahalanobis 距离时，我们不必担心协变量的量纲问题——无论是用米还是厘米，用摄氏度还是华氏度，M 的数值都不会改变。

有限总体中心极限定理（Li and Ding, 2017）保证：当 n 足够大时，在 CRE 下 M 近似服从 χ^2_K 分布，渐近期望为 K，方差为 $2K$。因此，在 CRE 下 M 的实现值很可能很大。

重随机化的核心思想是：**拒绝那些协变量不平衡的治疗分配**。

2.3. ReM 的正式定义.

定义 6.2 (重随机化（ReM）). *从 CRE 中抽取治疗向量* $\mathbf{Z}$*，当且仅当* $M \le a$ *时接受，其中* $a > 0$ *是预定阈值。*

阈值 a 的选择类似于 SRE 中层数的选择——这是一个在实践中非平凡的问题。当 $a = \infty$ 时，ReM 退化为普通 CRE；当 $a = 0$ 时，只有极少数可行的治疗分配，实验几乎没有随机性。折中方案是选择一个小但不极小的 a，例如取 χ^2_K 分布的上分位数（如 0.001 分位数）。

几何直观：ReM 通过约束协变量均值之差 $\hat{\tau}_X$ 的大小，在设计阶段”塑造”了实验的几何结构。Morgan 和 Rubin (2012) 的研究表明，这种设计具有优雅的数学理论和良好的几何性质。

3. ReM 下的统计推断

3.1. Fisher 随机化检验在 ReM 下的推广. 一个关键问题是：如何在 ReM 下分析数据？Bruhn and McKenzie (2009) 和 Morgan and Rubin (2012) 论证，只要我们在约束 $M \le a$ 下模拟治疗向量 $\mathbf{Z}$，就仍然可以使用 FRT。这在 sharp null 假设下总是给出有限样本精确 p 值。

在非 null 假设下，推导 ReM 的有限样本性质是一个具有挑战性的问题。Li et al. (2018b) 在正则条件（Condition 6.1）下推导了 $\hat{\tau}$ 的渐近分布。

条件 6.1 (条件 6.1). *当* $n \to \infty$ *时：*

(1) n_1/n *和* n_0/n *具有正极限；*
(2) $\{X_i, Y_i(1), Y_i(0), \tau_i\}$ *的有限总体协方差具有有限极限；*
(3) $\max_{1\le i\le n}\{Y_i(1)-\bar{Y}(1)\}^2/n \to 0$, $\max_{1\le i\le n}\{Y_i(0)-\bar{Y}(0)\}^2/n \to 0$, *且* $\max_{1\le i\le n} X_i^\top X_i/n \to 0$。

3.2. ReM 的渐近分布：几何解释. 下面给出 ReM 的主要渐近结果。为符号简洁，记

$$L_{K,a} \sim D_1 \mid \mathbf{D}^\top \mathbf{D} \le a,$$

其中 $\mathbf{D} = (D_1, \ldots, D_K)$ 服从 K 维标准正态分布；ε 服从一元标准正态分布；$L_{K,a} \perp \varepsilon$。

定理 6.3 (定理 6.1 —ReM 下的渐近分布). *在 ReM（$M \le a$）和条件* **??** *下，*

$$\hat{\tau} - \tau \dot{\sim} \sqrt{var(\hat{\tau})}\left\{\sqrt{R^2}L_{K,a} + \sqrt{1-R^2}\varepsilon\right\},$$

其中

$$var(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$$

是 *Neyman (1923)* 的方差公式（见第 [illegible]章），且

$$R^2 = corr^2(\hat{\tau}, \hat{\tau}_X)$$

是 $\hat{\tau}$ 与 $\hat{\tau}_X$ 在 *CRE* 下的平方多重相关系数。

[illegible] Li et al. (2018b) [illegible] $\hat{\tau}$ [illegible] $\hat{\tau}_X$ [illegible] $\hat{\tau}_X$ [illegible] $\cos^2\theta = R^2$ [illegible] θ [illegible] $\hat{\tau}$ [illegible] $\hat{\tau}_X$ [illegible] ReM [illegible] $L_{K,a}$ [illegible] ReM [illegible]

[illegible]

- [illegible] $a = \infty$ [illegible] CRE [illegible] $\hat{\tau} - \tau \dot{\sim} \sqrt{\text{var}(\hat{\tau})}\varepsilon$ [illegible]
- [illegible] $a \to 0$ [illegible] $\hat{\tau} - \tau \dot{\sim} \sqrt{\text{var}(\hat{\tau})(1-R^2)}\varepsilon$

[illegible] a [illegible] ReM [illegible] R^2 [illegible] R^2 [illegible][2]

[illegible] 6.1 [illegible] R^2 [illegible]）在 *CRE* 下，

$$R^2 = corr^2(\hat{\tau}, \hat{\tau}_X) = \frac{n_1^{-1}S^2(1 \mid X) + n_0^{-1}S^2(0 \mid X) - n^{-1}S^2(\tau \mid X)}{n_1^{-1}S^2(1) + n_0^{-1}S^2(0) - n^{-1}S^2(\tau)},$$

其中 $\{S^2(1), S^2(0), S^2(\tau)\}$ 是 $\{Y_i(1), Y_i(0), \tau_i\}$ 的有限总体方差，$\{S^2(1 \mid x), S^2(0 \mid x), S^2(\tau \mid x)\}$ 是它们在 $(1, X_i)$ 上线性投影的相应有限总体方差。在常因果效应假设（$\tau_i = \tau$）下，R^2 退化为 $S^2(0 \mid X)/S^2(0)$，即 $Y_i(0)$ 与 X_i 之间的有限总体平方多重相关系数。

[illegible] R^2 [illegible] ReM [illegible]

[illegible] ReM [illegible] Neyman (1923) [illegible] overcover τ [illegible] Li et al. (2018b) [illegible]$_1$ [illegible]

[illegible]

[illegible]$_2$ [illegible]

[illegible] [illegible] **X** [illegible] H_{0F} [illegible] $T = T(\mathbf{Z}, \mathbf{Y}, \mathbf{X})$ [illegible] p [illegible] FRT [illegible]$_1$ [illegible]

Zhao and Ding (2021a) [illegible]

[illegible] 6.5 ([illegible] FRT [illegible]). 基于拟合统计模型后的残差构造检验统计量。将 Y_i 对 X_i 回归得到残差 $\hat{\varepsilon}_i$，然后将 $\hat{\varepsilon}_i$ 视为伪结果来构造检验统计量。

[illegible] 6.6 ([illegible] FRT [illegible]). 使用回归系数作为检验统计量。将 Y_i 对 (Z_i, X_i) 回归，取 Z_i 的系数作为检验统计量。

[illegible] Y_i [illegible] X_i [illegible] Z_i [illegible] H_{0F} [illegible] H_{0F} [illegible] Y_i [illegible] (Z_i, X_i) [illegible] Z_i [illegible] Z_i [illegible] H_{0F} [illegible] ”[illegible]” [illegible] logistic [illegible]$_2$ [illegible] H_{0F} [illegible]

---

[2] [illegible] ??

[illegible]

[illegible] Li et al. (2018b) [illegible] ReM [illegible] Lin (2013) [illegible] a [illegible] ReM [illegible] $\hat{\tau}$ [illegible] CRE [illegible] $\hat{\tau}_{\mathrm{L}}$ [illegible] ReM [illegible] Lin (2013) [illegible] a [illegible] ReM [illegible]

[illegible] C_i [illegible] K [illegible] $K-1$ [illegible]

$$X_i = \left(\mathbb{I}(C_i = 1) - \pi_{[1]}, \ldots, \mathbb{I}(C_i = K-1) - \pi_{[K-1]}\right),$$

[illegible] $\pi_{[k]}$ [illegible] $C_i = k$ [illegible] Lin (2013) [illegible]

[illegible] 6.3 [illegible]）基于 X_i 的 $\hat{\tau}_{\mathrm{L}}$ 在数值上与基于 C_i 的后分层估计量 $\hat{\tau}_{\mathrm{PS}}$（见第 [illegible] 章）相同。

[illegible]

[illegible] X [illegible] Y [illegible] X [illegible] Y [illegible] X [illegible] X [illegible] $\beta_1 = \beta_0 = 1$ [illegible]

$$\hat{\tau}(1,1) = n_1^{-1}\sum_{i=1}^{n} Z_i(Y_i - X_i) - n_0^{-1}\sum_{i=1}^{n}(1 - Z_i)(Y_i - X_i).$$

[illegible] $g_i = Y_i - X_i$ [illegible] $\hat{\tau}(1,1)$ [illegible] $\beta_1 = \beta_0 = 1$ [illegible] Lin (2013) [illegible] β [illegible] $\hat{\tau}(1,1)$ [illegible] Lin (2013) [illegible] Lin (2013) [illegible] $\hat{\tau}(1,1)$ [illegible]

[illegible]

[illegible] C [illegible] X [illegible] Lin (2013) [illegible] $\hat{\tau}_{\mathrm{L},[k]}$ [illegible]

$$\hat{\tau}_{\mathrm{L,S}} = \sum_{k=1}^{K} \pi_{[k]}\hat{\tau}_{\mathrm{L},[k]}.$$

[illegible]

$$\hat{V}_{\mathrm{L,S}} = \sum_{k=1}^{K} \pi_{[k]}^2 \hat{V}_{\mathrm{EHW},[k]},$$

[illegible] $\hat{V}_{\mathrm{EHW},[k]}$ [illegible] k [illegible] OLS [illegible] EHW [illegible]

[illegible]

Li and Ding (2020) [illegible] Lin (2013) [illegible] EHW [illegible]

[illegible] Neyman (1923) [illegible] Li and Ding (2020) [illegible]

TABLE 1. [illegible]

	[illegible]	
[illegible]	$\hat{\tau}$ (Neyman, 1923)	$\hat{\tau}_{\mathrm{L}}$ (Lin, 2013)
CRE	[illegible]	[illegible]
ReM	$\hat{\tau}$ (Li et al., 2018b)	$\hat{\tau}_{\mathrm{L}}$ (Li and Ding, 2020)

[illegible]

- [illegible] 1[illegible] CRE [illegible] $\hat{\tau}_{\mathrm{L}}$ [illegible] $\hat{\tau}$ [illegible]
- [illegible] 2[illegible] ReM [illegible] ReM [illegible] $\hat{\tau}$ [illegible] CRE [illegible] $\hat{\tau}$ [illegible]
- [illegible] 3 [illegible] 4[illegible] ReM CRE [illegible]

R [illegible]

Angrist et al. (2009) [illegible] 8 [illegible] GPA [illegible] GPA

[illegible] 6.1 ([illegible]). 我们使用 *Angrist et al. (2009)* 数据的一个子集，评估四种设计与分析策略的表现。

代码实现（省略数据读取和缺失值填补）：

```
# 未调整估计量
fit_unadj = lm(y ~ z)
ace_unadj = coef(fit_unadj)[2]
se_unadj = sqrt(hccm(fit_unadj, type = "hc2"))[2, 2]

# 回归调整
x = scale(x)
fit_adj = lm(y ~ z*x)
ace_adj = coef(fit_adj)[2]
se_adj = sqrt(hccm(fit_adj, type = "hc2"))[2, 2]
```

模拟结果（*2000* 次蒙特卡洛复制，*ReM* 阈值 $a = 0.05$）：

估计量	估计值	标准误	t 统计量	p 值
Neyman	*0.054*	*0.076*	*0.719*	*0.472*
Lin	*0.075*	*0.072*	*1.036*	*0.300*

调整后的估计量有更小的标准误，尽管它给出了与未调整估计量相同的（不显著）结论。

[illegible] ReM [illegible] 0.1 [illegible] 0.25 [illegible]

[illegible] ReM [illegible]

[illegible]

[illegible] ReM[illegible] Mahalanobis [illegible] **Z** [illegible] **X** [illegible] $X_i = (x_{i1}, \dots, x_{iK})^\top$ [illegible] $|\hat{\tau}_{x_k}/\sqrt{n/(n_1 n_0) S^2_{x_k}}| \le a$ [illegible] $k = 1, \dots, K$ [illegible] **Z** [illegible] Zhao and Ding (2021b) [illegible]

[illegible] Lin (2013) [illegible] logistic [illegible] logistic [illegible] logistic [illegible] Lin (2013) [illegible] CRE [illegible] Guo and Basse (2023) [illegible] Lin (2013) [illegible]

[illegible] Bloniarz et al. (2016) [illegible] LASSO [illegible] OLS [illegible] Lei and Ding (2021) [illegible] Lin (2013) [illegible] Wager et al. (2016) [illegible] ReM [illegible]

[illegible]

[illegible] 6.1. *FRT under ReM* 描述在 *ReM* 下的 *FRT*。

[illegible] 6.2. *Mahalanobis* 距离的不变性 证明引理 { 。{

[illegible] 6.3. 重随机化下差值均值估计量的无偏性 假设从 *CRE* 中抽取 $\mathbf{Z} = (Z_1, \dots, Z_n)$，当且仅当 $\phi(\mathbf{Z}, \mathbf{X}) = 1$ 时接受，其中 ϕ 是预定的平衡准则。证明：如果 $n_1 = n_0$ 且 $\phi(\mathbf{Z}, \mathbf{X}) = \phi(\mathbf{1}_n - \mathbf{Z}, \mathbf{X})$，则 $\hat{\tau}$ 对 τ 无偏。验证当 $n_1 = n_0$ 时，使用 *Mahalanobis* 距离的重随机化满足此条件。给出当这两个条件不满足时 $\hat{\tau}$ 对 τ 有偏的反例。

[illegible] 6.4. R^2 的等价形式 证明命题 { 。{

[illegible] 6.5. 协变量调整 *FRT* 在协变量调整 *FRT* 中，分别在伪结果策略和模型输出策略下推导 p 值的计算方法。

[illegible] 6.6. *Lin* 估计量的偏 使用蒙特卡洛模拟研究 *Lin* 估计量在小样本下的有限样本偏。将样本量从 $n = 20$ 变化到 $n = 500$，重复 *1000* 次，报告平均偏和均方误差。

[illegible] 6.7. *Lin* 估计量的 *OLS* 表示 证明命题 { {(这是纯线性代数事实)。

[illegible] 6.8. 预测估计量的等价性 证明 $\hat{\tau}_{pred} = \hat{\tau}_{\mathrm{L}}$。

[illegible] 6.9. 投影估计量的等价性 证明 $\hat{\tau}_{proj} = \hat{\tau}_{\mathrm{L}}$。

[illegible] 6.10. 协变量不平衡调整形式 证明 $\hat{\tau}(\beta_1, \beta_0) = \hat{\tau} - \gamma^\top \hat{\tau}_X$ 和 $\hat{\tau}_{\mathrm{L}} = \hat{\tau} - \hat{\gamma}^\top \hat{\tau}_X$。

ǘ 6.11. 差分估计量的方差 推导 $\hat{\tau}(1,1)$ 的保守方差估计量 $\hat{V}(1,1)$。

ǘ 6.12. 回归调整与后分层的等价性 证明命题 { 。{

ǘ 6.13. 协变量不平衡与效率 模拟研究当协变量与结果的相关性从 0.1 变化到 0.9 时，回归调整的效率增益如何变化。

ǘ 6.14. 高维协变量的 *LASSO* 调整 使用 *LASSO* 代替 *OLS* 进行回归调整，比较两者在协变量稀疏时的表现。

ǘ 6.15. 缺失数据的填补 在 { {的模拟中，评估不同缺失值填补策略对估计量的影响。

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: (i;j) VU» i Ï¥» j ñ†í Ï $i = 1;\dots;n$ $j = 1;2$b†í (i;j) ¥. ² T¹ $Y_{ij}(1)$ ¤sÍ „ $Y_{ij}(0)$ ¤s vb ÄB = Û s¥Bñ†í¤sÍ 6Bñ¤s vbçl

UjjV
$$Z_i = \begin{cases} 1; & \text{Ï»Bñ†í¤sÍ}; \\ 0; & \text{Ï»=ñ†í¤sÍ}: \end{cases}$$

çl dX P¥ Ŵ L UJS1XáVÌµ $(Z_i)_{i=1}^n$ AA·"2`MQ(1/2)H:HB

' Í ·UM $(Z_1;\dots;Z_n)$ ÿë]sƒ ›V•”¹ 1/2 ¥umæsƒb

» i =¥4©²T¹

$$Y_{i1} = Z_i Y_{i1}(1) + (1 \quad Z_i) Y_{i1}(0); \quad Y_{i2} = Z_i Y_{i2}(0) + (1 \quad Z_i) Y_{i2}(1):$$

yN 4©” ¹ $(Z_i; Y_{i1}; Y_{i2})_{i=1}^n$ b

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Uj9V
$$H_{0F} : Y_{ij}(1) = Y_{ij}(0) \quad \text{îµ} \quad i = 1;\dots;n \ \text{„} \ j = 1;2:$$

6_hÏ áÌ³1V \\ E $(Z_1;\dots;Z_n)$ ¥sƒb[/^ ¿¥ =Í FÐ vF²T-µ¥+ÕÜÅ_ d9 b

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[R] ¥ ØE¥W%=¸n \b

3.1. 配对内差异. 定义配对 i 内的**配对差异**:

$$\hat{\tau}_i = \text{治疗组结果} - \text{对照组结果} = (2Z_i - 1)(Y_{i1} - Y_{i2}) = S_i(Y_{i1} - Y_{i2}), \tag{35}$$

其中 $S_i = 2Z_i - 1$ 是独立同分布的随机符号，均值为 0，方差为 1。由于 $\hat{\tau}_i = 0$ 的配对对随机化分布没有贡献，我们在 FRT 讨论中剔除这些配对。

配对差异的均值为

$$\hat{\tau} = n^{-1} \sum_{i=1}^{n} \hat{\tau}_i. \tag{36}$$

3.2. 配对 t 统计量.

例 7.1 (配对 t 统计量). 在 H_{0F} 下，$\hat{\tau}_i$ 是固定的，而 S_i 是随机的。因此

$$\mathbb{E}(\hat{\tau}) = 0, \quad var(\hat{\tau}) = n^{-2} \sum_{i=1}^{n} var(S_i)(Y_{i1} - Y_{i2})^2 = n^{-2} \sum_{i=1}^{n} \hat{\tau}_i^2.$$

基于独立随机变量之和的中心极限定理，我们有正态近似：[2]

$$\frac{\hat{\tau}}{\sqrt{n^{-2}\sum_{i=1}^{n} \hat{\tau}_i^2}} \to \mathrm{N}(0,1) \tag{37}$$

依分布收敛。许多标准教科书建议在 MPE 中使用以下配对 t 统计量:

$$t_{pair} = \frac{\hat{\tau}}{\sqrt{\{n(n-1)\}^{-1}\sum_{i=1}^{n}(\hat{\tau}_i - \hat{\tau})^2}}. \tag{38}$$

在 H_{0F} 下，当 n 较大且 $\hat{\tau}$ 较小时，t_{pair} 与 **??** 几乎相同。

在经典统计学中，使用 t_{pair} 的动机基于不同框架。当 $\hat{\tau}_i \overset{IID}{\sim} \mathrm{N}(0,\sigma^2)$ 时，可以证明 $t_{pair} \sim t(n-1)$，即 t_{pair} 的精确分布是自由度为 $n-1$ 的 t 分布，在大 n 下接近 $\mathrm{N}(0,1)$。R 函数 `t.test` 可以实现此检验。**??** 的讨论给出了经典配对 t 检验的另一种证明，无需假设数据的正态性。

3.3. Wilcoxon 符号秩统计量.

例 7.2 (Wilcoxon 符号秩统计量). 基于 $(|\hat{\tau}_1|, \ldots, |\hat{\tau}_n|)$ 的秩 $(R_1, \ldots, R_n)$，我们可以定义检验统计量

$$W = \sum_{i=1}^{n} \mathbb{I}(\hat{\tau}_i > 0) R_i. \tag{39}$$

在 H_{0F} 下，$|\hat{\tau}_i|$ 是固定的，因此 R_i 也是固定的。由此可得

$$\mathbb{E}(W) = \frac{1}{2}\sum_{i=1}^{n} R_i = \frac{1}{2}\sum_{i=1}^{n} i = \frac{n(n+1)}{4},$$

$$var(W) = \frac{1}{4}\sum_{i=1}^{n} R_i^2 = \frac{1}{4}\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{24}.$$

---

[2] 推导见附录 **??**

独立随机变量之和的中心极限定理保证以下正态近似：[3]

$$\frac{W - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \to \mathrm{N}(0,1) \tag{40}$$

依分布收敛。R 函数 *wilcox.test* 可以实现精确和渐近检验。

∼wXw dpgjpZp p ǎ 7、jwa l p”

ǎ 7.3 (Kolmogorov–Smirnov Ω). 在 H_{0F} 下，绝对值 $(|\hat\tau_1|, \ldots, |\hat\tau_n|)$ 是固定的，但符号是随机的。因此 $(\hat\tau_1, \ldots, \hat\tau_n)$ 和 $-(\hat\tau_1, \ldots, \hat\tau_n)$ 应具有相同分布。

设

$$\hat F(t) = n^{-1} \sum_{i=1}^{n} \mathbb{I}(\hat\tau_i \le t)$$

是 $(\hat\tau_1, \ldots, \hat\tau_n)$ 的经验分布函数，

$$1 - \hat F(-t-) = n^{-1} \sum_{i=1}^{n} \mathbb{I}(-\hat\tau_i \le t)$$

是 $-(\hat\tau_1, \ldots, \hat\tau_n)$ 的经验分布函数，其中 $\hat F(-t-)$ 是 $\hat F(\cdot)$ 在 $-t$ 处的左极限。*Kolmogorov–Smirnov* 型统计量定义为

$$D = \max_t |\hat F(t) + \hat F(-t-) - 1|. \tag{41}$$

{ 提出了这个检验统计量并推导了其精确和渐近分布。标准软件包中未实现此统计量，但我们可以基于 *FRT* 模拟其精确分布并计算 p 值。

∼wWw w

ǎ 7.4 (Åῷ Ω). 符号统计量仅使用配对内差异的符号：

$$\Delta = \sum_{i=1}^{n} \mathbb{I}(\hat\tau_i > 0). \tag{42}$$

在 H_{0F} 下，

$$\mathbb{I}(\hat\tau_i > 0) \overset{IID}{\sim} Bernoulli(1/2),$$

因此 $\Delta \sim Binomial(n, 1/2)$。这给出了一个基于二项分布的精确检验，$R$ 函数 *binom.test* 可以实现。利用中心极限定理，我们还可以基于以下正态近似进行检验：[4]

$$\frac{\Delta - n/2}{\sqrt{n/4}} \to \mathrm{N}(0,1) \tag{43}$$

依分布收敛。

---

[3] þ‖Òh ??

[4] þ‖Òh ??

[illegible] MPE [illegible] 1[5] [illegible] 2×2 [illegible]

	[illegible] =1	[illegible] =0
[illegible] =1	m_{11}	m_{10}
[illegible] =0	m_{01}	m_{00}

[illegible] 7.5 (McNemar [illegible]). 在 H_{0F} 下，一致对的数量 m_{11} 和 m_{00} 是固定的，$m_{10}+m_{01}$ 也是固定的。因此唯一的随机分量是 m_{10}，其分布为

$$m_{10} \sim Binomial(m_{10}+m_{01}, 1/2).$$

这给出了一个基于二项分布的精确检验。*R* 函数 *mcnemar.test* 给出基于二项分布正态近似的渐近检验：

$$\frac{m_{10}-(m_{10}+m_{01})/2}{\sqrt{(m_{10}+m_{01})/4}} = \frac{m_{10}-m_{01}}{\sqrt{m_{10}+m_{01}}} \rightarrow \mathrm{N}(0,1) \tag{44}$$

依分布收敛。精确 *FRT* 和渐近检验都不依赖于 m_{11} 或 m_{00}，只有不一致对的数量在这些检验中起作用。

[illegible]

[illegible] i [illegible]

$$\tau_i = \frac{1}{2}\{Y_{i1}(1)+Y_{i2}(1)-Y_{i1}(0)-Y_{i2}(0)\}, \tag{45}$$

[illegible]

$$\tau = n^{-1}\sum_{i=1}^{n}\tau_i = (2n)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{2}\{Y_{ij}(1)-Y_{ij}(0)\}. \tag{46}$$

[illegible] $\hat{\tau}_i$ [illegible] τ_i [illegible] $\hat{\tau}$ [illegible] τ [illegible][5] [illegible] MPE [illegible] SRE [illegible] $\hat{\tau}$ [illegible] i [illegible] $\hat{\tau}_i$ [illegible] MPE [illegible] $\hat{\tau}$ [illegible] MPE [illegible] $\hat{\tau}_i$ [illegible] μ [illegible] σ^2 [illegible] $\hat{\tau} = n^{-1}\sum_{i=1}^{n}\hat{\tau}_i$ [illegible] σ^2/n σ^2 [illegible] $(n-1)^{-1}\sum_{i=1}^{n}(\hat{\tau}_i-\hat{\tau})^2$ [illegible] $\mathrm{var}(\hat{\tau})$ [illegible]

$$\hat{V} = \{n(n-1)\}^{-1}\sum_{i=1}^{n}(\hat{\tau}_i-\hat{\tau})^2. \tag{47}$$

[illegible] MPE [illegible] $\hat{V}$ [illegible] $\hat{\tau}_i$ [illegible] $\hat{\tau}$ [illegible] MPE [illegible]

[illegible] **(**[illegible]**)** 在 *MPE* 下，由 [illegible] 定义的 $\hat{V}$ 是 $\hat{\tau}$ 真实方差的一个保守估计：

$$\mathbb{E}(\hat{V}) - var(\hat{\tau}) = \{n(n-1)\}^{-1}\sum_{i=1}^{n}(\tau_i-\tau)^2 \geq 0. \tag{48}$$

---

[5] [illegible] **??**

当 τ_i 在各配对间恒定时，$\mathbb{E}(\hat{V}) = var(\hat{\tau})$。

[illegible] MPE [illegible] $\hat{V}$ [illegible] $\hat{V}$ [illegible] $\hat{\tau}_i$ [illegible] $\mathrm{var}(\hat{\tau})$ [illegible] $\hat{\tau}_i$ [illegible][6]

[illegible] Neymanian [illegible]

$$\frac{\hat{\tau} - \tau}{\sqrt{\mathrm{var}(\hat{\tau})}} \to \mathrm{N}(0,1) \tag{49}$$

$n \to \infty$ [illegible] Wald [illegible]

$$\hat{\tau} \pm z_{1-\alpha/2}\sqrt{\hat{V}} \tag{50}$$

[illegible] $1-\alpha$ [illegible] τ

[illegible] $\hat{\tau}$ [illegible] $\hat{V}$ [illegible] OLS [illegible]

[illegible] **w**(OLS [illegible])**w** $\hat{\tau}$ 和 $\hat{V}$ 等于将向量 $(\hat{\tau}_1, \ldots, \hat{\tau}_n)^\top$ 仅对截距进行 *OLS* 拟合的系数和方差估计量。

$\hat{\tau}$ [illegible] $\hat{V}$ [illegible] OLS [illegible] R [illegible] `lm(tau_hat ~ 1)` [illegible][7]

Ww[illegible]

[illegible] $X_{i1} \neq X_{i2}$ [illegible][8]

[illegible] (i,j) [illegible] X_{ij} [illegible] MPE [illegible]

Ww q w [illegible] **w** FRT [illegible] OLS [illegible] $\hat{\varepsilon}_{ij}$ [illegible] $\hat{\varepsilon}_{ij}$ [illegible] MPE [illegible]

Ww [illegible] **w** **w** [illegible] τ [illegible] $\hat{\tau}_{X,i}$ [illegible] $\hat{\tau}_X$ [illegible]

$$\mathbb{E}(\hat{\tau}_{X,i}) = 0, \quad \mathbb{E}(\hat{\tau}_X) = 0,$$

[illegible]

$$\mathrm{cov}(\hat{\tau}_X) = n^{-2}\sum_{i=1}^{n} \hat{\tau}_{X,i}\hat{\tau}_{X,i}^\top.$$

[illegible] MPE [illegible] $\hat{\tau}_{X,i}$ [illegible] $\mathrm{cov}(\hat{\tau}_X)$ [illegible] $(Z_1, \ldots, Z_n)$ [illegible] $\hat{\tau}_X$ [illegible]

[illegible] γ [illegible]

$$\hat{\boldsymbol{\tau}}(\gamma) = \hat{\boldsymbol{\tau}} - \boldsymbol{\gamma}^\top \hat{\boldsymbol{\tau}}_X. \tag{51}$$

[illegible] γ [illegible] γ [illegible] $\hat{\tau}(\gamma)$ [illegible]

---

[6] [illegible] **??**

[7] [illegible] **??**

[8] [illegible] **??**

[illegible] $cov(\hat{\tau}_X, \hat{\tau})$ 的一个无偏估计为

$$\hat{\theta} = \{n(n-1)\}^{-1} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}). \tag{52}$$

[illegible]

$$\hat{\gamma} = \left(n^{-2} \sum_{i=1}^{n} \hat{\tau}_{X,i} \hat{\tau}_{X,i}^{\top} \right)^{-1} \hat{\theta} \approx \left(\sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^{\top} \right)^{-1} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}),$$

[illegible] $\hat{\tau}_i$ [illegible] $\hat{\tau}_{X,i}$ [illegible] OLS [illegible] $\hat{\tau}_{X,i}$ [illegible]

$$\hat{\tau}_{\text{adj}} = \hat{\tau}(\hat{\gamma}) = \hat{\tau} - \hat{\gamma}^{\top} \hat{\tau}_X. \tag{53}$$

OLS [illegible] $\hat{\tau}_{\text{adj}}$ [illegible] $\hat{\tau}_i$ [illegible] 1 [illegible] $\hat{\tau}_{X,i}$ [illegible] OLS [illegible] $\hat{\tau}_{\text{adj}}$ [illegible]

$$\hat{V}_{\text{adj}} = \hat{V} + \hat{\gamma}^{\top} \text{cov}(\hat{\tau}_X) \hat{\gamma} - 2\hat{\gamma}^{\top} \hat{\theta} = \hat{V} - \hat{\theta}^{\top} \text{cov}(\hat{\tau}_X)^{-1} \hat{\theta}.$$

[illegible] OLS [illegible] 在 *MPE* 下，协变量调整估计量 $\hat{\tau}_{adj}$ 和相关方差估计量 $\hat{V}_{adj}$ 可以方便地近似为将 $\hat{\tau}_i$ 的向量对 1 的向量和 $\hat{\tau}_{X,i}$ 的矩阵进行 *OLS* 拟合的截距和相关方差估计量。

[illegible] EHW [illegible] Eicker–Huber–White [illegible] MPE [illegible] CRE [illegible] Lin (2013) [illegible]

[illegible]

[illegible] 15 [illegible] R [illegible] HistData [illegible] cross [illegible] self [illegible] diff [illegible] MPE [illegible] $2^{15} = 32768$ [illegible] $\hat{\tau}$ [illegible] p [illegible]

```
> difference = ZeaMays$diff
> n.pairs = length(difference)
> abs.diff = abs(difference)
> t_obs = mean(difference)
> t_ran = sapply(1:2^15,
+ function(x) {
+ sum(MP_enumerate(x, 15)*abs.diff)
+ })/n.pairs
> pvalue = mean(t_ran >= t_obs)
> pvalue
[1] 0.02633667
```

[illegible] p [illegible] 0.026[illegible] 5% [illegible] H_{0F} [illegible] $\hat{\tau}$ [illegible]

Figure 1. Darwin [illegible] $\hat{\tau}$ [illegible]

[illegible]

[illegible] Workshop [illegible] 10 [illegible] 8 [illegible]

[illegible]	[illegible]			[illegible]		
	[illegible] 1	[illegible] 2	[illegible]	[illegible] 1	[illegible] 2	[illegible]
1	24	25	-1	51	56	-5
2	42	40	2	64	65	-1
3	52	52	0	75	72	3
4	71	70	1	79	77	2
5	29	27	2	54	49	5
6	50	50	0	75	78	-3
7	69	69	0	91	92	-1
8	36	34	2	64	55	9

1 Neymanian [illegible][9]

[illegible]

[illegible] MPE [illegible]

(1) [illegible] MPE [illegible] SRE [illegible]

(2) [illegible] $Z_i \overset{\text{IID}}{\sim} \text{Bernoulli}(1/2)$

(3) [illegible] t [illegible] Wilcoxon [illegible] Kolmogorov–Smirnov [illegible]

(4) [illegible] $\hat{V}$ [illegible] OLS [illegible]

(5) [illegible] $\hat{\tau}_{\text{adj}}$ [illegible] $\hat{V}_{\text{adj}}$ [illegible] OLS [illegible]

[illegible]

[illegible] 7.1. *MPE* 的方差 证明在 *MPE* 下，$\hat{\tau}$ 的方差为

$$var(\hat{\tau}) = n^{-2} \sum_{i=1}^{n} var(\hat{\tau}_i).$$

[illegible] 7.2. 配对 t 统计量的渐近分布 证明 [illegible] 在 H_{0F} 下，

$$\frac{\hat{\tau}}{\sqrt{n^{-2} \sum_{i=1}^{n} \hat{\tau}_i^2}} \to \text{N}(0, 1)$$

依分布收敛。

[illegible] 7.3. *Wilcoxon* 符号秩统计量的矩 证明 [illegible] $\mathbb{E}(W)$ 和 $var(W)$ 的公式。

[illegible] 7.4. *McNemar* 统计量的精确分布 证明在 H_{0F} 下，$m_{10} \sim Binomial(m_{10}+m_{01}, 1/2)$。

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[9] [illegible] **??**

ů 7.5. 方差估计量的偏 证明 { :{ 在 *MPE* 下，

$$\mathbb{E}(\hat{V}) - var(\hat{\tau}) = \{n(n-1)\}^{-1} \sum_{i=1}^{n} (\tau_i - \tau)^2 \geq 0.$$

ů 7.6. *OLS* 解释 证明 { :{ $\hat{\tau}$ 和 $\hat{V}$ 等于将 $(\hat{\tau}_1, \ldots, \hat{\tau}_n)^\top$ 仅对截距进行 *OLS* 拟合的系数和方差估计量。

ů 7.7. 协方差估计量 证明 { :{ $cov(\hat{\tau}_X, \hat{\tau})$ 的无偏估计为

$$\hat{\theta} = \{n(n-1)\}^{-1} \sum_{i=1}^{n} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}).$$

ů 7.8. 调整估计量的 *OLS* 解释 证明 { :{ 协变量调整估计量 $\hat{\tau}_{adj}$ 可以近似等于将 $\hat{\tau}_i$ 对 1 和 $\hat{\tau}_{X,i}$（带截距）进行 *OLS* 拟合的截距。

ů 7.9. *Darwin* 数据的 *Neymanian* 分析 使用 *Neymanian* 方法重新分析 *Darwin* 数据。报告点估计、方差估计和 *95%* 置信区间。

ů 7.10. 儿童电视 *Workshop* 实验的 *FRT* 分析 使用 *FRT* 重新分析儿童电视 *Workshop* 实验数据，使用配对 t 统计量作为检验统计量。

CHAPTER 8

[illegible]

[illegible]

[illegible] Fisher [illegible] FRT [illegible] sharp null [illegible] $H_{0F}: Y_i(1) = Y_i(0)$ [illegible] FRT [illegible] p [illegible]

$$\mathbb{P}(p_{\text{frt}} \le u) \le u \quad \text{for all} \quad 0 \le u \le 1.$$

[illegible] FRT [illegible] H_{0F} [illegible] $H_{0N}: \tau = 0$ [illegible] $\hat{\tau}$ [illegible] FRT [illegible] H_{0N} [illegible] anti-conservative [illegible] dual guarantees [illegible] $t = \hat{\tau}/\sqrt{\hat{V}}$ [illegible] H_{0F} [illegible] H_{0N} [illegible] Stratified Randomized Experiment, SRE [illegible]

[illegible]

[illegible]

[illegible] CRE [illegible] n_1 [illegible] n_0 [illegible] $n = n_1 + n_0$ [illegible]

(1) [illegible]

$$H_{0F}: Y_i(1) = Y_i(0) \qquad i = 1, \ldots, n.$$

(2) [illegible]

$$H_{0N}: \tau = 0 \iff \bar{Y}(1) = \bar{Y}(0).$$

H_{0F} [illegible] FRT [illegible] p [illegible] H_{0N} [illegible] $\hat{\tau}$ [illegible] H_{0F} [illegible] studentized [illegible]

$$t = \frac{\hat{\tau}}{\sqrt{\hat{V}}},$$

[illegible] $\hat{\tau}$ [illegible] $\hat{V}$ [illegible][1]

---

[1] [illegible] ?? [illegible] ??

[illegible]

[illegible] Ding and Dasgupta (2017) [illegible] 在 *CRE* 中，使用 *studentized* 统计量 t 的 *FRT* 具有以下双重保证：

(1) H_{0F} ：p_{frt} 是有限样本精确的；
(2) H_{0N} ：p_{frt} 是渐近保守的（*asymptotically conservative*）。

[illegible] $\hat{\tau}$ [illegible] t [illegible] H_{0N} [illegible]

H_{0N} [illegible] $\hat{\tau}$ [illegible]

$$\hat{\tau} \doteq \mathrm{N}\left(0, \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}\right).$$

[illegible] FRT pretends [illegible] Science Table [illegible] H_{0F} [illegible] $(Y_i, Y_i)_{i=1}^n$ [illegible] $\hat{\tau}$ [illegible]

$$\hat{\tau}^{\pi} \doteq \mathrm{N}\left(0, \frac{s^2}{n_1} + \frac{s^2}{n_0}\right),$$

[illegible] s^2 [illegible] FRT [illegible] H_{0N} [illegible] p

[illegible] studentized [illegible] $t = \hat{\tau}/\sqrt{\hat{V}}$ [illegible] H_{0N} [illegible]

$$t \doteq \mathrm{N}(0, C^2),$$

[illegible] $C^2 \leq 1$ [illegible] $Y_i(1) - Y_i(0) = \tau$ [illegible] constant treatment effect [illegible]

[illegible] FRT [illegible]

$$t^{\pi} \doteq \mathrm{N}(0, 1).$$

[illegible] C^2 FRT [illegible] 1 [illegible] p [illegible]

[illegible] studentization [illegible] FRT [illegible]

8.1. [illegible] 证明了这个双重保证是 *studentized* 统计量独有的性质。其他统计量（如原始 $\hat{\tau}$）的 *FRT* 在 H_{0N} 下可能不再控制第一类错误率。

[illegible]

[illegible] SRE [illegible] SRE [illegible] K [illegible] strata [illegible] k [illegible] $\tau_{[k]}$ [illegible] $\tau = \sum_{k=1}^K \pi_{[k]} \tau_{[k]}$ [illegible]

$$\hat{\tau}_{\mathrm{SRE}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]},$$

$$\hat{V}_{\mathrm{SRE}} = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}^2_{[k]}(1)}{n_{[k]1}} + \frac{\hat{S}^2_{[k]}(0)}{n_{[k]0}}\right).$$

[illegible] SRE [illegible] studentized [illegible]

$$t_{\mathrm{SRE}} = \frac{\hat{\tau}_{\mathrm{SRE}}}{\sqrt{\hat{V}_{\mathrm{SRE}}}}.$$

[illegible] **(SRE FRT [illegible])** 在 *SRE* 中，使用 *studentized* 统计量 t_{SRE} 的 *FRT* 具有以下双重保证：

(1) H_{0F} ：p_{frt} 是有限样本精确的；
(2) H_{0N} ：p_{frt} 是渐近保守的。

SRE FRT [illegible] CRE [illegible] X [illegible]

[illegible]

[illegible] [illegible]

[illegible] studentized Lin [illegible]

$$t_{\mathrm{L}} = \frac{\hat{\tau}_{\mathrm{L}}}{\sqrt{\hat{V}_{\mathrm{L}}}},$$

[illegible] $\hat{\tau}_{\mathrm{L}}$ [illegible] $\hat{V}_{\mathrm{L}}$ [illegible] OLS [illegible] t_{L} [illegible] Y_i on $(1, Z_i, X_i, Z_iX_i)$ [illegible] Z_i [illegible] studentized [illegible]

[illegible] **(Zhao and Ding (2021a) [illegible])** 在 *CRE* 中，使用 *studentized Lin* 统计量 t_{L} 的 *FRT* 具有以下多重保证：

(1) H_{0F} ：p_{frt} 是有限样本精确的；
(2) H_{0N} ：p_{frt} 是渐近保守的；
(3) [illegible] H_{0N} [illegible] ：p_{frt} 渐近地比使用 t 的 *FRT* 更高效（功效更高）；
(4) [illegible]。

[illegible] $\hat{\tau}_{\mathrm{L}}$ [illegible] FRT with t_{L} [illegible] t_{L} [illegible] FRT [illegible] FRT [illegible] H_{0N} [illegible] t_{L}

[illegible] SRE [illegible]

$$t_{\mathrm{L,SRE}} = \frac{\hat{\tau}_{\mathrm{L,SRE}}}{\sqrt{\hat{V}_{\mathrm{L,SRE}}}}.$$

[illegible] Lin [illegible]

8X Eùî

\ „ \ YV E £ bim/2MidB9x2¥/µK"'ŸÉbáÌ Nº BñÅ˜!
Âb

InBñ c $n = 100$ ñ†í¥µK98 Í F $n_1 = 20$ vF $n_0 = 80$ b Äñ
†í i 3îxM X_i lMB(7;1) . ²T¹

$$Y_i(1) \quad L(X_i^3; 1); \quad Y_i(0) \quad L(\ X_i^3; 0{:}5^2):$$

$Y_i(1)$ „ $Y_i(0)$ É›Ï•Ä[' =0 '¡@ H_{0N} b
E1 RkÕ_ d9

URLV2vK M URN¥kØVÕ PGaLC Í F"”a ¿ÜÅS ¥ t d9 a
¿ 1>qS ¥ t d9
UkÀV²T¦{ \ n5 Y_i $(1; X_i)$ —µÄ –ª‹¨ •ØÕd9
UjV6Bb?2` UR¥NkP8GVaLC n» \\c
U9GVBM Uky¥RPjVGaLC n» \\c KÔØÕ bim/2MidB9x2b/

²T 3XXR H_{0N} / ºµ¹Õ bim/2MidB9x2/ GB¥MØÕ„ 1>q¥BÕ ?
û l»BËp q OL=p qÊ® ob ÅÖÕd9 ¥L=p q(Ú¿
l£Ü yN,a†¨¿_ Ð,L!b

ƒñ E²TÐ \\¥Ø,ç©›†BÁb

eXLlyp

¿ •Ø,„ E²T áÌó [/Llyp

eXRX *¥1yp X

URÂVTº1• H_{0F} '^Ži ©…)Ør‹ P¨©…_ d9 ¥ 6_hûó
µK"'Ú' T´bV[Ê4ÿr Ú¥d9 Â ^b
UkÂVT1• H_{0N} 'Ü(yTr‹^Ž¹, A¶P¨ bim/2MiBdx92/ w
iP¨ t t_Gb
UjVÂTµxM V¨ P¨ t_G YÈ1 t ÷Úrb

eXkX a¥1yp X

URíVxM P¨ t_{a_1}
UkµVxM P¨ t_{G,a_1}b

eXjXÛ Ä _2J ¥+Yªü X ¿P¨ J ? H MQ@ÕEÉb×Û Ä¥L
_2J \ ?CP¨ðS t d9 ¥ 6_h, µ \\Ï¥ × £bŒP¨ t_G ¥ 6_h
– ûƒt £bƒjA xM Øœ„ bim/2MiBx iB_Q2MJ Ï¥×1Ÿ b

eX9ÄX¥ L JS1 ¥yp X ¿Ä¥ L JS1 ypÜ›¿ a_1¥yp
íxM P¨ t d9 PGaE† $^\wedge_i$ Q1Mª¥© [
µxM P¨ t d9 PGaE† $^\wedge_i$ Q1M„ $^\wedge_{X;i}$ ª¥© [b

eX8,X², XP¨ bim/2MiBdx92/¥ 6_h ^÷½†¥Ê4b 'v"'žÿÍ
» 'H ¿ LQ`KÍH»„ ¿ 6_h¥ T´+¬M]b'žÿÍ», 'H 6_h
À ? £ b? `T MLm/H HT´^µr¥bƒ^'€¥wiSEb

[illegible]

[illegible] " [illegible] " vs [illegible] " [illegible] " vs [illegible]$_1$ [illegible]

[illegible]

- **I** ~ **P** [illegible] t [illegible] t_{L} [illegible] FRT[illegible]
- [illegible] ~ **P** [illegible] t_{SRE} [illegible] $t_{\mathrm{L,SRE}}$ [illegible] FRT

[illegible] soccer vs control [illegible] physician vs control [illegible]

Table 1. [illegible] soccer vs control

	est	s.e.	p_{Normal}	p_{frt}
class 1 (N)	0.051	0.502	0.919	0.924
class 1 (L)	0.050	0.489	0.919	0.929
class 2 (N)	−0.158	0.451	0.726	0.722
class 2 (L)	−0.176	0.452	0.698	0.700
class 3 (N)	0.005	0.403	0.990	0.989
class 3 (L)	−0.096	0.385	0.803	0.806
class 4 (N)	−0.492	0.447	0.271	0.288
class 4 (L)	−0.511	0.447	0.253	0.283
class 5 (N)	0.390	0.369	0.291	0.314
class 5 (L)	0.443	0.318	0.164	0.186
all (N)	−0.051	0.204	0.802	0.800
all (L)	−0.074	0.200	0.712	0.712

N[illegible] Neyman (1923) [illegible] L[illegible] Lin (2013) [illegible]

Table 2. [illegible] physician vs control

	est	s.e.	p_{Normal}	p_{frt}
class 1 (N)	0.567	0.426	0.183	0.192
class 1 (L)	0.588	0.418	0.160	0.174
class 2 (N)	0.193	0.438	0.659	0.666
class 2 (L)	0.265	0.409	0.517	0.523
class 3 (N)	1.305	0.494	0.008	0.012
class 3 (L)	1.501	0.462	0.001	0.003
class 4 (N)	−0.273	0.413	0.508	0.515
class 4 (L)	−0.313	0.417	0.454	0.462
class 5 (N)	−0.050	0.379	0.895	0.912
class 5 (L)	−0.067	0.279	0.811	0.816
all (N)	0.406	0.202	0.045	0.047
all (L)	0.463	0.190	0.015	0.017

N[illegible] Neyman (1923) [illegible] L[illegible] Lin (2013) [illegible]

[illegible]

(1) [illegible] t_{L} [illegible] t [illegible]

(2) [illegible] FRT p [illegible] p [illegible]

(3) [illegible] soccer [illegible] physician [illegible] $p_{\mathrm{frt}} \approx 0.017$ [illegible] class 3 [illegible] $p_{\mathrm{frt}} \approx 0.003$ [illegible]

(4) [illegible] class 2 [illegible] class 4 [illegible]

[illegible] studentized [illegible] N(0, 1) [illegible]

Figure 1. [illegible] N(0, 1) [illegible] N(0, 1) [illegible]

[illegible]

[illegible] Fisher [illegible]

(1) [illegible] studentized [illegible] $t = \hat{\tau}/\sqrt{\hat{V}}$ [illegible] FRT [illegible] H_{0F} [illegible] H_{0N} [illegible]

(2) [illegible] $\hat{\tau}$ [illegible] FRT [illegible] H_{0N} [illegible] FRT [illegible] Studentization [illegible]

(3) [illegible] studentized Lin [illegible] t_{L} [illegible] FRT [illegible] H_{0F} [illegible] H_{0N} [illegible]

(4) [illegible] SRE [illegible] t_{SRE} [illegible] SRE [illegible] $t_{\mathrm{L,SRE}}$ [illegible]

(5) [illegible] studentized [illegible] FRT [illegible] p [illegible] sharp null [illegible]

[illegible]

[illegible] 8.1. *8.1 —Angrist and Lavy (2009) 数据重新分析 这是第 7 章 Problem 7.8 的 Fisher counterpart。报告使用 studentized 统计量的 FRT 所得到的* p_{frt}。

[illegible] 8.2. *8.2 —Zhao and Ding (2021a) Figure 1 的复制 Zhao and Ding (2021a) 使用模拟来评估各种检验统计量的 FRT 的有限样本性质。基于他们的 Figure 1，他们推荐使用* $t_{\mathrm{L},SRE}$ *来分析 SRE。复制他们的 Figure 1。*

[illegible] 8.3. *8.3 —Studentization 的直觉 解释为什么 studentization 能够修复* $\hat{\tau}$ *在* H_{0N} *下的反保守问题。从方差结构的角度阐述你的理由。*

[illegible] 8.4. *8.4 —推导* t *在* H_{0N} *下的渐近分布 在 CRE 中，推导 studentized 统计量* $t = \hat{\tau}/\sqrt{\hat{V}}$ *在* H_{0N} *下的渐近分布。证明其渐近方差不超过 1。*

[illegible] 8.5. *8.5 —Lin 估计量的性质 描述 Lin (2013) 协变量调整估计量* $\hat{\tau}_{\mathrm{L}}$ *的构造。证明它在线性结果模型正确设定时是高效的。*

Ǘ 8.6. *8.6* —推荐阅读 { 和 { 在 *permutation tests* 中使用了 *studentized* 统计量。在设计-*based* 框架下，{ 和 { 给出了 *multi-armed experiments* 的推荐，{ 将推荐推广到协变量调整情形。阅读这些文献并撰写一页摘要。

[illegible] $\hat{\tau}$ [illegible] $\mathrm{var}(\hat{\tau} \mid \mathbf{Z})$ [illegible] X [illegible]

[illegible]

[illegible] ANCOVA [illegible]

$$Y(1) = \gamma_1 + \beta_1^T X + \varepsilon(1), \qquad E\{\varepsilon(1)\} = 0,$$

$$Y(0) = \gamma_0 + \beta_0^T X + \varepsilon(0), \qquad E\{\varepsilon(0)\} = 0.$$

[illegible]

$$\tau = E\{Y(1) - Y(0)\} = (\gamma_1 - \gamma_0) + (\beta_1 - \beta_0)^T E(X).$$

[illegible] $E(X)$ [illegible] $\varepsilon(1)$ [illegible] $\varepsilon(0)$ [illegible] τ [illegible]

[illegible] $\hat{\gamma}_1, \hat{\beta}_1, \hat{\gamma}_0, \hat{\beta}_0$ [illegible] OLS [illegible] τ [illegible]

$$\hat{\tau}_{\mathrm{adj}} = \hat{\gamma}_1 - \hat{\gamma}_0 + (\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X}.$$

[illegible] $\bar{X} = 0$ [illegible]

$$\hat{\tau}_L = \hat{\gamma}_1 - \hat{\gamma}_0.$$

[illegible] OLS [illegible]

$$Y = \alpha_0 + \alpha_Z Z + \alpha_X^T X + \alpha_{ZX}^T (Z \cdot X) + \varepsilon,$$

[illegible] $\hat{\tau}_L = \hat{\alpha}_Z$[2] [illegible] $\hat{\tau}_L$ [illegible] τ [illegible]

[illegible] OLS [illegible] $\hat{\tau}_L$ [illegible] OLS [illegible] EHW [illegible] $\bar{X}$ [illegible] OLS [illegible][3] [illegible] EHW [illegible]

$$\hat{V}_L = \hat{V}_{\mathrm{EHW}} + \frac{1}{n} (\hat{\beta}_1 - \hat{\beta}_0)^T S_X^2 (\hat{\beta}_1 - \hat{\beta}_0),$$

[illegible] - $\hat{V}_{\mathrm{EHW}}$ [illegible] OLS [illegible] EHW [illegible] - S_X^2 [illegible] X [illegible] - $(\hat{\beta}_1 - \hat{\beta}_0)$ [illegible] $Z \cdot X$ [illegible] 95% [illegible][4]

---

[2] [illegible] OLS [illegible] ??

[3] [illegible] ??

[4] [illegible] ?? [illegible]

~[illegible]

[illegible] $n = 500$[illegible] 2000 [illegible] 0 [5] [illegible]

(1) **f a** [illegible] -0.0001[illegible]
(2) **P]** [illegible] EHW [illegible] 0.139[illegible] 0.151[illegible] 95% [illegible] 92.7%[illegible]
(3) [illegible] 0.153[illegible] 95% [illegible] 95.3%

[illegible]

Xw 、[illegible]**P**

[illegible] SRE[illegible] $X_i \in \{1, \dots, K\}$ [illegible]

[illegible] 9.2 (SRE [illegible]). 在超级总体框架下，分层随机实验满足：

$$Z \perp \{Y(1), Y(0)\} \mid X.$$

[illegible] $X = k$ [illegible]

$$\tau_{[k]} = E\{Y(1) - Y(0) \mid X = k\} = E(Y \mid Z = 1, X = k) - E(Y \mid Z = 0, X = k).$$

[illegible]

$$\tau = \sum_{k=1}^{K} \mathrm{pr}(X = k)\tau_{[k]}.$$

SRE [illegible] CRE[illegible] CRE [illegible] Lin (2013) [illegible][6] [illegible] 2 [illegible] $\hat{V}_S$ [illegible] $\mathrm{var}(\hat{\tau}_S)$ [illegible]

1

Ww[illegible]

[illegible]

(1) [illegible] $Z \perp \{Y(1), Y(0), X\}$ [illegible] τ [illegible]
(2) [illegible] X [illegible]
(3) **f a l t** [illegible]**q ~ u** [illegible] OLS [illegible]
(4) [illegible] EHW [illegible]
(5) 、 ~ **P**[illegible]

---

[5] R [illegible] **??**

[6] [illegible] **??**

[illegible]

[illegible]

[illegible] 9.1. *CRE* 的协变量平衡 在超级总体框架下，证明 [illegible] 平均因果效应可以写为 $E(Y \mid Z=1) - E(Y \mid Z=0)$。

[illegible] 9.2. *OLS* 分解与 *pooled regression* 基于 [illegible] 和 [illegible] 证明观测结果的 *OLS* 分解为：

$$Y = \alpha_0 + \alpha_Z Z + \alpha_X^T X + \alpha_{ZX}^T (Z \cdot X) + \varepsilon,$$

其中

$$\alpha_0 = \gamma_0, \quad \alpha_Z = \gamma_1 - \gamma_0, \quad \alpha_X = \beta_0, \quad \alpha_{ZX} = \beta_1 - \beta_0.$$

[illegible] 9.3. *Lin (2013)* 估计量的性质 证明在正则条件下，*Lin (2013)* 估计量 $\hat{\tau}_L$ 是 τ 的半参数有效估计量。

[illegible] 9.4. 方差估计量的比较 在超级总体框架下，模拟 $\{X_i, Z_i, Y_i(1), Y_i(0)\}_{i=1}^n$，其中 $\beta_1 \neq \beta_0$。比较以下四种方差估计量的表现：

(1) 标准的 *EHW* 稳健方差估计量；
(2) 带校正项 [illegible] 的 *EHW* 估计量；
(3) *Bootstrap* 方差估计量（协变量在开始时中心化，但不每次重新中心化）；
(4) *Bootstrap* 方差估计量（协变量在每次 *bootstrap* 样本中重新中心化）。

[illegible] 9.5. *SRE* 中的协变量调整 在 *SRE* 的超级总体框架下，推导 τ 的协变量调整估计量及其方差估计量。

[illegible] 9.6. 协变量调整与后分层的关系 证明：当各层的层别倾向得分 $e_{[k]} = n_{[k]1}/n_{[k]}$ 相等时，协变量调整与后分层给出相同的点估计量。

[illegible] 9.7. 文献阅读 阅读 [illegible] 的原文，总结其核心贡献和主要结论。

[illegible] 9.8. 回归调整的几何解释 从几何角度解释协变量调整：为什么减去 $(\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X}$ 可以提高估计精度？

CHAPTER 10

CRE

Part III

earnings

LaLonde

NHANES 2005-2006

homocysteine

female

age3 ed3 BMI bmi3 pov2

BMI

NHANES 2007-2008

BMI

[illegible]

[illegible] i [illegible] X_i [illegible] $Z_i \in \{0, 1\}$ [illegible] Y_i [illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible]

$$\{X_i, Z_i, Y_i(1), Y_i(0)\}_{i=1}^{n} \overset{\text{iid}}{\sim} \{X, Z, Y(1), Y(0)\}.$$

[illegible]

- [illegible]

$$\tau = \mathbb{E}\{Y(1) - Y(0)\}.$$

- [illegible]

$$\tau_{\mathrm{T}} = \mathbb{E}\{Y(1) - Y(0) \mid Z = 1\}.$$

- [illegible]

$$\tau_{\mathrm{C}} = \mathbb{E}\{Y(1) - Y(0) \mid Z = 0\}.$$

[illegible]

$$\tau_{\mathrm{T}} = \mathbb{E}(Y \mid Z = 1) - \mathbb{E}\{Y(0) \mid Z = 1\}.$$

[illegible] $\mathbb{E}(Y \mid Z = 1)$ [illegible] $\mathbb{E}\{Y(0) \mid Z = 1\}$ [illegible]

[illegible] τ_{C} [illegible]

$$\tau_{\mathrm{C}} = \mathbb{E}\{Y(1) \mid Z = 0\} - \mathbb{E}(Y \mid Z = 0),$$

[illegible] $\mathbb{E}\{Y(1) \mid Z = 0\}$ [illegible]

$$\tau_{\mathrm{PF}} = \mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0).$$

[illegible]

$$\tau_{\mathrm{PF}} - \tau_{\mathrm{T}} = \mathbb{E}\{Y(0) \mid Z = 1\} - \mathbb{E}\{Y(0) \mid Z = 0\},$$

$$\tau_{\mathrm{PF}} - \tau_{\mathrm{C}} = \mathbb{E}\{Y(1) \mid Z = 1\} - \mathbb{E}\{Y(1) \mid Z = 0\}.$$

[illegible] τ_{PF} [illegible] CRE [illegible] $Z \perp \{Y(1), Y(0)\}$ [illegible]

[illegible]

[illegible]

1 [illegible][8]

$$\hat{\tau} = n^{-1}\sum_{i=1}^{n}\left\{\frac{e^{\hat{\beta}_0+\hat{\beta}_z+\hat{\beta}_x X_i}}{1+e^{\hat{\beta}_0+\hat{\beta}_z+\hat{\beta}_x X_i}} - \frac{e^{\hat{\beta}_0+\hat{\beta}_x X_i}}{1+e^{\hat{\beta}_0+\hat{\beta}_x X_i}}\right\}.$$

[illegible] Z [illegible]

[illegible] ” [illegible]

” 5 ” [illegible] ”

[illegible]

[illegible] Part III [illegible] ”Four Pillars” [illegible]

(1) [illegible]

(2) [illegible] $Y(z) \perp Z \mid X$ [illegible] X [illegible]

(3) [illegible]

(4) [illegible] $\mathbb{E}(Y \mid Z, X)$ [illegible]

(5) [illegible]

[illegible]

[illegible]

(1) [illegible] $\tau = \Pr(Z=1)\tau_{\mathrm{T}} + \Pr(Z=0)\tau_{\mathrm{C}}$

(2) [illegible]

$$Y(1) = g_1(X, V_1), \quad Y(0) = g_0(X, V_0), \quad Z = \mathbb{I}\{g(X, V) \geq 0\},$$

[illegible] $(V_1, V_0) \perp V$ [illegible] V [illegible] U [illegible]

(3) [illegible] Y [illegible]

$$\mathbb{E}(Y \mid Z, X) = \frac{e^{\beta_0+\beta_z Z+\beta_x X+\beta_{xz} XZ}}{1+e^{\beta_0+\beta_z Z+\beta_x X+\beta_{xz} XZ}}.$$

[illegible] τ [illegible]

(4) [illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]
- [illegible] g-formula [illegible]
- [illegible]

---

[8] [illegible] **??**

CHAPTER 11

[illegible]

[illegible]

[illegible] randomization [illegible] treatment [illegible]

[illegible] X [illegible]

[illegible]

[illegible] $\Pr(Z = 1 \mid X)$ [illegible] X [illegible]

[illegible] 11.1 ([illegible] (Propensity Score)). 倾向得分定义为给定协变量 X 时接受治疗的概率：

$$e(X) = \Pr(Z = 1 \mid X).$$

[illegible] $[0, 1]$ [illegible] $e(X_i) = 0.8$ [illegible] X_i [illegible] 80% [illegible]

[illegible] $e(X) = \Pr(Z = 1)$ [illegible] randomization [illegible] X [illegible]

[illegible] X [illegible] $e(X)$ [illegible] X [illegible]

[illegible] (Rosenbaum & Rubin, 1983) 在强可忽略性（*strong ignorability*）假设下，如果

$$\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid X,$$

则

$$\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid e(X).$$

[illegible] $e(X)$ [illegible] X [illegible]

¹I¹ƒñ²T×1$ ðSxM X V[^©i»”¥ M-alÆa•âa¯<ll Œ`_¤s e(X) ^Bñ [0;1] uW¥S bYVØœƒñB»M áÌ©N¿Øœ îµðSxM iiƒü^ _Qb2M#„ m_Km#BěM¹ ô†»ý ô¥clb[R] ÐsªÛ L ¥ó" » \\ cÏ áÌ), sªÛ L a_1bÂTáÌ ¿`_¤s e(X) ¥|´ †íÉ›sª *¹ÄBª=¥4©Ÿ” Ð a_1 µM]¥ ² iiy¹óçM]¥`_¤s Í s¥Ð. ²THqÿëb

jX`_¤ssª

jXRX©ÖÛ`_¤s XL!`_¤s e(X) X© OÇ| K ñ,]¥´ $f e_1;:::;e_K g$b ® \\ ÄBª e_k = Í s¥Ð. ²THqÿëbyN áÌµ K ñÿë¥›†Û L *_1 iiƒž^» \\ c),¥ a_1² b
98Ü(yTr‹V[YVsª 9 É› 9

$$\hat{}_{Sa} = \sum_{k=1}^{K} {}_{[k]}\hat{}_{[k]};$$

Ï $_{[k]}$ ^» k ª¥981è $\hat{}_{[k]}$ ^¾ª=¥µ´(´b ^{k} LlÏ¥Ù5 K ¥Ê4³1ƒ™bÂT K þl sªª¥HqÿëŸV?,îë ÂTK þv Äª=¥"' ,@ ¸ ªV?º cÍ F vF¥†íbô \ ¥yp K =5 ¸ Æß/? h"v†sÊµiiƒñ ô* ¥”3 8ô| Q C 'c¥),Ïb

jXkÃ©õÃ`_¤s X L=ffƒÏ `_¤sYÈ^Ã©¥ áÌ³1YVd9 ˜É› 9bKÈn¥ZE^E†Bñ HQ;BbBB†

$$\mathrm{HQ} f_{;} e(X i) g = {}_0 + {}_1X_1 + \quad + {}_pX_p:$$

9¤ž¥`_¤s ê(X) V[| r n ñ,]¥´ ’Äñ†í¥xM û,›† M]Hb¹ ‹¨sªZE áÌ³1 ê(X) É›ÖÛÄbBÕÈ¨¦{^¿ sÊ” É›sª[j]

è RRXR UL>”L1¥a`_¤ssª VXln \\ Ï¥ L> L1a” báÌP ¨ HQ;Bb˜iB9+`_¤s –ª¿sÊ”|"'s¹ K ªb \\ ZU ,] K ´ K =5;10;30 / Í F„ vF`_¤ssƒ¥°Zmb

> 6B;m`2 RX L1a” "Ï 9`_¤s¥°Zm âä¹ vF ç ä¹Í Fb

’`_¤sÃ©OõÃH `_¤ssª 9 ¥µrŸG ¿[/ ñHq
URžV!ç ˜ S(Z =1 j X) ž’!ç P¤ ê(X) ? } ‡L¥`_¤sb
Uk¥Vß Hq Pp2`H T +QM/B0i<Be(X M) < 1 îµ†íîëiiƒ’ Äñ[9] †íûµ¤sÍ „ v¥V?Ÿb

---

[R] w•n¸c \\
[k] W%w•n¸c \\
[j] W%),n¸c \\
[9] ×ßHq¥W%),n¸c \\

稳健性观察：倾向得分分层估计量只需要 $\hat{e}(X)$ 的排序正确，而不需要其精确值——这使得它相对于其他方法更加稳健。这一性质在许多数值例子中得到验证，但其严格量化理论仍在研究中。

4. 逆概率加权估计量

4.1. Horvitz–Thompson 估计量. 基于倾向得分，我们可以构造平均因果效应的估计量。一个直接的想法是使用**逆概率加权（Inverse Probability Weighting, IPW）**：

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{Z_iY_i}{e(X_i)} - \frac{(1-Z_i)Y_i}{1-e(X_i)}\right\}.$$

?? 被称为 **Horvitz–Thompson（HT）估计量**。? 在抽样调查中首次提出这一估计量，? 将其引入因果推断领域。

为什么这个估计量有效？直观上，对于治疗组的单元，我们用 $1/e(X_i)$ 作为权重进行加权——倾向得分越小的单元（即越”不像”会接受治疗的单元），其权重越大。这相当于对这些单元进行”过度代表”，以消除由选择性偏差造成的治疗组和对照组之间的系统性差异。[5]

稳定性问题：当某些单元的倾向得分接近 0 或 1 时，权重 $1/e(X_i)$ 或 $1/(1-e(X_i))$ 会变得非常大，导致估计量不稳定。HT 估计量在有限样本中可能表现极差——某些估计值可能远离其他所有方法的结果。[6]

4.2. 标准化逆概率加权估计量. 为了解决 HT 估计量的不稳定性问题，一个自然的修正是对权重进行**标准化**：

$$\hat{\tau}_{\mathrm{IPW}} = \frac{1}{\hat{\mathbf{1}}_T}\sum_{i=1}^{n}\frac{Z_iY_i}{e(X_i)} - \frac{1}{\hat{\mathbf{1}}_C}\sum_{i=1}^{n}\frac{(1-Z_i)Y_i}{1-e(X_i)},$$

其中

$$\hat{\mathbf{1}}_T = \sum_{i=1}^{n}\frac{Z_i}{e(X_i)}, \quad \hat{\mathbf{1}}_C = \sum_{i=1}^{n}\frac{1-Z_i}{1-e(X_i)}.$$

直观理解：标准化权重的思想是将权重之和调整为 1——即 $\hat{\mathbf{1}}_T$ 和 $\hat{\mathbf{1}}_C$ 分别代表治疗组和对照组的”有效样本量”。标准化后的估计量对常数平移不敏感，而原始 HT 估计量则不具备这一性质。[7]

例 11.2 (NHANES 数据的逆概率加权估计). *延续* **??**，*我们对 NHANES 数据应用不同截断水平的逆概率加权估计量。*

TABLE 1. 不同截断水平下的 IPW 估计量

截断水平	点估计	标准误
$(0,1)$（无截断）	-52.3	18.2
$(0.01, 0.99)$	-8.7	4.3
$(0.05, 0.95)$	-6.2	3.8
$(0.10, 0.90)$	-5.1	3.5

---

[5] HT 估计量的期望推导见附录 ??

[6] HT 估计量的方差分析见附录 ??

[7] 标准化估计量的性质证明见附录 ??

[illegible] HT [illegible] IPW [illegible]

[illegible]

[illegible] 倾向得分 $e(X)$ 满足以下性质：

(1) 条件独立性：$X \perp\!\!\!\perp Z \mid e(X)$；

(2) 加权平衡：对于任何协变量函数 $g(X)$，有

$$\mathbb{E}\left\{\frac{Zg(X)}{e(X)}\right\} = \mathbb{E}\{g(X)\}.$$

[illegible] X [illegible] Z [illegible][8] [illegible] Z [illegible] X [illegible]

[illegible] $\hat{e}(X)$ [illegible] $\hat{e}(X)$ [illegible]

$$\mathrm{Bias}_{[k]} = \frac{1}{n_{[k]}} \sum_{i:\hat{e}\ (X_i)=e_k} Z_i g(X_i) - \frac{1}{n_{[k]}} \sum_{i:\hat{e}\ (X_i)=e_k} (1 - Z_i) g(X_i) \approx 0.$$

[illegible]

[illegible]

Table 2. [illegible] vs. [illegible]

	[illegible]	[illegible]
[illegible]	$\Pr(Z = 1 \mid X)$	[illegible] $\mathbb{E}(Y \mid Z, X)$
[illegible]		[illegible]
[illegible]		[illegible]
[illegible]		[illegible]

---

[8] [illegible] ??

[illegible]

[illegible] $0 < e(X) < 1$ [illegible]

[illegible] $e(X_i) = 1$ [illegible] i [illegible] $Y_i(0)$ [illegible]

[illegible]

(1) [illegible]

(2) [illegible] 0 ⁵ 1 [illegible]

[illegible] 0 ⁵ 1 [illegible]

[illegible]

[illegible]

(1) [illegible] $e(X) = \Pr(Z = 1 \mid X)$ [illegible]

(2) [illegible]

(3) [illegible]

(4) [illegible] Horvitz–Thompson [illegible]

(5) [illegible]

(6) [illegible] $0 < e(X) < 1$ [illegible]

[illegible]

[illegible] 11.1. 倾向得分的降维性质 证明 [illegible] 在强可忽略性假设下，如果 $\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid X$，则 $\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid e(X)$。

提示：利用概率论中的推论——如果 $A \perp\!\!\!\perp B \mid C$，且 D 是 C 的函数，则 $A \perp\!\!\!\perp B \mid D$。

[illegible] 11.2. *HT* 估计量的期望 证明 *HT* 估计量 [illegible] 是总体平均因果效应 τ 的无偏估计。

提示：分别计算 $\mathbb{E}\left(\frac{ZY(1)}{e(X)}\right)$ 和 $\mathbb{E}\left(\frac{(1-Z)Y(0)}{1-e(X)}\right)$，利用 $e(X) = \Pr(Z = 1 \mid X)$。

[illegible] 11.3. 标准化权重的性质 验证 [illegible] 中标准化权重的性质：$\hat{\mathbf{1}}_T$ 和 $\hat{\mathbf{1}}_C$ 在使用真实倾向得分时，其期望均为 n。

注：这也是 [illegible] 中使用特殊记号 $\hat{1}_T$ 和 $\hat{1}_C$ 的原因——这两个量的期望都是 *1*。

[illegible] 11.4. 倾向得分的平衡性质证明 证明 [illegible] 的第二个结论：对于任意协变量函数 $g(X)$，

$$\mathbb{E}\left\{\frac{Zg(X)}{e(X)}\right\} = \mathbb{E}\{g(X)\}.$$

提示：使用 *tower property* 和条件概率的定义。

ǘ 11.5. 协变量平衡的数值检验 使用 { {中的 *NHANES* 数据，检验不同分层数量 K 下的协变量平衡。报告治疗组和对照组在各层内关键协变量（如年龄、*BMI*）均值之差的绝对值。

ǘ 11.6. 截断水平的敏感性分析 对 { {中的结果进行敏感性分析：比较不同截断水平对点估计和标准误的影响，讨论截断水平选择的实际准则。

ǘ 11.7. *IPW* 与结果回归的比较 使用第 { {章中的模拟数据或真实数据，分别应用 *IPW* 估计量和结果回归估计量，比较两者的点估计和置信区间。

ǘ 11.8. 极端反事实的影响 { 讨论了极端反事实问题。构造一个例子，其中某些单元的倾向得分接近 *1*，讨论这种情况下因果推断面临的挑战。

CHAPTER 12

[illegible]

[illegible]

[illegible] outcome regression [illegible]

$$\tau = E\{\mu_1(X)\} - E\{\mu_0(X)\},$$

[illegible] $\mu_1(X) = E(Y \mid Z = 1, X)$ [illegible] $\mu_0(X) = E(Y \mid Z = 0, X)$ [illegible] X [illegible] $\mu_1(X)$ [illegible] $\mu_0(X)$ [illegible] τ [illegible] IPW [illegible]

$$\tau = E\left\{\frac{ZY}{e(X)}\right\} - E\left\{\frac{(1-Z)Y}{1-e(X)}\right\},$$

[illegible] $e(X) = \Pr(Z = 1 \mid X)$ [illegible] $e(X)$ [illegible] James Robins [illegible] doubly robust estimator [illegible]

[illegible]

[illegible] working models [illegible]

- [illegible] $\mu_1(X, \beta_1)$ [illegible] $\mu_0(X, \beta_0)$ [illegible] $E(Y(1) \mid X)$ [illegible] $E(Y(0) \mid X)$ [illegible] $\mu_1(X, \beta_1) = \mu_1(X)$
- [illegible] $e(X, \alpha)$ [illegible] $\Pr(Z = 1 \mid X)$ [illegible] $e(X, \alpha) = e(X)$

[illegible] ” [illegible] ” $_1$ [illegible]

$$\tilde{\mu}_1^{\text{dr}} = E\left[\frac{Z\{Y - \mu_1(X, \beta_1)\}}{e(X, \alpha)} + \mu_1(X, \beta_1)\right], \tag{60}$$

$$\tilde{\mu}_0^{\text{dr}} = E\left[\frac{(1-Z)\{Y - \mu_0(X, \beta_0)\}}{1 - e(X, \alpha)} + \mu_0(X, \beta_0)\right]. \tag{61}$$

[illegible]

$$\tilde{\mu}_1^{\mathrm{dr}} = E\left[\frac{ZY}{e(X,\alpha)} - \frac{Z - e(X,\alpha)}{e(X,\alpha)}\mu_1(X,\beta_1)\right], \tag{62}$$

$$\tilde{\mu}_0^{\mathrm{dr}} = E\left[\frac{(1-Z)Y}{1-e(X,\alpha)} - \frac{e(X,\alpha) - Z}{1-e(X,\alpha)}\mu_0(X,\beta_0)\right]. \tag{63}$$

[illegible]

- [illegible]
- [illegible] IPW [illegible]

[illegible] augmented inverse propensity score weighting, AIPW[illegible]

[illegible]

[illegible]

[illegible] [illegible]假设可忽略性 $Z \perp \{Y(1), Y(0)\} \mid X$ 和 *overlap* $0 < e(X) < 1$。

(1) 若 $e(X,\alpha) = e(X)$ 或 $\mu_1(X,\beta_1) = \mu_1(X)$ 之一成立，则 $\tilde{\mu}_1^{\mathrm{dr}} = E\{Y(1)\}$。
(2) 若 $e(X,\alpha) = e(X)$ 或 $\mu_0(X,\beta_0) = \mu_0(X)$ 之一成立，则 $\tilde{\mu}_0^{\mathrm{dr}} = E\{Y(0)\}$。
(3) 若 $e(X,\alpha) = e(X)$ 或 $\{\mu_1(X,\beta_1) = \mu_1(X), \mu_0(X,\beta_0) = \mu_0(X)\}$ 之一成立，则 $\tilde{\mu}_1^{\mathrm{dr}} - \tilde{\mu}_0^{\mathrm{dr}} = \tau$。

[illegible] $\tilde{\mu}_1^{\mathrm{dr}} - \tilde{\mu}_0^{\mathrm{dr}}$ [illegible] τ [illegible][1]

[illegible]

[illegible] analog [illegible]

[illegible] 12.2 ([illegible]). 基于数据 $(X_i, Z_i, Y_i)_{i=1}^n$，按以下步骤构造 τ 的双重稳健估计量：

(1) 拟合倾向得分：得到 $e(X_i, \hat{\alpha})$；
(2) 拟合结果均值：得到 $\mu_1(X_i, \hat{\beta}_1)$ 和 $\mu_0(X_i, \hat{\beta}_0)$；
(3) 构造双重稳健估计量 $\hat{\tau}^{\mathrm{dr}} = \hat{\mu}_1^{\mathrm{dr}} - \hat{\mu}_0^{\mathrm{dr}}$，其中

$$\hat{\mu}_1^{\mathrm{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_i\{Y_i - \mu_1(X_i,\hat{\beta}_1)\}}{e(X_i,\hat{\alpha})} + \mu_1(X_i,\hat{\beta}_1)\right],$$

$$\hat{\mu}_0^{\mathrm{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1-Z_i)\{Y_i - \mu_0(X_i,\hat{\beta}_0)\}}{1-e(X_i,\hat{\alpha})} + \mu_0(X_i,\hat{\beta}_0)\right].$$

[illegible][2]

$$\hat{\tau}^{\mathrm{dr}} = \hat{\tau}^{\mathrm{reg}} + \frac{1}{n}\sum_{i=1}^{n}\frac{Z_i\{Y_i - \mu_1(X_i,\hat{\beta}_1)\}}{e(X_i,\hat{\alpha})} - \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)\{Y_i - \mu_0(X_i,\hat{\beta}_0)\}}{1-e(X_i,\hat{\alpha})}.$$

---

[1] [illegible] **??** [illegible] **??**
[2] [illegible] **??**

[illegible] $\hat{\tau}^{\text{reg}}$ [illegible][3]

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{ipw}} - \frac{1}{n}\sum_{i=1}^{n}\frac{Z_i - e(X_i,\hat{\alpha})}{e(X_i,\hat{\alpha})}\mu_1(X_i,\hat{\beta}_1) + \frac{1}{n}\sum_{i=1}^{n}\frac{e(X_i,\hat{\alpha}) - Z_i}{1 - e(X_i,\hat{\alpha})}\mu_0(X_i,\hat{\beta}_0).$$

[illegible] bootstrap [illegible] $\hat{\tau}^{\text{dr}}$ [illegible]

[illegible]

[illegible] IPW [illegible]

[illegible] [illegible] **[illegible]** IPW [illegible] $\mu_1 = E\left\{\frac{ZY}{e(X)}\right\}$ [illegible] Y [illegible]

[illegible] $\mu_1(X,\beta_1)$ [illegible][4]

$$\mu_1 = E\{Y(1)\} = E\{Y(1) - \mu_1(X,\beta_1)\} + E\{\mu_1(X,\beta_1)\} = E\left\{\frac{Z\{Y - \mu_1(X,\beta_1)\}}{e(X)} + \mu_1(X,\beta_1)\right\}.$$

[illegible]

[illegible] [illegible] $\tilde{\mu}_1 = E\{\mu_1(X,\beta_1)\}$ [illegible] μ_1 [illegible] $E\{\mu_1(X,\beta_1) - Y(1)\}$ [illegible] IPW [illegible][5]

$$B = E\left\{\frac{Z\{\mu_1(X,\beta_1) - Y\}}{e(X)}\right\},$$

[illegible] $\tilde{\mu}_1 - B$ [illegible]

[illegible]

[illegible][6]

(1) [illegible]
(2) [illegible]
(3) [illegible]
(4) [illegible]

[illegible] 1-3 [illegible] DR [illegible] 4 [illegible] DR [illegible][7] ” [illegible] ” [illegible]

---

[3] [illegible] ??

[4] [illegible] ??

[5] [illegible] ??

[6] [illegible] ??

[7] [illegible] ??

7. 进一步讨论：为何脆弱？

回顾 ?? 的证明，双重稳健性的关键在于以下乘积结构：[8]

$$\tilde{\mu}_1^{\mathrm{dr}} - E\{Y(1)\} = E\left[\frac{e(X) - e(X,\alpha)}{e(X,\alpha)} \times \{\mu_1(X) - \mu_1(X,\beta_1)\}\right].$$

只要 $e(X) = e(X,\alpha)$ 或 $\mu_1(X) = \mu_1(X,\beta_1)$ 之一成立，上式就为零。然而，当两个模型都错误时，两个误差的乘积可能放大偏差——这就是”双重脆弱性”的数学本质。模拟研究证实了这一点。

尽管 ? 基于模拟研究批评了双重稳健估计量，但自 ? 的开创性工作以来，双重稳健估计量一直是因果推断的标准策略。近年来，它在理论统计和计量经济学文献中以更响亮的名字”双机器学习”（double machine learning）重新兴起。? 的基本思想是用更灵活的机器学习工具替代传统的参数工作模型。

8. 本章小结

本章介绍了双重稳健估计量，主要内容如下：

(1) **核心思想**：结合结果回归和 IPW，只要其中一个模型正确就能保证一致性。
(2) **公式结构（??–??）**：增强逆概率加权估计量（AIPW）。
(3) **双重稳健性（??）**：只要倾向得分模型或结果模型之一正确，$\tilde{\mu}_1^{\mathrm{dr}} - \tilde{\mu}_0^{\mathrm{dr}} = \tau$。
(4) **两种直觉视角**：从 IPW 降方差（??）和从结果回归降偏倚（??）。
(5) **双重脆弱性**：当两个模型都错误时，两个误差的乘积可能放大偏差。

9. 习题

练习 12.1. *验证双重稳健性 考虑协变量离散的情况* $X \in \{1, \ldots, K\}$*。证明在这种情况下，分层估计量、结果回归估计量、Horvitz-Thompson 估计量、Hajek 估计量和双重稳健估计量在数值上是完全相同的。(参见 ?, Theorem 12.1)*

练习 12.2. *DR 估计量的另一种形式 基于 ??，给出* $\mu_1 = E\{Y(1)\}$ *的双重稳健估计量的另一种形式：*

$$\tilde{\mu}_1^{\mathrm{dr2}} = \frac{E\left[\frac{Z\{Y - \mu_1(X,\beta_1)\}}{e(X,\alpha)}\right]}{E\left[\frac{Z}{e(X,\alpha)}\right]} + E\{\mu_1(X,\beta_1)\}.$$

证明若 $e(X,\alpha) = e(X)$ *或* $\mu_1(X,\beta_1) = \mu_1(X)$ *之一成立，则* $\tilde{\mu}_1^{\mathrm{dr2}} = \mu_1$*。给出估计* μ_0 *的类似公式，并给出基于这些公式的* τ *的双重稳健估计量的样本 analog。(参见 ?, 式 (12.7))*

练习 12.3. *DR 估计量的方差 证明 ?? 和 ?? 的等价性。(参见 ?, 式 (12.9))*

---

[8] 详细推导见附录 ??

[illegible]

[illegible] ignorability [illegible] overlap [illegible] $\tilde{\mu}_1^{\mathrm{dr}} = E\{Y(1)\}$ [illegible] [9]

$$
\begin{aligned}
\tilde{\mu}_1^{\mathrm{dr}} - E\{Y(1)\} &= E\left[\frac{Z\{Y(1)-\mu_1(X,\beta_1)\}}{e(X,\alpha)} - \{Y(1)-\mu_1(X,\beta_1)\}\right] \\
&= E\left[\frac{Z-e(X,\alpha)}{e(X,\alpha)}\{Y(1)-\mu_1(X,\beta_1)\}\right] \\
&= E\left(E\left[\frac{Z-e(X,\alpha)}{e(X,\alpha)}\{Y(1)-\mu_1(X,\beta_1)\} \mid X\right]\right) \quad \text{(tower property)} \\
&= E\left[E\left\{\frac{Z-e(X,\alpha)}{e(X,\alpha)} \mid X\right\} \times E\left\{Y(1)-\mu_1(X,\beta_1) \mid X\right\}\right] \quad \text{(ignorability)} \\
&= E\left[\frac{e(X)-e(X,\alpha)}{e(X,\alpha)} \times \{\mu_1(X)-\mu_1(X,\beta_1)\}\right].
\end{aligned}
$$

[illegible] $e(X) = e(X,\alpha)$ [illegible] $\mu_1(X) = \mu_1(X,\beta_1)$ [illegible] □

[illegible]

[illegible]

$$
\hat{\tau}^{\mathrm{dr}} = \hat{\mu}_1^{\mathrm{dr}} - \hat{\mu}_0^{\mathrm{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\mu_1(X_i,\hat{\beta}_1) - \mu_0(X_i,\hat{\beta}_0)\right] + \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_i\{Y_i-\mu_1(X_i,\hat{\beta}_1)\}}{e(X_i,\hat{\alpha})} - \frac{(1-Z_i)\{Y_i-\mu_0(X_i,\hat{\beta}_0)\}}{1-e(X_i,\hat{\alpha})}\right]
$$

[illegible] $\hat{\tau}^{\mathrm{reg}}$ [illegible]

[illegible]

$$
\frac{Z\{Y-\mu_1(X,\beta_1)\}}{e(X,\alpha)} = \frac{ZY}{e(X,\alpha)} - \frac{Z}{e(X,\alpha)}\mu_1(X,\beta_1),
$$

[illegible]

$$
\frac{(1-Z)\{Y-\mu_0(X,\beta_0)\}}{1-e(X,\alpha)} = \frac{(1-Z)Y}{1-e(X,\alpha)} - \frac{1-Z}{1-e(X,\alpha)}\mu_0(X,\beta_0),
$$

[illegible]

[illegible] **M** [illegible]

[illegible]

$$
\mu_1 = E\{Y(1)\} = E\{Y(1)-\mu_1(X,\beta_1)\} + E\{\mu_1(X,\beta_1)\}
$$

[illegible] IPW [illegible] $Y(1)-\mu_1(X,\beta_1)$ [illegible]

$$
E\{Y(1)-\mu_1(X,\beta_1)\} = E\left\{\frac{Z\{Y-\mu_1(X,\beta_1)\}}{e(X)}\right\},
$$

[illegible]

$$
\mu_1 = E\left\{\frac{Z\{Y-\mu_1(X,\beta_1)\}}{e(X)}\right\} + E\{\mu_1(X,\beta_1)\}.
$$

---

[9] [illegible] (**?**, Theorem 12.1)

[illegible]

[illegible] $\tilde{\mu}_1 = E\{\mu_1(X, \beta_1)\}$ [illegible]

$$\text{bias} = E\{\mu_1(X, \beta_1) - Y(1)\}.$$

[illegible] IPW [illegible]

$$B = E\left\{\frac{Z\{\mu_1(X, \beta_1) - Y\}}{e(X)}\right\}.$$

[illegible] $\tilde{\mu}_1 - B$ [illegible]

[illegible]

[illegible]

$$\tilde{\mu}_1^{\mathrm{dr}} - E\{Y(1)\} = E\left[\frac{e(X) - e(X, \alpha)}{e(X, \alpha)} \times \{\mu_1(X) - \mu_1(X, \beta_1)\}\right].$$

[illegible] $e(X) - e(X, \alpha)$ [illegible] $\mu_1(X) - \mu_1(X, \beta_1)$ [illegible]

[illegible]

[illegible] $n = 500$ [illegible] 500 [illegible]

[illegible]

```
x = matrix(rnorm(n*2), n, 2)
x1 = cbind(1, x)
beta.z = c(0, 1, 1)
pscore = 1/(1 + exp(- as.vector(x1 %*% beta.z)))
z = rbinom(n, 1, pscore)
beta.y1 = c(1, 2, 1)
beta.y0 = c(1, 2, 1)
y1 = rnorm(n, x1 %*% beta.y1)
y0 = rnorm(n, x1 %*% beta.y0)
y = z*y1 + (1-z)*y0
```

[illegible]

[illegible]

[illegible]

[illegible] 4 [illegible]

[illegible]

Table 1. [illegible] $n = 500$[illegible] 500 [illegible]

		reg	HT	Hajek	DR
[illegible]	1[illegible]				
[illegible]		0.00	0.02	0.03	0.01
[illegible]		0.11	0.28	0.26	0.13
[illegible]		0.10	0.25	0.23	0.12
[illegible]	2[illegible]				
[illegible]		0.00	−0.76	−0.75	−0.01
[illegible]		0.12	0.59	0.47	0.18
[illegible]		0.13	0.50	0.38	0.18
[illegible]	3[illegible]				
[illegible]		−0.05	0.00	−0.01	0.00
[illegible]		0.11	0.15	0.14	0.14
[illegible]		0.11	0.14	0.13	0.14
[illegible]	4[illegible]				
[illegible]		−0.08	0.11	−0.07	0.16
[illegible]		0.13	0.32	0.20	0.41
[illegible]		0.13	0.25	0.16	0.26

CHAPTER 13

[1]

##

"Block what you can and randomize what you cannot." — (page 103)

SRE

SRE X

treatment control

$e(X) = \Pr(Z = 1)$

$e(X) = \Pr(Z = 1 \mid X)$ X X

X X

X curse of dimensionality

SRE

confounding

$e_{[k]} = n_{[k]1}/n_{[k]}$

13.1 ( ). 倾向得分定义为给定协变量条件下治疗分配的条件概率：

$$e(X) = \Pr(Z = 1 \mid X).$$

$[0, 1]$

X ignorability X

( )如果 $Z \perp \{Y(1), Y(0)\} \mid X$，则有

$$Z \perp \{Y(1), Y(0)\} \mid e(X).$$

ignorability X

$e(X)$

X X $e(X)$

[1]

---

[1] **??**

$\Pr\{Z=1 \mid Y(1),Y(0),e(X)\} = \Pr\{Z=1 \mid e(X)\}$ tower property ignorability

$$\Pr\{Z=1 \mid Y(1),Y(0),e(X)\} = E\{Z \mid Y(1),Y(0),e(X)\} = E\big[E\{Z \mid Y(1),Y(0),X\} \mid Y(1),Y(0),e(X)\big].$$

ignorability $E\{Z \mid Y(1),Y(0),X\} = E\{Z \mid X\} = e(X)$

$$\Pr\{Z=1 \mid Y(1),Y(0),e(X)\} = E\{e(X) \mid Y(1),Y(0),e(X)\} = e(X).$$

$\Pr\{Z=1 \mid e(X)\} = E\{Z \mid e(X)\} = E\{e(X) \mid e(X)\} = e(X)$

Rosenbaum and Rubin 1983 *Biometrika*

Biometrika

ignorability X $e(X)$

$e(X)$ K $\{e_1,\ldots,e_K\}$ K n

$$Z \perp \{Y(1),Y(0)\} \mid e(X)=e_k \quad (k=1,\ldots,K).$$

k K CRE SRE

$$\tau_{[k]} = E(Y \mid Z=1, e(X)=e_k) - E(Y \mid Z=0, e(X)=e_k),$$

$$\tau = \sum_{k=1}^{K} \pi_{[k]}\tau_{[k]},$$

$\pi_{[k]} = \Pr\{e(X)=e_k\}$ k

$e(X) = \Pr(Z=1 \mid X)$ X logistic $\hat{e}(X)$ n

$$\hat{e}'(X_i) = e_k, \quad \hat{e}(X_i) \quad (k-1)/K \quad k/K \quad .$$

$$Z \perp \{Y(1),Y(0)\} \mid \hat{e}'(X)=e_k \quad (k=1,\ldots,K).$$

[illegible] ignorability [illegible] $\hat{e}'(X)$ [illegible] { {[illegible] SRE [illegible]

[illegible]

[illegible] K [illegible]

[illegible] K [illegible]

- [illegible] K [illegible] $Z \perp \{Y(1), Y(0)\} \mid \hat{e}'(X)$ [illegible]
- [illegible] K [illegible]

Rosenbaum and Rubin[illegible] 1983, 1984[illegible] Cochran[illegible] 1968[illegible] $K = 5$ [illegible] $K = 5$ [illegible]

[illegible] $K = 5$ [illegible] { [illegible] K [illegible] { [illegible] K [illegible]

[illegible]

[illegible] { {[illegible] SRE[illegible] SRE [illegible]

(1) [illegible] SRE [illegible] X [illegible] CRE[illegible] $e(X)$ [illegible] ” [illegible] ” [illegible]

(2) [illegible] SRE [illegible] $e_{[k]} = n_{[k]1}/n_{[k]}$ [illegible] e_k [illegible]

(3) [illegible]

[illegible] SRE [illegible] Neyman [illegible] Fisher [illegible]

[illegible]

[illegible] $\hat{e}'(X)$ [illegible] SRE [illegible]

[illegible] **B p p** [illegible]

[illegible]

[illegible]

[illegible] NHANES [illegible]

[illegible] Schoolmeal [illegible] BMI [illegible]

[illegible] $Z_i = 1$ [illegible] $Z_i = 0$ [illegible]

[illegible] BMI [illegible]

[illegible] logistic [illegible]

$$\hat{e}(X_i) = \mathrm{logit}^{-1}(\hat{\alpha}_0 + \hat{\alpha}_1 X_{i1} + \cdots + \hat{\alpha}_p X_{ip}).$$

[illegible] $K = 5, 10, 30$ [illegible]

FIGURE 1. NHANES [illegible]

[illegible] K [illegible] $K \in \{5, 10, 20, 50, 80\}$ [illegible][2]

TABLE 1. [illegible] K [illegible]

	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 80$
[illegible]	-0.116	-0.178	-0.200	-0.265	$-$
[illegible]	0.283	0.282	0.279	0.272	NA

[illegible]

(1) [illegible] K [illegible] 5 [illegible] 50 [illegible] -0.2[illegible]

(2) [illegible] $K = 80$ [illegible] NA [illegible]

(3) [illegible] BMI [illegible]

[illegible]

TABLE 2. [illegible]

	naive	Fisher	Lin
[illegible]	0.534	0.061	-0.017
[illegible]	0.225	0.227	0.226

[illegible]

- [illegible] 0.534 [illegible]
- [illegible] OLS [illegible] 0.061 [illegible]
- [illegible] OLS [illegible] -0.017 [illegible]
- [illegible] -0.17 [illegible] -0.27 [illegible] Lin [illegible] K [illegible]

---

[2] [illegible] ??

[illegible] SRE [illegible]

[illegible]

[illegible]

(1) [illegible] ignorability [illegible] $e(X)$ [illegible] X [illegible]

(2) [illegible] K [illegible] ” [illegible] ” [illegible]

(3) [illegible] SRE [illegible]

(4) K [illegible] $K = 5$ [illegible]

(5) [illegible]

[illegible] doubly robust [illegible]

CHAPTER 14

[illegible]

[illegible]

[illegible] IPW[illegible] Hajek [illegible]

$$\hat{\tau}^{\text{hajek}} = \frac{\sum_{i=1}^{n} \frac{Z_i Y_i}{\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{Z_i}{\hat{e}(X_i)}} - \frac{\sum_{i=1}^{n} \frac{(1-Z_i)Y_i}{1-\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{1-Z_i}{1-\hat{e}(X_i)}} \tag{64}$$

[illegible] Imbens, 2004[illegible] { [illegible] WLS[illegible] $\hat{\tau}^{\text{hajek}}$ [illegible] WLS [illegible] Z_i [illegible]

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} w_i (Y_i - \alpha - \beta Z_i)^2$$

[illegible]

$$w_i = \frac{Z_i}{\hat{e}(X_i)} + \frac{1-Z_i}{1-\hat{e}(X_i)} = \begin{cases} \frac{1}{\hat{e}(X_i)} & \text{if } Z_i = 1; \\ \frac{1}{1-\hat{e}(X_i)} & \text{if } Z_i = 0. \end{cases} \tag{65}$$

[illegible] IPW [illegible]

[illegible]

[illegible]

[illegible] 14.1 [illegible] Hajek [illegible] WLS)[illegible]在 [illegible]的记号下，$\hat{\tau}^{hajek}$ 严格等于上述 *WLS* 回归中 Z_i 的系数 $\hat{\beta}$。

[illegible] CRE[illegible] $e(X_i) \equiv p$ [illegible] w_i [illegible] WLS [illegible] OLS [illegible] CRE OLS [illegible] τ [illegible] $1/\hat{e}(X_i)$ [illegible] $1/\{1-\hat{e}(X_i)\}$ [illegible] " [illegible] " [illegible] Hajek [illegible] OLS [illegible] treatment [illegible] { [illegible] WLS [illegible] $\hat{\tau}^{\text{hajek}}$ [illegible] WLS [illegible] $\hat{\tau}^{\text{hajek}}$ [illegible] Bootstrap [illegible][1]

[illegible]

---

[1] [illegible] **??**

∼ [illegible] f a l t ‖ [illegible] q ∼ u

CRE [illegible] Lin (2013) [illegible] OLS [illegible] treatment [illegible]

$$Y_i \sim (1, Z_i, X_i, Z_i X_i)$$

[illegible] $\bar{X} = 0$ [illegible] OLS [illegible] WLS [illegible] { [illegible]
1 $\hat{\tau}$ [illegible] WLS [illegible]

$$Y_i \sim (1, Z_i, X_i, Z_i X_i)$$

Z_i [illegible] Hirano and Imbens (2001) [illegible]

∼ [illegible] [illegible]

$$\mathbb{E}(Y \mid Z = 1, X) = \beta_{10} + \beta_{1x}^{\top} X, \quad \mathbb{E}(Y \mid Z = 0, X) = \beta_{00} + \beta_{0x}^{\top} X$$

[illegible]

(1) [illegible] WLS [illegible] OLS [illegible] τ [illegible] Z [illegible] τ

(2) [illegible] WLS Z [illegible] τ [illegible]

[illegible] N p ‘ B g Q p B ‘ 。 — l Q 。 。

[illegible] Robins et al. (2007) [illegible] M. Joffe [illegible]

∼ [illegible] $\hat{e}(X_i, \hat{\alpha})$ [illegible] $(\mu_1(X_i, \hat{\beta}_1), \mu_0(X_i, \hat{\beta}_0))$
1 WLS [illegible]

$$\hat{\tau}_{\text{wls}}^{\text{reg}} = \frac{1}{n} \sum_{i=1}^{n} \mu_1(X_i, \hat{\beta}_1) - \frac{1}{n} \sum_{i=1}^{n} \mu_0(X_i, \hat{\beta}_0)$$

[illegible]

$$\hat{\tau}_{\text{wls}}^{\text{dr}} = \hat{\tau}_{\text{wls}}^{\text{reg}} + \frac{1}{n} \sum_{i=1}^{n} \frac{Z_i\{Y_i - \mu_1(X_i, \hat{\beta}_1)\}}{\hat{e}(X_i, \hat{\alpha})} - \frac{1}{n} \sum_{i=1}^{n} \frac{(1 - Z_i)\{Y_i - \mu_0(X_i, \hat{\beta}_0)\}}{1 - \hat{e}(X_i, \hat{\alpha})}$$

[illegible] WLS [illegible] Z_i [illegible]

[illegible] q X w [illegible] 14.2 [illegible]) 若 $\bar{X} = 0$ 且

$$(\mu_1(X_i, \hat{\beta}_1), \mu_0(X_i, \hat{\beta}_0)) = (\hat{\beta}_{10} + \hat{\beta}_{1x}^{\top} X_i,\ \hat{\beta}_{00} + \hat{\beta}_{0x}^{\top} X_i)$$

基于权重 { 对 $Y_i \sim (1, Z_i, X_i, Z_i X_i)$ 的 WLS 拟合，则

$$\hat{\tau}_{wls}^{dr} = \hat{\tau}_{wls}^{reg} = \hat{\beta}_{10} - \hat{\beta}_{00}$$

即 Z_i 在 WLS 回归中的系数。

[illegible] $\hat{s}$[2]
WLS [illegible]

$$\sum_{i=1}^{n} \frac{Z_i(Y_i - \hat{\beta}_{10} - \hat{\beta}_{1x}^{\top} X_i)}{\hat{e}(X_i)} = 0, \quad \sum_{i=1}^{n} \frac{(1 - Z_i)(Y_i - \hat{\beta}_{00} - \hat{\beta}_{0x}^{\top} X_i)}{1 - \hat{e}(X_i)} = 0.$$

[illegible] $\hat{\tau}^{\mathrm{dr}}_{\mathrm{wls}} - \hat{\tau}^{\mathrm{reg}}_{\mathrm{wls}} = 0$ [illegible] $\bar{X} =$ [illegible] $\hat{\beta}_{10} - \hat{\beta}_{00}$

[illegible] Freedman and Berk (2008)[1] [illegible]

[illegible] WLS [illegible] WLS [illegible] OLS [illegible]

[illegible] WLS [illegible] OLS [illegible] Freedman and Berk (2008) [illegible] WLS [illegible] $\hat{s}$ WLS [illegible] Bootstrap [illegible]

14.1. [illegible] ***WLS*** [illegible]：当（*1*）倾向得分模型可信且（*2*）误差方差可能与倾向得分相关（异方差）时，*WLS* 是更好的选择。当结局模型可信时，简单的 *OLS* 可能更稳定。

[illegible]

[illegible] Average Causal Effect on the Treated, τ_T [illegible]

[illegible] Hajek [illegible]

$$\hat{\tau}_T^{\mathrm{hajek}} = \hat{\bar{Y}}(1) - \frac{\sum_{i=1}^{n} \hat{o}(X_i)(1 - Z_i) Y_i}{\sum_{i=1}^{n} \hat{o}(X_i)(1 - Z_i)}$$

[illegible] $\hat{o}(X_i) = \hat{e}(X_i)/\{1 - \hat{e}(X_i)\}$

[illegible] 14.2 [illegible] Hajek [illegible] WLS) $\hat{\tau}_T^{hajek}$ 严格等于如下 *WLS* 回归中 Z_i 的系数：

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} w_{T_i}(Y_i - \alpha - \beta Z_i)^2$$

其中权重为

$$w_{T_i} = Z_i + (1 - Z_i)\hat{o}(X_i) = \begin{cases} 1 & \text{if } Z_i = 1; \\ \hat{o}(X_i) & \text{if } Z_i = 0. \end{cases} \tag{66}$$

[illegible] $\hat{\bar{X}}(1) =$ [illegible] τ_T [illegible] WLS [illegible]

$$Y_i \sim (1, Z_i, X_i, Z_i X_i)$$

[illegible] Z_i [illegible]

---

[2] [illegible] ??

[illegible] 14.3 [illegible]）若 $\hat{\hat{X}}(1)=0$ 且 $\mu_0(X_i,\hat{\beta}_0)=\hat{\beta}_{00}+\hat{\beta}_{0x}^{\top}X_i$ 基于权重 [illegible] 对 $Y_i \sim (1, Z_i, X_i, Z_iX_i)$ 的 *WLS* 拟合，则

$$\hat{\tau}^{dr}_{T,wls}=\hat{\tau}^{reg}_{T,wls}=\hat{\beta}_{10}-\hat{\beta}_{00}$$

即 Z_i 在 *WLS* 回归中的系数。

[illegible][3]

[illegible]

[illegible] CRE [illegible]

Table 1. [illegible] CRE [illegible]

	CRE	[illegible]
[illegible]	$Y_i \sim (1, Z_i)$	$Y_i \sim (1, Z_i)$ [illegible] w_i [illegible]
[illegible]	$Y_i \sim (1, Z_i, X_i, Z_iX_i)$	$Y_i \sim (1, Z_i, X_i, Z_iX_i)$ [illegible] w_i [illegible]

[illegible] w_i [illegible] CRE [illegible] OLS [illegible] OLS [illegible] WLS [illegible]

[illegible]

[illegible] WLS [illegible]

(1) [illegible] Hajek [illegible] WLS [illegible] treatment [illegible]
(2) [illegible] WLS [illegible] X_i [illegible] Z_iX_i [illegible] Lin, 2013[illegible]
(3) [illegible] WLS [illegible] Robins et al., 2007[illegible]
(4) [illegible] WLS [illegible] OLS [illegible]
(5) [illegible] τ_T [illegible]

WLS [illegible] IPW [illegible]

[illegible]

[illegible] 14.1. [illegible]命题 *14.1* 和 *14.2* 的证明 证明 [illegible]和 [illegible]
[illegible]：这些是附录 *B.3* 中一元 *WLS* 问题的特例。利用 *WLS* 的正规方程即可证明。

[illegible] 14.2. [illegible]预测估计量与双重稳健估计量 另一种结局回归估计量是预测估计量

$$\hat{\tau}^{pred}=\hat{\mu}_1^{pred}-\hat{\mu}_0^{pred}$$

其中

$$\hat{\mu}_1^{pred}=\frac{1}{n}\sum_{i=1}^{n}\{Z_iY_i+(1-Z_i)\mu_1(X_i,\hat{\beta}_1)\},\quad \hat{\mu}_0^{pred}=\frac{1}{n}\sum_{i=1}^{n}\{Z_i\mu_0(X_i,\hat{\beta}_1)+(1-Z_i)Y_i\}.$$

证明：当 $(\mu_1(X_i,\hat{\beta}_1),\mu_0(X_i,\hat{\beta}_1))=(\hat{\beta}_{10}+\hat{\beta}_{1x}^{\top}X_i,\hat{\beta}_{00}+\hat{\beta}_{0x}^{\top}X_i)$ 来自基于权重

$$w_i=Z_i/\hat{o}(X_i)+(1-Z_i)\hat{o}(X_i)$$

---

[3] [illegible] ??

对 $Y_i \sim (1, X_i)$ 的 *WLS* 拟合时，双重稳健估计量等于 $\hat{\tau}^{pred}$。
[illegible] ：*Cao et al. (2009)* 和 *Vermeulen and Vansteelandt (2015)* 从其他理论角度 *motivation* 了上述权重。

[illegible] 14.3. { {*WLS* 的效率性质 讨论在何种情形下 *WLS* 估计量比 *OLS* 估计量更有效。在 *Freedman and Berk (2008)* 的模拟设置中，为什么 *OLS* 表现更好？这与本章讨论的异方差情形有何关联？

[illegible] 14.4. { {推荐阅读 [illegible] ：{ 是关于因果推断中加权方法的权威综述。{ 讨论了协变量调整在随机化实验中的应用。

[illegible] ：*WLS* 与 *IPW* 的等价性最早由 *Imbens (2004)* 明确指出。双重稳健性的概念在 *Robins et al. (2007)* 中得到了系统阐述。

CHAPTER 15

[illegible]

[illegible]

“Matching is a way of making observational studies look like randomized experiments.” — ([illegible])

[illegible] **j ; — J ^ a l Z Q — a j ; — p** [illegible] Rubin [illegible]

[illegible]

[illegible] n_1 [illegible] n_0 [illegible] $i = 1, \ldots, n_1$ [illegible] $m(i)$ [illegible] $X_i = X_{m(i)}$ [illegible] matched-pairs experiment, MPE [illegible]

$$\Pr\{i \text{ [illegible]}, m(i) \text{ [illegible]}\} = \frac{e(X_i)\{1 - e(X_{m(i)})\}}{\sum_j \{e(X_j) + e(X_{m(j)})\}}.$$

$X_i = X_{m(i)}$ [illegible] $e(X_i) = e(X_{m(i)})$ [illegible] $e(X_i)/\{e(X_i) + e(X_i)\} = 1/2$ [illegible]

[illegible] $X_i = X_{m(i)}$ [illegible] FRT [illegible] $|\hat{e}(X_i) - \hat{e}(X_{m(i)})|$ [illegible] $|\mathrm{logit}\{\hat{e}(X_i)\} - \mathrm{logit}\{\hat{e}(X_{m(i)})\}|$ [illegible]

[illegible]

[illegible] Abadie–Imbens (AI) [illegible][1]

$$\{X_i, Z_i, Y_i(1), Y_i(0)\}_{i=1}^{n} \overset{\text{IID}}{\sim} \{X, Z, Y(1), Y(0)\}.$$

---

[1] [illegible] **??**

[illegible] P [illegible] $1-M$ [illegible] i [illegible] $Y_i(1)$ [illegible] Y_i[illegible]

$$\hat{Y}_i(1) = Y_i.$$

[illegible] $Y_i(0)$ [illegible]

$$\hat{Y}_i(0) = M^{-1}\sum_{j=1}^{M} Y_{m_j(i)},$$

[illegible] $m_j(i)$ [illegible] j [illegible]

$$\hat{\tau}^{\mathrm{m}} = n^{-1}\sum_{i=1}^{n} Z_i\{\hat{Y}_i(1)-\hat{Y}_i(0)\} + n^{-1}\sum_{i=1}^{n}(1-Z_i)\{\hat{Y}_i(0)-\hat{Y}_i(1)\}.$$

[illegible] $J(i)$ [illegible] i [illegible] $K_i = |J(i)\cap\{j : Z_j = 1\}|$ [illegible]

[illegible] ([illegible]) 在 M 固定的情况下，当 $n\to\infty$ 时，

$$\sqrt{n}(\hat{\tau}^{\mathrm{m}}-\tau) \Rightarrow \mathcal{N}\{0, V^{\mathrm{m}}(\tau)\},$$

其中偏差来自不精确匹配。[2]

[illegible] bootstrap [illegible] bootstrap[illegible] $\hat{V}^{\mathrm{mbc}}$ [illegible]

$$\hat{\tau}^{\mathrm{mbc}} = \hat{\tau}^{\mathrm{m}} + n^{-1}\sum_{i=1}^{n}\hat{\psi}_i,$$

[illegible] $\hat{\psi}_i$ [illegible][3] [illegible] $\hat{\tau}^{\mathrm{mbc}}$ [illegible] $\hat{\psi}_i$ [illegible] bootstrap [illegible] bootstrap [illegible] $\hat{V}^{\mathrm{mbc}}$[illegible]

[illegible]

[illegible] 1 [illegible]

[illegible] ([illegible]ACE [illegible]) 偏差校正匹配估计量满足

$$\hat{\tau}^{\mathrm{mbc}} = \hat{\tau}^{\mathrm{DR}},$$

其中 $\hat{\tau}^{\mathrm{DR}}$ 是第 [illegible]章定义的双稳健估计量。[4]

---

[2] [illegible] ??

[3] [illegible] ??

[4] [illegible] ??

[illegible] $1+K_i/M$ [illegible] $1/\hat{e}(X_i)$ [illegible] $e(X_i)$ [illegible] $1/\hat{e}(X_i)$ [illegible] K_i [illegible] $1+K_i/M$ [illegible] M [illegible] $1+K_i/M$ [illegible] $1/e(X_i)$ [illegible] M [illegible] M [illegible]

[illegible]

[illegible]

$$\tau_{\mathrm{T}} = \mathbb{E}\{Y_i(1) - Y_i(0) \mid Z_i = 1\}.$$

ATT [illegible]

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{m}} = n_1^{-1} \sum_{i:Z_i=1} \{\hat{Y}_i(1) - \hat{Y}_i(0)\},$$

[illegible] $\hat{Y}_i(1) = Y_i$ [illegible] $\hat{Y}_i(0) = M^{-1}\sum_{j=1}^{M} Y_{m_j(i)}$ [illegible]

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{mbc}} = \hat{\tau}_{\mathrm{T}}^{\mathrm{m}} + n_1^{-1} \sum_{i:Z_i=1} \hat{\psi}_i^{\mathrm{T}}.$$

[illegible] (ATT [illegible]) *ATT* 的偏差校正匹配估计量满足

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{mbc}} = \hat{\tau}_{\mathrm{T}}^{\mathrm{DR}},$$

其中 $\hat{\tau}_{\mathrm{T}}^{\mathrm{DR}}$ 是 *ATT* 的双稳健估计量。[5]

[illegible] K_i/M [illegible]

[illegible]

[illegible] M [illegible] M [illegible]

[illegible]

[illegible] OLS [illegible]

[illegible]

[illegible] ATT [illegible] \$1794 [illegible] \$623 [illegible][6]

[illegible] Matching [illegible] 1-1 [illegible] ATT [illegible] \$1658 [illegible] \$1794 [illegible]

---

[5] [illegible] ??

[6] [illegible] ?? ?? [illegible]

[illegible]

[illegible]

(1) [illegible]

(2) [illegible]

(3) [illegible]

(4) [illegible]

(5) [illegible] ATT [illegible]

(6) [illegible]

mw

[illegible] 15.1. 匹配与倾向得分 证明：在精确匹配下，倾向得分对所有配对单元相同，即 $e(X_i) = e(X_{m(i)})$。这如何支持”精确匹配模拟随机实验”的论断？

[illegible] 15.2. 一对一与一对多匹配的偏差比较 考虑 $1-1$ 匹配和 $1-M$ 匹配。当 M 增大时，匹配估计量的偏差如何变化？为什么？

[illegible] 15.3. 匹配估计量的偏差校正项 从 { {出发，推导偏差校正项 $\hat{\psi}_i$ 的具体形式。提示：参考 { 的推导。

[illegible] 15.4. *ATT* 匹配估计量的性质 证明 { :{ *ATT* 的偏差校正匹配估计量等于 *ATT* 的双稳健估计量。

[illegible] 15.5. 协变量平衡 对 *LaLonde* 数据进行匹配分析，检验匹配前后协变量的平衡性。使用不同的 M 值（一对一、一对二、一对四）比较结果。

[illegible] 15.6. 匹配与回归的比较 使用第 { {章 { {节的数据，分别用匹配估计量和结果回归估计量估计 *ATT*。比较两种方法的结果和效率。

[illegible] 15.7. 倾向得分匹配的理论性质 { 提供了基于倾向得分匹配的理论。阅读该论文，总结其主要结论及其与 { 建议的关联。

[illegible] 15.8. 高维协变量的挑战 当协变量维度很高时，直接匹配面临维数灾难。讨论三种可能的解决策略：降维、惩罚回归、机器学习方法。

CHAPTER 16

“Although we cannot see the wind, we can see its effects on the leaves.” —

$$Z \perp \{Y(1), Y(0)\} \mid X. \tag{16.1}$$

X Z

George Box “所有模型都是错的，但有些有用。”

X

U

Pearl (1995)

$$X \to Z \to Y$$

X Z Y X Z Z Y

$$\begin{cases} X \sim F_X(x), \\ Z = f_Z(X, \varepsilon_Z), \\ Y(z) = f_Y(X, z, \varepsilon_Y(z)), \end{cases} \tag{16.2}$$

$\varepsilon_Z \perp \varepsilon_Y(z)$ $Z \perp Y(z) \mid X$

U

$$U \to Z,\ U \to Y,\ X \to Z,\ X \to Y$$

[illegible]

$$\begin{cases} X \sim F_X(x), \\ U \sim F_U(u), \\ Z = f_Z(X, U, \varepsilon_Z), \\ Y(z) = f_Y(X, U, z, \varepsilon_Y(z)), \end{cases} \tag{16.3}$$

[illegible] $\varepsilon_Z \perp \varepsilon_Y(z)$ [illegible] $Z \perp Y(z) \mid (X, U)$ [illegible] $Z \not\perp Y(z) \mid X$ [illegible] (X, U) [illegible] X [illegible] U [illegible]

[illegible] Z [illegible] Y [illegible] backdoor path [illegible]

[illegible]

[illegible]

$$Z \perp Y(1) \mid X, \quad Z \perp Y(0) \mid X. \tag{16.4}$$

[illegible]

$$\Pr\{Y(1) \mid Z = 1, X\} = \Pr\{Y(1) \mid Z = 0, X\},$$

$$\Pr\{Y(0) \mid Z = 1, X\} = \Pr\{Y(0) \mid Z = 0, X\}. \tag{16.5}$$

[illegible] X [illegible] $\Pr\{Y(1) \mid Z = 0, X\}$ [illegible] $\Pr\{Y(0) \mid Z = 1, X\}$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible] $Z \perp Y(z) \mid X$ [illegible] $Z \perp Y^{\mathrm{n}}(z) \mid X$ [illegible] Y^{n} [illegible] Y [illegible] Z [illegible] Y^{n} [illegible] Y^{n} [illegible]

$$\tau(Z \to Y^{\mathrm{n}}) = \mathbb{E}\{Y^{\mathrm{n}}(1) - Y^{\mathrm{n}}(0)\} \tag{16.6}$$

[illegible] $\tau(Z \to Y^{\mathrm{n}}) = 0$ [illegible]

[illegible] Z [illegible] Y^{n} [illegible]

[illegible] 16.1 (Cornfield [illegible] (1959)[illegible]). *Cornfield* 等 *(1959)* 研究了吸烟对肺癌的因果效应。他们在分析中控制了众多重要的背景变量，但仍可能存在某些未测混杂因素使观察到的效应产生偏倚。

为了加强因果关系的证据，他们同时报告了吸烟对车祸的影响——基于生物学知识，吸烟对车祸应该没有因果效应。结果显示关联接近于零。因此，即使无法完全排除未测混杂，这个基于阴性结果的补充分析使吸烟对肺癌因果效应的证据更加有力。

[illegible] 16.2 (Jackson [illegible] (2006)[illegible]). *Jackson* 等 *(2006)* 对观测性研究的结果持怀疑态度。这些研究显示，在调整了测量的协变量后，流感疫苗接种显著降低了流感季节的肺炎/流感住院率和全因死亡率。

他们的补充分析考察了流感季节开始之前的结局指标。疫苗接种通常在秋季开始，但流感传播在冬季之前是极少的。基于生物学，疫苗效应在流感季节应该最为显著。但 *Jackson* 等发现，疫苗效应在流感季节之前反而更大——这暗示观察到的效应可能源于未测混杂。

[illegible] Z [illegible] U [illegible] Y[illegible] Z [illegible] U [illegible] Y [illegible]

Lipsitch [illegible] (2010) [illegible] Rosenbaum (1989) [illegible]

[illegible] Z^{n} [illegible] Z [illegible] Z^{n} [illegible] Y [illegible]

$$\tau(Z^{\mathrm{n}} \to Y) = \mathbb{E}\{Y(1^{\mathrm{n}}) - Y(0^{\mathrm{n}})\} \quad (16.7)$$

[illegible] 16.3 (Sanderson [illegible] (2017)[illegible]). *Sanderson* 等 *(2017)* 给出了利用阴性暴露的多个例子。他们建议比较母体暴露与后代结局的关联，以及父亲暴露与同样结局的关联。

在母体吸烟与后代结局的研究中：母体吸烟可能受到未测混杂（如社会经济地位）的影响，而父亲吸烟不太可能受到同样的混杂影响（因为父亲不经历怀孕）。因此，如果观察到的母体吸烟效应远大于父亲吸烟效应，这支持了因果解释；如果两者效应相似，则暗示可能存在混杂。

[illegible] Z [illegible] Z^{n} [illegible] Y [illegible]

[illegible]

[illegible]

[illegible]

Rosenbaum (2002b) [illegible] “没有理由避免调整描述研究对象治疗前特征的变量。”[illegible] Rubin (2007) [illegible] “通常，条件性越强的假设，越容易接受。”[illegible]

Pearl [illegible]

---

[1] [illegible]

[illegible] M [illegible]

$$U_1 \to X \leftarrow U_2,\ U_1 \to Z,\ U_2 \to Y$$

[illegible] M [illegible] M [illegible]

$$\begin{cases} U_1 \perp U_2, \\ X = f_X(U_1, U_2, \varepsilon_X), \\ Z = f_Z(U_1, \varepsilon_Z), \\ Y = Y(z) = f_Y(U_2, \varepsilon_Y), \end{cases} \tag{16.8}$$

[illegible] $(\varepsilon_X, \varepsilon_Z, \varepsilon_Y)$ [illegible] X [illegible] U_1 [illegible] U_2 [illegible] Z [illegible] Y [illegible] Z [illegible] Y [illegible] Z [illegible] Y [illegible]

[illegible] $Z \perp Y$ [illegible] Z [illegible] Y [illegible] X [illegible]

$$\tau_{\mathrm{PF}} = \mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0) = 0. \tag{16.9}$$

[illegible] X [illegible] X [illegible] U_1 [illegible] U_2 [illegible] $U_1 \not\perp U_2 \mid X$ [illegible] $Z \not\perp Y \mid X$ [illegible]

$$\int \{\mathbb{E}(Y \mid Z = 1, X = x) - \mathbb{E}(Y \mid Z = 0, X = x)\} f(x)\, dx \neq 0. \tag{16.10}$$

[illegible] X [illegible] U_1 [illegible] U_2 [illegible] U_1 [illegible] Z [illegible] U_2 [illegible] Y [illegible] X [illegible] X [illegible] Z [illegible] Y [illegible] X [illegible] U_1 [illegible] U_2 [illegible] Z [illegible] Y [illegible]

[illegible]

$$\begin{cases} X = aU_1 + bU_2 + \varepsilon_X, \\ Z = cU_1 + \varepsilon_Z, \\ Y = dU_2 + \varepsilon_Y, \end{cases} \tag{16.11}$$

[illegible] $(U_1, U_2, \varepsilon_X, \varepsilon_Z, \varepsilon_Y) \overset{\mathrm{IID}}{\sim} \mathrm{N}(0,1)$ [illegible]

$$\mathrm{cov}(Z, Y) = \mathrm{cov}(cU_1 + \varepsilon_Z, dU_2 + \varepsilon_Y) = 0, \tag{16.12}$$

[illegible] X [illegible] Z [illegible] Y [illegible]

$$\rho_{ZY|X} \propto -abcd, \tag{16.13}$$

[illegible] $abcd$ [illegible]

[illegible] 16.4 (M-[illegible]). 考虑样本量 $n = 10^6$ 的大规模蒙特卡洛模拟：

```
U1 = rnorm(n)
U2 = rnorm(n)
X = U1 + U2 + rnorm(n)
Y = U2 + rnorm(n)
Z = U1 + rnorm(n)

# 连续治疗
```

```
summary(lm(Y ~ Z))$coef[2, 1]  # [1] 0.001
summary(lm(Y ~ Z + X))$coef[2, 1]  # [1] -0.20

# 二值治疗
Z = (Z >= 0)
summary(lm(Y ~ Z))$coef[2, 1]  # [1] 0.002
summary(lm(Y ~ Z + X))$coef[2, 1]  # [1] -0.42
```

未调整的简单估计量接近真实值 (*0*)，但调整 X 后的估计量产生了显著偏倚 (约 -0.2 到 -0.4)。这就是 M-偏倚。

M-[illegible] U_1 [illegible] U_2 [illegible] X [illegible] U_1 [illegible] U_2 [illegible]

[illegible] M-[illegible]

$$Z \leftarrow X \rightarrow U \rightarrow Y$$

[illegible] X [illegible] Z [illegible] U [illegible] U [illegible] Z [illegible] Y [illegible]

$$\begin{cases} Z = aX + bU + \varepsilon_Z, \\ Y(z) = \tau z + cU + \varepsilon_Y, \end{cases} \tag{16.14}$$

[illegible] $(U, X, \varepsilon_Z, \varepsilon_Y) \overset{\text{IID}}{\sim} \mathrm{N}(0,1)$ [illegible] $X \perp U$ [illegible] X [illegible] Z [illegible] Y [illegible] X [illegible] Z [illegible] Y [illegible] U [illegible] Z [illegible] X [illegible] U [illegible] X [illegible] Z [illegible] U [illegible] U [illegible]

[illegible] OLS [illegible]

$$\tau_{\text{unadj}} = \tau + \frac{c\ \mathrm{cov}(Z, U)}{\mathrm{var}(Z)} = \tau + \frac{cb}{a^2 + b^2 + 1}, \tag{16.15}$$

[illegible] $bc/(a^2 + b^2 + 1)$ [illegible] OLS [illegible] Y [illegible] (Z, X) [illegible]

$$\tau_{\text{adj}} = \tau + \frac{bc}{b^2 + 1}. \tag{16.16}$$

[illegible] $a \neq 0$ [illegible] $a^2 + b^2 + 1 > b^2 + 1$ [illegible]

$$\left|\frac{bc}{a^2 + b^2 + 1}\right| < \left|\frac{bc}{b^2 + 1}\right|. \tag{16.17}$$

[illegible] X [illegible] Z [illegible] a [illegible]

[illegible] 16.5 (Z-[illegible]). 考虑蒙特卡洛模拟:

```
n = 10^6
X = rnorm(n)
U = rnorm(n)
Z = X + U + rnorm(n)
Y = U + rnorm(n)

# Z ~ X + U
```

```
summary(lm(Y ~ Z))$coef[2, 1]  # [1] 0.333
summary(lm(Y ~ Z + X))$coef[2, 1]  # [1] 0.501

# X-Z 关联更强时
Z = 2*X + U + rnorm(n)
summary(lm(Y ~ Z))$coef[2, 1]  # [1] 0.167
summary(lm(Y ~ Z + X))$coef[2, 1]  # [1] 0.501

# X-Z 关联更强时
Z = 10*X + U + rnorm(n)
summary(lm(Y ~ Z))$coef[2, 1]  # [1] 0.01
summary(lm(Y ~ Z + X))$coef[2, 1]  # [1] 0.50
```

可以看到：随着 X 与 Z 的关联增强，未调整估计量越来越接近真实值 $\tau = 0$，而调整后的估计量始终为 0.5——因为调整反而引入了更大的偏倚。

Z-[illegible] Problem [illegible] Pearl (2010a, 2011) [illegible] Ding [illegible] (2017b) [illegible]

[illegible] Pearl [illegible] Z [illegible] X [illegible] X [illegible] Z [illegible] Z-[illegible] [illegible]

[illegible] [illegible]

$$X_Y \to Y,\, X_Y \to Z,\, X \to Z,\, X \to Y,\, X_I \to Z,\, X_I \to Y$$

[illegible]

(1) X [illegible] X [illegible] X

(2) X_R [illegible]

(3) X_Z [illegible] Z-[illegible] X_Z [illegible]

(4) X_Y [illegible]

(5) X_I [illegible]

[illegible] X [illegible] X_Y [illegible]

[illegible]

[illegible] 16.1. 验证正文中的公式 [illegible]和 [illegible]

TABLE 1. [illegible]

[illegible]	Z [illegible]	Y [illegible]	
X [illegible]	[illegible]	[illegible]	[illegible]
X_R [illegible]	[illegible]	[illegible]	[illegible]
X_Z [illegible]	[illegible]	[illegible]	[illegible]
X_Y [illegible]	[illegible]	[illegible]	[illegible]
X_I [illegible]	[illegible]	[illegible]	[illegible]

[illegible] 16.2. 考虑正文中的 M-偏倚正态线性模型。证明给定 X 时 Z 和 Y 的偏相关系数公式 [illegible]

[illegible] 16.3. 找出一个你自己的研究领域中的阴性结果或阴性暴露的例子。讨论为什么它符合阴性控制的条件，以及它如何帮助评估不可混淆性假设。

[illegible] 16.4. [illegible] 和 *econometricians* 提出了以下 ***Cochran*** [illegible] （也称为遗漏变量偏倚公式）。设 $(y_i, x_{1i}, x_{2i})_{i=1}^n$ 为 *IID* 样本，其中 y_i 是标量，x_{1i} 维数为 K，x_{2i} 维数为 L。考虑 *OLS* 分解：

$$\begin{aligned} y_i &= \beta_1^{\mathrm{T}} x_{1i} + \beta_2^{\mathrm{T}} x_{2i} + \varepsilon_i \quad \text{（长回归）}, \\ y_i &= \gamma^{\mathrm{T}} x_{1i} + e_i \quad \text{（短回归）}, \\ x_{2i} &= \delta^{\mathrm{T}} x_{1i} + v_i. \end{aligned}$$

证明：$\gamma = \beta_1 + \delta\beta_2$。解释为什么 $\delta\beta_2$ 被称为遗漏变量偏倚。

[illegible] 16.5. 设计一个模拟研究，演示以下情形：

(1) 调整混杂变量 X 可以减少偏倚；
(2) 调整与未测混杂相关的变量会引入 M-偏倚；
(3) 调整工具变量 X_Z 在存在未测混杂时会放大偏倚。

对于每种情形，使用大样本量（$n = 10^5$）计算未调整和调整后的估计量。

[illegible]

[illegible] Pearl (2000) [illegible]
VanderWeele [illegible] Shpitser (2011) [illegible]
Rosenbaum (2002b) [illegible]
[illegible]
Ding [illegible] (2017b) [illegible] Z-[illegible]

CHAPTER 17

[illegible]

[illegible]

"It is possible that the entire relationship between cigarette smoking and lung cancer is due to some common cause." — ([illegible])

Part III [illegible] outcome regression [illegible] [illegible] [illegible] [illegible]

$$Z \perp \{Y(1), Y(0)\} \mid X. \tag{67}$$

[illegible] X [illegible] Z [illegible]

[illegible] VanderWeele and Ding, 2017 [illegible] [1]

[illegible]

[illegible]

$$Z \perp \{Y(1), Y(0)\} \mid (X, U), \tag{68}$$

[illegible] U [illegible] (X, U) [illegible] Z [illegible] Y [illegible] U [illegible] U [illegible] Y [illegible] Ding and VanderWeele, 2016 [illegible] Y [illegible]

[illegible] $X = x$ [illegible]

$$\mathrm{RR}^{\text{true}}_{ZY|x} = \frac{\Pr\{Y(1) = 1 \mid X = x\}}{\Pr\{Y(0) = 1 \mid X = x\}}. \tag{69}$$

---

[1] Ding and VanderWeele (2016) [illegible] Rosenbaum [illegible]

$$\mathrm{RR}^{\mathrm{obs}}_{ZY|x} = \frac{\Pr(Y=1 \mid Z=1, X=x)}{\Pr(Y=1 \mid Z=0, X=x)}. \tag{70}$$

e U U U U X

$$Z \leftarrow U \rightarrow Y,$$

U Z Y common cause $Z \perp Y \mid (X, U)$ $Z \not\perp Y \mid X$ U Ding and VanderWeele (2016) U

$$\mathrm{RR}_{ZU|x} = \frac{\Pr(U=1 \mid Z=1, X=x)}{\Pr(U=1 \mid Z=0, X=x)}, \tag{71}$$

$$\mathrm{RR}_{UY|x} = \frac{\Pr(Y=1 \mid U=1, X=x)}{\Pr(Y=1 \mid U=0, X=x)}, \tag{72}$$

E-value

Cornfield-type 在假设 下，假设

$$RR^{\mathrm{obs}}_{ZY|x} > 1, \quad RR_{ZU|x} > 1, \quad RR_{UY|x} > 1. \tag{73}$$

则有

$$RR^{\mathrm{obs}}_{ZY|x} \le \frac{RR_{ZU|x} \cdot RR_{UY|x}}{RR_{ZU|x} + RR_{UY|x} - 1}. \tag{74}$$

$Z \leftarrow U \rightarrow Y$ $\mathrm{RR}_{ZU|x} \cdot \mathrm{RR}_{UY|x}/(\mathrm{RR}_{ZU|x} + \mathrm{RR}_{UY|x} - 1)$ $\mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ Gastwirth et al., 1998

$$\min(\mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x}) \ge \mathrm{RR}^{\mathrm{obs}}_{ZY|x}. \tag{75}$$

Cornfield et al. (1959)

$f_{1,x} = \Pr(U=1 \mid Z=1, X=x)$ $f_{0,x} = \Pr(U=1 \mid Z=0, X=x)$ $\mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ $Z \perp Y \mid (X, U)$ ( ) [1]

---

[1] ??

[illegible]

[illegible] (4)[illegible] $w = \max(\mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x})$[illegible]

$$\mathrm{RR}^{\mathrm{obs}}_{ZY|x} \le \frac{w^2}{2w-1}. \tag{76}$$

[illegible] ≤ 1[illegible]

$$w \ge \mathrm{RR}^{\mathrm{obs}}_{ZY|x} + \sqrt{\mathrm{RR}^{\mathrm{obs}}_{ZY|x}(\mathrm{RR}^{\mathrm{obs}}_{ZY|x} - 1)}. \tag{77}$$

[illegible] E-value [illegible]

[illegible] 17.2 (E-value). 给定观测条件风险比 $RR^{\mathrm{obs}}_{ZY|x}$，定义 ***E-value*** 为

$$\text{E-value} = RR^{\mathrm{obs}}_{ZY|x} + \sqrt{RR^{\mathrm{obs}}_{ZY|x}(RR^{\mathrm{obs}}_{ZY|x} - 1)}. \tag{78}$$

[illegible] E-value [illegible] $\max(\mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x})$ [illegible] R[illegible] $R <$ E-value [illegible]

[illegible] p [illegible] Fisher p [illegible] p [illegible] p [illegible] E-value [illegible]

[illegible] β[illegible])定义 $\beta(w_1, w_2) = w_1 w_2/(w_1 + w_2 - 1)$（$w_1 > 1, w_2 > 1$）。则：

(1) $\beta(w_1, w_2)$ 对称于 w_1 和 w_2；
(2) $\beta(w_1, w_2)$ 对 w_1 和 w_2 均单调递增；
(3) $\beta(w_1, w_2) \le w_1$ 且 $\beta(w_1, w_2) \le w_2$；
(4) $\beta(w_1, w_2) \le w^2/(2w-1)$，其中 $w = \max(w_1, w_2)$。

[illegible] (3) [illegible] Cornfield [illegible][2] [illegible] (4) [illegible] Cornfield [illegible] E-value [illegible]

[illegible]

[illegible] 17.1 (Hammond and Horn, 1958). 研究美国人群中吸烟与肺癌的关系。不调整协变量的数据可列为下表：

	肺癌	无肺癌
吸烟者	*397*	*78,557*
非吸烟者	*51*	*108,778*

基于该数据，风险比估计为 $\widehat{RR} = 10.73$，*95%* 置信区间为 $[8.02, 14.36]$。

- 解释点估计的 *E-value*：

$$\text{E-value} = 10.73 + \sqrt{10.73 \times 9.73} \approx 20.95.$$

- 解释下限的 *E-value*：

$$\text{E-value}_{\mathrm{CI}} = 8.02 + \sqrt{8.02 \times 7.02} \approx 15.52.$$

---

[2] [illegible] **??**

[illegible]：*要完全解释掉观察到的风险比（甚至其置信下限），未测混杂与治疗的关联以及与结果的关联，都必须达到至少 15.52。这意味着未测混杂必须是一个*[illegible]*的风险因子——它与吸烟的关联程度要接近吸烟本身与肺癌的关联程度（10 倍以上）。在领域知识上，这种强度的未知混杂似乎不太合理，因此该研究提供了较强的因果证据。*

[illegible] Hammond and Horn [illegible] $(\mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x})$ [illegible]

[illegible]

[illegible] E-value [illegible] Bradford Hill, 1965 [illegible] Bradford Hill [illegible]

[illegible]

[illegible] E-value [illegible]

[illegible] Bradford Hill [illegible] temporality [illegible] biological plausibility [illegible] consistency [illegible] dose-response [illegible] E-value [illegible]

[illegible] 17.4 (Bradford Hill [illegible]). *([illegible])* 提出的九条因果判定准则：

(1) 强度（*strength*）；
(2) 一致性（*consistency*）；
(3) 特异性（*specificity*）；
(4) 时间顺序（*temporality*）；
(5) 生物学梯度（*biological gradient*）；
(6) 生物学合理性（*plausibility*）；
(7) 一致性（*coherence*）；
(8) 实验证据（*experiment*）；
(9) 类比（*analogy*）。

E-value [illegible] (1) [illegible] E-value [illegible]

[illegible]

$$\Pr(Y_i = 1 \mid Z_i, X_i) = \frac{e^{\beta_0+\beta_1 Z_i+\beta_2\, X_i}}{1 + e^{\beta_0+\beta_1 Z_i+\beta_2\, X_i}}. \tag{79}$$

[illegible] β_1 [illegible] X [illegible] common odds ratio [illegible]

[illegible] $\Pr(Y = 1 \mid Z, X)$ [illegible] β_1 [illegible] $\hat{\beta}_1$ [illegible] E-value[illegible]

$$\widehat{\text{E-value}} = \exp(\hat{\beta}_1) + \sqrt{\exp(\hat{\beta}_1)(\exp(\hat{\beta}_1) - 1)}. \tag{80}$$

[illegible] E-value [illegible]

[illegible] 17.2 (NCHS 2003 [illegible]). 使用美国国家卫生统计中心 *2003* 年出生证明数据，研究高龄母亲（≥ 35 岁）对早产的影响。调整社会人口学协变量后，逻辑斯蒂回归得到的条件风险比为 1.31，其 *95%* 置信区间为 $[1.29, 1.33]$。

对应的 *E-value* 约为 1.94（点估计）和 1.91（置信下限）。这意味着要解释掉观察到的关联，未测混杂必须同时与治疗（高龄）和结果（早产）有至少 *1.9* 倍的关联——这个强度在流行病学上看起来是较大的，但远不如吸烟-肺癌例子中的 *20.95*。

[illegible] [illegible] Ding and VanderWeele (2016) [illegible]

[illegible] **[illegible]** [illegible] 修改 $RR_{UY|x}$ 的定义为：

$$RR_{UY|x} = \max_{z=0,1} \frac{\Pr(Y=1 \mid Z=z, U=1, X=x)}{\Pr(Y=1 \mid Z=z, U=0, X=x)}. \tag{81}$$

则在假设 [illegible] 下，

$$RR^{\text{true}}_{ZY|x} \geq \frac{RR^{\text{obs}}_{ZY|x}}{\frac{RR_{ZU\ x}\cdot RR_{UY\ x}}{RR_{ZU\ x}+RR_{UY\ x}-1}}. \tag{82}$$

[illegible] $\mathrm{RR}^{\text{true}}_{ZY|x}$ [illegible] Cornfield [illegible]

[illegible]

E-value [illegible] Ioannidis et al., 2019 [illegible]

[illegible] [illegible] E-value [illegible] $2 \cdot \mathrm{RR}^{\text{obs}}_{ZY|x}$ [illegible] E-value [illegible]

[illegible] E-value [illegible] E-value [illegible] 10.73 [illegible] E-value 20.95 [illegible]

[illegible] E-value [illegible] $\max(\mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x})$ [illegible] E-value [illegible] E-value [illegible] X [illegible] E-value [illegible] X [illegible] X_j [illegible] Z-X_j [illegible] X_j-Y [illegible]

[illegible] [illegible] Ding and VanderWeele (2016) [illegible] HR [illegible] E-value [illegible] outcome regression [illegible] IPW [illegible]

[illegible]

[illegible] E-value [illegible]

(1) [illegible]

(2) [illegible] $\mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ [illegible] $\mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ [illegible]

(3) [illegible] $\text{E-value} = \mathrm{RR}^{\mathrm{obs}}_{ZY|x} + \sqrt{\mathrm{RR}^{\mathrm{obs}}_{ZY|x}(\mathrm{RR}^{\mathrm{obs}}_{ZY|x} - 1)}$ [illegible]

(4) E-value [illegible]

(5) E-value [illegible] Bradford Hill [illegible]

[illegible]

[illegible]

17.1 (17.1). 证明函数 $f(x) = (ax+1)/(bx+1)$ 在 $a > b$ 时关于 x 单调递增，在 $a < b$ 时单调递减。

17.2 (17.2). 证明在假设 $Z \perp Y \mid (X, U)$ 下，如果 $RR_{ZU|x} > 1$ 但 $RR_{UY|x} < 1$，则 $RR^{\mathrm{obs}}_{ZY|x} < 1$。这说明 [illegible] 中的第三个条件在某种意义上是冗余的。

17.3 (17.3). 证明引理 [illegible]

17.4 (17.4). （*Schlesselman 1978* 公式）在二值治疗 Z、结果 Y 和未测混杂 U 下，假设层内（$U = 0$ 和 $U = 1$）治疗对结果的风险比相同，$RR_{ZY|U=0} = RR_{ZY|U=1}$，且混杂对结果的风险比在两组治疗下也相同（记为 γ）。证明：

$$\frac{RR^{\mathrm{obs}}_{ZY}}{RR^{\mathrm{true}}_{ZY}} = \frac{1 + (\gamma - 1)\Pr(U = 1 \mid Z = 1)}{1 + (\gamma - 1)\Pr(U = 1 \mid Z = 0)}. \tag{83}$$

并讨论此公式与定理 [illegible] 的关系。

17.5 (17.5). 使用与例 [illegible] 相同的数据集，计算结果为子痫前期（*preeclampsia*）时的 *E-value*。

17.6 (17.6). （风险差的 *Cornfield* 不等式）考虑二值 Z, Y, U。在潜在可忽略性 $Z \perp Y \mid U$ 下，证明风险差的 *Cornfield* 不等式：

$$RD^{\mathrm{obs}}_{ZY} = RD_{ZU} \times RD_{UY}, \tag{84}$$

并由此推出：

$$\min(RD_{ZU}, RD_{UY}) \geq RD^{\mathrm{obs}}_{ZY}, \quad \max(RD_{ZU}, RD_{UY}) \geq \sqrt{RD^{\mathrm{obs}}_{ZY}}. \tag{85}$$

[illegible]

- [illegible] Cornfield [illegible] E-value [illegible]
- [illegible] E-value [illegible]
- [illegible] Bradford Hill [illegible]
- [illegible] E-value [illegible]

CHAPTER 18

[illegible]

[illegible]

Part III [illegible] outcome regression[illegible] X [illegible] Z [illegible]

$$Z \perp \{Y(1), Y(0)\} \mid X.$$

[illegible] $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ [illegible]

[illegible] Part III [illegible]

[illegible] Manski (1990) [illegible]

[illegible] E-value [illegible] outcome regression [illegible] IPW [illegible]

[illegible]

[illegible]

[illegible] 18.1 ([illegible]). 如果观测数据分布与参数 θ 的多个值相容，则称参数 θ 是[illegible]的。

[illegible] θ [illegible]

[illegible] τ [illegible] τ [illegible]

[illegible] outcomes [illegible]

$$\underline{y} \le Y_i(1), Y_i(0) \le \overline{y}.$$

[illegible] $[\underline{y}, \overline{y}] = [0, 1]$ [illegible] $E\{Y(1)\}$ [illegible] $E\{Y(0)\}$ [illegible] $E\{Y(1)\}$ [illegible]

$$E\{Y(1)\} = E\{Y(1) \mid Z = 1\} \Pr(Z = 1) + E\{Y(1) \mid Z = 0\} \Pr(Z = 0).$$

$Y(1) \in [\underline{y}, \overline{y}]$ [illegible]

$$E\{Y(1) \mid Z = 0\} \in [\underline{y}, \overline{y}].$$

[illegible] $E\{Y(1)\}$ [illegible]

$$E\{Y \mid Z = 1\} \Pr(Z = 1) + \underline{y} \Pr(Z = 0),$$

[illegible]

$$E\{Y \mid Z = 1\} \Pr(Z = 1) + \overline{y} \Pr(Z = 0).$$

[1]

[illegible] $E\{Y(0)\}$ [illegible]

$$\underline{y} \Pr(Z = 1) + E\{Y \mid Z = 0\} \Pr(Z = 0),$$

[illegible]

$$\overline{y} \Pr(Z = 1) + E\{Y \mid Z = 0\} \Pr(Z = 0).$$

[illegible] $\tau = E\{Y(1)\} - E\{Y(0)\}$ [illegible]

$$E\{Y \mid Z = 1\} \Pr(Z = 1) + \underline{y} \Pr(Z = 0) - \overline{y} \Pr(Z = 1) - E\{Y \mid Z = 0\} \Pr(Z = 0),$$

[illegible]

$$E\{Y \mid Z = 1\} \Pr(Z = 1) + \overline{y} \Pr(Z = 0) - \underline{y} \Pr(Z = 1) - E\{Y \mid Z = 0\} \Pr(Z = 0).$$

[illegible] $\overline{y} - \underline{y}$ [illegible] $[\underline{y} - \overline{y}, \overline{y} - \underline{y}]$ [illegible] $2(\overline{y} - \underline{y})$ [illegible] 1

[illegible] Cochran (1953) [illegible] Manski (1990) [illegible] Manski (2003) [illegible]

[illegible]

[illegible] Manski [illegible]

[illegible]

$$\varepsilon_1(X) = \frac{E\{Y(1) \mid Z = 1, X\}}{E\{Y(1) \mid Z = 0, X\}}, \quad \varepsilon_0(X) = \frac{E\{Y(0) \mid Z = 1, X\}}{E\{Y(0) \mid Z = 0, X\}}. \tag{86}$$

[illegible] $\varepsilon_1(X)$ [illegible] X [illegible] $\varepsilon_1(X) = 1$ [illegible] $Y(1)$ [illegible] $Y(1)$ [illegible] X [illegible] $\varepsilon_1(X) = \varepsilon_1$ [illegible] $\varepsilon_0(X) = \varepsilon_0$ [illegible]

[illegible] $\mu_1(X) = E(Y \mid Z = 1, X)$ [illegible] $\mu_0(X) = E(Y \mid Z = 0, X)$ [illegible]

---

[1] [illegible] **??**

[illegible]

[illegible] outcome regression)**，**在已知 $\varepsilon_1(X)$ 和 $\varepsilon_0(X)$ 的条件下，我们有

$$E\{Y(1) \mid Z=0\} = E\left\{\frac{\mu_1(X)}{\varepsilon_1(X)} \mid Z=0\right\},$$
$$E\{Y(0) \mid Z=1\} = E\left\{\mu_0(X)\varepsilon_0(X) \mid Z=1\right\}.$$

因此，

$$\tau = E\left[ZY + (1-Z)\frac{\mu_1(X)}{\varepsilon_1(X)}\right] - E\left[Z\mu_0(X)\varepsilon_0(X) + (1-Z)Y\right] \tag{87}$$
$$= E\left[Z\mu_1(X) + (1-Z)\frac{\mu_1(X)}{\varepsilon_1(X)}\right] - E\left[Z\mu_0(X)\varepsilon_0(X) + (1-Z)\mu_0(X)\right]. \tag{88}$$

[illegible] $\varepsilon_1(X) = \varepsilon_0(X) = 1$ [illegible][2] [illegible] outcome regression [illegible]

[illegible][1]

[illegible]

$$\hat{\tau}^{\text{pred}} = \left\{n^{-1}\sum_{i=1}^{n} Z_iY_i + n^{-1}\sum_{i=1}^{n}(1-Z_i)\frac{\hat{\mu}_1(X_i)}{\varepsilon_1(X_i)}\right\} \tag{89}$$
$$- \left\{n^{-1}\sum_{i=1}^{n} Z_i\hat{\mu}_0(X_i)\varepsilon_0(X_i) + n^{-1}\sum_{i=1}^{n}(1-Z_i)Y_i\right\}. \tag{90}$$

(91)

[illegible]

$$\hat{\tau}^{\text{proj}} = \left\{n^{-1}\sum_{i=1}^{n} Z_i\hat{\mu}_1(X_i) + n^{-1}\sum_{i=1}^{n}(1-Z_i)\frac{\hat{\mu}_1(X_i)}{\varepsilon_1(X_i)}\right\} \tag{92}$$
$$- \left\{n^{-1}\sum_{i=1}^{n} Z_i\hat{\mu}_0(X_i)\varepsilon_0(X_i) + n^{-1}\sum_{i=1}^{n}(1-Z_i)\hat{\mu}_0(X_i)\right\}. \tag{93}$$

(94)

[illegible] "predictive" [illegible] "projective" [illegible] Firth and Bennett, 1998; Ding and Li, 2018[illegible]

[illegible]

[illegible] IPW)**，**在已知 $\varepsilon_1(X)$ 和 $\varepsilon_0(X)$ 的条件下，

$$E\{Y(1)\} = E\left\{w_1(X)\frac{Z}{e(X)}Y\right\}, \quad E\{Y(0)\} = E\left\{w_0(X)\frac{1-Z}{1-e(X)}Y\right\},$$

其中

$$w_1(X) = e(X) + \frac{1-e(X)}{\varepsilon_1(X)}, \quad w_0(X) = e(X)\varepsilon_0(X) + 1 - e(X). \tag{95}$$

---

[2] [illegible] ??

[illegible] IPW [illegible] $w_1(X)$ [illegible] $w_0(X)$ [illegible] $\varepsilon_1(X) = \varepsilon_0(X) = 1$ [illegible] $w_1(X) = w_0(X) = 1$ [illegible] IPW[3] [illegible] Horvitz-Thompson [illegible]

$$\hat{\tau}^{\text{ht}} = n^{-1} \sum_{i=1}^{n} \frac{\{\hat{e}(X_i)\varepsilon_1(X_i) + 1 - \hat{e}(X_i)\} Z_i Y_i}{\varepsilon_1(X_i)\hat{e}(X_i)} \tag{96}$$

$$- n^{-1} \sum_{i=1}^{n} \frac{\{\hat{e}(X_i)\varepsilon_0(X_i) + 1 - \hat{e}(X_i)\}(1 - Z_i) Y_i}{1 - \hat{e}(X_i)}, \tag{97}$$

(98)

[illegible] Hajek [illegible]

$$\hat{\tau}^{\text{haj}} = \frac{\sum_{i=1}^{n} \frac{\{\hat{e}(X_i)\varepsilon_1(X_i)+1-\hat{e}(X_i)\}Z_iY_i}{\varepsilon_1(X_i)\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{Z_i}{\hat{e}(X_i)}} \tag{99}$$

$$- \frac{\sum_{i=1}^{n} \frac{\{\hat{e}(X_i)\varepsilon_0(X_i)+1-\hat{e}(X_i)\}(1-Z_i)Y_i}{1-\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{1-Z_i}{1-\hat{e}(X_i)}}. \tag{100}$$

(101)

[illegible] outcome [illegible]

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{ht}} \tag{102}$$

$$- n^{-1} \sum_{i=1}^{n} \{Z_i - \hat{e}(X_i)\} \left\{ \frac{\hat{\mu}_1(X_i)}{\hat{e}(X_i)\varepsilon_1(X_i)} + \frac{\hat{\mu}_0(X_i)\varepsilon_0(X_i)}{1 - \hat{e}(X_i)} \right\}. \tag{103}$$

(104)

[illegible] $\varepsilon_1(X_i)$ [illegible] $\varepsilon_0(X_i)$ [illegible] outcome [illegible] $\hat{\tau}^{\text{dr}}$ [illegible] τ [illegible] $\varepsilon_1(X_i) = \varepsilon_0(X_i) = 1$ [illegible] Part III [illegible] predictive [illegible] IPW [illegible]

X w [illegible]

X w q w [illegible] **w** R [illegible]

```
OS_est_ta = function(z, y, x, out.family = gaussian,
                     truncps = c(0, 1), e1 = 1, e0 = 1) {
    # fitted propensity score
    pscore = glm(z ~ x, family = binomial)$fitted.values
    pscore = pmax(truncps[1], pmin(truncps[2], pscore))

    # fitted potential outcomes
    outcome1 = glm(y ~ x, weights = z,
                   family = out.family)$fitted.values
    outcome0 = glm(y ~ x, weights = (1 - z),
                   family = out.family)$fitted.values
```

---

[3] [illegible] ??

```
    # outcome regression estimator
    ace.reg = mean(z*y) + mean((1-z)*outcome1/e1) -
              mean(z*outcome0*e0) - mean((1-z)*y)

    # IPW estimators
    w1 = pscore + (1-pscore)/e1
    w0 = pscore*e0 + (1-pscore)

    ace.ipw0 = mean(z*y*w1/pscore) - mean((1-z)*y*w0/(1-pscore))
    ace.ipw = mean(z*y*w1/pscore)/mean(z/pscore) -
              mean((1-z)*y*w0/(1-pscore))/mean((1-z)/(1-pscore))

    # doubly robust estimator
    aug = outcome1/pscore/e1 + outcome0*e0/(1-pscore)
    ace.dr = ace.ipw0 + mean((z-pscore)*aug)

    return(c(ace.reg, ace.ipw0, ace.ipw, ace.dr))
}
```

[illegible] NHANES [illegible] BMI [illegible]

$$\varepsilon_1(X) = \varepsilon_0(X) \in \{1/2, 1/1.7, 1/1.5, 1/1.3, 1, 1.3, 1.5, 1.7, 2\},$$

[illegible]
[illegible]

TABLE 1. [illegible] BMI [illegible]

ε	1/2	1/1.7	1/1.5	1/1.3	1	1.3	1.5	1.7	2
1/2	11.62	10.44	9.40	8.03	4.96	0.97	−1.69	−4.35	−8.34
1/1.7	9.22	8.05	7.00	5.64	2.57	−1.42	−4.08	−6.75	−10.74
1/1.5	7.63	6.45	5.41	4.05	0.97	−3.02	−5.68	−8.34	−12.33
1/1.3	6.03	4.86	3.81	2.45	−0.62	−4.61	−7.27	−9.94	−13.93
1	3.64	2.47	1.42	0.06	−3.01	−7.01	−9.67	−12.33	−16.32
1.3	1.80	0.63	−0.42	−1.78	−4.85	−8.85	−11.51	−14.17	−18.16
1.5	0.98	−0.19	−1.24	−2.60	−5.67	−9.66	−12.33	−14.99	−18.98
1.7	0.36	−0.82	−1.86	−3.23	−6.30	−10.29	−12.95	−15.61	−19.60
2	−0.35	−1.52	−2.57	−3.93	−7.00	−10.99	−13.65	−16.32	−20.31

[illegible] $\varepsilon_1(X) > 1$ [illegible] $\varepsilon_0(X) > 1$ [illegible] BMI [illegible] BMI [illegible] $\varepsilon_1(X) < 1$ [illegible] $\varepsilon_0(X) < 1$ [illegible]

[illegible]

[illegible]

(1) [illegible]

(2) i ; l 。 e a [illegible] outcomes [illegible]

(3) [illegible] $\varepsilon_1(X)$ [illegible] $\varepsilon_0(X)$ [illegible]

(4) [illegible] outcome regression~ IPW [illegible]

(5) [illegible] predictive~ projective~ IPW [illegible]

(6) [illegible] ε [illegible]

[illegible]

18.1 (18.1). 证明定理 [illegible]

18.2 (18.2). 证明定理 [illegible]

18.3 (18.3). 第 [illegible]节只给出了点估计。报告相应的 *Bootstrap* 标准误差。

18.4 (18.4). （处理组平均因果效应的灵敏度分析）本习题将第 [illegible]章的结果推广到允许未测混杂的情形，估计

$$\tau_{\mathrm{T}} = \mathbb{E}\{Y(1) - Y(0) \mid Z = 1\} = \mathbb{E}(Y \mid Z = 1) - \mathbb{E}\{Y(0) \mid Z = 1\}.$$

我们容易用样本矩估计 $\mathbb{E}(Y \mid Z = 1)$：$\hat{\mu}_{\mathrm{T1}} = \sum_{i=1}^n Z_i Y_i / \sum_{i=1}^n Z_i$。唯一的反事实项是 $\mathbb{E}\{Y(0) \mid Z = 1\}$，因此我们只需要灵敏度参数 $\varepsilon_0(X)$。在已知 $\varepsilon_0(X)$ 的条件下，我们有以下两个识别公式。

[illegible] 18.4 ([illegible]). 在已知 $\varepsilon_0(X)$ 的条件下，

$$\mathbb{E}\{Y(0) \mid Z = 1\} = \frac{\mathbb{E}\{Z\mu_0(X)\varepsilon_0(X)\}}{e} = \frac{\mathbb{E}\left\{e(X)\varepsilon_0(X)\frac{1-Z}{1-e(X)}Y\right\}}{e},$$

其中 $e = \Pr(Z = 1)$。

证明定理 [illegible]

[illegible]：定理 [illegible]启发我们使用 $\hat{\tau}_{\mathrm{T}}^* = \hat{\mu}_{\mathrm{T1}} - \hat{\mu}_{\mathrm{T0}}^*$ 估计 τ_{T}，其中

$$\hat{\mu}_{\mathrm{T0}}^{\mathrm{reg}} = n_1^{-1} \sum_{i=1}^n Z_i \varepsilon_0(X_i) \hat{\mu}_0(X_i),$$

$$\hat{\mu}_{\mathrm{T0}}^{\mathrm{ht}} = n_1^{-1} \sum_{i=1}^n \varepsilon_0(X_i) \hat{o}(X_i)(1 - Z_i) Y_i,$$

$$\hat{\mu}_{\mathrm{T0}}^{\mathrm{haj}} = \frac{\sum_{i=1}^n \varepsilon_0(X_i) \hat{o}(X_i)(1 - Z_i) Y_i}{\sum_{i=1}^n \hat{o}(X_i)(1 - Z_i)},$$

其中 $\hat{o}(X_i) = \hat{e}(X_i)/\{1 - \hat{e}(X_i)\}$ 是给定协变量条件下治疗的条件比数比估计量。此外，我们可以构造 τ_{T} 的双重稳健估计量

$$\hat{\tau}_{\mathrm{T}}^{\mathrm{dr}} = \hat{\mu}_{\mathrm{T1}} - \hat{\mu}_{\mathrm{T0}}^{\mathrm{dr}},$$

其中

$$\hat{\mu}_{\mathrm{T}0}^{\mathrm{dr}} = \hat{\mu}_{\mathrm{T}0}^{\mathrm{ht}} - n_1^{-1} \sum_{i=1}^{n} \varepsilon_0(X_i) \frac{\hat{e}(X_i) - Z_i}{1 - \hat{e}(X_i)} \hat{\mu}_0(X_i).$$

详见 *Lu and Ding (2023)*。

[illegible]

- [illegible] Manski [illegible]
- [illegible]
- [illegible] E-value [illegible]
- [illegible]
- [illegible]

CHAPTER 19

[illegible]

[illegible]

[illegible] matching [illegible] 11 [illegible]

[illegible] omitted variable bias [illegible]

Paul Rosenbaum 1987 [illegible] sensitivity analysis [illegible]

[illegible] Rosenbaum [illegible] p [illegible] Rosenbaum [illegible]

[illegible]

[illegible]

[illegible] n [illegible] i [illegible] $j = 1$ [illegible] $j = 0$ [illegible] $(i, 1)$ [illegible] i [illegible] $(i, 0)$ [illegible]

[illegible] (i, j) [illegible] $Y_{ij}(1)$ [illegible] $Y_{ij}(0)$ [illegible] sharp null hypothesis [illegible]

$$H_{0F} : Y_{ij}(1) = Y_{ij}(0) \quad \text{for all } i, j.$$

[illegible] $\pi_{i1} = 1/2$

[illegible]

[illegible] Rosenbaum (1987b) [illegible]

[illegible] 19.1 (Rosenbaum [illegible]). 对于第 i 对中的两个单位，处理分配的 *odds* 比被一个有界常数 $\Gamma \geq 1$ 控制：

$$\frac{\pi_{i1}/\pi_{i0}}{(1-\pi_{i1})/(1-\pi_{i0})} \leq \Gamma, \quad i = 1, \ldots, n,$$

其中 $\pi_{i1} = \Pr\{Z_{i1} = 1 \mid pair\ i\}$ 是配对 i 中处理组接受处理的概率。

[illegible] Γ [illegible]

- [illegible] $\Gamma = 1$ [illegible] $\pi_{i1} = \pi_{i0} = 1/2$ [illegible] MPE [illegible]

- [illegible] $\Gamma > 1$ [illegible] Γ [illegible]
- Γ [illegible]

[illegible] odds [illegible] odds [illegible] Γ [illegible] p [illegible]

[illegible]

[illegible] sharp null hypothesis H_{0F} [illegible] ” ” [illegible] Fisher randomization test [illegible] q_i [illegible] i [illegible] choices [illegible]

(1) [illegible] $q_i = Z_{i1} - Z_{i0}$ [illegible] vs [illegible]
(2) [illegible] $q_i = Y_{i1} - Y_{i0}$ [illegible]
(3) [illegible] $|Y_{i1} - Y_{i0}|$ [illegible]

[illegible]

$$T = \sum_{i=1}^{n} S_i q_i,$$

[illegible] $S_i \in \{0, 1\}$ [illegible] i [illegible] $S_i = 1$ [illegible] q_i [illegible] i [illegible]

[illegible] S_i [illegible] S_i [illegible]

[illegible]

[illegible] p [illegible] 在 [illegible] 下，零假设 H_{0F} 成立时，配对 i 的处理指示 S_i 服从最坏情况的 *Bernoulli* 分布：

$$S_i \sim \mathit{Bernoulli}\left(\frac{\Gamma}{1+\Gamma}\right),$$

且各配对的 S_i 之间相互独立。

[illegible] odds ratio [illegible] Γ [illegible] S_i [illegible] $[\pi_{\min}, \pi_{\max}]$ [illegible]

$$\pi_{\max} = \frac{\Gamma}{1+\Gamma}, \quad \pi_{\min} = \frac{1}{1+\Gamma}.$$

[illegible] p [illegible] S_i [illegible] $\pi_{\max}$ [illegible] ” [illegible] ” [illegible] p[1]

[illegible]

[illegible] p [illegible] T^{obs} [illegible] p [illegible]

$$\mathrm{p}_\Gamma = \Pr_\Gamma\left(|T - T^{\text{obs}}| \geq |T^{\text{obs}}|\right),$$

[illegible] T [illegible] S_i [illegible]

---

[1] [illegible] ?? [illegible] ??

[illegible] $q_i \geq 0$ [illegible]

$$T = \sum_{i=1}^{n} S_i q_i, \quad S_i \sim \text{Bernoulli}\left(\frac{\Gamma}{1+\Gamma}\right).$$

q_i [illegible] T [illegible] n [illegible] Bernoulli [illegible]

[illegible]

[illegible] LaLonde (1986) [illegible] National Supported Work Demonstration [illegible]

[illegible]

$$T = \sum_{i=1}^{n} S_i |\hat{\tau}_i|,$$

[illegible] $\hat{\tau}_i$ [illegible] i [illegible]

- [illegible] MPE [illegible] $\Gamma = 1$ [illegible] p [illegible] 0.002 [illegible]
- [illegible] $\Gamma = 1.1$ [illegible] 10% [illegible] odds [illegible] p [illegible] 0.011
- [illegible] $\Gamma = 1.3$ [illegible] p [illegible] 0.05 [illegible]

[illegible] $\Gamma = 1.233$ [illegible] odds [illegible] 1.233 [illegible] 0.05 [illegible]

[illegible] 19.1 (LaLonde [illegible]). 图 [illegible] 展示了 p 值随 Γ 变化的灵敏度曲线。曲线的陡峭程度告诉我们结论对未测量混杂的敏感程度——曲线越平缓，结论越稳健。

[illegible] Rosenbaum [illegible] R [illegible] `sensitivity2x2` [illegible] `sensitivitymv` [illegible]

[illegible] 2×2 [illegible] Γ [illegible]

[illegible]

[illegible]

[illegible] 19.1. *Rosenbaum 灵敏度模型的基本性质* 证明：当 $\Gamma \to \infty$ 时，[illegible] 中的约束变得无效，此时 S_i 的最坏情况分布趋近于 *Bernoulli*$(1/2)$。这对应着什么实际含义？

[illegible] 19.2. *最坏情况 p 值的单调性* 设 T^{obs} 为观测到的检验统计量。证明：最坏情况 p 值 p_Γ 是 Γ 的单调递增函数。解释这在直觉上意味着什么。

[illegible] 19.3. *Sign statistic 的灵敏度分析* 对于 *sign statistic* $q_i = 1$（所有配对权重相等），推导 T 在 [illegible] 下的分布，并给出 p 值的显式计算公式。

[illegible] 19.4. *LaLonde 数据的计算* 使用 *LaLonde* 数据集，计算在不同 Γ 值$(1, 1.1, 1.2, 1.3, 1.5)$下的最坏情况 p 值，并绘制灵敏度曲线。

[illegible]

[illegible] Rosenbaum [illegible] [illegible]

[illegible]

1. (1) [illegible] odds ratio [illegible] $\Gamma \geq 1$ [illegible]
2. (2) $\Gamma = 1$ [illegible] $\Gamma > 1$ [illegible]
3. (3) [illegible] S_i [illegible] p
4. (4) [illegible] ” [illegible] ” [illegible]

[illegible] Rosenbaum (1987b) [illegible] Rosenbaum [illegible] (2002, 2010) [illegible]

[illegible] 11 [illegible] 3 Fisher randomization test [illegible] 12 [illegible] ” [illegible] ” [illegible]

CHAPTER 20

[illegible]

[illegible]

“If we could intervene on a unit and set its treatment, then we would know its potential outcome under that treatment. But what if the treatment is deterministic for some units? Can we still conceive their counterfactual outcomes?” — (, adapted from King and Zeng, 2006)

[illegible] ** [illegible] ** [illegible] ** [illegible] ** [illegible] Rubin [illegible] Rubin, 1974, 1975 [illegible] ” [illegible] ” [illegible] { {[illegible] **[illegible]**

$$0 < e(X) < 1. \tag{20.1}$$

[illegible] $e(X) = 0$ [illegible] $e(X) = 1$ [illegible] ” [illegible] ” [illegible] **[illegible]**

[illegible]

[illegible] { {[illegible] { [illegible] D’Amour [illegible] 2021 [illegible]

- [illegible] ignorability [illegible] { { [illegible] M-bias [illegible]
- [illegible]

[illegible]

[illegible] 20.1 ([illegible]). 存在某个 $\eta \in (0, 1/2)$ 使得

$$\eta \le e(X) \le 1 - \eta.$$

D’Amour [illegible] 2021 [illegible] p [illegible][1] [illegible]

[illegible] (D’Amour et al., 2021) 在 { {下，设 $e = \Pr(Z = 1)$ 为治疗组比例，λ_1 和 λ_0 分别为 $X|Z = 1$ 和 $X|Z = 0$ 条件下协变量矩阵的最大特征值。则

$$\frac{1}{p}\sum_{k=1}^{p}\bigl|\mathbb{E}(X_k \mid Z = 1) - \mathbb{E}(X_k \mid Z = 0)\bigr| \le p^{-1/2}C^{1/2}\bigl\{e\lambda_1^{1/2} + (1-e)\lambda_0^{1/2}\bigr\}, \tag{20.2}$$

---

[1] [illegible] **??**

其中 $C = \dfrac{(e-\eta)(1-e-\eta)}{e^2(1-e)^2\eta(1-\eta)}$ 是仅依赖于 (e,η) 的正常数。

[illegible]

[illegible]

[illegible], 2009; Yang and Ding, 2018b[illegible]

[illegible] D’Amour [illegible] 2021[illegible]

[illegible] X [illegible] $r(X)$[illegible]

$$\mathbb{E}\{Y(z) \mid X\} = f_z(r(X)), \quad (z = 0, 1), \tag{20.3}$$

[illegible] $r(X)$ [illegible] $r(X)$ [illegible] X [illegible][2] [illegible]

[illegible]

[illegible] X [illegible]

$$Z = \mathbb{I}(X \geq x_0), \tag{20.4}$$

[illegible] x_0 [illegible]

$$Z \perp \{Y(1), Y(0)\} \mid X, \tag{20.5}$$

Z [illegible] X [illegible]

$$e(x) = \Pr(Z = 1 \mid X = x) = \mathbb{I}(x \geq x_0) = \begin{cases} 1 & \text{[illegible]} \ x \geq x_0, \\ 0 & \text{[illegible]} \ x < x_0. \end{cases} \tag{20.6}$$

[illegible]

---

[2] [illegible] ??

[illegible] [illegible] Thistlethwaite [illegible] Campbell [illegible] 1960 [illegible] "[illegible]" [illegible] "[illegible]" [illegible]

[illegible] HIV [illegible] Bor [illegible] 2014 [illegible] HIV [illegible] CD4 [illegible] 200 cells/μL [illegible]

[illegible] Carpenter [illegible] Dobkin [illegible] 2009 [illegible]

[illegible] X [illegible] x_0 [illegible]

$$\tau(x_0) = \mathbb{E}\{Y(1) - Y(0) \mid X = x_0\}. \tag{20.7}$$

[illegible] (RD [illegible]) 假设治疗由 $Z = \mathbb{I}(X \geq x_0)$ 确定，其中 x_0 是预设阈值。假设 $\mathbb{E}\{Y(1) \mid X = x\}$ 在 x_0 处右连续，$\mathbb{E}\{Y(0) \mid X = x\}$ 在 x_0 处左连续。则在 $X = x_0$ 处的局部平均因果效应由下式识别：

$$\tau(x_0) = \lim_{\varepsilon \to 0^+} \mathbb{E}(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \to 0^+} \mathbb{E}(Y \mid Z = 0, X = x_0 - \varepsilon). \tag{20.8}$$

[illegible] $\tau(x_0)$ [illegible][3]

[illegible] $\mathbb{E}(Y \mid Z = 1, X = x) = \gamma_1 + \beta_1 x$ [illegible] $\mathbb{E}(Y \mid Z = 0, X = x) = \gamma_0 + \beta_0 x$ [illegible] x [illegible] x_0 [illegible]

$$\hat{\tau}(x_0) = (\hat{\gamma}_1 - \hat{\gamma}_0) + (\hat{\beta}_1 - \hat{\beta}_0)x_0. \tag{20.9}$$

[illegible] $\hat{\tau}(x_0)$ OLS [illegible]

$$Y_i \sim \{1, Z_i, X_i - x_0, Z_i(X_i - x_0)\} \tag{20.10}$$

[illegible] Z_i [illegible][4]

[illegible] `rdrobust` [illegible] "[illegible]" [illegible]

[illegible] RD [illegible]

[illegible] 21 [illegible]

McCrary [illegible] 2008 [illegible]

[illegible]

[illegible]

(1) [illegible]

(2) [illegible]

---

[3] [illegible] **??**

[4] [illegible] **??**

(3) [illegible] RD [illegible]
[illegible]

RD [illegible]
[illegible] ” [illegible] ” [illegible]
[illegible]

[illegible]

[illegible] [illegible][5]
[illegible] $\Sigma_1 = \mathrm{Cov}(X \mid Z = 1)$ [illegible] $\Sigma_0 = \mathrm{Cov}(X \mid Z = 0)$ [illegible]

$$\Sigma = \mathbb{E}\{\mathrm{Cov}(X \mid Z)\} + \mathrm{Cov}\{\mathbb{E}(X \mid Z)\}. \tag{20.11}$$

[illegible] Σ_1 [illegible] Σ_0 [illegible] C [illegible] Cauchy-Schwarz [illegible]
[illegible]

$$\frac{1}{p}\sum_{k=1}^{p} |\mathbb{E}(X_k \mid Z = 1) - \mathbb{E}(X_k \mid Z = 0)| \le \frac{1}{\sqrt{p}}\|\Sigma^{1/2}\| \cdot \|d\|, \tag{20.12}$$

[illegible] $d = \mathbb{E}(X \mid Z = 1) - \mathbb{E}(X \mid Z = 0)$ [illegible] $\|\cdot\|$ [illegible]
[illegible] ☐

[illegible] [illegible] $\tau = \mathbb{E}\{Y(1) - Y(0)\}$ [illegible]

$$\tau = \mathbb{E}\big\{\mathbb{E}(Y \mid Z = 1, r(X)) - \mathbb{E}(Y \mid Z = 0, r(X))\big\} \tag{20.13}$$

[illegible]

$$\tau = \mathbb{E}\left\{\frac{ZY}{e(r(X))}\right\} - \mathbb{E}\left\{\frac{(1-Z)Y}{1-e(r(X))}\right\}, \tag{20.14}$$

[illegible] $e(r(X)) = \Pr\{Z = 1 \mid r(X)\}$[6]

[illegible] [illegible]

$$\mathbb{E}\{Y(1) \mid X = x_0\} = \lim_{\varepsilon \to 0^+} \mathbb{E}\{Y(1) \mid X = x_0 + \varepsilon\}. \tag{20.15}$$

Z [illegible] $X = x_0 + \varepsilon$ [illegible] $Z = 1$ [illegible]

$$\mathbb{E}\{Y(1) \mid X = x_0 + \varepsilon\} = \mathbb{E}(Y \mid Z = 1, X = x_0 + \varepsilon). \tag{20.16}$$

---

[5] [illegible] D’Amour et al. (2021, Corollary 1)

[6] [illegible] $r(X)$ [illegible]

[illegible] $\mathbb{E}\{Y(0) \mid X = x_0\}$ [illegible] { { □

WwXw [illegible] **w**[illegible] $R_i = \max(X_i - x_0, 0)$ [illegible] $L_i = \min(X_i - x_0, 0)$

$$Z_i(X_i - x_0) = Z_iR_i - (1 - Z_i)L_i = R_i - L_i - (1 - Z_i)R_i + (1 - Z_i)L_i. \tag{20.17}$$

[illegible] $Y_i \sim \{1, Z_i, X_i - x_0, Z_i(X_i - x_0)\}$ [illegible] Z_i [illegible]

$$\hat{\tau} = (\hat{\gamma}_1 - \hat{\gamma}_0) + (\hat{\beta}_1 - \hat{\beta}_0)x_0. \tag{20.18}$$

{ { [illegible] $Y_i \sim \{1, Z_i, R_i, L_i\}$. [illegible] □

CHAPTER 21

CRE

(1) Jobs treatment

(2)

(3)

Z D D

Z D Y D

D Z

“如果我无法做随机实验，能否找到一个自然的随机化来源？”

- $Z \in \{0,1\}$ $Z = 1$
- $D \in \{0,1,2,\ldots\}$
- Y

21.1 ( ). 治疗分配 Z 与所有潜在结果独立：

$$Z \perp\!\!\!\perp \{D(1), D(0), Y(1), Y(0)\},$$

---

[1] ??

[2] ??

其中 $D(z)$ 是在分配 z 下的潜在治疗接受情况，$Y(z)$ 是相应的潜在结局。

[illegible] Z [illegible] D [illegible] τ_Y [illegible] ITT [illegible][3] [illegible] ITT [illegible]

$$\hat{\tau}_Y = \bar{Y}_1 - \bar{Y}_0$$

[illegible] IV [illegible]

[illegible]

[illegible] $D_i(1)$ [illegible] $D_i(0)$ [illegible] $U_i = (D_i(1), D_i(0))$ [illegible]

(1) [illegible] $U_i = a$ [illegible] $D_i(1) = D_i(0) = 1$ [illegible]

(2) [illegible] $U_i = c$ [illegible] $D_i(1) = 1, D_i(0) = 0$ [illegible]

(3) [illegible] $U_i = d$ [illegible] $D_i(1) = 0, D_i(0) = 1$ [illegible]

(4) [illegible] $U_i = n$ [illegible] $D_i(1) = D_i(0) = 0$ [illegible]

[illegible] $D(1)$ [illegible] $D(0)$ [illegible] U_i [illegible]

[illegible][4]

$$
\begin{aligned}
\tau_Y &= \mathbb{E}\{Y(1) - Y(0)\} && (105)\\
&= \mathbb{E}\{Y(1) - Y(0) \mid U = a\} \Pr(U = a) && (106)\\
&\quad + \mathbb{E}\{Y(1) - Y(0) \mid U = c\} \Pr(U = c) && (107)\\
&\quad + \mathbb{E}\{Y(1) - Y(0) \mid U = d\} \Pr(U = d) && (108)\\
&\quad + \mathbb{E}\{Y(1) - Y(0) \mid U = n\} \Pr(U = n). && (109)
\end{aligned}
$$

[illegible] complier [illegible]

[illegible]

[illegible] complier [illegible] CACE [illegible]

---

[3] [illegible] ITT [illegible] **??**

[4] [illegible] **??**

[illegible]

[illegible] 21.2 ([illegible]). 没有 *defier*: $\Pr(U_i = d) = 0$，或者等价地，$D_i(1) \geq D_i(0)$ 对所有 i 成立。

[illegible] Defier [illegible]

$$\Pr(D = 1 \mid Z = 1) \geq \Pr(D = 1 \mid Z = 0), \tag{110}$$

[illegible][5] [illegible] $D_i(1) \geq D_i(0)$ [illegible] $D_i(0) = 0$ [illegible]

[illegible]

[illegible] 21.3 ([illegible]). 对 *always taker* 和 *never taker*，治疗分配不影响结局：

$$Y_i(1) = Y_i(0) \quad 当 \quad U_i = a \text{ 或} U_i = n.$$

[illegible] always taker [illegible] never taker [illegible] Z [illegible] Z [illegible] Z [illegible] Y [illegible] $D_i(1) = D_i(0)$ [illegible] $Y_i(1) = Y_i(0)$ [illegible] i [illegible] Z [illegible] D [illegible] Y

[illegible] Z [illegible]

[illegible]

[illegible] always taker [illegible] never taker [illegible] $Y(1) = Y(0)$ [illegible] defier [illegible]

$$\tau_Y = \mathbb{E}\{Y(1) - Y(0) \mid U = c\} \Pr(U = c). \tag{111}$$

[illegible]

$$\tau_D = \mathbb{E}\{D(1) - D(0)\} = \Pr(U = c), \tag{112}$$

[illegible] complier [illegible] $Z = 1$ [illegible] $Z = 0$ [illegible]

---

[5] [illegible] ?? [illegible] ??

[illegible] CACE [illegible]) [illegible],

$$\tau_c \overset{def}{=} \mathbb{E}\{Y(1) - Y(0) \mid U = c\} = \frac{\tau_Y}{\tau_D}. \tag{113}$$

[illegible] CACE/LATE [illegible]) [illegible],

$$\tau_c = \frac{\mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0)}{\mathbb{E}(D \mid Z = 1) - \mathbb{E}(D \mid Z = 0)}. \tag{114}$$

[illegible] ITT [illegible] ITT [illegible] τ_Y/τ_D [illegible] complier average causal effect [illegible] CACE [illegible][6] [illegible]

[illegible] CACE [illegible] complier [illegible] complier [illegible] $D(1) > D(0)$ [illegible] $D(1) = 1, D(0) = 0$ [illegible] complier [illegible] $D = Z$ [illegible] CACE [illegible] complier [illegible] CACE [illegible] complier [illegible] always taker [illegible] never taker[5] [illegible] defier [illegible] IV [illegible]

[illegible]

1 [illegible] τ_c [illegible]

$$\hat{\tau}_c = \frac{\hat{\tau}_Y}{\hat{\tau}_D} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0}, \tag{115}$$

[illegible] $\bar{Y}_z$ [illegible] $\bar{D}_z$ [illegible] $Z = z$ [illegible] 5 [illegible] [7] $\tau_c \neq 0$ [illegible]

$$\hat{V}(\hat{\tau}_c) = \frac{\hat{V}_Y}{\hat{\tau}_D^2},$$

[illegible] $\hat{V}_Y$ [illegible] $\hat{\tau}_Y$ [illegible] Neyman [illegible]

$$\left(\frac{\hat{\tau}_Y}{\hat{\tau}_D} - z_{1-\alpha/2}\sqrt{\frac{\hat{V}_Y}{\hat{\tau}_D^2}},\ \frac{\hat{\tau}_Y}{\hat{\tau}_D} + z_{1-\alpha/2}\sqrt{\frac{\hat{V}_Y}{\hat{\tau}_D^2}} \right).$$

[illegible] $\tau_c = 0$ [illegible] $\hat{V}_Y/\hat{\tau}_D^2$ [illegible] Wald [illegible] ITT [illegible] $1/\hat{\tau}_D > 1$ [illegible]

[illegible]

[illegible] IV [illegible] X

---

[6] [illegible] ??

[7] [illegible] ??

[illegible]

$$\tau_c(x) = \frac{\mathbb{E}(Y \mid Z=1, X=x) - \mathbb{E}(Y \mid Z=0, X=x)}{\mathbb{E}(D \mid Z=1, X=x) - \mathbb{E}(D \mid Z=0, X=x)}. \tag{116}$$

[illegible] X [illegible] CACE [illegible] CRE [illegible] $X = x$ [illegible] IV [illegible]

[illegible] X [illegible]

$$Z \perp\!\!\!\perp \{D(1), D(0), Y(1), Y(0)\} \mid X,$$

[illegible] X [illegible]

[illegible]

[illegible] $\tau_D > 0$ [illegible] complier [illegible] τ_D [illegible] 0 [illegible] Wald [illegible] $\tau_D > 0$ [illegible] $\hat{\tau}_D = 0$ [illegible] $\hat{\tau}_c = \hat{\tau}_Y / \hat{\tau}_D$ [illegible][8] [illegible] IV [illegible]

(1) $\hat{\tau}_c$ [illegible]
(2) $\hat{\tau}_c$ [illegible]
(3) Wald [illegible]

[illegible] Y [illegible] $\tau_Y \in [-1, 1]$ [illegible] $\hat{\tau}_c = \hat{\tau}_Y / \hat{\tau}_D$ [illegible]

[illegible] IV [illegible][9] FAR [illegible] τ_c [illegible] τ_c [illegible] IV [illegible] CRE [illegible] FAR [illegible]

$$(\tau - \hat{\tau}_Y)^2 \leq z_{1-\alpha/2}^2 \hat{V}_Y,$$

[illegible]

[illegible]

notation $\{D(1), D(0), Y(1), Y(0)\}$ [illegible] Z [illegible] notation [illegible] notation [illegible] τ_c [illegible] complier [illegible] complier [illegible] $D(1) = 1$ [illegible] $D(0) = 0$ [illegible] $D = Z$ [illegible] τ_c [illegible] complier [illegible] $Y_i(z, d)$ [illegible] i [illegible] z [illegible] d [illegible] notation [illegible]

[illegible] 21.4 ([illegible] Angrist notation[illegible]). $Y_i(z, d) = Y_i(d)$ 对所有 i, z, d 成立，即潜在结局只是 d 的函数。

---

[8] [illegible] $\hat{\tau}_D$ [illegible] 0 [illegible] ??

[9] [illegible] FAR [illegible] ??

notation [illegible] Z [illegible] Y [illegible] Z [illegible] D [illegible]
[illegible]

$$\tau_c = \frac{\mathbb{E}\{Y(D(1)) - Y(D(0))\}}{\mathbb{E}\{D(1) - D(0)\}},$$

[illegible] CACE [illegible]
[illegible] τ_c [illegible] complier [illegible] ” [illegible] ”[illegible]
[illegible] Z [illegible] Z [illegible] complier [illegible]
[illegible] LATE [illegible]

21.1. [illegible] ***CACE*** [illegible] “[illegible]” [illegible]**：** 因为它是总体中一个特定子群体（*complier*）的平均因果效应。这个子群体恰好是那些“被 Z 影响而改变治疗状态”的人。不同的工具变量可能识别不同的子群体。

[illegible]

[illegible] 21.1. [illegible]*Wald* 估计量的方差 证明 *Wald* 估计量 $\hat{\tau}_c = \hat{\tau}_Y/\hat{\tau}_D$ 的渐近方差为 $\hat{V}_Y/\tau_D^2$，其中 $\hat{V}_Y$ 是 *ITT* 估计量 $\hat{\tau}_Y$ 的方差。

[illegible] 21.2. [illegible]强单调性下的 *IV* 估计 在强单调性（所有人都是 *complier* 或 *never taker*，且 $\Pr(U = n) = 0$）下，证明 *IV* 估计量 $\hat{\tau}_c$ 的一致性。

[illegible] 21.3. [illegible]单侧非依从性 在单侧非依从性（*never taker* 存在，但没有 *always taker*，且 $D_i(0) = 0$ 对所有人成立）下，推导 *CACE* 的识别公式，并解释其含义。

[illegible] 21.4. [illegible]违反排除限制的后果 假设排除限制不成立，即存在 *always taker* 使得 $Y_i(1) \neq Y_i(0)$。证明此时 [illegible]估计量是有偏的，并讨论偏误的方向。

CHAPTER 22

[illegible]

[illegible]

[illegible]

“The essential problem of causal inference is that we can observe at most one of the potential outcomes for any unit.”

[illegible]

[illegible]

[illegible]

- $Z_i \in \{0,1\}$ [illegible] $1 =$ [illegible] $0 =$ [illegible]
- $D_i \in \{0,1\}$ [illegible] $1 =$ [illegible] $0 =$ [illegible]
- Y_i [illegible]

[illegible] $Z_i \neq D_i$ [illegible]

[illegible]

$$\{D_i(1), D_i(0), Y_i(1), Y_i(0)\},$$

[illegible] $D_i(z)$ [illegible] z [illegible] $Y_i(z)$ [illegible] z [illegible]

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0), \quad Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

[illegible]

[illegible] 22.1 ([illegible]). 治疗分配 Z 与潜在结果独立：

$$Z \perp\!\!\!\perp \{D(1), D(0), Y(1), Y(0)\}.$$

[illegible] D [illegible] Y [illegible][1]

$$\tau_D = \mathbb{E}\{D(1) - D(0)\} = \mathbb{E}(D \mid Z = 1) - \mathbb{E}(D \mid Z = 0),$$

---

[1] [illegible] **??**

$$\tau_Y = \mathbb{E}\{Y(1) - Y(0)\} = \mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0).$$

[illegible] $\hat{\tau}_D$ [illegible] $\hat{\tau}_Y$ [illegible] τ_D [illegible] τ_Y

[illegible] $\hat{\tau}_Y$ [illegible][2] [illegible]
ITT [illegible]

$$\underbrace{\hat{\tau}_Y}_{\text{ITT [illegible]}} \xrightarrow{p} \underbrace{\tau_Y}_{\text{[illegible]}},$$

[illegible]

$$\underbrace{\tau_{\text{treatment}}}_{\text{[illegible]}} \neq \underbrace{\tau_Y}_{\text{[illegible]}}.$$

[illegible]

[illegible]

[illegible] $\{D_i(1), D_i(0)\}$ [illegible] D [illegible]

22.1 ([illegible]). 令 $U_i = \{D_i(1), D_i(0)\}$，则：

$$U_i = \begin{cases} a \text{（}\textit{always taker}\text{）}, & \text{若} D_i(1) = 1 \text{ 且} D_i(0) = 1; \\ c \text{（}\textit{complier}\text{）}, & \text{若} D_i(1) = 1 \text{ 且} D_i(0) = 0; \\ d \text{（}\textit{defier}\text{）}, & \text{若} D_i(1) = 0 \text{ 且} D_i(0) = 1; \\ n \text{（}\textit{never taker}\text{）}, & \text{若} D_i(1) = 0 \text{ 且} D_i(0) = 0. \end{cases}$$

[illegible]

- [illegible]
- [illegible]
- [illegible]
- [illegible]

[illegible] $D_i(1)$ [illegible] $D_i(0)$ [illegible] U_i [illegible]

[illegible] U [illegible] τ_Y [illegible][3]

$$\begin{aligned} \tau_Y = {} & \mathbb{E}\{Y(1) - Y(0) \mid U = \mathrm{a}\} \Pr(U = \mathrm{a}) \\ & + \mathbb{E}\{Y(1) - Y(0) \mid U = \mathrm{c}\} \Pr(U = \mathrm{c}) \\ & + \mathbb{E}\{Y(1) - Y(0) \mid U = \mathrm{d}\} \Pr(U = \mathrm{d}) \\ & + \mathbb{E}\{Y(1) - Y(0) \mid U = \mathrm{n}\} \Pr(U = \mathrm{n}). \end{aligned}$$

[illegible] τ_Y [illegible]

---

[2] ITT [illegible]

[3] [illegible] **??**

[illegible] τ_D [illegible][4]

$$\begin{aligned}\tau_D = {} & \mathbb{E}\{D(1) - D(0) \mid U = \mathrm{a}\} \Pr(U = \mathrm{a}) \\ & + \mathbb{E}\{D(1) - D(0) \mid U = \mathrm{c}\} \Pr(U = \mathrm{c}) \\ & + \mathbb{E}\{D(1) - D(0) \mid U = \mathrm{d}\} \Pr(U = \mathrm{d}) \\ & + \mathbb{E}\{D(1) - D(0) \mid U = \mathrm{n}\} \Pr(U = \mathrm{n}) \\ = {} & 0 \cdot \Pr(U = \mathrm{a}) + 1 \cdot \Pr(U = \mathrm{c}) + (-1) \cdot \Pr(U = \mathrm{d}) + 0 \cdot \Pr(U = \mathrm{n}) \\ = {} & \Pr(U = \mathrm{c}) - \Pr(U = \mathrm{d}).\end{aligned}$$

X w [illegible]

[illegible] τ_Y [illegible] τ_Y [illegible] τ_D [illegible]

X w [illegible] w

[illegible] 22.2 ([illegible]). $\Pr(U = d) = 0$，或等价地，$D_i(1) \geq D_i(0)$ 对所有单元成立。即，[illegible] ***defiers***。

[illegible] $D_i(0) = 0$ [illegible] $D_i(1) \geq D_i(0)$ [illegible][5]

$$\Pr(D = 1 \mid Z = 1) \geq \Pr(D = 1 \mid Z = 0).$$

[illegible] $D_i(1)$ [illegible] $D_i(0)$ [illegible]

$$\tau_D = \Pr(U = \mathrm{c}) \equiv \pi_c.$$

compliers [illegible] π_c [illegible]

X w [illegible] w

[illegible] 22.3 ([illegible]). 对于 *always takers* ($U = a$) 和 *never takers* ($U = n$)，有 $Y_i(1) = Y_i(0)$。

[illegible]

$$\tau_Y = \mathbb{E}\{Y(1) - Y(0) \mid U = \mathrm{c}\} \Pr(U = \mathrm{c}).$$

[illegible]

---

[4] [illegible] ??

[5] [illegible] ??

[illegible]

[illegible] (CACE/LATE [illegible]) 在 [illegible] 和 [illegible] 下，如果 $\tau_D \neq 0$，则

$$\mathbb{E}\{Y(1) - Y(0) \mid U = c\} = \frac{\tau_Y}{\tau_D}.$$

[illegible][6] [illegible] τ_Y/τ_D [illegible] compliers [illegible] compliers [illegible] ” [illegible] ” [illegible] always takers [illegible] never takers [illegible] defiers [illegible]

[illegible]

[illegible]

[illegible] 22.3 (CACE [illegible] LATE). [illegible] ***complier average causal effect, CACE*** 或 [illegible] ***local average treatment effect, LATE*** 定义为：

$$\tau_c = \mathbb{E}\{Y(1) - Y(0) \mid U = c\}.$$

CACE [illegible]

$$\tau_c = \mathbb{E}\{Y(1) - Y(0) \mid D(1) = 1, D(0) = 0\} = \mathbb{E}\{Y(1) - Y(0) \mid D(1) > D(0)\}.$$

1 [illegible]

$$\tau_c = \frac{\tau_Y}{\tau_D}.$$

[illegible] **[illegible]** [illegible]

$$\tau_c = \frac{\mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0)}{\mathbb{E}(D \mid Z = 1) - \mathbb{E}(D \mid Z = 0)}.$$

[illegible] **[illegible]** [illegible]

[illegible]

[illegible]

[illegible] CACE [illegible]

$$\hat{\tau}_c = \frac{\hat{\tau}_Y}{\hat{\tau}_D},$$

[illegible]

$$\hat{\tau}_Y = \frac{1}{n_1} \sum_{i:Z_i=1} Y_i - \frac{1}{n_0} \sum_{i:Z_i=0} Y_i, \quad \hat{\tau}_D = \frac{1}{n_1} \sum_{i:Z_i=1} D_i - \frac{1}{n_0} \sum_{i:Z_i=0} D_i.$$

[illegible] Z [illegible] D [illegible]

[illegible] Wald [illegible] ” [illegible] ” ITT [illegible] $\hat{\tau}_Y$ [illegible] $\hat{\tau}_D$ [illegible] ” [illegible] ” [illegible] compliers [illegible] ” [illegible] ” [illegible] compliers [illegible]

---

[6] [illegible] ??

[illegible] $\hat{\tau}_c$ [illegible][7] [illegible] $\hat{\tau}_c$ [illegible][8] [illegible]

$$\hat{\tau}_c - \tau_c \approx \frac{\hat{\tau}_Y - \tau_c \hat{\tau}_D}{\tau_D}.$$

[illegible] $A_i = Y_i - \tau_c D_i$ [illegible] $\hat{\tau}_A / \tau_D$ [illegible] $\hat{\tau}_A$ [illegible]

[illegible] $\hat{\tau}_c$ [illegible] $\hat{V}_A / \tau_D^2$ [illegible] $\hat{V}_A$ [illegible] Neyman [illegible]

$$\hat{V}_A = \frac{\hat{S}_A^2(1)}{n_1} + \frac{\hat{S}_A^2(0)}{n_0},$$

[illegible] $\hat{S}_A^2(1)$ [illegible] $\hat{S}_A^2(0)$ [illegible]

$$\hat{V}_{\hat{\tau}_c} = \frac{\hat{V}_A}{\hat{\tau}_D^2}$$

$\hat{\tau}_c$ [illegible] τ_c [illegible] $1 - \alpha$ [illegible]

$$\hat{\tau}_c \pm z_{1-\alpha/2} \sqrt{\hat{V}_{\hat{\tau}_c}}.$$

[illegible]

[illegible] 5% [illegible] Wald [illegible]

[illegible] $\tau_D > 0$ [illegible] compliers [illegible] $\hat{\tau}_D$ [illegible] Wald [illegible] $\tau_D \to 0$ [illegible] $\operatorname{var}(\hat{\tau}_c) \to \infty$[9] [illegible]

- $\hat{\tau}_c$ [illegible]
- [illegible]
- [illegible]

[illegible] F [illegible] 5 [illegible] $\hat{\tau}_D$ [illegible] $\hat{\tau}_D$ [illegible] Wald [illegible]

[illegible] [illegible] $\tau_c = \tau_Y / \tau_D$ [illegible] $\tau_c = 0$ [illegible] $\tau_Y = 0$ [illegible][10] [illegible] τ_c [illegible] b [illegible] $H_0(b) : \tau_c = b$ [illegible] $\tau_Y - b\tau_D = 0$ [illegible] $A(b)_i = Y_i - bD_i$ [illegible]

---

[7] [illegible] ??

[8] [illegible] ??

[9] [illegible] Fieller-Creasy [illegible]

[10] FAR [illegible] ??

[illegible] Wald [illegible]

$$\left\{ b : \frac{\hat{\tau}_A^2(b)}{\hat{V}_A(b)} \le z_{1-\alpha/2}^2 \right\},$$

[illegible] $\hat{\tau}_A(b)$ [illegible] $\hat{V}_A(b)$ [illegible] $A(b)_i = Y_i - bD_i$ [illegible] FAR [illegible] Wald [illegible]

[illegible]

[illegible]

[illegible] [illegible]

$$Z \perp \{D(1), D(0), Y(1), Y(0), X\}.$$

[illegible]

$$D_i = \beta_0 + \beta_1 Z_i + \beta_2 X_i + \beta_3 Z_i X_i + \epsilon_i,$$

$$Y_i = \alpha_0 + \alpha_1 Z_i + \alpha_2 X_i + \alpha_3 Z_i X_i + \eta_i,$$

[illegible] $\hat{\tau}_{D,\mathrm{L}} = \hat{\beta}_1$ [illegible] $\hat{\tau}_{Y,\mathrm{L}} = \hat{\alpha}_1$ [illegible] CACE [illegible]

$$\hat{\tau}_{c,\mathrm{L}} = \frac{\hat{\tau}_{Y,\mathrm{L}}}{\hat{\tau}_{D,\mathrm{L}}}.$$

[illegible] Bootstrap [illegible]

[illegible]

[illegible] JOBS II [illegible] jobsdata.csv [illegible]

- `treat` [illegible] Z [illegible]
- `comply` [illegible] D [illegible]
- `jobseek` [illegible] Y [illegible] 1-5 [illegible]

[illegible][11]

[illegible] 22.1 (JOBS II [illegible] IV [illegible]). 我们使用 *Wald* 估计量分析 *JOBS II* 数据。结果如下：

方法	估计量	标准误	*95% CI* 下限	*95% CI* 上限
Delta 方法	*0.109*	*0.081*	*-0.050*	*0.268*
Bootstrap	*0.109*	*0.083*	*-0.054*	*0.271*
含协变量调整	*0.118*	*0.082*	*-0.042*	*0.278*

[illegible]：点估计表明，对于 *compliers*，参加培训项目使求职自我效能得分提高约 *0.11* 分（满分 *5* 分）。但 *95%* 置信区间包含零，这意味着在常规显著性水平下，我们无法拒绝”培训对 *compliers* 无效果”的原假设。

---

[11] [illegible] **??**

[illegible]

[illegible] CACE [illegible] CACE [illegible]

[illegible] τ_c [illegible]

(1) [illegible] $\tau_c = E\{Y(1) - Y(0) \mid U = c\}$ [illegible] compliers [illegible]
(2) [illegible] compliers [illegible] $D(1) = 1$ [illegible] $D(0) = 0$ [illegible] $D = Z$ [illegible] τ_c [illegible] compliers [illegible]

[illegible]

[illegible] CACE [illegible] compliers [illegible] always takers、never takers [illegible] defiers [illegible] compliers [illegible] CACE [illegible]

[illegible]

[illegible]

(1) [illegible]
(2) [illegible] always takers、compliers、defiers、never takers
(3) [illegible] defiers [illegible]
(4) [illegible] CACE $= \tau_Y / \tau_D$
(5) [illegible] Wald [illegible] $\hat{\tau}_c = \hat{\tau}_Y / \hat{\tau}_D$
(6) [illegible] compliers [illegible] Wald [illegible] FAR [illegible]
(7) [illegible]

[illegible]

[illegible] 22.1. *ITT 与 CACE 的关系* 证明在 [illegible]下，$\tau_Y = \tau_c \cdot \pi_c$，其中 $\pi_c = \Pr(U = c)$。

[illegible] 22.2. *单调性的可检验含义* 证明在完全随机化下，[illegible]蕴含 [illegible]。

[illegible] 22.3. *Wald 估计量的方差* 证明当 $\tau_D = 0$ 时，$var(\hat{\tau}_c) = \infty$。

[illegible] 22.4. *单侧依从性* 考虑单侧依从性情形：对照组被试无法获得治疗，即 $D_i(0) = 0$ 对所有 i 成立。

(1) 证明此时 [illegible]自动成立。
(2) 在这种情况下，*CACE* 如何简化为 *ITT* 的函数？

[illegible] 22.5. *FAR 置信区间的形状* 在 [illegible]中，*FAR* 置信集可能是单个闭区间、两个不相连的区间、空集或整个实数线。找出每种情况对应的精确条件。

[illegible] 22.6. *协变量调整的 CACE* 使用 *JOBS II* 数据进行以下分析：

(1) 计算原始 *Wald* 估计量及其标准误。
(2) 加入协变量（*sex, age, marital, nonwhite, educ, income*）后重新估计。
(3) 比较两种方法的估计量和标准误。

Ǘ 22.7. *Always takers* 与 *Never takers* 假设我们想估计 *always takers* 的平均因果效应。

(1) 在当前框架下，这个量能否被识别？为什么？
(2) 如果加入额外假设”*always takers* 和 *never takers* 的潜在结果分布相同”，结果会如何改变？

Ǘ 22.8. 模拟研究 进行模拟研究，比较以下三种情况：

(1) 强工具变量（$\pi_c = 0.5$）
(2) 中等强度工具变量（$\pi_c = 0.2$）
(3) 弱工具变量（$\pi_c = 0.05$）

对每种情况，报告 *Wald* 估计量的偏差、均方误差和置信区间的覆盖概率。

CHAPTER 23

[illegible]

[illegible]

[illegible] just-identified [illegible] IV [illegible] Z [illegible] D [illegible] $E(ZY) = E(ZD^\top)\beta$ [illegible] IV [illegible] over-identified [illegible]

[illegible] Angrist and Krueger (1991) [illegible]

[illegible] ” [illegible] ” [illegible] Theil (1953) [illegible] Basmann (1957) [illegible] Two-Stage Least Squares, TSLS [illegible] TSLS [illegible]

[illegible]

[illegible]

$$Y = D^\top\beta + \varepsilon, \tag{117}$$

[illegible] Y [illegible] D [illegible] p [illegible] β [illegible] p [illegible] q [illegible] Z [illegible] restrictions [illegible] $E(\varepsilon Z) = 0$ [illegible] $q > p$ [illegible]

$$n^{-1}\sum_{i=1}^{n} Z_iY_i = n^{-1}\sum_{i=1}^{n} Z_iD_i^\top\beta \tag{118}$$

[illegible] q [illegible] p [illegible] $q > p$ [illegible] ” [illegible] ” [illegible]

[illegible]

TSLS [illegible] [[illegible]] [illegible] Z [illegible] D [illegible] β [illegible] TSLS [illegible]

[illegible] D [illegible] Z OLS [illegible] $\hat{D}_i$ [illegible] $i = 1, \ldots, n$ [illegible] D_i [illegible] OLS [illegible] $\hat{D}$ [illegible] i [illegible] $\hat{D}_i^\top$ [illegible]

[illegible] Y [illegible] $\hat{D}$ OLS [illegible] $\hat{\beta}_{\text{TSLS}}$

[illegible]

[illegible] TSLS [illegible][1] [illegible] OLS [illegible]

$$\hat{D}_i = \hat{\Gamma}^\top Z_i, \tag{119}$$

[illegible] $\hat{\Gamma} = \left(\sum_{i=1}^n Z_i Z_i^\top\right)^{-1} \sum_{i=1}^n Z_i D_i^\top$ [illegible] OLS [illegible]

$$\hat{\beta}_{\text{TSLS}} = \left(\sum_{i=1}^n \hat{D}_i \hat{D}_i^\top\right)^{-1} \sum_{i=1}^n \hat{D}_i Y_i. \tag{120}$$

[illegible] $Y_i = D_i^\top \beta + \varepsilon_i$ [illegible]

$$\hat{\beta}_{\text{TSLS}} = \beta + \left\{\hat{\Gamma}^\top \left(n^{-1} \sum_{i=1}^n Z_i Z_i^\top\right) \hat{\Gamma}\right\}^{-1} \hat{\Gamma}^\top \left(n^{-1} \sum_{i=1}^n Z_i \varepsilon_i\right). \tag{121}$$

[illegible] TSLS [illegible] $E(Z\varepsilon) = 0$ [illegible] 0 [illegible] $\hat{\beta}_{\text{TSLS}}$ [illegible] β [illegible] $q = p$ [illegible] TSLS [illegible] IV [illegible] $\hat{\beta}_{\text{IV}}$ [illegible]

[illegible]

[illegible] TSLS [illegible] $\hat{\varepsilon}_i = Y_i - \hat{\beta}_{\text{TSLS}}^\top D_i$ [illegible] Eicker-Huber-White [illegible]

$$\hat{V}_{\text{TSLS}} = \left(\sum_{i=1}^n \hat{D}_i \hat{D}_i^\top\right)^{-1} \left(\sum_{i=1}^n \hat{\varepsilon}_i^2 \hat{D}_i \hat{D}_i^\top\right) \left(\sum_{i=1}^n \hat{D}_i \hat{D}_i^\top\right)^{-1}. \tag{122}$$

[illegible] $\hat{\varepsilon}_i$ [illegible] Y_i on D_i [illegible] 而不是 [illegible] OLS [illegible] Y_i on $\hat{D}_i$ [illegible] $\hat{V}_{\text{TSLS}}$ [illegible]

[illegible]

[illegible] Z [illegible] D [illegible]

$$\begin{cases} Y_i = \beta_0 + \beta_1 D_i + \beta_2^\top X_i + \varepsilon_i, \\ D_i = \gamma_0 + \gamma_1 Z_i + \gamma_2^\top X_i + \varepsilon_{2i}, \end{cases} \tag{123}$$

[illegible] D_i [illegible] Z_i [illegible] $E(\varepsilon_i Z_i) = 0$ [illegible] X_i [illegible] $E(\varepsilon_i X_i) = 0$ [illegible]

---

[1] [illegible] **??**

[illegible] [illegible] TSLS [illegible]

[illegible] TSLS] [illegible] TSLS [illegible]

[illegible] D [illegible] $(1, Z, X)$ OLS [illegible] $\hat{D}_i$ [illegible] $i = 1, \ldots, n$ [illegible]

[illegible] Y [illegible] $(1, \hat{D}, X)$ OLS [illegible] $\hat{\beta}_{\text{TSLS}}$ [illegible] $\hat{\beta}_{1,\text{TSLS}}$ [illegible] $\hat{D}$ [illegible]

[illegible]

[illegible] [illegible] reduced form [illegible]

$$
\begin{cases} Y_i = \Gamma_0 + \Gamma_1 Z_i + \Gamma_2^\top X_i + \varepsilon_{1i}, \\ D_i = \gamma_0 + \gamma_1 Z_i + \gamma_2^\top X_i + \varepsilon_{2i}, \end{cases} \tag{124}
$$

[illegible] $\Gamma_0 = \beta_0 + \beta_1\gamma_0$ [illegible] $\Gamma_1 = \beta_1\gamma_1$ [illegible] $\Gamma_2 = \beta_2 + \beta_1\gamma_2$ [illegible] $\varepsilon_{1i} = \varepsilon_i + \beta_1\varepsilon_{2i}$ [illegible] (Z, X) [illegible] OLS [illegible] $\hat{\Gamma}_1$ [illegible] $\hat{\gamma}_1$ [illegible] β_1 [illegible]

$$
\beta_1 = \frac{\Gamma_1}{\gamma_1}. \tag{125}
$$

[illegible] Indirect Least Squares, ILS [illegible]

$$
\hat{\beta}_{1,\text{ILS}} = \frac{\hat{\Gamma}_1}{\hat{\gamma}_1}. \tag{126}
$$

[illegible] ILS [illegible] TSLS [illegible] [TSLS ILS [illegible]] [illegible]

$$
\hat{\beta}_{1,\text{ILS}} = \hat{\beta}_{1,\text{TSLS}}. \tag{127}
$$

[illegible] TSLS [illegible] ILS [illegible][2] [illegible]

[illegible] TSLS [illegible] γ_1 [illegible]

[illegible]

[illegible] TSLS [illegible] $\hat{\gamma}_1$ [illegible] $\hat{\Gamma}_1$ [illegible] γ_1 [illegible]

[illegible] Fieller-Anderson-Rubin (FAR) [illegible] b [illegible] $H_0(b) : \beta_1 = b$ [illegible] $Y_i(b) = Y_i - bD_i$ [illegible] $Y_i(b)$ [illegible] Z_i [illegible] b [illegible] $E(Z\varepsilon) = 0$ [illegible]

[illegible] FAR [illegible]

$$
\left\{ b : |t_Z(b)| \le z_{1-\alpha/2} \right\}, \tag{128}
$$

---

[2] [illegible] **??** [illegible] **??**

[illegible] $t_Z(b)$ [illegible] Z [illegible] $Y_i(b)$ on $(1, Z, X)$ [illegible] t [illegible] EHW [illegible] $z_{1-\alpha/2}$ [illegible] $1-\alpha/2$ [illegible]

[illegible] FAR [illegible] 1 [illegible] TSLS [illegible] Wald [illegible] TSLS [illegible]

[illegible]

[illegible] TSLS [illegible] Card (1993) [illegible]

[illegible] Z [illegible] ” [illegible] ” [illegible] D [illegible] Y [illegible] 1976 [illegible] X [illegible]

[illegible] TSLS [illegible] 95% [illegible]

	TSLS [illegible]	[illegible]	[illegible]
[illegible]	0.132	0.026	0.237

[illegible] 13.2% [illegible] FAR [illegible] $[0.028, 0.282]$ [illegible] TSLS [illegible]

[illegible]

[illegible] TSLS [illegible] TSLS [illegible] OLS [illegible]

[illegible] TSLS [illegible] IV [illegible] TSLS [illegible] ILS [illegible] FAR [illegible] TSLS [illegible]

[illegible]

[illegible] 23.1 (TSLS [illegible] IV [illegible]). 证明在恰好识别情形下（Z 与 D 维度相同），*TSLS* 估计量 $\hat{\beta}_{\text{TSLS}}$ 与 *IV* 估计量 $\hat{\beta}_{\text{IV}}$ 数值相同。

[illegible]：利用第一阶段 *OLS* 的正交性条件 $\sum_{i=1}^n \hat{D}_i \check{D}_i^\top = 0$。

[illegible] 23.2 (TSLS [illegible] ILS [illegible]). 证明定理 [illegible] 在单一内生处理变量和单一工具变量情形下，*TSLS* 估计量等于约化形式系数比值 $\hat{\Gamma}_1/\hat{\gamma}_1$。

[illegible]：利用第一阶段拟合值的表达式 $\hat{D}_i = \hat{\gamma}_0 + \hat{\gamma}_1 Z_i + \hat{\gamma}_2^\top X_i$，验证第二阶段 *OLS* 中 $\hat{D}$ 的系数恰好等于比值形式。

[illegible] 23.3 ([illegible]). 考虑 *TSLS* 估计量 $\hat{\beta}_{1,\text{TSLS}} = \hat{\Gamma}_1/\hat{\gamma}_1$。当工具变量较弱（即 γ_1 接近于零）时，讨论：

(1) 为什么 $\hat{\gamma}_1$ 的小误差会导致 $\hat{\beta}_{1,\text{TSLS}}$ 的巨大偏差？
(2) *FAR* 置信区间相比 *Wald* 型置信区间的优势是什么？

CHAPTER 24

[illegible]

[illegible]

[illegible] Regression Discontinuity Design, RDD[illegible] Instrumental Variables, IV[illegible]

[illegible] Fuzzy Regression Discontinuity, FRDD[illegible] ” [illegible] ”[illegible]

[illegible]

[illegible] [illegible] Sharp RDD[illegible] $Z_i = I(X_i \geq x_0)$ [illegible] running variable[illegible] X_i [illegible] x_0 [illegible]

[illegible] RDD [illegible] $\{Y_i(1), Y_i(0)\}$ [illegible]

$$\tau_Y(x_0) = \lim_{\varepsilon \to 0+} \mathbb{E}(Y \mid X = x_0 + \varepsilon) - \lim_{\varepsilon \to 0+} \mathbb{E}(Y \mid X = x_0 - \varepsilon). \tag{129}$$

[illegible] [illegible] RDD [illegible] $Z_i = I(X_i \geq x_0)$ [illegible]

(1) [illegible] Z_i [illegible] D_i [illegible] $X_i \geq x_0$ [illegible]
(2) [illegible] Z_i [illegible] D_i [illegible] Y_i[illegible]

[illegible]

[illegible]

[illegible] 24.1 ([illegible]). [illegible] 研究了印度政府 *2000* 年启动的”总理乡村道路计划”。该计划根据 *2001* 年人口普查的阈值来优先考虑规模较大的村庄。

关键点：村庄人口规模与阈值的差并不能 [illegible] 是否获得道路——它只改变了获得道路的[illegible]。在断点附近，获得道路的概率发生了跳跃，但不是从 *0* 到 *1* 的确定性跳跃。

[illegible] 24.2 ([illegible]). [illegible] 研究了意大利大学助学金对辍学率的影响。学生如果标准化家庭收入低于 *15,000* 欧元，则有资格获得助学金。

为获得助学金，学生必须先申请。因此，资格状态和申请状态共同决定了最终治疗状态。运行变量（*15,000* 减去标准化家庭收入）单独并不能决定治疗状态，但它在断点处改变了治疗概率。

Figure 1. [illegible]

[illegible]

[illegible] X_i [illegible] x_0 [illegible]

$$Z_i = I(X_i \ge x_0).$$

[illegible] D_i [illegible] Z_i [illegible] $\mathbb{P}(D_i = 1 \mid X_i = x)$ [illegible] x_0 [illegible] RDD [illegible] RDD [illegible]

[illegible]

[illegible] Y_i [illegible] Z_i [illegible] $\{D_i(1), D_i(0), Y_i(1), Y_i(0)\}$ [illegible] RDD [illegible]

$$\tau_D(x_0) = \mathbb{E}\{D(1) - D(0) \mid X = x_0\}$$

[illegible]

$$\tau_Y(x_0) = \mathbb{E}\{Y(1) - Y(0) \mid X = x_0\}. \tag{130}$$

[illegible] Z [illegible] D [illegible] $X = x_0$ [illegible] Local Complier Average Causal Effect, LATE[illegible]

[illegible] **[illegible]** RDD [illegible]) 设治疗由 $Z = I(X \ge x_0)$ 决定，其中 x_0 是预定的阈值。

假设[illegible]：

$$D_i(1) \ge D_i(0), \tag{131}$$

和[illegible]：

$$D_i(1) = D_i(0) \Longrightarrow Y_i(1) = Y_i(0), \tag{132}$$

在 x_0 的无穷小邻域内成立。

[illegible]：$E\{D(1) \mid X = x\}$ 和 $E\{Y(1) \mid X = x\}$ 在 $x = x_0$ 处右连续，$E\{D(0) \mid X = x\}$ 和 $E\{Y(0) \mid X = x\}$ 在 $x = x_0$ 处左连续。

局部依从者平均因果效应（*Local Average Treatment Effect, LATE*）定义为：

$$\tau_{\text{LATE}}(x_0) = \mathbb{E}\{Y(1) - Y(0) \mid D(1) > D(0), X = x_0\}, \tag{133}$$

可以由下式识别：

$$\tau_{\text{LATE}}(x_0) = \frac{\mathbb{E}\{Y(1) - Y(0) \mid X = x_0\}}{\mathbb{E}\{D(1) - D(0) \mid X = x_0\}}. \tag{134}$$

如果 $E(D \mid X = x)$ 在 $x = x_0$ 处有非零跳跃，则局部依从者平均因果效应可以由下式识别：

$$\tau_{\text{LATE}}(x_0) = \frac{\lim_{\varepsilon \to 0+} \mathbb{E}(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \to 0+} \mathbb{E}(Y \mid Z = 0, X = x_0 - \varepsilon)}{\lim_{\varepsilon \to 0+} \mathbb{E}(D \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \to 0+} \mathbb{E}(D \mid Z = 0, X = x_0 - \varepsilon)}. \tag{135}$$

[illegible] RDD [illegible] ({) [illegible] ({) [illegible] RDD [illegible] Z [illegible] IV [illegible] IV [illegible]

[illegible]

[illegible]

[illegible] local randomization [illegible] [illegible]

[illegible] RDD [illegible] h [illegible]

[illegible]

[illegible]

[illegible]

(1) [illegible] RDD [illegible]

(2) [illegible] $Z = I(X \geq x_0)$ [illegible] LATE

(3) [illegible] ILS [illegible] TSLS [illegible]

(4) [illegible]

(5) [illegible] ~**MM** [illegible] RDD [illegible] RDD [illegible] IV [illegible]

[illegible] RDD [illegible]

[illegible]

[illegible] 24.1. [illegible]模糊 *RDD* 识别定理的证明 证明 [illegible]中的识别公式。

[illegible] 24.2. [illegible]清晰 *RDD* 与模糊 *RDD* 的比较

(1) 在清晰 *RDD* 中，治疗分配 D 和工具变量 Z 有什么关系？

(2) 在模糊 *RDD* 中，这种关系如何变化？

(3) 给出清晰 *RDD* 和模糊 *RDD* 各一个现实生活中的例子。

[illegible] 24.3. [illegible]带宽敏感性分析 对于 [illegible]中的印度道路数据，使用不同的带宽 h 进行敏感性分析。报告点估计和标准误如何随 h 变化。

[illegible] 24.4. [illegible]*ILS* 与 *TSLS* 的等价性 证明在模糊 *RDD* 中，间接最小二乘估计器（比率形式）和两阶段最小二乘估计器在数值上是等价的。

CHAPTER 25

[illegible]

[illegible]

[illegible] IV [illegible] IV [illegible]

[illegible] 20 [illegible] 80 [illegible] Katan [illegible] ([illegible]) [illegible] law of random assortment [illegible]

[illegible] 2003 [illegible] Davey Smith [illegible] Ebrahim [illegible] ([illegible]) [illegible] MR [illegible] ([illegible]) [illegible]

[illegible]

[illegible] MR [illegible] 20 [illegible] 80 [illegible]

(1) [illegible]

(2) [illegible]

Katan ([illegible]) [illegible] Apolipoprotein E [illegible] IV [illegible]

[illegible] MR [illegible] MR [illegible]

[illegible]

[illegible] MR [illegible] IV [illegible] MR [illegible]

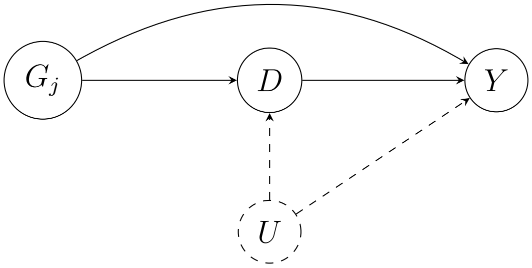

Figure 1. MR [illegible] G_j [illegible] D [illegible] Y [illegible] U [illegible] G_j [illegible]

[illegible]

[illegible] 25.1 ([illegible] IV [illegible]). 结构形式为：

$$Y = \beta_0 + \beta D + \beta_u U + \varepsilon_Y, \tag{25.1}$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D. \tag{25.2}$$

约简形式为：

$$Y = (\beta_0 + \beta\gamma_0) + \beta\gamma_1 G_1 + \cdots + \beta\gamma_p G_p + (\beta_u + \beta\gamma_u)U + \varepsilon_Y, \tag{25.3}$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D. \tag{25.4}$$

[illegible] G_j [illegible] Y [illegible] $\Gamma_j = \beta\gamma_j$[illegible] $j = 1, \dots, p$[illegible]

[illegible] MR [illegible]

[illegible] 25.2 ([illegible]). 线性模型为：

$$Y = \beta_0 + \beta D + \alpha_1 G_1 + \cdots + \alpha_p G_p + \beta_u U + \varepsilon_Y, \tag{25.5}$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D. \tag{25.6}$$

约简形式为：

$$Y = (\beta_0 + \beta\gamma_0) + (\alpha_1 + \beta\gamma_1)G_1 + \cdots + (\alpha_p + \beta\gamma_p)G_p + (\beta_u + \beta\gamma_u)U + \varepsilon_Y, \tag{25.7}$$

$$D = \gamma_0 + \gamma_1 G_1 + \cdots + \gamma_p G_p + \gamma_u U + \varepsilon_D. \tag{25.8}$$

[illegible] $\Gamma_j = \alpha_j + \beta\gamma_j$[illegible] $j = 1, \dots, p$[illegible] $\alpha_j \neq 0$[illegible] G_j [illegible]

25.1. 定义 [illegible]和 [illegible]与第 [illegible]章的线性 *IV* 模型略有不同——我们显式包含了未测量混杂因子 U。但这一差异本质上不影响讨论，因为 U 在约简形式中通过 $\beta_u + \beta\gamma_u$ 被合并到系数中。

[illegible]

[illegible] IV [illegible] MR [illegible] genome-wide association studies, GWAS[illegible] MR [illegible]

[illegible] 25.1 ([illegible] MR [illegible]). (1) 我们有治疗变量对遗传工具变量的回归系数 $\hat{\gamma}_1, \dots, \hat{\gamma}_p$ 及其标准误 $\text{se}_{D1}, \dots, \text{se}_{Dp}$。假设

$$\hat{\gamma}_1 \xrightarrow{p} \gamma_1, \dots, \hat{\gamma}_p \xrightarrow{p} \gamma_p \tag{25.9}$$

并忽略标准误的不确定性。

(2) 我们有结果变量对遗传工具变量的回归系数 $\hat{\Gamma}_1, \dots, \hat{\Gamma}_p$ 及其标准误 $\text{se}_{Y1}, \dots, \text{se}_{Yp}$。假设

$$\hat{\Gamma}_1 \xrightarrow{p} \Gamma_1, \dots, \hat{\Gamma}_p \xrightarrow{p} \Gamma_p \tag{25.10}$$

并忽略标准误的不确定性。

(3) 假设 $\hat{\gamma}_1, \dots, \hat{\gamma}_p, \hat{\Gamma}_1, \dots, \hat{\Gamma}_p$ 联合服从正态分布且相互独立。

[illegible] $\hat{\gamma}_j$ [illegible] $\hat{\Gamma}_j$ [illegible]

[illegible] $\alpha_j = 0$ [illegible] $\beta = \Gamma_j/\gamma_j$ [illegible] j [illegible] $\hat{\beta}_j = \hat{\Gamma}_j/\hat{\gamma}_j$ [illegible] β [illegible]

[illegible] delta [illegible] $\hat{\beta}_j$ [illegible]

$$\text{se}_j^2 = (\text{se}_{Yj}^2 + \hat{\beta}_j^2 \text{se}_{Dj}^2)/\hat{\gamma}_j^2. \tag{25.11}$$

[illegible] β [illegible] Fisher [illegible]

$$\hat{\beta}_{\text{fisher0}} = \frac{\sum_{j=1}^p \hat{\beta}_j/\text{se}_j^2}{\sum_{j=1}^p 1/\text{se}_j^2}, \tag{25.12}$$

[illegible] $\left(\sum_{j=1}^p 1/\text{se}_j^2\right)^{-1}$ [illegible] $\hat{\gamma}_j$ [illegible] se_{Dj} [illegible]

$$\hat{\beta}_{\text{fisher1}} = \frac{\sum_{j=1}^p \hat{\beta}_j \hat{\gamma}_j^2/\text{se}_{Yj}^2}{\sum_{j=1}^p \hat{\gamma}_j^2/\text{se}_{Yj}^2} = \frac{\sum_{j=1}^p \hat{\Gamma}_j \hat{\gamma}_j/\text{se}_{Yj}^2}{\sum_{j=1}^p \hat{\gamma}_j^2/\text{se}_{Yj}^2}, \tag{25.13}$$

[illegible] $\left(\sum_{j=1}^p \hat{\gamma}_j^2/\text{se}_{Yj}^2\right)^{-1}$ [illegible] $\hat{\beta}_{\text{fisher1}}$ [illegible] ([illegible]) [illegible]

$$\hat{\beta}_{\text{fisher1}} \xrightarrow{p} \frac{\sum_{j=1}^p \Gamma_j \gamma_j/\text{se}_{Yj}^2}{\sum_{j=1}^p \gamma_j^2/\text{se}_{Yj}^2} = \beta + \frac{\sum_{j=1}^p \alpha_j \gamma_j/\text{se}_{Yj}^2}{\sum_{j=1}^p \gamma_j^2/\text{se}_{Yj}^2}. \tag{25.14}$$

[illegible] j [illegible] $\alpha_j = 0$ [illegible] $\hat{\beta}_{\text{fisher1}}$ [illegible] α_j [illegible] γ_j [illegible] $1/\text{se}_{Yj}^2$ [illegible] $\hat{\beta}_{\text{fisher1}}$ [illegible]

25.2. 当有许多遗传工具变量，且违反排他性限制（由 α_j 捕获）是来自均值为零的分布的独立随机抽样时，上述条件近似成立。

[illegible]

$$\Gamma_j = \beta\gamma_j \quad (j = 1, \dots, p). \tag{25.15}$$

[illegible]

$$\hat{\Gamma}_j \approx \beta\hat{\gamma}_j \quad (j = 1, \dots, p). \tag{25.16}$$

[illegible] $\{\hat{\Gamma}_j\}_{j=1}^{p}$ [illegible] $\{\hat{\gamma}_j\}_{j=1}^{p}$ [illegible] $\hat{\Gamma}_j$ [illegible] $\hat{\gamma}_j$ [illegible] WLS [illegible] w_j [illegible] β [illegible] WLS [illegible] $\hat{\gamma}_j$ [illegible]

$$\hat{\beta}_{\text{egger1}} = \frac{\sum_{j=1}^{p} \hat{\gamma}_j \hat{\Gamma}_j w_j}{\sum_{j=1}^{p} \hat{\gamma}_j^2 w_j}, \tag{25.17}$$

[illegible] $w_j = 1/\mathrm{se}_{Yj}^2$ [illegible] $\hat{\beta}_{\text{fisher1}}$ WLS [illegible] $\hat{\gamma}_j$ [illegible]

$$\hat{\beta}_{\text{egger0}} = \frac{\sum_{j=1}^{p} (\hat{\gamma}_j - \hat{\gamma}_w)(\hat{\Gamma}_j - \hat{\Gamma}_w) w_j}{\sum_{j=1}^{p} (\hat{\gamma}_j - \hat{\gamma}_w)^2 w_j}, \tag{25.18}$$

[illegible] $\hat{\gamma}_w = \sum_{j=1}^{p} \hat{\gamma}_j w_j / \sum_{j=1}^{p} w_j$ [illegible] $\hat{\Gamma}_w = \sum_{j=1}^{p} \hat{\Gamma}_j w_j / \sum_{j=1}^{p} w_j$ [illegible] $\hat{\gamma}_j$ [illegible] $\hat{\Gamma}_j$ [illegible]

[illegible] γ_j [illegible]

$$\hat{\beta}_{\text{egger0}} \xrightarrow{p} \frac{\sum_{j=1}^{p} (\gamma_j - \gamma_w)(\Gamma_j - \Gamma_w) w_j}{\sum_{j=1}^{p} (\gamma_j - \gamma_w)^2 w_j} = \beta + \frac{\sum_{j=1}^{p} (\gamma_j - \gamma_w)(\alpha_j - \alpha_w) w_j}{\sum_{j=1}^{p} (\gamma_j - \gamma_w)^2 w_j}, \tag{25.19}$$

[illegible] γ_w, Γ_w [illegible] α_w [illegible] $\hat{\beta}_{\text{egger0}}$ [illegible] β [illegible] α_j [illegible] γ_j [illegible] WLS [illegible] $\alpha_j = 0$ [illegible] j [illegible] γ_j [illegible] α_j [illegible] **MP** ( ) [illegible] Egger [illegible]

$$\hat{\alpha}_{\text{egger0}} = \hat{\Gamma}_w - \hat{\beta}_{\text{egger0}} \hat{\gamma}_w, \tag{25.20}$$

InSIDE [illegible]

$$\Gamma_w - \beta\gamma_w = \alpha_w, \tag{25.21}$$

[illegible]

[illegible]

[illegible] `mr.raps` [illegible] `bmi.sbp` [illegible] ( ) [illegible] Fisher [illegible] Egger [illegible]

[illegible] Fisher [illegible]

```
fisher.weight = function(est, se)
{
    n = sum(est/se^2)
    d = sum(1/se^2)
    res = c(n/d, sqrt(1/d))
    names(res) = c("est", "se")
    res
}
```

[illegible] BMI-SBP [illegible] $\hat{\beta}_{\text{fisher0}}$ [illegible] $\hat{\beta}_{\text{fisher1}}$ [illegible]

```
> fisher.weight(bmisbp$iv, bmisbp$se.iv)
      est        se
0.31727680 0.05388827
> fisher.weight(bmisbp$iv, bmisbp$se.iv1)
      est        se
0.31576007 0.05893783
```

[illegible] Egger [illegible]

Fisher [illegible]

```
> mr.egger = lm(beta.output ~ 0 + beta.exposure,
+ data = bmisbp, weights = 1/se.output^2)
> summary(mr.egger)$coef
               Estimate Std. Error t value Pr(>|t|)
beta.exposure 0.3172768  0.1105994  2.868704 0.004681659

> mr.egger.w = lm(beta.output ~ beta.exposure,
+ data = bmisbp, weights = 1/se.output^2)
> summary(mr.egger.w)$coef
               Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.0001133328 0.002079418 0.05450217 0.956603931
beta.exposure 0.3172989306 0.110948506 2.85987564 0.004811017
```

[illegible] Egger [illegible] $\hat{\gamma}_j \hat{\Gamma}_j$ [illegible]

FIGURE 2. [illegible] Egger [illegible]

[illegible]

MR [illegible] IV [illegible]

[illegible] MR [illegible]

[illegible] BMI [illegible] SUTVA [illegible]

[illegible] IV [illegible]
[illegible]
[illegible] $_1$ [illegible]
[illegible]
[illegible]

[illegible] MR [illegible] IV [illegible] $\hat{\gamma}_j$ [illegible] $\hat{\Gamma}_j$
[illegible] IV [illegible]
[illegible] $_1$ [illegible] $_1$ [illegible]
[illegible]

VanderWeele [illegible] (2014) [illegible] MR [illegible]

[illegible]

[illegible] 25.1. [illegible] *Fisher* 加权估计的推导 证明第 [illegible] 节中 $\hat{\beta}_{\text{fisher0}}$ 的方差为 $\left(\sum_{j=1}^{p} 1/\text{se}_j^2\right)^{-1}$。提示：利用 $\hat{\beta}_j$ 的独立性。

[illegible] 25.2. [illegible] *InSIDE* 假设的含义 解释 *InSIDE*（工具强度独立于直接效应）假设在什么意义上弱于 $\alpha_j = 0$（对所有 j）的排他性限制假设。给出一个满足 *InSIDE* 但不满足 $\alpha_j = 0$ 的数值例子。

[illegible] 25.3. [illegible] *Egger* 截距的解释 在第 [illegible] 节中，我们看到 *Egger* 回归的截距 $\hat{\alpha}_{\text{egger0}}$ 收敛到直接效应的加权平均 α_w。讨论为什么这个性质在实践中有用——即为什么我们可能关心估计 α_w 而不仅仅是 β。

[illegible] 25.4. [illegible] 数据分析 分析 *mr.raps* 包中的 *bmi.bmi* 数据。参见 [illegible] 第 *7.2* 节 了解更多细节。

[illegible]

[illegible] MR [illegible]
[illegible] Egger [illegible]
[illegible] $_1$ [illegible] robust adjusted profile score, RAPS [illegible]
[illegible] MR [illegible]

CHAPTER 26

[illegible]

[illegible]

[illegible] 2020 [illegible] [illegible] [illegible] 2019 [illegible] 1000 [illegible] 2021 [illegible] 1200 [illegible] 20% [illegible] ** [illegible] ** [illegible] [illegible] ** [illegible] **[illegible] panel data [illegible]

[illegible]

[illegible] ** [illegible] ** [illegible] ** [illegible] ** [illegible]

[illegible]

[illegible] DiD [illegible]

[illegible]

- $G_i \in \{0,1\}$ [illegible] i [illegible] $G_i = 1$ [illegible] $G_i = 0$ [illegible]
- $T_t \in \{0,1\}$ [illegible] $T_t = 0$ [illegible] $T_t = 1$ [illegible]
- Y_{it} [illegible] i [illegible] t [illegible]
- $D_i = G_i$ [illegible]

[illegible] N [illegible] $i = 1, \ldots, N$ [illegible] N_1 [illegible] N_0 [illegible] $N_1 + N_0 = N$ [illegible]

[illegible] i [illegible] t [illegible]

$$Y_{it}(1) \quad [illegible] \quad Y_{it}(0)$$

[illegible] i [illegible] t [illegible]

$$Y_{it} = Y_{it}(0) + D_i\big(Y_{it}(1) - Y_{it}(0)\big).$$

[illegible] t [illegible] Average Treatment effect on the Treated, ATT [illegible]

$$\tau_t = \mathbb{E}\big(Y_{it}(1) - Y_{it}(0) \mid G_i = 1\big).$$

[illegible]

$$\bar{Y}_{00} = \mathbb{E}(Y_{it} \mid G_i = 0, T_t = 0) \text{ [illegible]}$$

$$\bar{Y}_{01} = \mathbb{E}(Y_{it} \mid G_i = 0, T_t = 1) \text{ [illegible]}$$

$$\bar{Y}_{10} = \mathbb{E}(Y_{it} \mid G_i = 1, T_t = 0) \text{ [illegible]}$$

$$\bar{Y}_{11} = \mathbb{E}(Y_{it} \mid G_i = 1, T_t = 1) \text{ [illegible]}$$

[illegible]

[illegible] $\bar{Y}_{11} - \bar{Y}_{10}$ [illegible]

$$\bar{Y}_{11} - \bar{Y}_{10} = \underbrace{\tau_1}_{\text{[illegible]}} + \underbrace{\left[\bar{Y}_{10}(0) - \bar{Y}_{11}(0)\right]}_{\text{[illegible]}},$$

[illegible] $\bar{Y}_{1t}(0) = \mathbb{E}(Y_{it}(0) \mid G_i = 1, T_t = t)$ [illegible] t [illegible] $\bar{Y}_{01} - \bar{Y}_{00}$ [illegible]

$$\bar{Y}_{01} - \bar{Y}_{00} = \bar{Y}_{10}(0) - \bar{Y}_{11}(0).$$

[illegible]

$$\underbrace{(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})}_{\text{[illegible]}} = \tau_1 + \underbrace{\left[\bar{Y}_{10}(0) - \bar{Y}_{11}(0)\right]}_{\text{[illegible]}} - \underbrace{\left[\bar{Y}_{10}(0) - \bar{Y}_{11}(0)\right]}_{\text{[illegible]}}.$$

[illegible]

$$\text{DiD} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) = \tau_1.$$

[illegible] ATT

DiD [illegible]

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) = (\bar{Y}_{11} - \bar{Y}_{01}) - (\bar{Y}_{10} - \bar{Y}_{00}).$$

> [illegible]
>
> $$\hat{\tau}_{\text{DiD}} = \underbrace{(\bar{Y}_{11} - \bar{Y}_{01})}_{\text{[illegible]}} - \underbrace{(\bar{Y}_{10} - \bar{Y}_{00})}_{\text{[illegible]}}.$$
>
> [illegible]

[illegible]

DiD [illegible]

[illegible] 26.1 ([illegible]). 在没有处理的情况下，处理组和对照组的结果随时间变化的趋势相同，即

$$\mathbb{E}\big(Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1\big) = \mathbb{E}\big(Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0\big).$$

Equivalently,

$$\bar{Y}_{10}(0) - \bar{Y}_{11}(0) = \bar{Y}_{00} - \bar{Y}_{01}.$$

[illegible] ** [illegible] **

[illegible]

[illegible] DiD [illegible] " " [illegible]

$$Y_{it}(0) = \alpha + \beta \cdot G_i + \gamma \cdot t + \varepsilon_{it},$$

[illegible]

[illegible] " " [illegible] DiD [illegible]

(1) [illegible] `"0300pre-trend test'[illegible]
(2) [illegible]
(3) [illegible] placebo test[illegible]

[illegible]

[illegible]

DiD [illegible] Two-Way Fixed Effects, TWFE[illegible]

$$Y_{it} = \alpha + \beta \cdot D_{it} + \gamma_i + \delta_t + \varepsilon_{it},$$

[illegible]

- $D_{it} = G_i \cdot T_t$ [illegible] × [illegible]
- γ_i [illegible]
- δ_t [illegible]

[illegible] β [illegible] DiD [illegible] $\hat{\tau}_{\text{DiD}}$ [illegible]

- [illegible] **I g ' 。—Q QN 、— ; l[illegible]N [illegible] N P p** 。 [illegible]
- **M a。J p g g[illegible] d** [illegible]
- **A p p—。**[illegible]

[illegible]

[illegible] TWFE [illegible] β [illegible] i [illegible] t [illegible] $\tau_{it} = \mathbb{E}(Y_{it}(1) - Y_{it}(0))$[illegible] i [illegible] t

Angrist and Pischke ([illegible]) [illegible] TWFE [illegible] ** [illegible] ** [illegible] ATT [illegible]

> [illegible] [illegible] TWFE [illegible]
> [illegible] β [illegible] ATT [illegible]
> [illegible]

[illegible] [illegible] DiD [illegible] ** [illegible] ** [illegible] $T_t \in \{0, 1, \ldots, T\}$ [illegible] $D_{it} = G_i \cdot \mathbb{I}(t \geq t^*)$ [illegible] t^* [illegible] DiD [illegible] TWFE [illegible]

$$Y_{it} = \alpha_i + \delta_t + \sum_{t=0}^{T} \beta_t \cdot D_{it} + \varepsilon_{it},$$

[illegible] β_t [illegible] $_1$ [illegible]

[illegible] **[illegible]** DiD [illegible] β_t [illegible]

[illegible]

[illegible] DiD [illegible]

DiD [illegible] pre-trend [illegible]

DiD [illegible] 2×2 [illegible] TWFE [illegible]

[illegible]

[illegible] DiD [illegible]

- [illegible] Mostly Harmless Econometrics [illegible] 5 [illegible] DiD [illegible]
- [illegible] DiD [illegible]
- [illegible] TWFE [illegible]
- [illegible] TWFE [illegible]

[illegible] DiD [illegible] Bootstrap [illegible]

CHAPTER 27

[illegible]

[illegible]

[illegible] “[illegible]” [illegible]
[illegible]
[illegible]
[illegible]

[illegible] transmission mechanism [illegible] [illegible] **[illegible]** [illegible]

[illegible] nested potential outcomes [illegible] mediation formula [illegible] **[illegible]** [illegible]

[illegible]

[illegible] Z [illegible] treatment [illegible] Y [illegible] outcome [illegible] M [illegible] mediator [illegible] X [illegible] covariates [illegible] DAG [illegible]

> [illegible] τ [illegible] M [illegible] M [illegible]

[illegible]

[illegible] 27.1 ([illegible]). (*2012*) 研究了染色体 *15q25.1* 上的遗传变异如何影响肺癌风险。[1] 该研究的核心问题是：遗传变异是通过影响吸烟行为（吸烟量）进而增加肺癌风险的，还是通过其他独立通路（如直接影响肺部细胞）直接致癌的？这就是一个典型的中介分析问题：遗传变异是处理 Z，吸烟是中介 M，肺癌是结果 Y。

[illegible] 27.2 ([illegible]). 等人 (*2018*) 研究了邻里贫困对青少年物质使用的影响，并探讨了这一影响是通过学校和同伴环境传导的。[2] 他们使用了美国国家共病调查（*National Comorbidity Survey*）的数据，将邻里贫困指数作为处理，青少年物质使用（如饮酒、吸毒）作为结果，而学校和同伴环境因素作为中介。

---

[1] **?**

[2] **?**

[illegible] 27.3 ([illegible]). *JOBS II* 是一个关于就业培训对失业工人影响的现场实验。[3] 该培训项目的目的不仅是提高再就业率，还希望改善求职者的心理健康。研究者感兴趣的是：培训对心理健康的效应中，有多少是通过提高工作搜寻效能这一中介传导的，又有多少是通过其他途径（如社会支持网络）直接产生的。[4]

[illegible]

[illegible] [illegible] nested potential outcomes [illegible] z [illegible]

$$\{M(z),\ Y(z) : z = 0,1\}.$$

[illegible] z [illegible] m [illegible]

$$\{Y(z,m) : z = 0,1,\ m \in \mathcal{M}\},$$

[illegible] $\mathcal{M}$ [illegible]

[illegible] nested interventions [illegible] z [illegible] m [illegible] $M_z \equiv M(z')$ [illegible]

$$\{Y(z, M_z) : z = 0,1,\ z' = 0,1\}.$$

[illegible] $Y(z, M_z)$ [illegible] z [illegible] z' [illegible] M_z [illegible] z [illegible] z' [illegible] “[illegible]” [illegible]

$$\{Y(1,M_1),\ Y(1,M_0),\ Y(0,M_1),\ Y(0,M_0)\}.$$

[illegible] $Y(1,M_1)$ [illegible] $z=1$ [illegible] $z=1$ [illegible] $Y(0,M_0)$ [illegible] $z=0$ [illegible] $z=0$ [illegible]

Composition [illegible]

[Composition [illegible]] $Y(z, M_z) = Y(z)$ [illegible] $z = 0,1$ [illegible]

Composition [illegible] $Y(1)$ [illegible] $Y(1,M_1)$ [illegible] $Y(0)$ [illegible] $Y(0,M_0)$ [illegible]

[illegible]

[illegible] 27.1 ([illegible]). 总效应（*total effect*）定义为

$$\tau = \mathbb{E}\{Y(1) - Y(0)\}.$$

自然直接效应（*natural direct effect, NDE*）定义为

$$= \mathbb{E}\{Y(1,M_0) - Y(0,M_0)\}.$$

自然间接效应（*natural indirect effect, NIE*）定义为

$$= \mathbb{E}\{Y(1,M_1) - Y(1,M_0)\}.$$

[illegible]

- [illegible] τ [illegible] 0 [illegible] 1 [illegible]
- **[illegible]** $\mathbb{E}\{Y(1,M_0) - Y(0,M_0)\}$ [illegible] 0 [illegible] 1 [illegible] M_0 [illegible] M [illegible]

---

[3] [illegible] ?? [illegible]

[4] [illegible] ?? [illegible] Baron–Kenny [illegible]

- **k ` P**$\mathbb{E}\{Y(1,M_1)-Y(1,M_0)\}$ [illegible] $z=1$ [illegible] M_0 [illegible] M_1 [illegible] M [illegible]

Composition [illegible]

[illegible] **w** [illegible] **w**在 [illegible] 和 [illegible] 下，有

$$\tau = +.$$

[illegible] **k ` P** [illegible] $Y(1,M_1)-Y(1,M_0)$ [illegible] $z=1$ [illegible] “[illegible]” [illegible] “[illegible]” [illegible] “[illegible]” [illegible]

[illegible] $Y(1,M_0)$ [illegible] **J p ‘ l — Q V ; J — ‘ ;** [illegible] cross-world counterfactual [illegible] $z=1$ [illegible] $m=M_0$ [illegible] $z=1$ [illegible] $Y(1,M_0)$ [illegible]

[width = 0.7]figures/crossworld.pdf

FIGURE 2. [illegible] counterfactual $Y(1,M_0)$ [illegible] $Y(0,M_1)$ [illegible]

[illegible] $Y(1,M_0)$ [illegible] $Y(0,M_1)$ [illegible] **J p ‘ l — Q V ; J — ‘ ; g** a priori counterfactuals [illegible] Popperian [illegible][5]

[illegible] $Y(1)$ [illegible] $Y(0)$ [illegible] $Y(1,M_0)$ [illegible] $\{Y(1),Y(0)\}$ [illegible]

X w [illegible]

X w q [illegible] **w** [illegible] X [illegible]

[[illegible]] [illegible] X [illegible]

$$Z \perp\!\!\!\perp Y(z,m) \mid X, \quad \forall z, m.$$

[[illegible]] [illegible] (X,Z) [illegible]

$$M \perp\!\!\!\perp Y(z,m) \mid (X,Z), \quad \forall z, m.$$

[[illegible]] [illegible] X [illegible]

$$Z \perp\!\!\!\perp M(z) \mid X, \quad \forall z.$$

---

[5] [illegible] **?** [illegible] **?**

[illegible] sequential ignorability [illegible] X [illegible] (X, Z) [illegible] X [illegible] (Z, M) [illegible]

$$(Z, M) \perp\!\!\!\perp Y(z, m) \mid X, \quad \forall z, m. \tag{136}$$

[illegible][6]

[illegible]

[[illegible]] [illegible] X [illegible]

$$Y(z, m) \perp\!\!\!\perp M(z') \mid X, \quad \forall z, z', m.$$

[illegible] $z \neq z'$ [illegible] $Y(z, m)$ [illegible] $M(z')$ [illegible] $Y(1, M_0)$ [illegible] cross-world counterfactual [illegible]

[illegible] 27.4 ([illegible]). 给定 X，我们按如下方式生成数据：

$$\begin{aligned} Z &= 1\{g_Z(X, \varepsilon_Z) \geq 0\}, \\ M(z) &= 1\{g_M(X, z, \varepsilon_M) \geq 0\}, \quad z = 0, 1, \\ Y(z, m) &= g_Y(X, z, m, \varepsilon_Y), \quad z, m = 0, 1, \end{aligned}$$

其中 $\varepsilon_Z, \varepsilon_M, \varepsilon_Y$ 相互独立。[7] 可以验证，在上述数据生成过程下，[illegible]均成立。如果允许 ε_M 和 ε_Y 依赖于处理水平（即改为 $\varepsilon_M(z)$ 和 $\varepsilon_Y(z, m)$），则这些假设可能被违反。

[illegible]

[illegible]（[illegible]）在[illegible]下，[illegible]有

$$\mathbb{E}\{Y(z, M_z) \mid X = x\} = \sum_m \mathbb{E}(Y \mid Z = z, M = m, X = x)\ \Pr(M = m \mid Z = z', X = x).$$

[illegible] counterfactual $Y(z, M_z)$ [illegible] $Y(z, M_z)$ [illegible] $Z = z, M = m, X = x$ [illegible] $Z = z', X = x$ [illegible]

[illegible] $\Pr(M = m \mid Z = z', X = x) = \Pr(M_z = m \mid X = x)$ [illegible] $Y(z, m)$ [illegible] M_z [illegible]

[illegible] X [illegible]

$$\mathbb{E}\{Y(z, M_z)\} = \sum_x \sum_m \mathbb{E}(Y \mid Z = z, M = m, X = x)\ \Pr(M = m \mid Z = z', X = x)\ \Pr(X = x).$$

[illegible] M [illegible] X [illegible] NDE [illegible] NIE [illegible]

---

[6] ??

[7] ?

[illegible] (NDE [illegible] NIE [illegible]) 在 [illegible] 下 [illegible], [illegible] 条件 *NDE* 和 *NIE* 由下式识别：

$$(x) = \sum_m \big[\mathbb{E}(Y \mid Z=1, M=m, X=x) - \mathbb{E}(Y \mid Z=0, M=m, X=x)\big] \Pr(M=m \mid Z=0, X=x), \tag{137}$$

$$(x) = \sum_m \mathbb{E}(Y \mid Z=1, M=m, X=x) \big[\Pr(M=m \mid Z=1, X=x) - \Pr(M=m \mid Z=0, X=x)\big]. \tag{138}$$

[illegible] (x) [illegible] 0 [illegible] 1 [illegible] m [illegible] 0 [illegible] 1 [illegible] $\mathbb{E}(Y \mid Z=1, M=m, X=x) - \mathbb{E}(Y \mid Z=0, M=m, X=x)$ [illegible] $Z=0$ [illegible] $\Pr(M=m \mid Z=0, X=x)$ [illegible]

[illegible] $M \in \{0,1\}$ [illegible] NIE [illegible]

[illegible] NIE [illegible]) 在 [illegible] 下 [illegible], [illegible] 对 [illegible] 二值中介 M，有

$$(x) = \tau_{Z\to M}(x) \cdot \tau_{M\to Y}(1, x),$$

其中

$$\tau_{Z\to M}(x) = \Pr(M=1 \mid Z=1, X=x) - \Pr(M=1 \mid Z=0, X=x)$$

是 Z 对 M 的条件平均因果效应，

$$\tau_{M\to Y}(z, x) = \mathbb{E}(Y \mid Z=z, M=1, X=x) - \mathbb{E}(Y \mid Z=z, M=0, X=x)$$

是 M 对 Y 的条件平均因果效应。

[illegible] Z [illegible] M [illegible] $\tau_{Z\to M}$ [illegible] M [illegible] Y [illegible] $\tau_{M\to Y}$ [illegible]

[illegible][8]

[illegible] A ; p l 7 d Q l l 》

[illegible] A ; p l 7 d Q l l 》 [illegible]

[illegible] Baron–Kenny [illegible] [Baron–Kenny [illegible]] [illegible]

$$\begin{cases} \mathbb{E}(M \mid Z, X) = \beta_0 + \beta_1 Z + \beta_2^\top X, \\ \mathbb{E}(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_4^\top X. \end{cases} \tag{27.1}$$

[illegible] NDE [illegible] NIE [illegible]

[illegible] (Baron–Kenny [illegible]) 在 [illegible] 下 [illegible], [illegible] 有 [illegible]

$$= \theta_1, \qquad = \theta_2 \beta_1. \tag{27.2}$$

---

[8] [illegible] ??

M : Y θ_1 $Z \to Y$ $\beta_1\theta_2$
$Z \to M \to Y$

X (x) (x) $X = x$
Baron–Kenny θ_1 $\theta_2\beta_1$ x

Baron–Kenny
OLS $\hat{\beta}_1$ $\hat{\theta}_2$ NDE NIE

$$\hat{} = \hat{\theta}_1, \qquad \hat{} = \hat{\theta}_2\hat{\beta}_1,$$

Baron–Kenny delta $\hat{}$

$$\hat{\mathrm{var}}(\hat{}) = \hat{\mathrm{var}}(\hat{\theta}_2)\hat{\beta}_1^2 + \hat{\theta}_2^2\hat{\mathrm{var}}(\hat{\beta}_1).$$

Sobel's test[9]

A ; p l 7 d Q l l

b o A、

R Baron–Kenny JOBS II[10]

27.5 (JOBS II). 我们关注就业培训（处理 Z）对心理健康（结果 Y，用抑郁量表得分度量）的影响，以及这一影响是否通过工作搜寻效能（中介 M）传导。背景协变量 X 包括经济困难程度、基线抑郁得分、性别、年龄、职业、婚姻状态、种族、教育程度和家庭收入。

通过以下 R 代码实现 Baron–Kenny 方法：

```
library("car")
BKmediation = function(Z, M, Y, X) {
  mediator.reg   = lm(M ~ Z + X)
  mediator.Zcoef = mediator.reg$coef[2]
  mediator.Zse   = sqrt(hccm(mediator.reg)[2, 2])

  outcome.reg    = lm(Y ~ Z + M + X)
  outcome.Zcoef  = outcome.reg$coef[2]
  outcome.Zse    = sqrt(hccm(outcome.reg)[2, 2])
  outcome.Mcoef  = outcome.reg$coef[3]
  outcome.Mse    = sqrt(hccm(outcome.reg)[3, 3])

  NDE = outcome.Zcoef
  NIE = outcome.Mcoef * mediator.Zcoef

  NDE.se = outcome.Zse
  NIE.se = sqrt(outcome.Mse^2 * mediator.Zcoef^2 +
                outcome.Mcoef^2 * mediator.Zse^2)
```

---

[9] Sobel ?

[10] JOBS II R `mediation`

```
  res = matrix(c(NDE, NIE, NDE.se, NIE.se,
                 NDE/NDE.se, NIE/NIE.se), 2, 3)
  rownames(res) = c("NDE", "NIE")
  colnames(res) = c("est", "se", "t")
  res
}
```

运行结果如下：

```
> res = BKmediation(Z, M, Y, X)
> round(res, 3)
        est      se       t
NDE -0.037  0.042  -0.885
NIE -0.014  0.009  -1.528
```

估计的直接效应 ˆ= −0.037（培训对心理健康的直接保护效应）和间接效应 ˆ= −0.014（通过提高工作搜寻效能而改善心理健康）均为负，但统计上均不显著。这表明 *JOBS II* 培训项目对心理健康的效应尚无充分证据支持按中介路径分解。

[illegible] $Z \to M$ [illegible] $\beta_1 > 0$ [illegible] $M \to Y$ [illegible] $\theta_2 > 0$ [illegible] Z [illegible] M [illegible] $\theta_3 < 0$ [illegible] $\beta_1(\theta_2 + \theta_3)$ [illegible] mediator paradox [illegible] surrogate endpoint paradox [illegible][a]

---

[a] [illegible] ?

[illegible]

[illegible] $Y(z, m)$ [illegible] $M(z')$ [illegible] $z \neq z'$ [illegible] sensitivity analysis [illegible]

- [illegible] Cornfield [illegible]
- [illegible] Baron–Kenny [illegible]

[illegible]

R [illegible]

[illegible]

(1) [illegible]

(2) [illegible] $Y(z, M_z)$ [illegible] counterfactual [illegible] NDE [illegible] NIE [illegible]

(3) [illegible] Composition [illegible] $\tau = +$

(4) [illegible] $+$ [illegible]

(5) [illegible] NDE [illegible] NIE [illegible]

(6) [illegible] $= \theta_1$ [illegible] $= \theta_2\beta_1$

(7) [illegible] Popperian [illegible]

(8) [illegible]

{ { Baron–Kenny {

{ {

27.1. 假设 (Z, M, Y) 均为二值变量。基于潜在结果

$$M(1), M(0), Y(1, M_1), Y(1, M_0), Y(0, M_1), Y(0, M_0)$$

的联合取值，单元可以分成多少类型？

若进一步假设单调性 $M(1) \geq M(0)$（Z 对 M 的单调性），则单元可以分成多少类型？

若再假设 $Y(1, m) \geq Y(0, m)$ 且 $Y(z, 1) \geq Y(z, 0)$ 对所有 z, m 成立（Z 和 M 对 Y 的单调性），则单元可以分成多少类型？

{ {

27.2. 证明 { {与 { { {等{价。

{ {

27.3. { 提出了另一组假设来推导中介公式：

$$\{Y(z, m), M(z')\} \perp\!\!\!\perp Z \mid X, \qquad Y(z, m) \perp\!\!\!\perp M(z') \mid (Z = z', X),$$

对所有 z, z', m 成立。

证明：在上述假设下，{ {中的中介公式同样成立。

{ {

27.4. 在 *Baron–Kenny* 线性模型（{ }{ 下，首先拟合：

$$\hat{M} = \hat{\beta}_0 + \hat{\beta}_1 Z + \hat{\beta}_2^\top X, \qquad \hat{Y} = \hat{\theta}_0 + \hat{\theta}_1 Z + \hat{\theta}_2 M + \hat{\theta}_4^\top X.$$

则 *NIE* 的估计量为乘积 $\hat{\theta}_2 \hat{\beta}_1$（乘积法）。

另一方面，首先拟合：

$$\hat{Y} = \hat{\alpha}_0 + \hat{\alpha}_1 Z + \hat{\alpha}_1^\top X,$$

再拟合：

$$\hat{Y} = \tilde{\theta}_0 + \tilde{\theta}_1 Z + \tilde{\theta}_2 M + \tilde{\theta}_4^\top X,$$

则 *NIE* 的另一估计量为差值 $\hat{\alpha}_1 - \tilde{\theta}_1$（差值法）。

证明：$\hat{\alpha}_1 - \tilde{\theta}_1 = \hat{\theta}_2 \hat{\beta}_1$。

{ {

27.5. 证明 { {(二值中介的 *NIE* 乘积公式)。

{ {

27.6. { 建议在结果模型中加入处理与中介的交互项：

$$\begin{cases} \mathbb{E}(M \mid Z, X) = \beta_0 + \beta_1 Z + \beta_2^\top X, \\ \mathbb{E}(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_3 ZM + \theta_4^\top X. \end{cases}$$

证明：在此模型下，

$$= \theta_1 + \theta_3\{\beta_0 + \beta_2^\top \mathbb{E}(X)\}, \qquad = (\theta_2 + \theta_3)\beta_1.$$

并说明如何在 *IID* 数据下估计 和 。

V̈ ɔB‖ȯɦ { {

ű 27.7. 考虑中介 M 服从正态线性模型、结果 Y 服从 *logistic* 模型：

$$\begin{cases} M \mid Z, X \sim \mathcal{N}(\beta_0 + \beta_1 Z + \beta_2^\top X, \sigma_M^2), \\ \text{logit}\{\Pr(Y = 1 \mid Z, M, X)\} = \theta_0 + \theta_1 Z + \theta_2 M + \theta_4^\top X. \end{cases}$$

请将 和 表示为模型参数和 X 分布的函数，并说明如何在 *IID* 数据下估计它们。

V̈ ɔB‖ȯɦ { {

ű 27.8. 考虑二值中介 M 服从 *logistic* 模型、结果 Y 服从线性模型：

$$\begin{cases} \text{logit}\{\Pr(M = 1 \mid Z, X)\} = \beta_0 + \beta_1 Z + \beta_2^\top X, \\ \mathbb{E}(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_4^\top X. \end{cases}$$

证明：

$$= \theta_1, \qquad = \theta_2 \, \mathbb{E}\Big\{ \text{expit}(\beta_0 + \beta_1 + \beta_2^\top X) - \text{expit}(\beta_0 + \beta_2^\top X) \Big\}.$$

并说明如何在 *IID* 数据下估计 和 。

V̈ ɔB‖ȯɦ { {

ű 27.9. 将嵌套潜在结果修改为

$$Y(z, F_{M_z \mid X}) = \int Y(z, m) \, f(M_z = m \mid X) \, \mathrm{d}m,$$

其中 $F_{M_z \mid X}$ 是条件分布而非确定值。相应地将 *NDE* 和 *NIE* 定义为

$$= \mathbb{E}\{Y(1, F_{M_0|X}) - Y(0, F_{M_0|X})\}, \quad = \mathbb{E}\{Y(1, F_{M_1|X}) - Y(1, F_{M_0|X})\}.$$

证明：在仅假设 { { { {(不要求跨世界独立性）下， 和 的识别公式与正文相同。

V̈ ɔB‖ȯɦ { {

ű 27.10. { 和 { 综述并比较了主分层与中介分析两种方法。请查阅这些文献，并撰写一页总结，阐述两种方法的异同与各自的优缺点。

V̈. ų ẅ ű ŏ { {

*> Sh1_ k3

yT?C

RX„ý Bñ¦•qq¥Ù5

-ë¥c«Ï áÌL!yT² X©ii'X©'tM ^ðya'tM ^² Tb–7 CLÏ áÌªªë Bñ÷¹[4¥Ù5 ?ŽV” ÏÐ M -W¥ yT1"$

ƒñÙ5¥×1Ÿ,ý7ÜbSÐ?C¥'Éü^V4³” Ïw yT² båN

C K2b GYBWM/vL ?CÇÍË£TV[Í ÊÓh Œ ¸ fƒ/ L Ä ç^,V› ,¡£¥báÌ?ŽÇV$î4³¥” Ï ?CM -W¥yT1"$

ƒñÙ5 }Ð ;$ª¹^,V?¥b óM1, cyT ôii ƒ ìý¡ d9 Ð¥.dÆðb–7 V€ØEMn£ BÆá ¥ê B" Ú•!9¥L!/ yT² V[VBe¥4³” ÏÐ b

ƒÆê ¥5z “ Cm/2 Sð?`ZH ¿m¥yTwØØ,[# S2i2` aTaB`i2b *H `F :HvK„Q_mB`+? `/ a+?ð2ÌByM2b S*©ØEV” Ï4|yT² b'c |º ƒB5×¥ '±X„ZEb

kXyT”:V•HqÐÑôŸ

), 8ØE-- áÌ³1Ø³yT?C¥V?ŸHqbInBñ cM $X_1;X_2;:::;X_p$ ¥"dbÂTƒtM -Wi yT1" áÌYÈ¨µ_íÌm . : ŸVU Äñ«Ä}VBñM ph X! Y VUX ^ Y ¥°¤ðy -Bb

Bñô'Ÿ¥4³^ ¸ †Ø¥yT ˜/ M -W¥HqÿëŸ c"yT ² ¥• b

(yT”:V•Hq) óçBñM ¥îµ°¤ðy ¾M ÿë¿îµd ª}¥M b

°4 ÂT X •Á Y 7 Y »•Á Z *¹ X© Y ¥fƒ/ X „ Z -W, ‹¾µ°¤¥ó"iiñÌ-W¥©…1óûXÜ$ Y ³d b

(ÑôŸHq) mÏi ¥HqÿëŸ1" z ‹¿” sƒÏi ¥HqÿëŸ 1"b

ÑôŸHq]¸ óF†ÿëŸ ôii' ñM ” ÏHqÿë Œ mÏ•YV BH^åMõbƒÕF† •”Ä ˜Ï^,© ¥b

ƒ ñL!/ áÌV[VM ¥HqÿëŸ² Ï |yT• b

jX S*ØE VHqÿëŸžyT O

S*ØE aTB`i2b 2i H^Xy-TRyCy5y×KÜÅ¥ØE-Bbñ¥,•±Xe± 7µï VHqÿëŸ ? ï„'çyT² b

jXR»XB„ ?C O XØEVBñ›†í_¥m ? YVHqÿëŸ_ ï„÷ "Hb

[illegible] X [illegible] Y [illegible] S [illegible]
[illegible]

$$X \perp\!\!\!\perp Y \mid S \tag{139}$$

[illegible] $\perp\!\!\!\perp$ [illegible] (X, Y) [illegible]
[illegible]

[illegible] 28.1 ([illegible]). 考虑三个变量 X, Y, Z。假设我们进行条件独立性检验:

- $X \not\perp\!\!\!\perp Y$（边缘不独立）
- $X \perp\!\!\!\perp Y \mid Z$（给定 Z 条件独立）

这告诉我们什么？在 *Markov* 条件下，这暗示 Z 是 X 和 Y 的共同原因，或者 Z 位于它们之间的中介路径上。

[illegible]

[illegible] [illegible] v-[illegible]
[illegible] X, Y, Z [illegible] $X - Z - Y$ [illegible] X [illegible] Y [illegible] Z [illegible] $X \not\perp\!\!\!\perp Y \mid \emptyset$ [illegible] $X \perp\!\!\!\perp Y \mid Z$ [illegible]
Markov [illegible] $X \to Z \to Y$ [illegible] Z [illegible] Z [illegible] X [illegible] Y [illegible] $X \perp\!\!\!\perp Y \mid Z$
[illegible] Z [illegible] X [illegible] Y [illegible] $Z \to X$ [illegible] $Z \to Y$ [illegible] X [illegible] Y [illegible] Z [illegible]
PC [illegible]
[v-[illegible]] [illegible] (X, Z, Y) [illegible] v-[illegible]

(1) X [illegible] Y [illegible]
(2) Z [illegible] X [illegible] Z [illegible] Y [illegible]
(3) [illegible] Z [illegible] X [illegible] Y [illegible]

[illegible] v-[illegible] $X \to Z \leftarrow Y$ [illegible] $X \to Z \to Y$ [illegible] $X \leftarrow Z \to Y$ [illegible] Z [illegible] X [illegible] Y [illegible]
PC [illegible]

(1) [illegible] v-[illegible]
(2) [illegible] v-[illegible]

[illegible] [illegible]
[illegible] $X - Y$ [illegible] $Y - Z$ [illegible] X [illegible] Z [illegible] $X \to Y \to Z$ [illegible] $X \leftarrow Y \leftarrow Z$ [illegible]
[illegible]

[illegible]

[illegible] **[illegible]** [illegible] DAG [illegible] PAG [illegible]

- [illegible] $\circ - \circ$ [illegible]
- [illegible] $\to$ [illegible] $\leftarrow$ [illegible]

[illegible] **R** [illegible]) 在因果马尔可夫条件和忠诚性条件下，观察分布唯一确定一个 *PAG*。

[illegible] PC [illegible] PAG [illegible]

[illegible]

PC [illegible] nontrivial [illegible]

W [illegible] S [illegible] X [illegible] Y [illegible] $\rho_{XY\cdot S}$ [illegible] $H_0 : \rho_{XY\cdot S} = 0$ [illegible] p

28.1. 偏相关系数的计算需要 S 中变量的条件协方差矩阵。当 $|S|$ 较大时，估计精度会下降，这是 *PC* 算法的一个 *practical limitation*。

W [illegible]

$$I(X;Y \mid S) = \sum_{x,y,s} p(x,y,s) \log \frac{p(x,y \mid s)}{p(x \mid s)p(y \mid s)}. \tag{140}$$

[illegible]

[illegible]

[illegible] PC [illegible]

(1) [illegible]
(2) [illegible] (X,Y) [illegible] S [illegible] $X \perp\!\!\!\perp Y \mid S$ [illegible] (X,Y)
(3) [illegible] (X,Z,Y) [illegible] v-[illegible] $X \to Z \leftarrow Y$
(4) [illegible]
(5) [illegible] PAG

[illegible] p [illegible] $\sum_{k=0}^{p-2}\binom{p-2}{k}$ [illegible] PC [illegible]

[illegible]

PC [illegible]

W [illegible] GIES [illegible] Greedy Equivalence [illegible] Search [illegible] DAG [illegible] BIC [illegible] BD [illegible]

[illegible]

W [illegible] DAG [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] NP [illegible]

[illegible]

[illegible]

[illegible] Pearl (2000) [illegible] Causality [illegible] Spirtes et al. (2000) [illegible] Causation, Prediction, and Search [illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible]

[illegible]

(1) [illegible]
(2) PC [illegible] v-[illegible]
(3) [illegible] PAG [illegible]
(4) [illegible]

[illegible]

[illegible]

[illegible] 28.1. 证明：在马尔可夫条件和忠诚性条件下，如果 X 和 Y 在骨架中无边，但都与 Z 相连，且 $X \not\perp\!\!\!\perp Y \mid \emptyset$ 但 $X \perp\!\!\!\perp Y \mid Z$，则 (X, Z, Y) 形成 v-结构。

[illegible] 28.2. 考虑四个变量 X_1, X_2, X_3, X_4，假设真实因果结构是 $X_1 \to X_2 \to X_4$ 和 $X_1 \to X_3 \to X_4$。说明 PC 算法在什么条件下能够恢复这个结构，以及在什么条件下可能失败。

[illegible] 28.3. 证明：对于 n 个变量的 DAG，完全图（有 $\binom{n}{2}$ 条边）和完全无向图在骨架发现步骤后的边数关系。

ǘ 28.4. 讨论因果发现与因果推断其他方法（如第 *10-12* 章的倾向得分方法）的关系。它们分别适合什么场景？

CHAPTER 29

[illegible]

[illegible]

[illegible]

[illegible] Z_1 [illegible] X_1 [illegible] Z_2 [illegible] Y [illegible] 5 [illegible]

- Z_1 [illegible] X_1 [illegible]
- X_1 [illegible] Z_2 [illegible] Y [illegible]
- [illegible] X_1 [illegible]

[illegible] X_0 [illegible] X_1 [illegible] Z_2 [illegible] Y [illegible] Z_1 [illegible] Z_1 [illegible] Y [illegible]

[illegible]

[illegible]

[illegible]

$$X_0 \to Z_1 \to X_1 \to Z_2 \to Y$$

[illegible]

- X_0 [illegible]
- Z_1 [illegible] 1 [illegible]
- X_1 [illegible] 1 [illegible] 2 [illegible]
- Z_2 [illegible] 2 [illegible]
- Y [illegible]

[illegible] (Z_1, Z_2) [illegible]

$$Y(z_1, z_2), \quad z_1, z_2 \in \{0, 1\}$$

[illegible]

$$Y = Y(Z_1, Z_2) = \sum_{z_1=0}^{1} \sum_{z_2=0}^{1} \mathbf{1}(Z_1 = z_1)\mathbf{1}(Z_2 = z_2)Y(z_1, z_2)$$

[illegible] $Z \perp Y(z) \mid X$ [illegible]

[illegible]

[illegible] 29.1 ([illegible]). *(1)* Z_1 在给定 X_0 条件下随机化:

$$Z_1 \perp Y(z_1, z_2) \mid X_0, \quad \forall z_1, z_2 \in \{0,1\}$$

(2) Z_2 在给定 (Z_1, X_1, X_0) 条件下随机化:

$$Z_2 \perp Y(z_1, z_2) \mid (Z_1, X_1, X_0), \quad \forall z_1, z_2 \in \{0,1\}$$

[illegible] Z_1 [illegible] X_0 [illegible] X_0 [illegible] Z_1 [illegible] Z_2 [illegible] (Z_1, X_1, X_0) [illegible] X_0 [illegible] Z_1 [illegible] X_1

[illegible] X_1 [illegible] Y [illegible] U [illegible] X_1 [illegible] Z_1 [illegible] Z_1 [illegible] Y [illegible] U [illegible] X_1 [illegible] Y [illegible] Z_1, Z_2 [illegible]

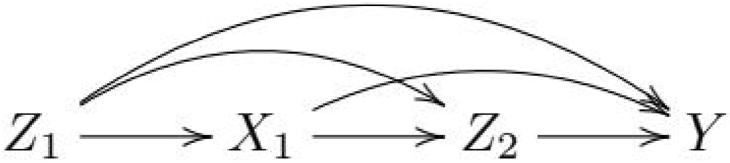

Figure 1. Assumption [illegible] X_1 [illegible] Y [illegible] U [illegible]

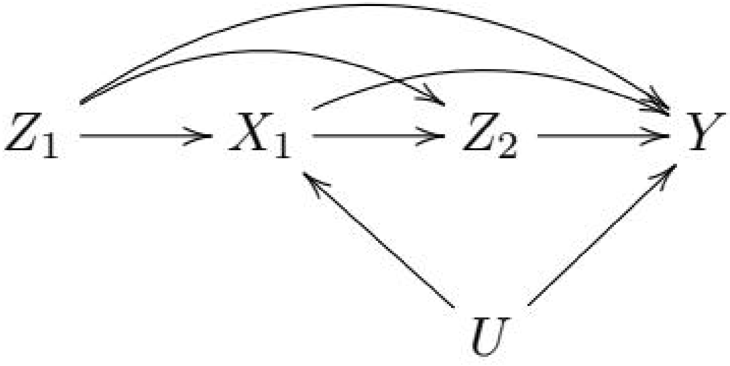

Figure 2. Assumption [illegible] X_1 [illegible] Y [illegible] U [illegible] (Z_1, Z_2) [illegible] Y [illegible] X_1 [illegible] Y [illegible]

[illegible]

[illegible]

$$E\{Y(z)\} = E\{E(Y \mid Z = z, X)\}$$

[illegible] X [illegible]

$$E\{Y(z)\} = \sum_{x} E(Y \mid Z = z, X = x)\Pr(X = x)$$

[illegible] X [illegible]

$$E\{Y(z)\} = \int E(Y \mid Z = z, X = x) f(x)\,dx$$

[illegible]

[illegible] 在 *Assumption* [illegible] 下，

$$E\{Y(z_1, z_2)\} = E\Big[E\{E(Y \mid z_2, z_1, X_1, X_0) \mid z_1, X_0\}\Big] \tag{29.1}$$

[illegible] $E(Y \mid z_2, z_1, X_1, X_0)$ [illegible] (z_1, X_0) [illegible] X_1 [illegible] z_2 [illegible] $E\{\cdot \mid z_1, X_0\}$ [illegible] X_1 [illegible] $\Pr(X_1 \mid z_1, X_0)$ $Z_1 = z_1$ [illegible] $E[\cdot]$ [illegible] X_0 [illegible]

[illegible] X_0, X_1 [illegible]

$$E\{Y(z_1, z_2)\} = \sum_{x_0}\sum_{x_1} E(Y \mid z_2, z_1, x_1, x_0)\Pr(x_1 \mid z_1, x_0)\Pr(x_0) \tag{29.2}$$

[illegible] X_0, X_1 [illegible]

$$E\{Y(z_1, z_2)\} = \iint E(Y \mid z_2, z_1, x_1, x_0) f(x_1 \mid z_1, x_0) f(x_0)\,dx_1\,dx_0 \tag{29.3}$$

[illegible]

$$\begin{aligned} E(Y) = &\sum_{x_0}\sum_{z_1}\sum_{x_1}\sum_{z_2} E(Y \mid z_2, z_1, x_1, x_0) \\ &\Pr(z_2 \mid z_1, x_1, x_0)\Pr(x_1 \mid z_1, x_0)\Pr(z_1 \mid x_0)\Pr(x_0) \end{aligned} \tag{29.4}$$

[illegible] $\Pr(z_2 \mid \cdots)$ [illegible] $\Pr(z_1 \mid \cdots)$ [illegible] $Y(z_1, z_2)$ [illegible] $Z_1 = z_1$ [illegible] $Z_2 = z_2$ [illegible] Robins [illegible][1]

[illegible]

[illegible] $E(Y \mid z_2, z_1, x_1, x_0)$ [illegible] $\Pr(x_1 \mid z_1, x_0)$ [illegible] $\Pr(x_0)$ [illegible] ” [illegible] ” [illegible] 29.1 [illegible]

[illegible] 29.1 ([illegible] 29.1 [illegible]). 假设线性结果模型：

$$E(Y \mid z_2, z_1, x_1, x_0) = \beta_0 + \beta_1 z_2 + \beta_2 z_1 + \beta_3 x_1 + \beta_4 x_0 \tag{29.5}$$

---

[1] [illegible] **??**

可以验证：

$$
\begin{aligned}
E\{Y(z_1,z_2)\} =& \sum_{x_0}\sum_{x_1}(\beta_0+\beta_1 z_2+\beta_2 z_1+\beta_3 x_1+\beta_4 x_0) \\
& \Pr(x_1 \mid z_1, x_0)\Pr(x_0) \\
=&\beta_0+\beta_1 z_2+\beta_2 z_1+\beta_3\sum_{x_0}E(X_1 \mid z_1,x_0)\Pr(x_0)+\beta_4 E(X_0) \\
=&\beta_0+\beta_1 z_2+\beta_2 z_1+\beta_3 E\{X_1(z_1)\}+\beta_4 E(X_0)
\end{aligned}
\tag{29.6}
$$

其中我们定义了 X_1 在治疗 $Z_1 = z_1$ 下的潜在结局：

$$E\{X_1(z_1)\} = \sum_{x_0} E(X_1 \mid z_1, x_0)\Pr(x_0) \tag{29.7}$$

定义 $\tau_{Z_1\to X_1} = E\{X_1(1) - X_1(0)\}$，则：

$$E\{Y(1,0)-Y(0,0)\} = \beta_2 + \beta_3\tau_{Z_1\to X_1} \tag{29.8}$$

$$E\{Y(0,1)-Y(0,0)\} = \beta_1 \tag{29.9}$$

$$E\{Y(1,1)-Y(0,0)\} = \beta_1 + \beta_2 + \beta_3\tau_{Z_1\to X_1} \tag{29.10}$$

[illegible]（参见图 [illegible]：[illegible]中 Z_1 的效应等于路径 $Z_1 \to Y$（系数 β_2）和路径 $Z_1 \to X_1 \to Y$（系数 $\beta_3\tau_{Z_1\to X_1}$）之和；[illegible]中 Z_2 的效应等于路径 $Z_2 \to Y$（系数 β_1）；[illegible]中 (Z_1, Z_2) 的总效应等于三条路径系数之和。

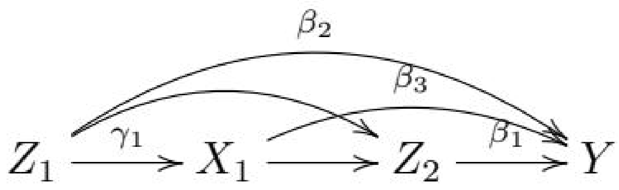

FIGURE 3. [illegible] X_0

[illegible] Robins [illegible] Wasserman [illegible] 1997 [illegible] null [illegible] (Z_1, Z_2) [illegible] Y [illegible] 29.1 [illegible] g-null [illegible] 29.1 [illegible]

[illegible] 29.3.1 [illegible] X_1 [illegible] g-null [illegible]

$$\tilde{Y}_2(z_1,z_2) = E(Y \mid Z_2 = z_2, Z_1 = z_1, X_1, X_0)$$

[illegible] $(Z_2 = z_2, Z_1 = z_1)$ [illegible] Y [illegible] (X_1, X_0) [illegible] $\hat{Y}_{2i}(z_1, z_2)$ [illegible]

$$\tilde{Y}_1(z_1,z_2) = E\{\tilde{Y}_2(z_1,z_2) \mid Z_1 = z_1, X_0\}$$

[illegible] $Z_1 = z_1$ [illegible] $\hat{Y}_2(z_1, z_2)$ [illegible] X_0 [illegible] $\hat{Y}_{1i}(z_1, z_2)$ [illegible] $E\{Y(z_1, z_2)\}$ [illegible]

$$\hat{E}\{Y(z_1, z_2)\} = n^{-1} \sum_{i=1}^{n} \hat{Y}_{1i}(z_1, z_2) \tag{29.11}$$

[illegible] X_1 [illegible] g-null [illegible] 29.2[illegible] [illegible] $\tilde{Y}_2(z_1, z_2)$

X [illegible]

[illegible] IPW [illegible]

$$E\{Y(z)\} = E\left\{\frac{\mathbf{1}(Z = z)Y}{\Pr(Z = z \mid X)}\right\}$$

[illegible]

$$e(z_1, X_0) = \Pr(Z_1 = z_1 \mid X_0) \tag{29.12}$$

$$e(z_2, Z_1, X_1, X_0) = \Pr(Z_2 = z_2 \mid Z_1, X_1, X_0) \tag{29.13}$$

[illegible] 1 [illegible] 2 [illegible]

[illegible] **mw**(IPW [illegible]) **w**[illegible] *Assumption* [illegible],

$$E\{Y(z_1, z_2)\} = E\left\{\frac{\mathbf{1}(Z_1 = z_1)\mathbf{1}(Z_2 = z_2)Y}{e(z_1, X_0)\, e(z_2, Z_1, X_1, X_0)}\right\} \tag{29.14}$$

[illegible]

$$0 < e(z_1, X_0) < 1, \quad 0 < e(z_2, Z_1, X_1, X_0) < 1 \tag{29.15}$$

[illegible] z_1, z_2 [illegible] 0 [illegible] 1 [illegible] IPW [illegible] Z_2 [illegible] Z_1 [illegible] $\mathbf{1}(Z_1 = z_1)\mathbf{1}(Z_2 = z_2)$ [illegible]

[illegible] IPW [illegible] Z_1 [illegible] X_0 [illegible] $\hat{e}_1(z_1, X_{0i})$ [illegible] Z_2 [illegible] (Z_1, X_1, X_0) [illegible] $\hat{e}_2(z_2, Z_{1i}, X_{1i}, X_{0i})$ [illegible] IPW [illegible] Horvitz-Thompson [illegible]

$$\hat{E}^{\text{ht}}\{Y(z_1, z_2)\} = n^{-1} \sum_{i=1}^{n} \frac{\mathbf{1}(Z_{1i} = z_1)\mathbf{1}(Z_{2i} = z_2)Y_i}{\hat{e}_1(z_1, X_{0i})\hat{e}_2(z_2, Z_{1i}, X_{1i}, X_{0i})} \tag{29.16}$$

[illegible] 11 [illegible] HT [illegible] Hajek [illegible] $\hat{E}^{\text{haj}}\{Y(z_1, z_2)\} = \hat{E}^{\text{ht}}\{Y(z_1, z_2)\}/\hat{1}^{\text{ht}}(z_1, z_2)$ [illegible]

$$\hat{1}^{\text{ht}}(z_1, z_2) = n^{-1} \sum_{i=1}^{n} \frac{\mathbf{1}(Z_{1i} = z_1)\mathbf{1}(Z_{2i} = z_2)}{\hat{e}_1(z_1, X_{0i})\hat{e}_2(z_2, Z_{1i}, X_{1i}, X_{0i})} \tag{29.17}$$

[illegible]

[illegible] 29.2 [illegible] 29.3 [illegible] K [illegible] K [illegible] $2^5 = 32$ [illegible] $2^{10} = 1024$ [illegible] 29.2 [illegible] 29.3 [illegible] IPW [illegible]

[illegible] Robins [illegible]

[illegible] [illegible] Z_1 [illegible] Robins et al., 2000; Hernán et al., 2000 [illegible] $K = 2$ [illegible] MSM [illegible]

[illegible] 29.3 (MSM [illegible]). $Y(z_1, z_2)$ 的边际均值为：

$$E\{Y(z_1, z_2)\} = f(z_1, z_2; \beta) \tag{29.18}$$

[illegible] $E\{Y(z_1, z_2)\} = \beta_0 + \beta_1 z_1 + \beta_2 z_2$ [illegible] Z_1 [illegible]

[illegible] 29.4 ([illegible] MSM). 给定 X_0 条件下 $Y(z_1, z_2)$ 的均值为：

$$E\{Y(z_1, z_2) \mid X_0\} = f(z_1, z_2, X_0; \beta) \tag{29.19}$$

[illegible]

$$E\{Y(z_1, z_2) \mid X_0\} = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3^\top X_0 \tag{29.20}$$

[illegible] β [illegible]

$$\beta = \arg\min_b \sum_{z_2} \sum_{z_1} E\left\{Y(z_1, z_2) - f(z_1, z_2, X_0; b)\right\}^2 \tag{29.21}$$

[illegible] β

[illegible] (MSM [illegible] IPW) 在 *Assumption* [illegible] 和 *Definition* [illegible] 下，[illegible] 中的 β 等于：

$$\beta = \arg\min_b \sum_{z_2} \sum_{z_1} E\left[\frac{\mathbf{1}(Z_1 = z_1)\mathbf{1}(Z_2 = z_2)}{e(z_1, X_0)\, e(z_2, Z_1, X_1, X_0)}\{Y - f(z_1, z_2, X_0; b)\}^2\right] \tag{29.22}$$

[illegible] Y_i [illegible] $(1, Z_{1i}, Z_{2i}, X_{0i})$ [illegible] WLS [illegible] $\hat{e}_1^{-1}(Z_{1i}, X_{0i})\hat{e}_{2i}^{-1}(Z_{2i}, Z_{1i}, X_{1i}, X_{0i})$ [illegible] MSM [illegible] IPW [illegible] $E(Y \mid Z, X)$ [illegible] MSM [illegible] IPW [illegible] $E\{Y(z) \mid X_0\}$ [illegible] MSM [illegible]

[illegible] X_0 [illegible] $\beta_3^\top X_0$ [illegible] β_3 [illegible] LASSO [illegible] Tibshirani, 1996 [illegible] β [illegible] Bloniarz et al., 2016; Wager et al., 2016 [illegible]

[illegible] IPW [illegible] IPW [illegible] Robins [illegible]

[illegible] 29.6 ([illegible] SNM [illegible]). 时间点 *1* 的条件效应为：

$$E\{Y(z_1, 0) - Y(0, 0) \mid Z_1 = z_1, X_0\} = g_1(z_1, X_0; \beta) \tag{29.23}$$

时间点 2 的条件效应为：

$$E\{Y(z_1,z_2)-Y(z_1,0)\mid Z_2=z_2,Z_1=z_1,X_1,X_0\}=g_2(z_2,z_1,X_1,X_0;\beta) \tag{29.24}$$

Definition [illegible]

$$g_1(0,X_0;\beta)=0,\quad g_2(0,z_1,X_1,X_0;\beta)=0,\quad \forall z_1 \tag{29.25}$$

[illegible]

[illegible] 29.2 ([illegible] 29.2 [illegible] SNM). 假设：

$$\begin{cases} g_1(z_1,X_0;\beta)=\beta_1 z_1 \\ g_2(z_2,z_1,X_1,X_0;\beta)=(\beta_2+\beta_3 z_1)z_2 \end{cases} \tag{29.26}$$

[illegible] 29.3 ([illegible] 29.3 [illegible] SNM). 假设：

$$\begin{cases} g_1(z_1,X_0;\beta)=(\beta_1+\beta_2^{\mathsf{T}}X_0)z_1 \\ g_2(z_2,z_1,X_1,X_0;\beta)=(\beta_3+\beta_4 z_1+\beta_5^{\mathsf{T}}X_1)z_2 \end{cases} \tag{29.27}$$

i、i 、k i [illegible] $_1$ [illegible] Definition [illegible] $_1$ [illegible]

$$U_2(\beta)=Y-g_2(Z_2,Z_1,X_1,X_0;\beta) \tag{29.28}$$

$$U_1(\beta)=Y-g_2(Z_2,Z_1,X_1,X_0;\beta)-g_1(Z_1,X_0;\beta) \tag{29.29}$$

[illegible]

[illegible] 29.1) 在 *Assumption* [illegible] 和 *Definition* [illegible] 下，

$$E\{U_2(\beta)\mid Z_2,Z_1,X_1,X_0\}=E\{U_2(\beta)\mid Z_1,X_1,X_0\}=E\{Y(Z_1,0)\mid Z_1,X_1,X_0\} \tag{29.30}$$

且

$$E\{U_1(\beta)\mid Z_1,X_0\}=E\{U_1(\beta)\mid X_0\}=E\{Y(0,0)\mid X_0\} \tag{29.31}$$

[illegible] $U_1(\beta)$ [illegible] $U_2(\beta)$ [illegible] 1 [illegible] 2 [illegible]

[illegible] 29.4 [illegible] SNM [illegible]) 在 *Assumption* [illegible] 和 *Definition* [illegible] 下，

$$E\left[h_2(Z_1,X_1,X_0)\{Z_2-e(1,Z_1,X_1,X_0)\}U_2(\beta)\right]=0 \tag{29.32}$$

且

$$E\left[h_1(X_0)\{Z_1-e(1,X_0)\}U_1(\beta)\right]=0 \tag{29.33}$$

对任何函数 h_1 和 h_2（只要矩存在）。

[illegible] h_1 [illegible] h_2 [illegible] β [illegible] 29.4 [illegible] 29.2

[illegible] 29.4 ([illegible] 29.4 [illegible] 29.2 [illegible]). 在例 *29.2* 下，我们可以选择 $h_1 = 1$ 和 $h_2 = (1, Z_1)$ 得到：

$$
\begin{aligned}
&E[\{Z_2 - e(1, Z_1, X_1, X_0)\}\{Y - (\beta_2 + \beta_3 Z_1)Z_2\}] = 0\\
&E[Z_1\{Z_2 - e(1, Z_1, X_1, X_0)\}\{Y - (\beta_2 + \beta_3 Z_1)Z_2\}] = 0 \qquad (29.34)\\
&E[\{Z_1 - e(1, X_0)\}\{Y - (\beta_2 + \beta_3 Z_1)Z_2 - \beta_1 Z_1\}] = 0
\end{aligned}
$$

然后我们可以从上述线性方程组求解 β（见习题 *29.6*）。

[illegible] β [illegible] (h_1, h_2) [illegible] (h_1, h_2) [illegible] GMM; Hansen, 1982 [illegible] Naimi et al. 2017 [illegible] Vansteelandt and Joffe 2014 [illegible]

[illegible]

[illegible] [illegible] X_0 [illegible] (Z_1, Z_2) [illegible] Y [illegible] (Z_1, Z_2) [illegible] Y [illegible]

FIGURE 4. 29.1 [illegible] X_1 [illegible] Y [illegible] U [illegible]

[illegible] 29.1 [illegible]

$$\beta_1 = \beta_2 = 0 \quad [illegible] \quad \beta_3 = 0 \qquad (29.35)$$

5

$$\beta_1 = \beta_2 = 0 \quad [illegible] \quad E\{X_1(z_1)\} \text{ [illegible] } z_1 \qquad (29.36)$$

[illegible] $E\{Y(z_1, z_2)\}$ [illegible] (z_1, z_2) [illegible] $\beta_3 = 0$ [illegible] Y [illegible] X_1 [illegible] X_1 [illegible] Y [illegible] U [illegible] $E\{X_1(z_1)\}$ [illegible] z_1 [illegible] X_1 [illegible] Z_1 [illegible] Z_1 [illegible] X_1 [illegible] X_1 [illegible] Y [illegible] U [illegible] Z_1 [illegible] X_1 [illegible] 29.1 $E\{Y(z_1, z_2)\}$ [illegible] (z_1, z_2) [illegible] (Z_1, Z_2) [illegible] Y [illegible] g-null [illegible]

[illegible] [illegible] 29.3.2 [illegible] 29.1 [illegible] $_1$ [illegible] 0

[illegible]

[illegible] [illegible] 29.2 [illegible]

$$\operatorname{logit}[\Pr\{Y(z_1, z_2) = 1 \mid X_0\}] = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3^\intercal X_0 \qquad (29.37)$$

[illegible] β [illegible] B.6.2 [illegible]

$$\beta = \arg\min_b \sum_{z_2}\sum_{z_1} E\{\log(1 + e^{\ell}) - Y(z_1, z_2)\ell\} \qquad (29.38)$$

[illegible] $\ell = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3^\intercal X_0$ [illegible] β

[illegible] Assumption [illegible] Definition [illegible] β [illegible]

$$\beta = \arg\min_{b} \sum_{z_2} \sum_{z_1} E\left[\frac{\mathbf{1}(Z_1 = z_1)\mathbf{1}(Z_2 = z_2)}{e(z_1, X_0)\, e(z_2, Z_1, X_1, X_0)}\{\log(1 + e^{\ell}) - Y(z_1, z_2)\ell\}\right] \tag{29.39}$$

[illegible] 29.5

[illegible] 29.5 [illegible] [illegible]
[illegible] Y_i [illegible] $(1, Z_{1i}, Z_{2i}, X_{0i})$ [illegible] logistic [illegible] $\hat{e}_1^{-1}(Z_{1i}, X_{0i})\hat{e}_{2i}^{-1}(Z_{2i}, Z_{1i}, X_{1i}, X_{0i})$

[illegible] IID [illegible] $\{X, Z, Y(1), Y(0)\}$ [illegible] $e(X) = \Pr(Z = 1 \mid X)$ [illegible]

$$Z \perp Y(0) \mid X \tag{29.40}$$

[illegible]

[illegible]

$$E\{Y(z) - Y(0) \mid Z = z, X\} = g(z, X; \beta) \tag{29.41}$$

[illegible] 29.4 [illegible] $g(0, X; \beta) = 0$ [illegible]

[illegible]

$$E\{Y - g(Z, X; \beta) \mid X, Z\} = E\{Y - g(Z, X; \beta) \mid X\} = E\{Y(0) \mid X\} \tag{29.42}$$

[illegible]

$$E[h(X)\{Z - e(X)\}\{Y - g(Z, X; \beta)\}] = 0 \tag{29.43}$$

[illegible] h [illegible]
[illegible] 29.4 [illegible] $g(z, X; \beta) = \beta z$ [illegible] $h(X) = 1$ [illegible]

$$E[\{Z - e(X)\}\{Y - \beta Z\}] = 0 \tag{29.44}$$

[illegible] 11 [illegible]

[illegible]

Robins [illegible] Robins [illegible] 1986 [illegible] g-formula [illegible] Robins [illegible] 1998 [illegible] Robins [illegible] Hernán [illegible] 2009 [illegible] g-[illegible] Robins [illegible] Waterman [illegible] 1994 [illegible] MSM [illegible] SNM [illegible] g-null [illegible] Robins [illegible] Wasserman [illegible] 1997 [illegible] McGrath et al. [illegible] 2021 [illegible]

[illegible] Bloniarz et al. [illegible] 2016 [illegible] LASSO [illegible] Wager et al [illegible] 2016 [illegible] Lei [illegible] Ding [illegible] 2021 [illegible] Lin [illegible] 2013 [illegible] D'Amour et al. [illegible] 2021 [illegible]

APPENDIX A

[illegible]

[illegible]

[illegible] 1975 [illegible]

$$2S(1,0) = S^2(1) + S^2(0) - S^2(\tau).$$

[illegible] Lemma 4.1 [illegible] Neyman (1923) [illegible]

- $S^2(1) = (n-1)^{-1}\sum_{i=1}^{n}\{Y_i(1)-\bar{Y}(1)\}^2$ [illegible] $Y_i(1)$ [illegible]
- $S^2(0) = (n-1)^{-1}\sum_{i=1}^{n}\{Y_i(0)-\bar{Y}(0)\}^2$ [illegible] $Y_i(0)$ [illegible]
- $S(1,0) = (n-1)^{-1}\sum_{i=1}^{n}\{Y_i(1)-\bar{Y}(1)\}\{Y_i(0)-\bar{Y}(0)\}$ [illegible] $Y_i(1)$ [illegible] $Y_i(0)$ [illegible]
- $\tau_i = Y_i(1) - Y_i(0)$ [illegible]
- $\tau = n^{-1}\sum_{i=1}^{n}\tau_i = \bar{Y}(1) - \bar{Y}(0)$ [illegible]
- $S^2(\tau) = (n-1)^{-1}\sum_{i=1}^{n}(\tau_i - \tau)^2$ [illegible]

[illegible]

$$2S(1,0) = S^2(1) + S^2(0) - S^2(\tau).$$

[illegible]

[illegible] $S^2(\tau)$ [illegible]

$$\begin{aligned}
S^2(\tau) &= (n-1)^{-1}\sum_{i=1}^{n}(\tau_i - \tau)^2 \\
&= (n-1)^{-1}\sum_{i=1}^{n}\left[(Y_i(1)-Y_i(0)) - (\bar{Y}(1)-\bar{Y}(0))\right]^2 \\
&= (n-1)^{-1}\sum_{i=1}^{n}\left[(Y_i(1)-\bar{Y}(1)) - (Y_i(0)-\bar{Y}(0))\right]^2.
\end{aligned}$$

[illegible]

$$\begin{aligned}
S^2(\tau) &= (n-1)^{-1}\sum_{i=1}^{n}\Big[(Y_i(1)-\bar{Y}(1))^2+(Y_i(0)-\bar{Y}(0))^2\\
&\quad -2(Y_i(1)-\bar{Y}(1))(Y_i(0)-\bar{Y}(0))\Big]\\
&= (n-1)^{-1}\sum_{i=1}^{n}(Y_i(1)-\bar{Y}(1))^2+(n-1)^{-1}\sum_{i=1}^{n}(Y_i(0)-\bar{Y}(0))^2\\
&\quad -2(n-1)^{-1}\sum_{i=1}^{n}(Y_i(1)-\bar{Y}(1))(Y_i(0)-\bar{Y}(0))\\
&= S^2(1)+S^2(0)-2S(1,0).
\end{aligned}$$

[illegible]

$$S^2(1)+S^2(0)-S^2(\tau)=2S(1,0).$$

[illegible]

$$\boxed{2S(1,0)=S^2(1)+S^2(0)-S^2(\tau)}.$$

[illegible] $S(1,0)$ [illegible] $S^2(\tau)$ [illegible] $S^2(\tau)=0$ [illegible] $2S(1,0)=S^2(1)+S^2(0)$ [illegible] $Y(1)$ [illegible] $Y(0)$ [illegible] 1 [illegible] $S^2(\tau)>0$ [illegible] $S(1,0)$ [illegible]

[illegible] Neyman [illegible]

$$\mathrm{var}(\hat{\tau})=\frac{S^2(1)}{n_1}+\frac{S^2(0)}{n_0}-\frac{S^2(\tau)}{n}=\frac{n_0}{n_1 n}S^2(1)+\frac{n_1}{n_0 n}S^2(0)+\frac{2}{n}S(1,0).$$

[illegible]

[illegible]

$$(n-1)s^2=\sum_{Z_i=1}\{Y_i-\hat{\bar{Y}}(1)\}^2+\sum_{Z_i=0}\{Y_i-\hat{\bar{Y}}(0)\}^2+\frac{n_1 n_0}{n}\hat{\tau}^2.$$

[illegible]

[illegible]

$$(n-1)s^2=\sum_{i=1}^{n}(Y_i-\bar{Y})^2=\sum_{i=1}^{n}\left[(Y_i-\bar{Y})\right]^2.$$

[illegible] $Y_i-\bar{Y}$ [illegible]

$$Y_i-\bar{Y}=\underbrace{(Y_i-\hat{\bar{Y}}(Z_i))}_{\text{[illegible]}}+\underbrace{(\hat{\bar{Y}}(Z_i)-\bar{Y})}_{\text{[illegible]}}.$$

[illegible]

$$(Y_i-\bar{Y})^2=(Y_i-\hat{\bar{Y}}(Z_i))^2+(\hat{\bar{Y}}(Z_i)-\bar{Y})^2+2(Y_i-\hat{\bar{Y}}(Z_i))(\hat{\bar{Y}}(Z_i)-\bar{Y}).$$

[illegible] [redacted] [illegible]

$$\sum_{i=1}^{n} 2(Y_i - \hat{\bar{Y}}(Z_i))(\hat{\bar{Y}}(Z_i) - \bar{Y}) = 2\sum_{i=1}^{n}(Y_i - \hat{\bar{Y}}(Z_i)) \cdot (\hat{\bar{Y}}(Z_i) - \bar{Y}).$$

[illegible] $Z_i = 1$ [illegible]

$$2\sum_{Z_i=1}(Y_i - \hat{\bar{Y}}(1)) \cdot (\hat{\bar{Y}}(1) - \bar{Y}) = 2(\hat{\bar{Y}}(1) - \bar{Y}) \underbrace{\sum_{Z_i=1}(Y_i - \hat{\bar{Y}}(1))}_{=0} = 0,$$

$\sum_{Z_i=1} Y_i = n_1\hat{\bar{Y}}(1)$ [illegible] $\sum_{Z_i=1}(Y_i - \hat{\bar{Y}}(1)) = 0$

[illegible] $Z_i = 0$ [illegible]

$$2\sum_{Z_i=0}(Y_i - \hat{\bar{Y}}(0)) \cdot (\hat{\bar{Y}}(0) - \bar{Y}) = 2(\hat{\bar{Y}}(0) - \bar{Y}) \underbrace{\sum_{Z_i=0}(Y_i - \hat{\bar{Y}}(0))}_{=0} = 0.$$

[illegible] [redacted] [illegible]

[illegible]

$$\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \underbrace{\sum_{Z_i=1}(Y_i - \hat{\bar{Y}}(1))^2 + \sum_{Z_i=0}(Y_i - \hat{\bar{Y}}(0))^2}_{\text{[illegible]}} + \underbrace{\sum_{i=1}^{n}(\hat{\bar{Y}}(Z_i) - \bar{Y})^2}_{\text{[illegible]}}.$$

[illegible]

$$\sum_{i=1}^{n}(\hat{\bar{Y}}(Z_i) - \bar{Y})^2 = n_1(\hat{\bar{Y}}(1) - \bar{Y})^2 + n_0(\hat{\bar{Y}}(0) - \bar{Y})^2.$$

[illegible]

(1) $\hat{\bar{Y}}(1) - \hat{\bar{Y}}(0) = \hat{\tau}$

(2) $n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0) = n\bar{Y}$ [illegible]

[illegible]

$$\begin{aligned}
&n_1(\hat{\bar{Y}}(1) - \bar{Y})^2 + n_0(\hat{\bar{Y}}(0) - \bar{Y})^2 \\
&= n_1\hat{\bar{Y}}(1)^2 - 2n_1\hat{\bar{Y}}(1)\bar{Y} + n_1\bar{Y}^2 + n_0\hat{\bar{Y}}(0)^2 - 2n_0\hat{\bar{Y}}(0)\bar{Y} + n_0\bar{Y}^2 \\
&= n_1\hat{\bar{Y}}(1)^2 + n_0\hat{\bar{Y}}(0)^2 - n\bar{Y}^2.
\end{aligned}$$

[illegible] $(n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0))^2 = n^2\bar{Y}^2$ [illegible]

$$n_1^2\hat{\bar{Y}}(1)^2 + 2n_1n_0\hat{\bar{Y}}(1)\hat{\bar{Y}}(0) + n_0^2\hat{\bar{Y}}(0)^2 = n^2\bar{Y}^2.$$

[illegible] $\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)$ [illegible]

$$n_1(\hat{\bar{Y}}(1) - \bar{Y})^2 + n_0(\hat{\bar{Y}}(0) - \bar{Y})^2 = \frac{n_1n_0}{n}\hat{\tau}^2.$$

[illegible] { {

[illegible] 1

[illegible] $H_{0\mathrm{F}}$

$$\operatorname{var}(W) = \frac{n_1 n_0 (n+1)}{12}.$$

[illegible]

[illegible] $W = \sum_{i=1}^n Z_i R_i$ [illegible] R_i [illegible] Y_i [illegible] Z_i [illegible]

[illegible]

$$\operatorname{var}(W) = \operatorname{var}\left(\sum_{i=1}^n Z_i R_i\right) = \sum_{i=1}^n R_i^2 \operatorname{var}(Z_i) + \sum_{i \neq j} R_i R_j \operatorname{cov}(Z_i, Z_j).$$

[illegible] (Z_i) [illegible] CRE [illegible] $Z_i \sim \mathrm{Bernoulli}(n_1/n)$ [illegible]

$$\operatorname{var}(Z_i) = \frac{n_1}{n} \cdot \frac{n_0}{n} = \frac{n_1 n_0}{n^2}.$$

[illegible] (Z_i, Z_j) [illegible] $i \neq j$ [illegible] $\sum_{i=1}^n Z_i = n_1$ [illegible]

$$\operatorname{cov}(Z_i, Z_j) = -\frac{n_1 n_0}{n^2(n-1)} \quad (i \neq j).$$

[illegible]

$$\begin{aligned}\operatorname{var}(W) &= \left(\sum_{i=1}^n R_i^2\right) \frac{n_1 n_0}{n^2} + \sum_{i \neq j} R_i R_j \left(-\frac{n_1 n_0}{n^2(n-1)}\right) \\ &= \frac{n_1 n_0}{n^2} \left[\sum_{i=1}^n R_i^2 - \frac{1}{n-1} \sum_{i \neq j} R_i R_j\right].\end{aligned}$$

$\sum_{i \neq j} R_i R_j = \left(\sum_{i=1}^n R_i\right)^2 - \sum_{i=1}^n R_i^2$ [illegible]

$$\operatorname{var}(W) = \frac{n_1 n_0}{n^2} \left[\sum_{i=1}^n R_i^2 - \frac{1}{n-1} \left(\left(\sum_{i=1}^n R_i\right)^2 - \sum_{i=1}^n R_i^2\right)\right].$$

[illegible] $\sum_{i=1}^n R_i$ [illegible] $\sum_{i=1}^n R_i^2$ [illegible] ties [illegible] $\{R_i\}$ [illegible] $1, 2, \ldots, n$ [illegible]

$$\sum_{i=1}^n R_i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

$$\sum_{i=1}^n R_i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

[illegible]

$$\begin{aligned}\operatorname{var}(W) &= \frac{n_1 n_0}{n^2} \left[\frac{n(n+1)(2n+1)}{6} - \frac{1}{n-1} \left(\frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6}\right)\right] \\ &= \frac{n_1 n_0 (n+1)}{12}.\end{aligned}$$

[illegible]

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8 ꞓẊȷ ‗ Ã ۡگ ɽ∏ ŝᴶÃ $K = 4$ Ã ” Ë ” ȷ $N = 8$ ŏăų‡

ɱ $n = 4$ Ãŏ X ȷ ËÃ

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ŝ

$$\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, 2, 3, 4.$$

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$$\sum_{k=0}^{4} \mathbb{P}(X = k) = \frac{1}{\binom{8}{4}} \sum_{k=0}^{4} \binom{4}{k}\binom{4}{4-k} = \frac{1}{70} \cdot \binom{8}{4} = 1,$$

ʕ α̋ Δ ăñɽ $\sum_k \binom{a}{k}\binom{b}{n-k} = \binom{a+b}{n}$

ÃÕ̃ƹȷ ŝ

$$\mathbb{P}(X = 0) = \frac{\binom{4}{0}\binom{4}{4}}{\binom{8}{4}} = \frac{1 \cdot 1}{70} = \frac{1}{70},$$

$$\mathbb{P}(X = 1) = \frac{\binom{4}{1}\binom{4}{3}}{\binom{8}{4}} = \frac{4 \cdot 4}{70} = \frac{16}{70},$$

$$\mathbb{P}(X = 2) = \frac{\binom{4}{2}\binom{4}{2}}{\binom{8}{4}} = \frac{6 \cdot 6}{70} = \frac{36}{70},$$

$$\mathbb{P}(X = 3) = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} = \frac{4 \cdot 4}{70} = \frac{16}{70},$$

$$\mathbb{P}(X = 4) = \frac{\binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = \frac{1 \cdot 1}{70} = \frac{1}{70}.$$

Ȍ ẆÄȷϚ ŝلὼ ” ɱų‡ ”ŋđۡگV^E ƏЫĬɵỷ ᵓBŌ̆ $X \sim$ Binomial$(4, 1/2)$ŏ^B

$$\mathbb{P}(X = 4) = \binom{4}{4}(1/2)^4 = 1/16 = 0.0625 > 0.014.$$

` ⅆ̃ΩẆÄŋăų‡Ōȷ æÕ̃ƹɛ̈ ŏk Fisher ȷ⌐ Ա∂ ɛ̈:§§

FRT” : ” ȷÃ

[illegible]

[illegible]

$$\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0).$$

[illegible] H_{0F} [illegible] $\hat{\tau}$ [illegible]

无偏性.

[illegible] CRE [illegible] **Z** [illegible] $\{\mathbf{Y}(1), \mathbf{Y}(0)\}$ [illegible] n_1/n [illegible]

$$\mathbb{E}(Z_i) = \frac{n_1}{n}, \quad \mathbb{E}(1 - Z_i) = \frac{n_0}{n}.$$

[illegible]

$$\begin{aligned}
\mathbb{E}(\hat{\tau}) &= \mathbb{E}\left(n_1^{-1}\sum_{i=1}^{n} Z_iY_i - n_0^{-1}\sum_{i=1}^{n}(1 - Z_i)Y_i\right)\\
&= n_1^{-1}\sum_{i=1}^{n}\mathbb{E}(Z_i)Y_i - n_0^{-1}\sum_{i=1}^{n}\mathbb{E}(1 - Z_i)Y_i\\
&= n_1^{-1}\sum_{i=1}^{n}\frac{n_1}{n}Y_i - n_0^{-1}\sum_{i=1}^{n}\frac{n_0}{n}Y_i\\
&= \frac{1}{n}\sum_{i=1}^{n}Y_i - \frac{1}{n}\sum_{i=1}^{n}Y_i = 0.
\end{aligned}$$

[illegible]

$$\hat{\tau} = n_1^{-1}\sum_{i=1}^{n} Z_iY_i - n_0^{-1}\sum_{i=1}^{n}(1 - Z_i)Y_i = \frac{n}{n_0}\cdot n_1^{-1}\sum_{i=1}^{n} Z_iY_i - n_0^{-1}\sum_{i=1}^{n} Y_i.$$

$\sum_{i=1}^{n} Y_i$ [illegible] **Z** [illegible]

$$\mathrm{var}(\hat{\tau}) = \frac{n^2}{n_0^2 n_1^2}\cdot \mathrm{var}\left(\sum_{i=1}^{n} Z_iY_i\right).$$

[illegible] $\mathrm{var}(Z_i)$ [illegible] $\mathrm{cov}(Z_i, Z_j)$ CRE [illegible]

$$Z_i \sim \text{Bernoulli}\left(\frac{n_1}{n}\right), \quad \mathrm{var}(Z_i) = \frac{n_1}{n}\cdot\frac{n_0}{n} = \frac{n_1n_0}{n^2}.$$

$\sum_{i=1}^{n} Z_i = n_1$ [illegible] $i \neq j$ [illegible]

$$\mathrm{cov}(Z_i, Z_j) = -\frac{n_1n_0}{n^2(n-1)}.$$

[illegible] $\mathrm{var}\left(\sum_{i=1}^{n} Z_iY_i\right)$ [illegible]

$$\begin{aligned}
\mathrm{var}\left(\sum_{i=1}^{n} Z_iY_i\right) &= \sum_{i=1}^{n} Y_i^2\,\mathrm{var}(Z_i) + \sum_{i\neq j} Y_iY_j\,\mathrm{cov}(Z_i, Z_j)\\
&= \left(\sum_{i=1}^{n} Y_i^2\right)\frac{n_1n_0}{n^2} + \sum_{i\neq j} Y_iY_j\left(-\frac{n_1n_0}{n^2(n-1)}\right).
\end{aligned}$$

$\sum_{i\neq j} Y_iY_j = (\sum_{i=1}^n Y_i)^2 - \sum_{i=1}^n Y_i^2 = n^2\bar{Y}^2 - \sum_{i=1}^n Y_i^2$ [illegible]

$$
\begin{aligned}
\operatorname{var}\left(\sum_{i=1}^n Z_iY_i\right) &= \frac{n_1n_0}{n^2}\sum_{i=1}^n Y_i^2 - \frac{n_1n_0}{n^2(n-1)}\left(n^2\bar{Y}^2 - \sum_{i=1}^n Y_i^2\right) \\
&= \frac{n_1n_0}{n^2(n-1)}\left[(n-1)\sum_{i=1}^n Y_i^2 - n^2\bar{Y}^2 + \sum_{i=1}^n Y_i^2\right] \\
&= \frac{n_1n_0}{n^2(n-1)}\left[n\sum_{i=1}^n Y_i^2 - n^2\bar{Y}^2\right] \\
&= \frac{n_1n_0}{n(n-1)}\left(\sum_{i=1}^n Y_i^2 - n\bar{Y}^2\right).
\end{aligned}
$$

[illegible] s^2 [illegible]

$$
s^2 = (n-1)^{-1}\sum_{i=1}^n (Y_i - \bar{Y})^2 = (n-1)^{-1}\left(\sum_{i=1}^n Y_i^2 - n\bar{Y}^2\right).
$$

[illegible]

$$
\operatorname{var}\left(\sum_{i=1}^n Z_iY_i\right) = \frac{n_1n_0}{n}\cdot s^2.
$$

[illegible] $\operatorname{var}(\hat{\tau})$ [illegible]

$$
\operatorname{var}(\hat{\tau}) = \frac{n^2}{n_0^2n_1^2}\cdot\frac{n_1n_0}{n}\cdot s^2 = \frac{n}{n_1n_0}\cdot s^2.
$$

[illegible]

$$
\boxed{\operatorname{var}(\hat{\tau}) = \frac{n}{n_1n_0}\cdot s^2}.
$$

[illegible]

- [illegible] n [illegible]
- [illegible] n_1n_0 [illegible] $n_1 = n_0 = n/2$ [illegible]
- [illegible]

[illegible]

[illegible] CRE [illegible] n_1 [illegible] n_0 [illegible] $n_{[k]1}$ [illegible] $n_{[k]0}$ [illegible]

[illegible]

- n [illegible]
- $n_{[k]}$ [illegible] k [illegible]
- n_1 [illegible]
- n_0 [illegible] $n_0 = n - n_1$ [illegible]
- $n_{[k]1}$ [illegible] k [illegible]
- $n_{[k]0}$ [illegible] k [illegible] $n_{[k]0} = n_{[k]} - n_{[k]1}$ [illegible]
- $\pi_{[k]} = n_{[k]}/n$ [illegible] k [illegible]

Y a ” Q l

- CRE **Z** $\binom{n}{n_1}$
- n_1/n

Y p g

$$\mathbb{E}\left(\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}\right) = 0.$$

MQ a ” ; — a p l 、 — Q s 。

$n_{[k]1}$

CRE n_1/n k

$X_i = k$

$$\mathbb{E}(Z_i) = \frac{n_1}{n}.$$

k

$$n_{[k]1} = \sum_{i:X_i=k} Z_i.$$

$n_{[k]}$ Z_i n_1

$$\mathbb{E}(n_{[k]1}) = \sum_{i:X_i=k} \mathbb{E}(Z_i) = n_{[k]} \cdot \frac{n_1}{n} = \pi_{[k]} \cdot n_1.$$

$n_{[k]0} = n_{[k]} - n_{[k]1}$

$$\mathbb{E}(n_{[k]0}) = n_{[k]} - \mathbb{E}(n_{[k]1}) = n_{[k]} - \pi_{[k]}n_1 = n_{[k]} - \frac{n_{[k]}}{n}n_1 = n_{[k]}\left(1 - \frac{n_1}{n}\right) = n_{[k]} \cdot \frac{n_0}{n} = \pi_{[k]} \cdot n_0.$$

$$\mathbb{E}\left(\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}\right) = \frac{\mathbb{E}(n_{[k]1})}{n_1} - \frac{\mathbb{E}(n_{[k]0})}{n_0} = \frac{\pi_{[k]}n_1}{n_1} - \frac{\pi_{[k]}n_0}{n_0} = \pi_{[k]} - \pi_{[k]} = 0.$$

$\mathbb{E}\left(n_{[k]1}/n_1 - n_{[k]0}/n_0\right) = 0$

$n_{[k]1}/n_1 - n_{[k]0}/n_0$

A ; J e Z N SRE

$$t_{\text{SRE}} = \frac{\hat{\tau}_{\text{SRE}}}{\sqrt{\hat{V}_{\text{SRE}}}}$$

Fisher sharp null $H_{0F} : Y_i(1) = Y_i(0)$ t_{SRE}

r ; ; j Q — Q MQ 【 l a — a p l 。

- $\hat{\tau}_{\text{SRE}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]}$
- $\hat{\tau}_{[k]} = \bar{Y}_{[k]}(1) - \bar{Y}_{[k]}(0)$ k
- $\hat{V}_{\text{SRE}} = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}}\right)$

- $\hat{S}^2_{[k]}(1)$ [illegible] $\hat{S}^2_{[k]}(0)$ [illegible] k [illegible]

[illegible] $\hat{\tau}_{\text{SRE}}$ [illegible] $\hat{V}_{\text{SRE}}$ [illegible] H_{0F} [illegible]
[illegible]

[illegible] $\hat{\tau}_{[k]}$ [illegible]

H_{0F} [illegible] $Y_i(1) = Y_i(0) = Y_i$ [illegible]

$$\hat{\tau}_{[k]} = \bar{Y}_{[k]}(1) - \bar{Y}_{[k]}(0) = \frac{\sum_{i:X_i=k} Z_i Y_i}{n_{[k]1}} - \frac{\sum_{i:X_i=k}(1-Z_i)Y_i}{n_{[k]0}}.$$

[illegible] $\mathbb{E}(Z_i) = n_1/n$ [illegible] $\mathbb{E}(1-Z_i) = n_0/n$ [illegible] Y_i [illegible] Z [illegible]

$$\mathbb{E}(\hat{\tau}_{[k]}) = \frac{n_{[k]}}{n_{[k]1}} \cdot \frac{n_1}{n} \cdot \bar{Y}_{[k]} - \frac{n_{[k]}}{n_{[k]0}} \cdot \frac{n_0}{n} \cdot \bar{Y}_{[k]} = \bar{Y}_{[k]} \left(\frac{n_{[k]}}{n_{[k]1}} \cdot \frac{n_1}{n} - \frac{n_{[k]}}{n_{[k]0}} \cdot \frac{n_0}{n} \right).$$

$n_{[k]1}/n_{[k]} = e_{[k]}$ [illegible] $n_{[k]0}/n_{[k]} = 1 - e_{[k]}$ [illegible] n_1/n [illegible] n_0/n [illegible]
[illegible] $\bar{Y}_{[k]} \cdot ($ [illegible] $-$ [illegible] $) = 0$
[illegible] $\mathbb{E}(Z_i) = n_1/n$ [illegible] H_{0F} [illegible]

$$\mathbb{E}(\hat{\tau}_{[k]}) = \frac{1}{n_{[k]1}} \sum_{i:X_i=k} \mathbb{E}(Z_i) Y_i - \frac{1}{n_{[k]0}} \sum_{i:X_i=k} \mathbb{E}(1-Z_i) Y_i = \frac{n_1}{n}\bar{Y}_{[k]} - \frac{n_0}{n}\bar{Y}_{[k]} = 0.$$

[illegible] $\hat{\tau}_{\text{SRE}}$ [illegible]
$\hat{\tau}_{\text{SRE}} = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]}$ [illegible] $\pi_{[k]}$ [illegible]

$$\mathbb{E}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^K \pi_{[k]} \mathbb{E}(\hat{\tau}_{[k]}) = \sum_{k=1}^K \pi_{[k]} \cdot 0 = 0.$$

[illegible] $\hat{\tau}_{[k]}$ [illegible]
[illegible] Neyman (1923) [illegible] k [illegible]

$$\text{var}(\hat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}.$$

H_{0F} [illegible] $S^2_{[k]}(\tau) = 0$ [illegible]

$$\text{var}(\hat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}}.$$

[illegible] $\hat{\tau}_{\text{SRE}}$ [illegible]
[illegible]

$$\text{var}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^K \pi^2_{[k]} \text{var}(\hat{\tau}_{[k]}) = \sum_{k=1}^K \pi^2_{[k]} \left(\frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} \right).$$

[illegible] $\hat{V}_{\text{SRE}}$ [illegible]
[illegible] $\mathbb{E}(\hat{S}^2_{[k]}(1)) = S^2_{[k]}(1)$ [illegible] $\mathbb{E}(\hat{S}^2_{[k]}(0)) = S^2_{[k]}(0)$ [illegible]

$$\mathbb{E}(\hat{V}_{\text{SRE}}) = \sum_{k=1}^K \pi^2_{[k]} \left(\frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} \right) = \text{var}(\hat{\tau}_{\text{SRE}}) + \sum_{k=1}^K \pi^2_{[k]} \frac{S^2_{[k]}(\tau)}{n_{[k]}}.$$

H_{0F} [illegible] $S^2_{[k]}(\tau) = 0$ [illegible] $\mathbb{E}(\hat{V}_{\text{SRE}}) = \text{var}(\hat{\tau}_{\text{SRE}})$

H_{0F} $\hat{\tau}_{\text{SRE}}$ $\hat{V}_{\text{SRE}}$ $t_{\text{SRE}} = \hat{\tau}_{\text{SRE}}/\sqrt{\hat{V}_{\text{SRE}}}$

SRE k

$\hat{\tau}_{[k]}$ $\hat{\tau}_{\text{SRE}}$

- $S^2_{[k]}(1)$ $S^2_{[k]}(0)$ k $Y(1)$ $Y(0)$
- $S^2_{[k]}(\tau)$ k $\tau_i = Y_i(1) - Y_i(0)$
- $\hat{\tau}_{[k]} = \bar{Y}_{[k]}(1) - \bar{Y}_{[k]}(0)$ k
- $\hat{\tau}_{\text{SRE}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]}$

(1) $\text{var}(\hat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}$

(2) $\text{var}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^{K} \pi^2_{[k]} \text{var}(\hat{\tau}_{[k]})$

Neyman (1923) k k CRE $n_{[k]}$ n Neyman

$$\text{var}(\hat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}.$$

CRE 1

1

SRE $\hat{\tau}_{[1]}, \ldots, \hat{\tau}_{[K]}$

$$\text{var}(\hat{\tau}_{\text{SRE}}) = \text{var}\left(\sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]}\right) = \sum_{k=1}^{K} \pi^2_{[k]} \text{var}(\hat{\tau}_{[k]}) + \sum_{k \neq j} \pi_{[k]} \pi_{[j]} \text{cov}(\hat{\tau}_{[k]}, \hat{\tau}_{[j]}).$$

$\text{cov}(\hat{\tau}_{[k]}, \hat{\tau}_{[j]}) = 0$ $k \neq j$

$$\text{var}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^{K} \pi^2_{[k]} \text{var}(\hat{\tau}_{[k]}).$$

$$\text{var}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^{K} \pi^2_{[k]} \left(\frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}\right).$$

[illegible] CRE [illegible] SRE [illegible]

[illegible] CRE [illegible] $\hat{\tau}$ [illegible] SRE [illegible]
1 $\hat{\tau}_{\text{SRE}}$ [illegible]

[illegible]

- $\hat{\tau} = \bar{Y}(1) - \bar{Y}(0)$ [illegible] CRE [illegible]
- $\hat{\tau}_{\text{SRE}} = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]}$ [illegible] SRE [illegible]
- $S^2(1)$ [illegible] $S^2(0)$ [illegible]
- $S^2_{[k]}(1)$ [illegible] $S^2_{[k]}(0)$ [illegible] k [illegible]
- $\bar{Y}(1)$ [illegible] $\bar{Y}(0)$ [illegible]
- $\bar{Y}_{[k]}(1)$ [illegible] $\bar{Y}_{[k]}(0)$ [illegible] k [illegible]

[illegible]

- [illegible] CRE [illegible] $\text{var}_{CRE}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$
- [illegible] SRE [illegible] $\text{var}_{SRE}(\hat{\tau}_{\text{SRE}}) = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}} \right)$
- [illegible] $S^2(1) = \sum_{k=1}^{K} \pi_{[k]} S^2_{[k]}(1) + \sum_{k=1}^{K} \pi_{[k]} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 -$ [illegible]

[illegible]

$$\sum_{k=1}^{K} \frac{\pi_{[k]}}{n} \left\{ \sqrt{\frac{n_0}{n_1}} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\} + \sqrt{\frac{n_1}{n_0}} \{\bar{Y}_{[k]}(0) - \bar{Y}(0)\} \right\}^2 \geq 0.$$

[illegible]

[illegible]

[illegible] $S^2(1)$ [illegible]

$$\begin{aligned} S^2(1) &= (n-1)^{-1} \sum_{i=1}^{n} \{Y_i(1) - \bar{Y}(1)\}^2 \\ &= \sum_{k=1}^{K} \frac{n_{[k]} - 1}{n - 1} \cdot \frac{1}{n_{[k]} - 1} \sum_{i:X_i = k} \{Y_i(1) - \bar{Y}_{[k]}(1)\}^2 \\ &\quad + \sum_{k=1}^{K} \frac{n_{[k]}}{n-1} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2. \end{aligned}$$

[illegible] $n_{[k]}$ [illegible] $(n_{[k]} - 1)/(n - 1) \approx n_{[k]}/n = \pi_{[k]}$ [illegible]

$$S^2(1) \approx \sum_{k=1}^{K} \pi_{[k]} S^2_{[k]}(1) + \sum_{k=1}^{K} \pi_{[k]} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2.$$

[illegible] $S^2(0)$ [illegible] $S^2(\tau)$ [illegible]

[illegible] CRE [illegible]

$$\text{var}_{CRE}(\hat{\tau}) = \frac{1}{n_1} \sum_{k=1}^{K} \pi_{[k]} S^2_{[k]}(1) + \frac{1}{n_0} \sum_{k=1}^{K} \pi_{[k]} S^2_{[k]}(0) - \frac{1}{n} \sum_{k=1}^{K} \pi_{[k]} S^2_{[k]}(\tau) + \text{[illegible]} \quad .$$

[illegible]

[illegible]

$$\frac{1}{n_1}\sum_{k=1}^{K}\pi_{[k]}\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\}^2+\frac{1}{n_0}\sum_{k=1}^{K}\pi_{[k]}\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\}^2-\frac{1}{n}\sum_{k=1}^{K}\pi_{[k]}(\tau_{[k]}-\tau)^2.$$

[illegible]
[illegible] $n_{[k]1}\approx\pi_{[k]}n_1$ [illegible] $n_{[k]0}\approx\pi_{[k]}n_0$ [illegible]

$$\sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left\{\sqrt{\frac{n_0}{n_1}}\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\}+\sqrt{\frac{n_1}{n_0}}\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\}\right\}^2.$$

[illegible] k [illegible]

$$\sqrt{\frac{n_0}{n_1}}\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\}+\sqrt{\frac{n_1}{n_0}}\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\}=0$$

[illegible] SRE [illegible]
CRE [illegible]

[illegible]

[illegible] $\mathbf{n}=\{n_{[k]1},n_{[k]0}\}_{k=1}^{K}$ [illegible] CRE [illegible] **Z** [illegible] SRE [illegible]

[illegible]

- $\mathbf{Z}=(Z_1,\ldots,Z_n)$ [illegible]
- $n_{[k]1}$ [illegible] k [illegible]
- $n_{[k]0}$ [illegible] k [illegible]
- $\mathbf{n}=\{n_{[k]1},n_{[k]0}\}_{k=1}^{K}$ [illegible]

[illegible]

- CRE [illegible] $\mathbb{P}(\mathbf{Z}=\mathbf{z})=\frac{1}{\binom{n}{n_1}}$ [illegible] $\sum Z_i=n_1$ [illegible] $\mathbf{z}$
- SRE [illegible] k [illegible] $\frac{1}{\binom{n_{[k]}}{n_{[k]1}}}$

[illegible]

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z}\mid\mathbf{n})=\frac{1}{\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}}.$$

[illegible]

[illegible]
CRE [illegible] **Z** [illegible]

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z})=\frac{1}{\binom{n}{n_1}}.$$

[illegible]
[illegible] **n** [illegible] $\mathbf{Z}=\mathbf{z}$ [illegible]

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z}\mid\mathbf{n})=\frac{\Pr_{CRE}(\mathbf{Z}=\mathbf{z},\mathbf{n})}{\Pr_{CRE}(\mathbf{n})}.$$

[illegible] $\{\mathbf{Z}=\mathbf{z},\mathbf{n}\}$ [illegible] k [illegible] $n_{[k]1}$ [illegible] [illegible]

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z},\mathbf{n})=\prod_{k=1}^{K}\frac{1}{\binom{n_{[k]}}{n_{[k]1}}}.$$

[illegible]

[illegible] $\Pr_{CRE}(\mathbf{n})$ [illegible] k [illegible] $n_{[k]1}$ [illegible] $\mathbf{z}$ [illegible]

$$\Pr_{CRE}(\mathbf{n})=\sum_{\mathbf{z}:n_{[k]1}(\mathbf{z})=n_{[k]1}}\Pr_{CRE}(\mathbf{Z}=\mathbf{z})=\sum_{\mathbf{z}:n_{[k]1}(\mathbf{z})=n_{[k]1}}\frac{1}{\binom{n}{n_1}}.$$

[illegible] $\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}$ [illegible]

$$\Pr_{CRE}(\mathbf{n})=\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}\cdot\frac{1}{\binom{n}{n_1}}.$$

[illegible]

$$\Pr_{CRE}(\mathbf{Z}=\mathbf{z}\mid\mathbf{n})=\frac{\prod_{k=1}^{K}\frac{1}{\binom{n_{[k]}}{n_{[k]1}}}}{\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}\cdot\frac{1}{\binom{n}{n_1}}}=\frac{1}{\prod_{k=1}^{K}\binom{n_{[k]}}{n_{[k]1}}}.$$

[illegible] CRE [illegible] SRE [illegible]

[illegible]

[illegible] Chapter 4 [illegible] Neymanian [illegible] OLS [illegible]

[illegible]

- $\hat{\tau}=\hat{\bar{Y}}(1)-\hat{\bar{Y}}(0)$ [illegible]
- $\hat{\beta}$ [illegible] OLS [illegible] $Y_i=\alpha+\beta Z_i+\varepsilon_i$ [illegible] treatment [illegible]
- $\hat{V}=\frac{\hat{S}^2(1)}{n_1}+\frac{\hat{S}^2(0)}{n_0}$ [illegible] Neyman [illegible]
- $\hat{V}_{\mathrm{OLS}}$ [illegible] OLS [illegible]
- $\hat{V}_{\mathrm{EHW}}$ [illegible] Eicker-Huber-White [illegible]

[illegible]

(1) [illegible] $\hat{\beta}=\hat{\tau}$

(2) [illegible] $\hat{V}_{\mathrm{OLS}}=\frac{n(n_1-1)}{(n-2)n_1n_0}\hat{S}^2(1)+\frac{n(n_0-1)}{(n-2)n_1n_0}\hat{S}^2(0)$

(3) [illegible] $\hat{V}_{\mathrm{EHW}}=\frac{\hat{S}^2(1)}{n_1}\frac{n_1-1}{n_1}+\frac{\hat{S}^2(0)}{n_0}\frac{n_0-1}{n_0}$

(4) [illegible] HC2 [illegible] $\hat{V}$

[illegible]

[illegible] $\hat{\beta} = \hat{\tau}$ [illegible] OLS [illegible] $Y_i = \alpha + \beta Z_i + \varepsilon_i$ [illegible] OLS [illegible]

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(Z_i - \bar{Z})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(Z_i - \bar{Z})^2}.$$

[illegible]

CRE [illegible] $\bar{Z} = n_1/n$ [illegible]

$$\sum_{i=1}^{n}(Z_i - \bar{Z})(Y_i - \bar{Y}) = \sum_{i=1}^{n} Z_iY_i - \bar{Z}\sum_{i=1}^{n} Y_i - \bar{Y}\sum_{i=1}^{n} Z_i + n\bar{Z}\bar{Y}.$$

$\sum_{i=1}^{n} Z_i = n_1$ [illegible] $\bar{Z} = n_1/n$ [illegible] $\bar{Z}\sum_{i=1}^{n} Z_i = n_1^2/n$ [illegible] $\bar{Y} = n^{-1}\sum_{i=1}^{n} Y_i$ [illegible]

$$\sum_{i=1}^{n} Z_iY_i - \frac{n_1}{n}\sum_{i=1}^{n} Y_i = \sum_{i=1}^{n} Z_iY_i - n_1\bar{Y}.$$

[illegible] $\sum_{i=1}^{n} Z_iY_i$ [illegible]

$$\sum_{i=1}^{n} Z_iY_i = \sum_{Z_i=1} Y_i + \sum_{Z_i=0} Y_i = n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0).$$

[illegible]

$$n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0) - n_1\bar{Y}.$$

[illegible] $\bar{Y} = (n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0))/n$ [illegible]

$$n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0) - \frac{n_1}{n}\left[n_1\hat{\bar{Y}}(1) + n_0\hat{\bar{Y}}(0)\right] = \frac{n_0n_1}{n}\hat{\bar{Y}}(1) - \frac{n_1n_0}{n}\hat{\bar{Y}}(0) = \frac{n_0n_1}{n}\hat{\tau}.$$

[illegible]

$$\sum_{i=1}^{n}(Z_i - \bar{Z})^2 = \sum_{i=1}^{n} Z_i^2 - 2\bar{Z}\sum_{i=1}^{n} Z_i + n\bar{Z}^2 = n_1 - 2\frac{n_1}{n}\cdot n_1 + n\cdot\frac{n_1^2}{n^2} = n_1 - \frac{n_1^2}{n} = \frac{n_0n_1}{n}.$$

[illegible] $\hat{\beta}$

$$\hat{\beta} = \frac{\frac{n_0n_1}{n}\hat{\tau}}{\frac{n_0n_1}{n}} = \hat{\tau}.$$

[illegible] $\hat{\beta} = \hat{\tau}$ [illegible] OLS [illegible]

[illegible] **o f** [illegible] **w** [illegible] OLS [illegible] Appendix B [illegible] (B.4) [illegible]

$$\hat{V}_{\text{OLS}} = \frac{\hat{\sigma}^2}{\sum_{i=1}^{n}(Z_i - \bar{Z})^2},$$

[illegible] $\hat{\sigma}^2 = (n-2)^{-1}\sum_{i=1}^{n}(Y_i - \hat{\alpha} - \hat{\beta}Z_i)^2$ [illegible]

[illegible]

$\sum_{i=1}^{n}(Z_i-\bar{Z})^2 = n_0n_1/n$[illegible]

$$\hat{V}_{\mathrm{OLS}} = \frac{n}{n_0n_1}\cdot\frac{1}{n-2}\sum_{i=1}^{n}(Y_i-\hat{\alpha}-\hat{\beta}Z_i)^2.$$

[illegible] ANOVA [illegible]
[illegible] ANOVA [illegible]

$$(n-1)s^2 = \sum_{Z_i=1}(Y_i-\hat{\bar{Y}}(1))^2 + \sum_{Z_i=0}(Y_i-\hat{\bar{Y}}(0))^2 + \frac{n_1n_0}{n}\hat{\tau}^2,$$

[illegible] $s^2=(n-1)^{-1}\sum_{i=1}^{n}(Y_i-\bar{Y})^2$
[illegible]
OLS [illegible] $\hat{\alpha}=\bar{Y}-\hat{\beta}\bar{Z}$ [illegible] $\hat{\beta}=\hat{\tau}$[illegible]

$$Y_i-\hat{\alpha}-\hat{\beta}Z_i = Y_i-\bar{Y}-\hat{\tau}(Z_i-\bar{Z}).$$

[illegible]

$$\sum_{i=1}^{n}(Y_i-\hat{\alpha}-\hat{\beta}Z_i)^2 = \sum_{i=1}^{n}(Y_i-\bar{Y})^2-\hat{\tau}^2\sum_{i=1}^{n}(Z_i-\bar{Z})^2 = (n-1)s^2-\hat{\tau}^2\cdot\frac{n_0n_1}{n}.$$

[illegible] $\hat{V}_{\mathrm{OLS}}$

$$\hat{V}_{\mathrm{OLS}} = \frac{n}{(n-2)n_0n_1}\left[(n-1)s^2-\frac{n_0n_1}{n}\hat{\tau}^2\right].$$

[illegible] $\hat{S}^2(1)$ [illegible] $\hat{S}^2(0)$ [illegible] $\hat{\tau}=\hat{\bar{Y}}(1)-\hat{\bar{Y}}(0)$[illegible]

$$\hat{V}_{\mathrm{OLS}} = \frac{n(n_1-1)}{(n-2)n_1n_0}\hat{S}^2(1)+\frac{n(n_0-1)}{(n-2)n_1n_0}\hat{S}^2(0).$$

[illegible] ([illegible])[illegible]

([illegible])[illegible] **P**] [illegible] **W**[illegible] EHW [illegible]
[illegible] $Y_i=\alpha+\beta Z_i+\varepsilon_i$[illegible] EHW [illegible] Appendix B [illegible] (B.3)[illegible]

$$\hat{V}_{\mathrm{EHW}} = \frac{1}{n-2}\sum_{i=1}^{n}\frac{(Y_i-\hat{\alpha}-\hat{\beta}Z_i)^2}{(1-h_i)^2},$$

[illegible] $h_i=n^{-1}+(Z_i-\bar{Z})^2/\sum_{j=1}^{n}(Z_j-\bar{Z})^2$ [illegible] leverage[illegible]
[illegible]
$\sum_{i=1}^{n}(Z_i-\bar{Z})^2=n_0n_1/n$[illegible] $Z_i=1$[illegible]

$$h_i=\frac{1}{n}+\frac{(1-n_1/n)^2}{n_0n_1/n}=\frac{1}{n}+\frac{(n_0/n)^2}{n_0n_1/n}=\frac{1}{n}+\frac{n_0}{n_1n}=\frac{n_1+n_0}{n_1n}=\frac{1}{n_1}.$$

[illegible] $Z_i=0$[illegible]

$$h_i=\frac{1}{n}+\frac{(0-n_1/n)^2}{n_0n_1/n}=\frac{1}{n}+\frac{n_1}{n_0n}=\frac{n_0+n_1}{n_0n}=\frac{1}{n_0}.$$

[illegible] $(1-h_i)^2$
[illegible] $(1-h_i)^2=(1-1/n_1)^2=((n_1-1)/n_1)^2$
[illegible] $(1-h_i)^2=(1-1/n_0)^2=((n_0-1)/n_0)^2$
[illegible] EHW [illegible]

$$\hat{V}_{\mathrm{EHW}} = \frac{1}{n-2}\left[\sum_{Z_i=1}\frac{(Y_i-\hat{\alpha}-\hat{\beta})^2}{((n_1-1)/n_1)^2} + \sum_{Z_i=0}\frac{(Y_i-\hat{\alpha})^2}{((n_0-1)/n_0)^2}\right].$$

[illegible] $Y_i - \hat{\alpha} - \hat{\beta} = Y_i - \hat{\bar{Y}}(1)$ [illegible] $Z_i = 1$ [illegible] $Y_i - \hat{\alpha} = Y_i - \hat{\bar{Y}}(0)$ [illegible] $Z_i = 0$ [illegible]

$$\hat{V}_{\mathrm{EHW}} = \frac{n_1^2}{(n-2)(n_1-1)^2}\sum_{Z_i=1}(Y_i-\hat{\bar{Y}}(1))^2 + \frac{n_0^2}{(n-2)(n_0-1)^2}\sum_{Z_i=0}(Y_i-\hat{\bar{Y}}(0))^2.$$

[illegible] $\hat{S}^2(1) = (n_1-1)^{-1}\sum_{Z_i=1}(Y_i-\hat{\bar{Y}}(1))^2$ [illegible] $\hat{S}^2(0) = (n_0-1)^{-1}\sum_{Z_i=0}(Y_i-\hat{\bar{Y}}(0))^2$ [illegible]

$$\hat{V}_{\mathrm{EHW}} = \frac{n_1}{(n-2)(n_1-1)}\hat{S}^2(1) + \frac{n_0}{(n-2)(n_0-1)}\hat{S}^2(0).$$

[illegible] n [illegible] $n-2 \approx n$ [illegible]

$$\hat{V}_{\mathrm{EHW}} \approx \frac{\hat{S}^2(1)}{n_1}\frac{n_1}{n_1-1} + \frac{\hat{S}^2(0)}{n_0}\frac{n_0}{n_0-1} \approx \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0} = \hat{V}.$$

[illegible] n_1, n_0 [illegible] $\hat{V}_{\mathrm{EHW}} \approx \hat{V}$

[illegible] $\hat{V}$ **[illegible]** HC2 [illegible] $(n-2)/(n-p)$ [illegible] $p = 2$ [illegible] $(1-h_i)^{-1}$ [illegible] $(1-h_i)^{-2}$[illegible]

$$\hat{V}_{\mathrm{HC2}} = \frac{1}{n-p}\sum_{i=1}^{n}\frac{(Y_i-\hat{\alpha}-\hat{\beta}Z_i)^2}{1-h_i}.$$

[illegible] $1-h_i$[illegible]

[illegible] $1-h_i = (n_1-1)/n_1$

[illegible] $1-h_i = (n_0-1)/n_0$

[illegible]

$$\hat{V}_{\mathrm{HC2}} = \frac{n_1}{n-2}\cdot\frac{1}{n_1-1}\sum_{Z_i=1}(Y_i-\hat{\bar{Y}}(1))^2 + \frac{n_0}{n-2}\cdot\frac{1}{n_0-1}\sum_{Z_i=0}(Y_i-\hat{\bar{Y}}(0))^2.$$

[illegible] $\hat{S}^2(1)$ [illegible] $\hat{S}^2(0)$ [illegible]

$$\hat{V}_{\mathrm{HC2}} = \frac{n_1}{n-2}\hat{S}^2(1) + \frac{n_0}{n-2}\hat{S}^2(0).$$

[illegible] n [illegible] $n-2 \approx n$ [illegible] $n_1/n \approx n_1/(n_1+n_0)$ [illegible]

$$\hat{V}_{\mathrm{HC2}} = \frac{n}{n-2}\left(\frac{n_1}{n}\hat{S}^2(1) + \frac{n_0}{n}\hat{S}^2(0)\right)\cdot\frac{n}{n_1 n_0}\times\frac{n_1 n_0}{n} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0} = \hat{V}.$$

[illegible] $\hat{V}_{\mathrm{HC2}} = \hat{V}$ [illegible] $n_1/(n-2)\cdot(n_1-1)^{-1} = 1/n_1$ [illegible]

[illegible] HC2 [illegible] Neyman [illegible] $\hat{V}$

q ‖ w MaM I ^ ; s—Q ‖ ˇ

A ; J e Z p l N { { DiD

TWFE DiD

r ; ; j Q—Q MQ 【 l a—a p l 。

- $G_i \in \{0,1\}$ i $G_i = 1$ $G_i = 0$
- $T_t \in \{0,1\}$ $T_t = 0$ $T_t = 1$
- $Y_{it}(d)$ i t $d \in \{0,1\}$
- $D_i = G_i$
- $\tau_t = \mathbb{E}(Y_{it}(1) - Y_{it}(0) \mid G_i = 1)$ t ATT

Y a ” Q l ´ $\bar{Y}_{00}, \bar{Y}_{01}, \bar{Y}_{10}, \bar{Y}_{11}$ { {

Y p g DiD $\hat{\tau}_{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$ τ_1

MQ a ” ; —a p l 、—Q s 。

$$\bar{Y}_{11} - \bar{Y}_{10} = \underbrace{\bar{Y}_{11}(1) - \bar{Y}_{10}(1)}_{} = \underbrace{\tau_1}_{\text{ATT}} + \underbrace{\left[\bar{Y}_{11}(0) - \bar{Y}_{10}(0)\right]}_{} .$$

$$\bar{Y}_{01} - \bar{Y}_{00} = \bar{Y}_{11}(0) - \bar{Y}_{10}(0).$$

$$\begin{aligned} \text{DiD} &= (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) \\ &= \left[\tau_1 + (\bar{Y}_{11}(0) - \bar{Y}_{10}(0))\right] - \left[\bar{Y}_{11}(0) - \bar{Y}_{10}(0)\right] \\ &= \tau_1. \end{aligned}$$

$\bar{Y}_{11}(0) - \bar{Y}_{10}(0) = \bar{Y}_{01} - \bar{Y}_{00}$ DiD : τ_1

U P MaM

$$Y_{it} = \alpha + \beta D_{it} + \gamma_i + \delta_t + \varepsilon_{it}, \quad D_{it} = G_i \cdot T_t.$$

OLS β

$$\hat{\beta} = \bar{Y}_{11} - \bar{Y}_{10} - \bar{Y}_{01} + \bar{Y}_{00} = \hat{\tau}_{\text{DiD}}.$$

γ_i δ_t $\hat{\beta}$

A p p—。 ; s

DiD Bootstrap

(1) cluster i t (Y_{it}, D_{it})
(2) $\hat{\tau}_{\text{DiD}}$
(3) B $B = 999$ Bootstrap $\{\hat{\tau}^{(b)}_{\text{DiD}}\}_{b=1}^{B}$
(4) $\hat{\sigma}_{\text{boot}} = \sqrt{(B-1)^{-1} \sum_{b=1}^{B} (\hat{\tau}^{(b)}_{\text{DiD}} - \bar{\hat{\tau}})^2}$

[illegible]

[illegible] TSLS [illegible]

[illegible]

- Y_i [illegible]
- D_i [illegible] p [illegible]
- Z_i [illegible] q [illegible] $q \geq p$ [illegible]
- β [illegible] p [illegible]
- ε_i [illegible] $E(\varepsilon_i Z_i) = 0$
- $\hat{D}_i$ [illegible] OLS [illegible]
- $\hat{\Gamma}$ [illegible] OLS [illegible]

[illegible] TSLS [illegible]

[illegible]

[illegible] D on Z [illegible] OLS [illegible]

$$\hat{\Gamma} = \left(\sum_{i=1}^{n} Z_i Z_i^\top \right)^{-1} \sum_{i=1}^{n} Z_i D_i^\top .$$

[illegible]

$$\hat{D}_i = \hat{\Gamma}^\top Z_i .$$

[illegible]

$$\sum_{i=1}^{n} \hat{D}_i \check{D}_i^\top = 0,$$

[illegible] $\check{D}_i = D_i - \hat{D}_i$ [illegible]

[illegible]

$$\sum_{i=1}^{n} \hat{D}_i D_i^\top = \sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top .$$

[illegible] Y on $\hat{D}$ [illegible] OLS [illegible]

$$\hat{\beta}_{\text{TSLS}} = \left(\sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_{i=1}^{n} \hat{D}_i Y_i .$$

[illegible] $Y_i = D_i^\top \beta + \varepsilon_i$ [illegible]

$$\begin{aligned}
\hat{\beta}_{\text{TSLS}} &= \left(\sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_{i=1}^{n} \hat{D}_i (D_i^\top \beta + \varepsilon_i) \\
&= \left(\sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_{i=1}^{n} \hat{D}_i D_i^\top \beta + \left(\sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_{i=1}^{n} \hat{D}_i \varepsilon_i .
\end{aligned}$$

[illegible]

$$\hat{\beta}_{\text{TSLS}} = \beta + \left(\sum_{i=1}^{n} \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_{i=1}^{n} \hat{D}_i \varepsilon_i .$$

[illegible] $\hat{D}_i = \hat{\Gamma}^\top Z_i$ [illegible]

$$\hat{\beta}_{\mathrm{TSLS}} = \beta + \left(\hat{\Gamma}^\top \sum_{i=1}^n Z_i Z_i^\top \hat{\Gamma}\right)^{-1} \hat{\Gamma}^\top \sum_{i=1}^n Z_i \varepsilon_i$$

$$= \beta + \left\{\hat{\Gamma}^\top \left(n^{-1}\sum_{i=1}^n Z_i Z_i^\top\right)\hat{\Gamma}\right\}^{-1} \hat{\Gamma}^\top \left(n^{-1}\sum_{i=1}^n Z_i \varepsilon_i\right).$$

[illegible] $E(Z_i\varepsilon_i) = 0$ [illegible] restriction [illegible]

$$n^{-1}\sum_{i=1}^n Z_i\varepsilon_i \xrightarrow{p} 0.$$

[illegible]

$$n^{-1}\sum_{i=1}^n Z_i Z_i^\top \xrightarrow{p} E(ZZ^\top),$$

$$\hat{\Gamma} \xrightarrow{p} \Gamma = E(ZZ^\top)^{-1}E(ZD^\top).$$

[illegible]

$$\hat{\beta}_{\mathrm{TSLS}} \xrightarrow{p} \beta.$$

[illegible] TSLS [illegible] β [illegible]

[illegible] $q = p$ [illegible] $\hat{\Gamma}$ [illegible]

$$\hat{\beta}_{\mathrm{TSLS}} = \left(\hat{\Gamma}^{-1}\sum_{i=1}^n Z_i Y_i\right) = \left(\sum_{i=1}^n Z_i D_i^\top\right)^{-1}\sum_{i=1}^n Z_i Y_i = \hat{\beta}_{\mathrm{IV}}.$$

[illegible] TSLS [illegible] IV [illegible]

[illegible]

[illegible] TSLS [illegible] ILS [illegible]

$$\begin{cases} Y_i = \beta_0 + \beta_1 D_i + \beta_2^\top X_i + \varepsilon_i, \\ D_i = \gamma_0 + \gamma_1 Z_i + \gamma_2^\top X_i + \varepsilon_{2i}. \end{cases}$$

[illegible] $\hat{\beta}_{1,\mathrm{TSLS}} = \hat{\Gamma}_1/\hat{\gamma}_1 = \hat{\beta}_{1,\mathrm{ILS}}$ [illegible]

[illegible] OLS [illegible]

$$\hat{\Gamma}_1 = \frac{\sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\sum_i (Z_i - \bar{Z})^2}, \quad \hat{\gamma}_1 = \frac{\sum_i (Z_i - \bar{Z})(D_i - \bar{D})}{\sum_i (Z_i - \bar{Z})^2}.$$

[illegible]

$$\frac{\hat{\Gamma}_1}{\hat{\gamma}_1} = \frac{\sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\sum_i (Z_i - \bar{Z})(D_i - \bar{D})}.$$

TSLS

$$\hat{D}_i = \hat{\gamma}_0 + \hat{\gamma}_1 Z_i + \hat{\gamma}_2^\top X_i.$$

Y on $(1, \hat{D}, X)$ $\hat{D}$

$$\hat{\beta}_{1,\mathrm{TSLS}} = \frac{\sum_i (Y_i - \bar{Y})(\hat{D}_i - \bar{\hat{D}})}{\sum_i (\hat{D}_i - \bar{\hat{D}})^2}.$$

$\hat{\gamma}_1 \sum_i (Z_i - \bar{Z})(\hat{D}_i - \bar{\hat{D}}) = \hat{\gamma}_1^2 \sum_i (Z_i - \bar{Z})^2$ $\hat{\gamma}_1 \sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})$

$$\hat{\beta}_{1,\mathrm{TSLS}} = \frac{\hat{\gamma}_1 \sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\hat{\gamma}_1^2 \sum_i (Z_i - \bar{Z})^2} = \frac{\sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\hat{\gamma}_1 \sum_i (Z_i - \bar{Z})^2} = \frac{\hat{\Gamma}_1}{\hat{\gamma}_1}.$$

$\hat{\beta}_{1,\mathrm{TSLS}} = \hat{\beta}_{1,\mathrm{ILS}}$ TSLS ILS TSLS ” ” ILS

Neyman (1923) $\mathrm{var}(\hat{\tau} \mid \mathbf{Z})$ ” ”

- n IID $\{Z_i, Y_i(1), Y_i(0), X_i\}_{i=1}^n$
- $Z \perp \{Y(1), Y(0), X\}$
- n_1 n_0

IID $S^2(1)$ $S^2(0)$ $\mathrm{var}\{Y(1)\}$ $\mathrm{var}\{Y(0)\}$

$$S_N^2(1) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}_N(1)\}^2,$$

$\bar{Y}_N(1) = n^{-1} \sum_{i=1}^n Y_i(1)$

$$\sigma^2(1) = E\{Y_i(1) - E(Y_i(1))\}^2.$$

IID $E(S_N^2(1)) = \sigma^2(1)$ IID Neyman Neyman

$$\hat{V}_N = \frac{S_N^2(1)}{n_1} + \frac{S_N^2(0)}{n_0}.$$

$$E(\hat{V}_N) = \frac{E\{S_N^2(1)\}}{n_1} + \frac{E\{S_N^2(0)\}}{n_0} = \frac{\sigma^2(1)}{n_1} + \frac{\sigma^2(0)}{n_0} = \mathrm{var}(\hat{\tau} \mid \mathbf{Z}).$$

♦ ŝ ` Ꝁ ⱯⲶ ŏ $\hat{V}_N$ ꙃ var($\hat{\tau} \mid \mathbf{Z}$) ļ ꞔꙆꙍ§§ħ ꙃ∆

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ѫfi ŝ { {ŏ ɞ Lin (2013) Ꙇꙍ $\hat{\tau}_L = \hat{\alpha}_Z$ ŗ‖ { { $\bar{X} = 0$ ļ
V̈●Ãȷȷ ļ OLS Ẇ ᵽ
Ꝩ Ḩ ŝ ɞꝛ X ω̃ŋđ $\bar{X} = 0$Ō ŏ

$$\hat{\tau}_L = \hat{\gamma}_1 - \hat{\gamma}_0 = \hat{\alpha}_Z.$$

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ᵖ ä ŝ OLS ω̃ţ
Ḧ ᾰΨ‡Ώ { ŏ{ OLS Ꙇꙍ ƅ ω̃ţ ŝ

$$\sum_{i=1}^{n}(Y_i - \hat{\alpha}_0 - \hat{\alpha}_Z Z_i - \hat{\alpha}_X^T X_i - \hat{\alpha}_{ZX}^T Z_i X_i) = 0,$$

$$\sum_{i=1}^{n}(Y_i - \hat{\alpha}_0 - \hat{\alpha}_Z Z_i - \hat{\alpha}_X^T X_i - \hat{\alpha}_{ZX}^T Z_i X_i)Z_i = 0,$$

$$\sum_{i=1}^{n}(Y_i - \hat{\alpha}_0 - \hat{\alpha}_Z Z_i - \hat{\alpha}_X^T X_i - \hat{\alpha}_{ZX}^T Z_i X_i)X_i = 0,$$

$$\sum_{i=1}^{n}(Y_i - \hat{\alpha}_0 - \hat{\alpha}_Z Z_i - \hat{\alpha}_X^T X_i - \hat{\alpha}_{ZX}^T Z_i X_i)Z_i X_i = 0.$$

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ᾰΩ ɞŝ

$$\hat{\alpha}_Z = (\hat{\gamma}_1 - \hat{\gamma}_0) + (\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X}.$$

ᵖä ŝ ω̃
$\bar{X} = 0$ŋ X ω̃Ō ŝ

$$\hat{\alpha}_Z = \hat{\gamma}_1 - \hat{\gamma}_0 = \hat{\tau}_L.$$

♦ ŝķ $\hat{\tau}_L = \hat{\alpha}_Z$ŏđ Lin (2013) Ꙇꙍ ŗ ᾰΨ‡Ώ treatment Ḩļ

q ¯ w mw∼ ţPꙆꙍ ļ 〔

ѫfi ŝ { {ŏ ɞ ʉ OLS ļ EHW Ḩ ă ļ V̈●
√ ŝ
` Ꝁ ⱯⲶ ŏ Ꝁ X ļ V̈´ $\bar{X}$ $_2$Ꝁ §§ ‰ ꝺ ɐ̧
Ḩ OLS ļ EHW ţ꙳Ꙇꙍ ∂ $\bar{X}$ ő̄ ļŋ ╔ X ļ ω̃Ō ᴮ ˇ
ȗ $\bar{X}$ ļꝺ Ꝁ
ɏ ɞ╟ $\hat{\tau}_L$ ŝ

$$\hat{\tau}_L = \hat{\gamma}_1 - \hat{\gamma}_0 = (\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X} + \text{Ö}\mho\quad,$$

‰Ǿ $\hat{\tau}_L$ ļţ꙳iĄ ŝ

$$\mathrm{var}(\hat{\tau}_L) = \underbrace{\mathrm{var}(\hat{\gamma}_1 - \hat{\gamma}_0 \mid X)}_{\text{EHW ˇȗļ}} + \underbrace{\mathrm{var}\{(\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X}\}}_{\text{EHW Ꙇꝕļ}}.$$

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ļ√ ŝ

[illegible]

$$\frac{1}{n}(\hat{\beta}_1 - \hat{\beta}_0)^T S_X^2 (\hat{\beta}_1 - \hat{\beta}_0).$$

[illegible] $\text{var}\{(\hat{\beta}_1 - \hat{\beta}_0)^T \bar{X}\}$ [illegible]

[illegible]

[illegible] Lin (2013) [illegible]

[illegible] $X_i \in \{1, \ldots, K\}$ [illegible] SRE [illegible]

$$Z \perp \{Y(1), Y(0)\} \mid X.$$

[illegible] τ [illegible] $\hat{\tau}_S$ [illegible] k [illegible] $Z \perp \{Y(1), Y(0)\} \mid X = k$ [illegible] $X = k$ [illegible] $Z \perp \{Y(1), Y(0)\}$ [illegible] k [illegible] Lin (2013) [illegible]

$$\hat{\tau}_{L,[k]} = \hat{\gamma}_{1,[k]} - \hat{\gamma}_{0,[k]} + (\hat{\beta}_{1,[k]} - \hat{\beta}_{0,[k]})^T \bar{X}_{[k]},$$

[illegible] $\bar{X}_{[k]} = n_{[k]}^{-1} \sum_{X_i = k} X_i$ [illegible]

$$\hat{\tau}_S = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{L,[k]},$$

[illegible] $\pi_{[k]} = \text{pr}(X = k)$ [illegible]

$$\text{var}(\hat{\tau}_S) = \sum_{k=1}^{K} \pi_{[k]}^2 \text{var}(\hat{\tau}_{L,[k]}).$$

[illegible] EHW [illegible] $\text{var}(\hat{\tau}_{L,[k]})$ [illegible] $\hat{V}_S$ [illegible] SRE [illegible] $\hat{\tau}_S$ [illegible] τ [illegible] $\hat{V}_S$ [illegible]

[illegible]

[illegible] R [illegible]

[illegible]

```
linestimator = function(Z, Y, X) {
  # 标准化协变量
  X = scale(X)

  n = dim(X)[1]
  p = dim(X)[2]
  # 完全交互的 OLS 回归
  linreg = lm(Y ~ Z * X)

  # Lin 估计量 (Z 的系数)
  est = coef(linreg)[2]

  # EHW 标准误
  vehw = hccm(linreg)[2, 2]

  # 超级总体校正项
  inter = coef(linreg)[(p+3):(2*p+2)]
  vsuper = vehw + sum(inter * (cov(X) %*% inter)) / n

  # 返回：估计量、EHW 标准误、校正标准误
  c(est, sqrt(vehw), sqrt(vsuper))
}
```

[illegible] $n = 500$[illegible] 2000 [illegible]

```
set.seed(123)
res = replicate(2000, {
  n = 500
  X = matrix(rnorm(n*2), n, 2)
  Y1 = X[,1] + X[,1]^2 + runif(n, -0.5, 0.5)
  Y0 = X[,2] + X[,2]^2 + runif(n, -1, 1)
  Z = rbinom(n, 1, 0.6)
  Y = Z*Y1 + (1-Z)*Y0
  linestimator(Z, Y, X)
})
```

[illegible]

- [illegible] `mean(res[1,])` ≈ -0.0001
- [illegible] `sd(res[1,])` ≈ 0.151
- [illegible] EHW [illegible] `mean(res[2,])` ≈ 0.139
- [illegible] `mean(res[3,])` ≈ 0.153

[illegible]

[illegible] Theorem 19.1 [illegible] Rosenbaum [illegible] S_i [illegible] p [illegible]

[illegible]

- n[illegible]
- $\Gamma \geq 1$[illegible] Rosenbaum [illegible] odds ratio [illegible]
- $\pi_{i1} = \Pr(Z_{i1} = 1 \mid \text{pair } i)$[illegible] i [illegible]
- $\pi_{i0} = 1 - \pi_{i1}$[illegible] i [illegible]
- $o_i = \frac{\pi_{i1}/\pi_{i0}}{(1-\pi_{i1})/(1-\pi_{i0})} = \frac{\pi_{i1}(1-\pi_{i0})}{\pi_{i0}(1-\pi_{i1})}$[illegible] i [illegible] odds [illegible]
- S_i[illegible] i [illegible] Fisher randomization test [illegible] S_i [illegible]

[illegible]

(1) Rosenbaum [illegible] Assumption 19.1 [illegible] $o_i \le \Gamma$ for all i
(2) Sharp null hypothesis $H_{0F}: Y_{ij}(1) = Y_{ij}(0)$ for all i, j
(3) [illegible] i [illegible] $Z_{i1} + Z_{i0} = 1$

[illegible] $o_i \le \Gamma$ [illegible] S_i [illegible] Bernoulli$\left(\frac{\Gamma}{1+\Gamma}\right)$ [illegible]

[illegible] π_{i1} [illegible] $o_i \le \Gamma$ [illegible]

$$\frac{\pi_{i1}(1-\pi_{i0})}{\pi_{i0}(1-\pi_{i1})} \le \Gamma.$$

$Z_{i1} + Z_{i0} = 1$ [illegible]

$$\pi_{i1} \le \frac{\Gamma}{\Gamma+1} \quad [illegible] \quad \pi_{i1} \ge \frac{1}{\Gamma+1}.$$

[illegible] Rosenbaum (1987b) [illegible] π_{i1} [illegible] Fisher randomization test [illegible] S_i [illegible] Fisher randomization test [illegible] i [illegible] ” [illegible] ” [illegible]

$$T_{i1} = q_i \cdot 1 \quad [illegible] \quad T_{i0} = q_i \cdot 0,$$

[illegible] sharp null [illegible] $T = \sum_{i=1}^n S_i q_i$ [illegible] S_i [illegible] i [illegible] T_{i1} [illegible] T_{i0} [illegible] T [illegible] ” [illegible] ” [illegible] S_i [illegible] S_i [illegible] $q_i > 0$ [illegible] T [illegible] Γ [illegible] S_i [illegible] Bernoulli$\left(\frac{\Gamma}{1+\Gamma}\right)$ [illegible]

$$\mathbb{E}_\Gamma(S_i) = \frac{\Gamma}{1+\Gamma} = 1 - \frac{1}{1+\Gamma}.$$

[illegible] i [illegible] $\pi_{i1} = \frac{\Gamma}{1+\Gamma}$ [illegible]

$$S_i \sim \text{Bernoulli}\left(\frac{\Gamma}{1+\Gamma}\right).$$

[illegible] Rosenbaum 1987b, Theorem 1 [illegible]

[illegible] Γ [illegible] $\frac{\Gamma}{1+\Gamma}$ [illegible] 1 [illegible] $q_i > 0$ [illegible] ” [illegible] ” [illegible]

[illegible] $\Gamma = 1$ [illegible] $\frac{\Gamma}{1+\Gamma} = \frac{1}{2}$ [illegible]

[illegible] Assumption 19.1 [illegible] H_{0F} [illegible] S_i [illegible] Bernoulli$\left(\frac{\Gamma}{1+\Gamma}\right)$

[illegible]

[illegible] $q_i = 1$ for all i [illegible] $T = \sum_{i=1}^n S_i$ [illegible] $B(n, \pi)$ [illegible] $\pi = \frac{\Gamma}{1+\Gamma}$

p ŝ

$$\mathrm{p}_{\Gamma} = 2\sum_{k=T}^{n} \binom{n}{k} \left(\frac{\Gamma}{1+\Gamma}\right)^{k} \left(\frac{1}{1+\Gamma}\right)^{n-k}.$$

a g J p p l a Z l Q N _ ; l — Wilcoxon-signed-rank statistic $q_i = \mathrm{rank}(|Y_{i1} - Y_{i0}|)$ T 5 4

k p

Rosenbaum R `sensitivity2x2`

P _ “ ; g ‘ Q

APPENDIX B

[illegible]

[illegible]

[illegible] (X, U) [illegible] Z [illegible] Y [illegible] U [illegible]

[illegible]

[illegible]

$$f_{1,x} = \Pr(U = 1 \mid Z = 1, X = x), \quad f_{0,x} = \Pr(U = 1 \mid Z = 0, X = x). \tag{141}$$

[illegible]

$$\mathrm{RR}_{ZU|x} = \frac{f_{1,x}}{f_{0,x}} \quad \Longrightarrow \quad f_{0,x} = \frac{f_{1,x}}{\mathrm{RR}_{ZU|x}}. \tag{142}$$

[illegible] $\gamma_x = \mathrm{RR}_{UY|x}$ [illegible]

[illegible]

[illegible]

$$\begin{aligned}
\Pr(Y = 1 \mid Z = 1, X = x) &= \Pr(Y = 1 \mid U = 1, X = x)\Pr(U = 1 \mid Z = 1, X = x) \\
&\quad + \Pr(Y = 1 \mid U = 0, X = x)\Pr(U = 0 \mid Z = 1, X = x) \\
&= \gamma_x f_{1,x} + (1 - f_{1,x}) \cdot 1 \\
&= (\gamma_x - 1) f_{1,x} + 1.
\end{aligned}$$

[illegible]

$$\begin{aligned}
\Pr(Y = 1 \mid Z = 0, X = x) &= \Pr(Y = 1 \mid U = 1, X = x)\Pr(U = 1 \mid Z = 0, X = x) \\
&\quad + \Pr(Y = 1 \mid U = 0, X = x)\Pr(U = 0 \mid Z = 0, X = x) \\
&= \gamma_x f_{0,x} + (1 - f_{0,x}) \\
&= (\gamma_x - 1) f_{0,x} + 1.
\end{aligned}$$

[illegible] $f_{0,x} = f_{1,x}/\mathrm{RR}_{ZU|x}$ [illegible]

$$\Pr(Y = 1 \mid Z = 0, X = x) = (\gamma_x - 1)\frac{f_{1,x}}{\mathrm{RR}_{ZU|x}} + 1.$$

[illegible]

$$\mathrm{RR}^{\mathrm{obs}}_{ZY|x} = \frac{(\gamma_x - 1) f_{1,x} + 1}{(\gamma_x - 1)\frac{f_{1,x}}{\mathrm{RR}_{ZU\ x}} + 1}. \tag{143}$$

X [illegible]

[illegible] $f_{1,x} \in [0,1]$ [illegible]

$$g(t) = \frac{(a-1)t+1}{(a-1)\frac{t}{b}+1}$$

$a > 1, b > 1$ [illegible] t [illegible] $f_{1,x}$ [illegible] $\mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ [illegible] $f_{1,x} = 1$ [illegible]

$$\mathrm{RR}^{\mathrm{obs}}_{ZY|x} \leq \frac{(\gamma_x - 1) + 1}{\frac{\gamma_x - 1}{\mathrm{RR}_{ZU\ x}} + 1} = \frac{\gamma_x \cdot \mathrm{RR}_{ZU|x}}{\gamma_x + \mathrm{RR}_{ZU|x} - 1}.$$

[illegible]

W [illegible]

[illegible] $(f_{1,x}, \mathrm{RR}_{ZU|x}, \mathrm{RR}_{UY|x})$ [illegible]

APPENDIX C

[illegible]

[illegible] $\beta(w_1, w_2) = w_1 w_2/(w_1 + w_2 - 1)$ [illegible] $w_1 > 1, w_2 > 1$

[illegible] $\beta(w_1, w_2) = \beta(w_2, w_1)$

[illegible]

$$\frac{\partial \beta}{\partial w_1} = \frac{w_2(w_1 + w_2 - 1) - w_1 w_2}{(w_1 + w_2 - 1)^2} = \frac{w_2(w_2 - 1)}{(w_1 + w_2 - 1)^2} > 0.$$

[illegible] $\partial\beta/\partial w_2 > 0$ [illegible] β [illegible]

[illegible]

$$\beta(w_1, w_2) = \frac{w_1 w_2}{w_1 + w_2 - 1} \leq \frac{w_1 w_2}{w_1} = w_2 \quad (\qquad w_2 - 1 \geq 0).$$

[illegible] $\beta \leq w_1$

[illegible] $w = \max(w_1, w_2)$ [illegible] $w_1 = w$ [illegible] $w_2 \leq w$ [illegible]

$$\beta(w_1, w_2) = \frac{w \cdot w_2}{w + w_2 - 1} \leq \frac{w \cdot w}{w + w - 1} = \frac{w^2}{2w - 1}.$$

APPENDIX D

[illegible]

[illegible] Y [illegible] Z [illegible] β_1 [illegible]

$$\beta_1 = \log \mathrm{OR}_{ZY|x} = \log \frac{\mathrm{odds}(Y=1 \mid Z=1, X=x)}{\mathrm{odds}(Y=1 \mid Z=0, X=x)}.$$

[illegible] $\Pr(Y=1 \mid Z=1, X=x) \approx 0$ [illegible] $\Pr(Y=1 \mid Z=0, X=x) \approx 0$ [illegible]

$$\mathrm{odds}(Y=1 \mid Z, X) \approx \frac{\Pr(Y=1 \mid Z, X)}{1-\Pr(Y=1 \mid Z, X)} \approx \Pr(Y=1 \mid Z, X).$$

[illegible]

$$\beta_1 \approx \log \mathrm{RR}^{\mathrm{obs}}_{ZY|x}.$$

[illegible] $\exp(\beta_1) \approx \mathrm{RR}^{\mathrm{obs}}_{ZY|x}$ [illegible] E-value [illegible]

[illegible]

APPENDIX E

[illegible]

[illegible]

[illegible] $\varepsilon_1(X)$ [illegible] $\varepsilon_0(X)$ [illegible]

[illegible]

[illegible] $\mu_1(X) = E(Y \mid Z = 1, X)$ [illegible] $\mu_0(X) = E(Y \mid Z = 0, X)$ [illegible]

$$\varepsilon_1(X) = \frac{E\{Y(1) \mid Z = 1, X\}}{E\{Y(1) \mid Z = 0, X\}} = \frac{\mu_1(X)}{E\{Y(1) \mid Z = 0, X\}},$$

[illegible]

$$E\{Y(1) \mid Z = 0, X\} = \frac{\mu_1(X)}{\varepsilon_1(X)}. \tag{144}$$

[illegible]

$$E\{Y(0) \mid Z = 1, X\} = \mu_0(X)\varepsilon_0(X). \tag{145}$$

[illegible]

[illegible] $E\{Y(1) \mid Z = 0\} = E\{\mu_1(X)/\varepsilon_1(X) \mid Z = 0\}$ [illegible]

$$E\{Y(1) \mid Z = 0\} = E\left[E\{Y(1) \mid Z = 0, X\} \mid Z = 0\right].$$

[illegible]

$$E\{Y(1) \mid Z = 0\} = E\left[\frac{\mu_1(X)}{\varepsilon_1(X)} \mid Z = 0\right].$$

[illegible] $E\{Y(0) \mid Z = 1\} = E\{\mu_0(X)\varepsilon_0(X) \mid Z = 1\}$ [illegible]

$$E\{Y(0) \mid Z = 1\} = E\left[E\{Y(0) \mid Z = 1, X\} \mid Z = 1\right].$$

[illegible]

$$E\{Y(0) \mid Z = 1\} = E\left[\mu_0(X)\varepsilon_0(X) \mid Z = 1\right].$$

[illegible] τ [illegible]

$$\tau = E\{Y(1)\} - E\{Y(0)\} = E\{Y \mid Z = 1\} - E\{Y(0) \mid Z = 1\}.$$

[illegible]

$$\tau = E\{Y \mid Z = 1\} - E\left[\mu_0(X)\varepsilon_0(X) \mid Z = 1\right].$$

[illegible] $Y = ZY(1) + (1-Z)Y(0)$ [illegible] $E\{Y \mid Z = 1\}$ [illegible]

$$E\{Y \mid Z = 1\} = E\{Y(1) \mid Z = 1\} = E\left[\frac{\mu_1(X)}{\varepsilon_1(X)} \mid Z = 1\right].$$

B

$$\tau = E\left[\frac{\mu_1(X)}{\varepsilon_1(X)} \mid Z = 1\right] - E\left[\mu_0(X)\varepsilon_0(X) \mid Z = 1\right].$$

[illegible] { [illegible] { [illegible]

APPENDIX F

[illegible]

[illegible]

[illegible] IPW [illegible]

[illegible]

[illegible]

$$E\{Y(1) \mid Z = 0\} = E\left\{\frac{\mu_1(X)}{\varepsilon_1(X)} \mid Z = 0\right\}.$$

[illegible] IPW [illegible]

$$E\{Y(1)\} = E\{Y(1) \mid Z = 1\}\Pr(Z = 1) + E\{Y(1) \mid Z = 0\}\Pr(Z = 0).$$

[illegible] $E\{Y(1) \mid Z = 1\} = \mu_1(X)/\varepsilon_1(X)$ [illegible] $E\{Y(1) \mid Z = 0\} = \mu_1(X)/\varepsilon_1(X)$ [illegible]

$$E\{Y(1)\} = E\left[\frac{\mu_1(X)}{\varepsilon_1(X)}\right].$$

[illegible] $e(X) = \Pr(Z = 1 \mid X)$ [illegible] IPW [illegible]

$$E\{Y(1)\} = E\left\{\frac{ZY}{e(X)}\right\} = E\left\{w_1(X)\frac{ZY}{e(X)}\right\},$$

[illegible] $w_1(X) = e(X) + (1 - e(X))/\varepsilon_1(X)$
[illegible] $E\{Y(0)\}$ [illegible]

APPENDIX G

[illegible]

[illegible]

[illegible] outcomes [illegible] $[\underline{y}, \overline{y}]$ [illegible] $E\{Y(1)\}$ [illegible] $E\{Y(0)\}$ [illegible]

[illegible]

[illegible] $p = \Pr(Z = 1)$ [illegible] $\Pr(Z = 0) = 1 - p$ [illegible] $\mu_1 = E(Y \mid Z = 1)$ [illegible] $\mu_0 = E(Y \mid Z = 0)$ [illegible]

[illegible] $E\{Y(1)\}$ [illegible]

[illegible]

$$E\{Y(1)\} = \mu_1 \cdot p + E\{Y(1) \mid Z = 0\} \cdot (1 - p).$$

[illegible] $Y(1) \in [\underline{y}, \overline{y}]$ [illegible]

$$\underline{y} \leq E\{Y(1) \mid Z = 0\} \leq \overline{y}.$$

[illegible]

$$\mu_1 \cdot p + \underline{y}(1 - p) \leq E\{Y(1)\} \leq \mu_1 \cdot p + \overline{y}(1 - p).$$

[illegible] $E\{Y(0)\}$ [illegible]

[illegible]

$$E\{Y(0)\} = E\{Y(0) \mid Z = 1\} \cdot p + \mu_0 \cdot (1 - p),$$

[illegible]

$$\underline{y} \leq E\{Y(0) \mid Z = 1\} \leq \overline{y}.$$

[illegible]

$$\underline{y} \cdot p + \mu_0(1 - p) \leq E\{Y(0)\} \leq \overline{y} \cdot p + \mu_0(1 - p).$$

[illegible] τ [illegible]

$\tau = E\{Y(1)\} - E\{Y(0)\}$ [illegible]

$$\underline{\tau} = \mu_1 p + \underline{y}(1 - p) - \overline{y}p - \mu_0(1 - p),$$

$$\overline{\tau} = \mu_1 p + \overline{y}(1 - p) - \underline{y}p - \mu_0(1 - p).$$

[illegible] $\overline{y} - \underline{y}$

[illegible]

[illegible] [illegible] **[illegible]** [illegible] $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$ [illegible] [illegible] [illegible]

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] \tag{146}$$

$$= \frac{1}{n}\sum_{i=1}^{n}[Y_i(1) - Y_i(0)] \quad \text{[illegible]} \tag{147}$$

$$= \frac{1}{n}\sum_{i=1}^{n} Y_i(1) - \frac{1}{n}\sum_{i=1}^{n} Y_i(0) \tag{148}$$

$$= \bar{Y}(1) - \bar{Y}(0) \tag{149}$$

[illegible]

[illegible] [illegible] **[illegible]** [illegible] τ_x [illegible] $X_i = x$ [illegible] [illegible] [illegible] [illegible]

$$\tau_x = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x] \tag{150}$$

$$= \frac{\sum_{i=1}^{n} \mathbb{I}(X_i = x)[Y_i(1) - Y_i(0)]}{\sum_{i=1}^{n} \mathbb{I}(X_i = x)} \quad \text{[illegible]} \tag{151}$$

[illegible] $\mathbb{I}(X_i = x)$ [illegible] $X_i = x$ [illegible] 1 [illegible] 0 [illegible]

[illegible] [illegible] **[illegible]** [illegible] [illegible] [illegible]

[illegible]

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\{Y_i(1) - Y_i(0)\} \tag{152}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\big[\mathbb{I}(X_i = 1) + \mathbb{I}(X_i = 0)\big]\{Y_i(1) - Y_i(0)\} \tag{153}$$

(154) [illegible] $\mathbb{I}(X_i = 1) + \mathbb{I}(X_i = 0) = 1$

$$= \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(X_i = 1)\{Y_i(1) - Y_i(0)\}}_{\text{treatment}} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(X_i = 0)\{Y_i(1) - Y_i(0)\}}_{\text{control}} \tag{155}$$

$$= \frac{n_1}{n}\cdot\underbrace{\frac{\sum_{i=1}^{n}\mathbb{I}(X_i = 1)\{Y_i(1) - Y_i(0)\}}{n_1}}_{\tau_1} + \frac{n_0}{n}\cdot\underbrace{\frac{\sum_{i=1}^{n}\mathbb{I}(X_i = 0)\{Y_i(1) - Y_i(0)\}}{n_0}}_{\tau_0} \tag{156}$$

$$= \pi_1\tau_1 + \pi_0\tau_0 \tag{157}$$

[illegible] $n_1 = \sum_{i=1}^{n}\mathbb{I}(X_i = 1)$ [illegible] $n_0 = \sum_{i=1}^{n}\mathbb{I}(X_i = 0)$ [illegible] $\pi_x = n_x/n$ [illegible]

[illegible] [illegible]

[illegible] $\delta_1 \neq \delta_2$ [illegible]

[illegible] $n = 4$ [illegible]

i	$Y_i(1)$	$Y_i(0)$
1	10	2
2	8	3
3	6	4
4	4	5

[illegible]

- $\delta_1 = \text{median}\{Y_i(1)\} - \text{median}\{Y_i(0)\} = 7 - 3.5 = 3.5$
- $\delta_2 = \text{median}\{Y_i(1) - Y_i(0)\} = \text{median}\{8, 5, 2, -1\} = 3.5$

[illegible] $\delta_1 = \delta_2$

[illegible] $n = 5$ [illegible]

i	$Y_i(1)$	$Y_i(0)$
1	100	1
2	90	2
3	50	3
4	10	4
5	5	5

[illegible]

- $\delta_1 = \text{median}\{Y_i(1)\} - \text{median}\{Y_i(0)\} = 50 - 3 = 47$
- $\delta_2 = \text{median}\{Y_i(1) - Y_i(0)\} = \text{median}\{99, 88, 47, 6, 0\} = 47$

[illegible] $\delta_1 > \delta_2$ [illegible]

i	$Y_i(1)$	$Y_i(0)$
1	100	0
2	1	0
3	1	0
4	1	0

[illegible]

- $\delta_1 = 50.5 - 0 = 50.5$
- $\delta_2 = \text{median}\{100, 1, 1, 1\} = 1$

[illegible] $\delta_1 = 50.5 > \delta_2 = 1$ [illegible] δ_1 [illegible] δ_2 [illegible]

[illegible]

[illegible] [illegible] $\hat{\tau}_X$ [illegible]
[illegible] $\text{cov}(\hat{\tau}_X) = \frac{n}{n_1 n_0} S_X^2$
[illegible] …[1]

[illegible] R^2 [illegible] [illegible] R^2 [illegible]
[illegible] { {
[illegible] …

[illegible] [illegible]
[illegible] $\hat{\tau}(\beta_1, \beta_0)$ [illegible]
[illegible] …

[illegible] [illegible] $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$ [illegible] OLS [illegible] Z_i [illegible]
[illegible] OLS [illegible]
[illegible] …

[illegible] [illegible] $\hat{\tau}_{\text{pred}} = \hat{\tau}_{\text{L}}$
[illegible] Lin [illegible]
[illegible] …

[illegible] [illegible] $\hat{\tau}(\beta_1, \beta_0)$ [illegible]
[illegible] $\hat{\tau}(\beta_1, \beta_0) = \hat{\tau} - \gamma^\top \hat{\tau}_X$
[illegible] …

---

[1] [illegible]

R w [illegible]

R w q w [illegible] $\sum_{i=1}^n(a_i - \bar{a})^2 = \sum_{i=1}^n a_i^2 - n\bar{a}^2$ [illegible] $\mathrm{var}(W) = \mathbb{E}(W^2) - (\mathbb{E}W)^2$ [illegible]

$$
\begin{aligned}
n(n-1)\mathbb{E}(\hat{V}) &= \mathbb{E}\left\{\sum_{i=1}^n(\hat{\tau}_i - \hat{\tau})^2\right\} \\
&= \mathbb{E}\left(\sum_{i=1}^n \hat{\tau}_i^2 - n\hat{\tau}^2\right) \\
&= \sum_{i=1}^n \left\{\mathrm{var}(\hat{\tau}_i) + \tau_i^2\right\} - n\left\{\mathrm{var}(\hat{\tau}) + \tau^2\right\} \\
&= \sum_{i=1}^n \mathrm{var}(\hat{\tau}_i) - n\mathrm{var}(\hat{\tau}) + \sum_{i=1}^n \tau_i^2 - n\tau^2 \\
&= n^2\mathrm{var}(\hat{\tau}) - n\mathrm{var}(\hat{\tau}) + \sum_{i=1}^n(\tau_i - \tau)^2.
\end{aligned}
$$

[illegible]

$$
\mathbb{E}(\hat{V}) = \mathrm{var}(\hat{\tau}) + \{n(n-1)\}^{-1}\sum_{i=1}^n(\tau_i - \tau)^2 \geq \mathrm{var}(\hat{\tau}).
$$

□

R w ‖ w [illegible] $(\hat{\tau}_1, \ldots, \hat{\tau}_n)^\top$ [illegible] OLS [illegible]

$$
\hat{\beta}_0 = \bar{\hat{\tau}} = n^{-1}\sum_{i=1}^n \hat{\tau}_i = \hat{\tau},
$$

[illegible] $\hat{\tau}$ [illegible] $\hat{\varepsilon}_i = \hat{\tau}_i - \hat{\tau}$ [illegible]

$$
\hat{\sigma}^2 = (n-1)^{-1}\sum_{i=1}^n \hat{\varepsilon}_i^2 = (n-1)^{-1}\sum_{i=1}^n(\hat{\tau}_i - \hat{\tau})^2 = \hat{V}.
$$

[illegible] $\hat{\tau}$ [illegible] $\hat{V}$ [illegible] OLS [illegible] □

m w [illegible]

m w q̈ w t [illegible] **w** H_{0F} [illegible] $\hat{\tau}_i = S_iD_i$ [illegible] $S_i = 2Z_i - 1 \in \{-1, 1\}$ [illegible] $D_i = Y_{i1} - Y_{i2}$ [illegible] S_i [illegible] 0 [illegible] 1 [illegible]

$$
\mathrm{var}(\hat{\tau}_i) = \mathbb{E}(S_i^2 D_i^2) - \mathbb{E}(S_i)^2\mathbb{E}(D_i)^2 = D_i^2 - 0 = D_i^2 = \hat{\tau}_i^2.
$$

[illegible]

$$
\frac{\hat{\tau} - \mathbb{E}\hat{\tau}}{\sqrt{\mathrm{var}(\hat{\tau})}} = \frac{\hat{\tau}}{\sqrt{n^{-2}\sum_{i=1}^n \hat{\tau}_i^2}} \to \mathrm{N}(0,1)
$$

[illegible] □

[illegible] H_{0F} [illegible] $|\hat{\tau}_i|$ [illegible] R_i [illegible] $U_i = \mathbb{I}(\hat{\tau}_i > 0)$ [illegible] $W = \sum_{i=1}^n U_i R_i$ [illegible] $U_i \overset{\text{IID}}{\sim} \text{Bernoulli}(1/2)$ [illegible] R_i [illegible]

$$\mathbb{E}(W) = \sum_{i=1}^n \mathbb{E}(U_i) R_i = \frac{1}{2}\sum_{i=1}^n i = \frac{n(n+1)}{4},$$

$$\text{var}(W) = \sum_{i=1}^n \text{var}(U_i) R_i^2 = \frac{1}{4}\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{24}.$$

[illegible] □

[illegible] H_{0F} [illegible] $\Delta = \sum_{i=1}^n \mathbb{I}(\hat{\tau}_i > 0) \sim \text{Binomial}(n, 1/2)$ [illegible]

$$\frac{\Delta - n/2}{\sqrt{n/4}} \to \text{N}(0,1)$$

[illegible] □

MPE [illegible] i [illegible] Z_i [illegible] $1 - Z_i$ [illegible]

$$\hat{\tau}_i = Z_i Y_{i1} + (1 - Z_i) Y_{i2} - [(1 - Z_i) Y_{i1} + Z_i Y_{i2}] = (2Z_i - 1)(Y_{i1} - Y_{i2}) = S_i D_i,$$

[illegible] $S_i = 2Z_i - 1$ [illegible] $D_i = Y_{i1} - Y_{i2}$

$Z_i \overset{\text{IID}}{\sim} \text{Bernoulli}(1/2)$ [illegible] $\mathbb{E}(Z_i) = 1/2$ [illegible] $\text{var}(Z_i) = 1/4$ [illegible] $\mathbb{E}(S_i) = 0$ [illegible] $\text{var}(S_i) = 1$ [illegible] S_i [illegible] D_i [illegible] D_i [illegible] S_i [illegible]

$$\mathbb{E}(\hat{\tau}_i) = \mathbb{E}(S_i)\mathbb{E}(D_i) = 0,$$

$$\text{var}(\hat{\tau}_i) = \mathbb{E}(S_i^2)\mathbb{E}(D_i^2) - \mathbb{E}(S_i)^2 \mathbb{E}(D_i)^2 = 1 \cdot \mathbb{E}(D_i^2) = \mathbb{E}(D_i^2).$$

$D_i = Y_{i1} - Y_{i2}$ [illegible]

$$\mathbb{E}(D_i^2) = \mathbb{E}\left[\{Z_i Y_{i1}(1) + (1 - Z_i) Y_{i1}(0) - Z_i Y_{i2}(0) - (1 - Z_i) Y_{i2}(1)\}^2\right].$$

[illegible] $Z_i^2 = Z_i$ [illegible] $(1 - Z_i)^2 = 1 - Z_i$ [illegible] $Z_i(1 - Z_i) = 0$ [illegible] $\mathbb{E}(Z_i) = 1/2$ [illegible] $\mathbb{E}(Z_i^2) = 1/2$ [illegible][2] [illegible] $\text{var}(\hat{\tau}_i) = \mathbb{E}(Y_{i1}(1) - Y_{i2}(1) - Y_{i1}(0) + Y_{i2}(0))^2/4 = \tau_i^2$ [illegible]

[illegible] $\hat{\tau} = n^{-1}\sum_{i=1}^n \hat{\tau}_i$ [illegible]

$$\text{var}(\hat{\tau}) = n^{-2}\sum_{i=1}^n \text{var}(\hat{\tau}_i) + [illegible] .$$

Z_i [illegible] $\hat{\tau}_i$ [illegible]

$$\text{var}(\hat{\tau}) = n^{-2}\sum_{i=1}^n \text{var}(\hat{\tau}_i).$$

□

---

[2] [illegible]

q q [illegible]

[illegible] (i, j) [illegible] X_{ij} [illegible]

$$\hat{\tau}_{X,i} = X_{i1} - X_{i2},$$

[illegible] $\hat{\tau}_X = n^{-1} \sum_{i=1}^{n} \hat{\tau}_{X,i}$ MPE [illegible] Z_i [illegible] X_{ij} [illegible] [illegible]

$$\mathbb{E}(\hat{\tau}_{X,i}) = \mathbb{E}(X_{i1} - X_{i2}) = \mathbb{E}(X_{i1}) - \mathbb{E}(X_{i2}) = 0,$$

[illegible] $\mathbb{E}(\hat{\tau}_X) = 0$ [illegible]

$$\operatorname{cov}(\hat{\tau}_X) = n^{-2} \sum_{i=1}^{n} \operatorname{cov}(\hat{\tau}_{X,i}) = n^{-2} \sum_{i=1}^{n} \hat{\tau}_{X,i} \hat{\tau}_{X,i}^{\top},$$

[illegible] $\hat{\tau}_{X,i}$ [illegible] $\hat{\tau}(\gamma) = \hat{\tau} - \gamma^{\top} \hat{\tau}_X$ [illegible]

$$\operatorname{var}(\hat{\tau}(\gamma)) = \operatorname{var}(\hat{\tau}) + \gamma^{\top} \operatorname{cov}(\hat{\tau}_X) \gamma - 2\gamma^{\top} \operatorname{cov}(\hat{\tau}_X, \hat{\tau}).$$

[illegible]

$$\tilde{\gamma} = \operatorname{cov}(\hat{\tau}_X)^{-1} \operatorname{cov}(\hat{\tau}_X, \hat{\tau}).$$

$\operatorname{cov}(\hat{\tau}_X)$ [illegible] $\operatorname{cov}(\hat{\tau}_X, \hat{\tau})$ [illegible] { { [illegible] $\hat{\theta}$ [illegible] [illegible] $\hat{\gamma}$ [illegible] $\hat{\tau}_{\text{adj}} = \hat{\tau} - \hat{\gamma}^{\top} \hat{\tau}_X$ [illegible] $\hat{V}_{\text{adj}}$ □

q [illegible] p e [illegible] p[illegible]s

1 8 [illegible]

[illegible]	[illegible] $\hat{\tau}_{X,i}$	[illegible] $\hat{\tau}_i$
1	-1	-5
2	2	-1
3	0	3
4	1	2
5	2	5
6	0	-3
7	0	-1
8	2	9

[illegible]

$$\hat{\tau} = \frac{1}{8} \sum_{i=1}^{8} \hat{\tau}_i = \frac{-5 + (-1) + 3 + 2 + 5 + (-3) + (-1) + 9}{8} = \frac{9}{8} = 1.125.$$

[illegible]

$$\hat{V} = \frac{1}{8 \times 7} \sum_{i=1}^{8} (\hat{\tau}_i - \hat{\tau})^2 = \frac{1}{56} \left[(-6.125)^2 + (-2.125)^2 + (1.875)^2 + \cdots \right] \approx 4.98.$$

95% [illegible] $\hat{\tau} \pm 1.96\sqrt{\hat{V}} = 1.125 \pm 1.96 \times 2.23 = [-3.25, 5.50]$

[illegible] $\hat{\gamma}$ [illegible] $\hat{\tau}_{\text{adj}}$ [3]

q～[illegible]

q～w q w { { [illegible] **w** [illegible] ignorability [illegible]

$$\Pr\{Z=1 \mid Y(1),Y(0),e(X)\} = \Pr\{Z=1 \mid e(X)\}.$$

[illegible] tower property[illegible]

$$\Pr\{Z=1 \mid Y(1),Y(0),e(X)\} = E\{Z \mid Y(1),Y(0),e(X)\}.$$

[illegible] tower property[illegible] X [illegible] $Y(1),Y(0),e(X)$ [illegible]

$$E\{Z \mid Y(1),Y(0),e(X)\} = E\big[E\{Z \mid Y(1),Y(0),X\} \mid Y(1),Y(0),e(X)\big].$$

[illegible] ignorability [illegible] $Z \perp \{Y(1),Y(0)\} \mid X$[illegible]

$$E\{Z \mid Y(1),Y(0),X\} = E\{Z \mid X\} = e(X).$$

[illegible]

$$E\{Z \mid Y(1),Y(0),e(X)\} = E\{e(X) \mid Y(1),Y(0),e(X)\} = e(X).$$

[illegible]

$$\Pr\{Z=1 \mid e(X)\} = E\{Z \mid e(X)\} = E\big[E\{Z \mid X\} \mid e(X)\big] = E\{e(X) \mid e(X)\} = e(X).$$

[illegible]

q～w ‖ w k] [illegible] **P、** ～ [illegible] **w** [illegible] R [illegible]

```
# 读取数据
nhanes'bmi = Q ; JN。("nhanes'bmi.csv")[, −1]
z = nhanes'bmiOSchoolmeal
y = nhanes'bmiOBMI
x = ; 。j ; — (nhanes'bmi[, −J(1, 2)])
x = 。 J ;(g)Q # 标准化协变量

# 估计倾向得分
pscore = Z g(jz > x, V ; j a=g》B a l p j) O y g a —.−valQNes

# 倾向得分分层
n.strata = J(5, 10, 20, 50, 80)
strat.res = 。 ; s s(n.g》strata, FUN = V ' l J −(nn) p {
   z.pscore = z ' ; l −(pascgoQe, (1:(nn−1)) ¨ nn)
   s。strata = J '(−pscore, breaks = J(0, z.pscore, 1),
                  g ; B Q=g 1:nn)
  Neyman'SRE(z, y, s。strata)
})

  p〕 l ;(jstQrat.res) = J("est", "se")
J p g l ;(jstQrat.res) = n.strata
```

---

[3] [illegible]

p ‘ (lsNrat.res, 3)

ʕ `Neyman_SRE` ŋ̇ṁ̄ ᴅ { {ŏ Ωᵩ SRE ⅧΣ ḽ Neyman ʽΙΩ ʰˢʕ

Ḫ

q Xὀẅ̄ ɹ ‖ mẅỵڨq

̄ ɹ { {ڨ ŏ

$$E\{Y(z_1,z_2)\} = E[E\{Y(z_1,z_2) \mid X_0\}]$$

ᴮ ۏ̇ʲ▌ $E\{Y(z_1,z_2) \mid X_0\}$ Assumption { (1) ῼ ŏ

$$\begin{aligned}E\{Y(z_1,z_2) \mid X_0\} &= E\{Y(z_1,z_2) \mid z_1, X_0\} \\ &= E[E\{Y(z_1,z_2) \mid z_1, X_1, X_0\} \mid z_1, X_0]\end{aligned}$$

Assumption { (2)ŏ

$$\begin{aligned}E\{Y(z_1,z_2) \mid X_0\} &= E[E\{Y \mid z_2, z_1, X_1, X_0\} \mid z_1, X_0] \\ &= E\{E(Y \mid z_2, z_1, X_1, X_0) \mid z_1, X_0\}\end{aligned}$$

ڨڤ ḏķ { { □

APPENDIX A

[illegible]

[illegible]

[illegible] [illegible]

[illegible] subgroup [illegible] X_i [illegible]

[illegible] $\tau = \pi_1\tau_1 + \pi_0\tau_0$ [illegible] $\tau = \pi_1\tau_1 + \pi_0\tau_0$ [illegible]

[illegible] $\mathbb{I}(X_i = 1) + \mathbb{I}(X_i = 0) = 1$ [illegible]

[illegible] H_{0F} $\mathbf{Y}$ [illegible] Fisher's sharp null ($H_{0F}: Y_i(1) = Y_i(0)$) [illegible] $\mathbf{Y}$ [illegible]

[illegible] H_{0F} [illegible] treatment [illegible] $Y_i(1) = Y_i(0) = Y_i$ [illegible] i [illegible] $\mathbf{Y}(1) = \mathbf{Y}(0) = \mathbf{Y}$ [illegible] FRT [illegible] condition on [illegible] $\mathbf{Y}$ [illegible] $\mathbf{Z}$ [illegible] $M = \binom{n}{n_1}$ [illegible] FRT" [illegible] " [illegible]

[illegible] [illegible] $\tilde{p}_{frt}$ [illegible]

[illegible]

$$\hat{p}_{frt} = \frac{1}{R}\sum_{r=1}^{R} \mathbb{I}\{T(\mathbf{z}^r, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}$$

[illegible] R [illegible] p [illegible]

[illegible]

$$\tilde{p}_{frt} = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\{T(\mathbf{z}^r, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}}{1 + R}$$

[illegible] 1 [illegible] " [illegible] " [illegible]

[illegible] $R = 99$ [illegible] 0 [illegible]

$$\hat{p}_{frt} = \frac{0}{99} = 0$$

[illegible] M [illegible] p [illegible] $1/M$ [illegible] 0

[illegible] 1 [illegible]

$$\tilde{p}_{frt} = \frac{1}{100}$$

[illegible]

(1) p [illegible] 1[illegible]
(2) [illegible] R [illegible] $\mathbb{P}(p \le u) \le u$
(3) [illegible] $R \to \infty$ [illegible] $\tilde{p}_{frt} \to \hat{p}_{frt}$

[illegible] [illegible] FRT [illegible] p [illegible] ” [illegible] ” [illegible]

$$\mathbb{P}(p_{frt} \le u) \le u \quad \text{for all } u \in [0,1]$$

[illegible] finite-sample validity [illegible] p_{frt} [illegible] FRT [illegible]

[illegible] [illegible] Fisher [illegible]

- [illegible] “[illegible]” [illegible]
- **[illegible]** “[illegible]” [illegible]

Fisher [illegible] sharp null [illegible] FRT [illegible] justify [illegible]

[illegible] [illegible] justify [illegible] FRT [illegible] FRT [illegible]

- [illegible] Fisher [illegible] $Y_i \sim \text{i.i.d. } \mathcal{N}(\mu, \sigma^2)$ [illegible]
- [illegible] CLT [illegible]

[illegible] FRT [illegible] ” [illegible] ” [illegible]

- [illegible] ” [illegible] ” [illegible]
- FRT [illegible] ” [illegible] ” [illegible]

[illegible] CRE [illegible] p [illegible] ” [illegible] ” [illegible] FRT [illegible]

[illegible] [illegible] FRT

[illegible]

- [illegible]
- [illegible] CLT [illegible]

[illegible] [illegible] + sharp null [illegible] Monte Carlo [illegible]

- [illegible]
- FRT [illegible] treatment [illegible]

FRT [illegible] CLT [illegible] FRT [illegible] p [illegible]

[illegible] FRT [illegible] FRT [illegible] p

[illegible] Fisher [illegible]

[illegible] Potential Outcomes [illegible] treatment [illegible]

- $Y_i(1)$ [illegible] i [illegible] treatment [illegible]
- $Y_i(0)$ [illegible] i [illegible] control [illegible]

[illegible] treatment [illegible] counterfactual [illegible]

$$Y_i^{obs} = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

[illegible] $Z_i = 1$ [illegible] treatment [illegible] $Y_i(1)$ [illegible] $Z_i = 0$ [illegible] $Y_i(0)$ [illegible] $Y_i(1) - Y_i(0)$ [illegible] i [illegible]

[illegible] Fisher [illegible]

[illegible] treatment [illegible] control [illegible] selection bias [illegible]

[illegible] randomization mechanism [illegible] p [illegible]

(1) [illegible] H_{0F} [illegible] treatment [illegible] $Y_i(1) = Y_i(0)$ [illegible]
(2) [illegible]
(3) [illegible] Monte Carlo [illegible]
(4) [illegible]

[illegible]

Fisher Randomization Test

Fisher

Fisher Randomization Test (FRT) Ronald Fisher 1935 Design of Experiments

(1) $H_{0F}: Y_i(1) = Y_i(0)$

Y

(2) **Z**

(3) CRE **Z** $\binom{n}{n_1}$

(4)

(5) p

FRT condition on **Y** **Z** condition on FRT condition on p FRT finite-sample exactness

$\{\mathbf{z}^1, \ldots, \mathbf{z}^M\}$

CRE **Z** $\{\mathbf{z}^1, \ldots, \mathbf{z}^M\}$

Z $\mathbf{Z} = (Z_1, \ldots, Z_n)$ $Z_i = 1$ i $Z_i = 0$ i

$\{\mathbf{z}^1, \ldots, \mathbf{z}^M\}$ $n = 3, n_1 = 1$

$$\mathbf{z}^1 = (1,0,0), \mathbf{z}^2 = (0,1,0), \mathbf{z}^3 = (0,0,1)$$

$M = \binom{3}{1} = 3$

CRE

$\mathbf{z}^m$

$$\mathbb{P}(\mathbf{Z} = \mathbf{z}^m) = \frac{1}{M} = \frac{1}{\binom{n}{n_1}}$$

$n = 100, n_1 = 50$ $M \approx 10^{29}$ 10^{-29}

CRE treatment control Fisher "randomization provides a valid basis for inference"

Z

Z **Z**

Z Fisher treatment

Z

Z n $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_n)$ n

- $Z_i = 1$ [illegible] i [illegible] treatment [illegible]
- $Z_i = 0$ [illegible] i [illegible] control [illegible]

[illegible] $n = 5, n_1 = 2$ [illegible] $\mathbf{Z} = (1,0,0,1,0)$ [illegible] 1 [illegible] 4 [illegible] 2 [illegible] 3 [illegible] 5 [illegible]

[illegible] CRE [illegible] $\mathbf{Z}$ [illegible] $M = \binom{n}{n_1}$ [illegible] $1/M$ [illegible]

- $n = 3, n_1 = 1$ [illegible] $M = 3$ [illegible] $(1,0,0)$ [illegible] $(0,1,0)$ [illegible] $(0,0,1)$ [illegible] $1/3$
- $n = 100, n_1 = 50$ [illegible] $M \approx 10^{29}$ [illegible] 10^{-29}

$\mathbf{z}^r$ [illegible] Monte Carlo [illegible] M [illegible] R [illegible] $\mathbf{z}^r$ [illegible] $\mathbf{z}^r$ [illegible] $\mathbf{Z}$ [illegible] ⋆ [illegible] $\mathbf{Z}$ [illegible] $\mathbf{Z}$ [illegible] indicator vector [illegible] Monte Carlo [illegible]

[illegible]

[illegible] Potential Outcomes [illegible] Potential Outcomes [illegible] ** [illegible] ** [illegible] i [illegible]

- $Y_i(1)$ [illegible] i [illegible] treatment [illegible]
- $Y_i(0)$ [illegible] i [illegible] control [illegible]

[illegible] ** [illegible] ** [illegible] counterfactual [illegible] Fundamental Problem of Causal Inference [illegible]

[illegible] First Fundamental Form [illegible]

[illegible]

[illegible] far confidence set [illegible]

[illegible]

[illegible] ITT [illegible] Intent-to-Treat [illegible]

[illegible] ITT [illegible] Intent-to-Treat [illegible]

[illegible] Instrumental Variable [illegible]

[illegible] Z [illegible] 1 [illegible] treatment [illegible] 2 [illegible] 3 [illegible] treatment [illegible] variation [illegible] variation [illegible] treatment [illegible]

[illegible] Monotonicity [illegible]

[illegible] $Y_i(1) \geq Y_i(0)$ [illegible]

Randomization Assumption

Z $(Y_i(0), Y_i(1))$ $(D_i(0), D_i(1))$

$Z \perp (Y_i(0), Y_i(1), D_i(0), D_i(1))$

Weak IV Problem

treatment F

F

Staiger-Stock 10

LATE ATE

Compliers” variation

Compliers $D_i(1) > D_i(0)$ Defiers Always-takers

Compliers ATE LATE

m Y

$m(X) = \mathbb{E}[Y|X]$

m Logit

m

Robins mediation formula latent groups

Y

cross-world independence

mediation analysis

Surrogate Sr Jr

当你在阅读过程中提出问题时，回答将被记录在此处。

在这里记录你的学习心得和思考。

A a B g a p Z ; s ^⟫

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